THE SAUDI ARABIAN MONETARY AGENCY, 1952–2016

Central Bank of Oil

Ahmed Banafe
Rory Macleod

FINANCIAL INSTITUTIONS, REFORMS, AND POLICIES IN MUSLIM COUNTRIES

Financial Institutions, Reforms, and Policies in Muslim Countries

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Although strong evidence points to the existence of a relationship between economic development and growth and a well-developed financial system that promotes efficient financial intermediation through a reduction in information, transaction and monitoring costs, this linkage and the direction of causation is not as simple and straightforward as it may at first appear. The form of financial intermediation, the level of economic development, macroeconomic policies, and the regulatory and legal framework are some of the factors that can complicate the design of an efficient financial system. Most, if not all, Middle Eastern countries could benefit from financial reforms that would enhance the effectiveness of their financial system while making them less susceptible to crises. While most Middle Eastern countries have adopted the conventional interest-based financial system, a few, most prominently a non-Middle Eastern country Malaysia, are looking into financial reforms that would embrace and introduce elements of Islamic finance. The Palgrave Macmillan series on Financial Institutions, Reforms, and Policies in Muslim Countries breaks new ground by proffering studies that broadly address financial reforms and financial policies, whether in the context of conventional or Islamic finance in the broader Middle East, instead of the general pattern of focusing on financial or economic development in various Middle Eastern countries. As such, the series will also serve as a guide for the adoption of fundamental financial reforms in the Middle East region.

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This book is dedicated to the staff and advisors – past, present and to come – of SAMA

Preface

This book has been 30 years in the making. It dates back to when we worked in SAMA's Investment Department and Rory Macleod prepared his own history of SAMA, which forms the basis for the first part. In 1993 Ahmed Banafe published his book on the Saudi financial system. The decades of working together since have deepened our understanding of the Saudi economy, the financial system and the central bank's role within them. The Department's internal language and archives are mainly in English so they lend themselves to this enterprise. We do not believe we are betraying any confidences.

It is our belief that almost all writing about Saudi Arabia suffers from a lack of knowledge about how the government system actually works. In particular, there are no administrative histories. Contrast this, for example, with the series of volumes published in London after World War I, which explained how the British government had coped with the challenges of the conflict – in war finance, food rationing and so on. These departmental histories proved invaluable reference documents during World War II as they showed what had been effective last time round. This tradition has been continued in Britain and the United States up to the present day. In the same way we hope that current and future officials at SAMA will benefit from our account of past events and how the central bank coped with them.

We also hope that we can demonstrate the administrative competence of that part of the Saudi government which we understand from the inside. We have done this by describing the policy challenges facing the central bank and how they are rationally analyzed and dealt with. Facts are the best disinfectant for conspiracy theories, so we also trust that we can lay to rest the species of analysis that peddles various myths about the Saudi government, as well as the type of academic writing that imposes the author's favorite theoretical framework on Saudi policy-making. In other words, we want to spread the idea that – in financial matters at least – the kingdom is a country just like any other, though to be sure with its own particular distinctiveness. Saudi policy-makers know that oil is a curse as well as a blessing, that it brings great vulnerabilities as well as great strengths, and they have handled this dilemma with skill.

The introductory chapter sets out the four paradoxes of Saudi finances; the first part of the book recounts where SAMA came from by telling the story of its founding and first sixty years; and the second part discusses current questions. Chapter 8 analyzes the likelihood of a Gulf Monetary Union. Managing the foreign reserves is the subject of Chapter 9, while Chapter 10 looks at how the domestic bond markets might develop. Monetary policy and the future of the currency peg are taken up in Chapter 11, while Chapter 12 examines Saudi Arabia's role in the multipolar monetary system that may be emerging. Chapter 13 reviews the Saudi banking system, while the final chapter considers what challenges lie ahead for SAMA and for the economy of the kingdom.

Just as we were finishing this book, SAMA changed its name to the Saudi Arabian Monetary Authority, and the government launched its most determined attempt yet to end the country's dependence on oil. Both events mark the end of the period which our work covers.

Riyadh and Oxford, May 2017

Ahmed Banafe
Rory Macleod

Acknowledgments

We are grateful to the following for their help in Riyadh, Oxford and New York: SAMA's Research & International Affairs Department for careful reading and corrections; Scott Urban for the initial draft of the chapters on Gulf Monetary Union, the currency regime and the international monetary system; Jeremy Smith for all the illustrations in the book; Gerry and Tabitha Butt for acting as editors and agents; Steffen Hertog and Andrew Cunningham for expert reading of an early draft; the team at Oxford Analytica for intellectual debate; Ricardo de Soares Oliveira for insights from other oil economies; and Rachel Crawford, Sarah Lawrence and Allison Neuburger, our editors at Palgrave Macmillan. Finally, we would like to thank family, friends and colleagues whose help, support and encouragement have enabled us to complete this book – especially our wives who have put up with all of this for a long time.

We acknowledge the support SAMA management and ex-Governors have given us, but the views and analysis in this book are our own and do not represent those of SAMA.

Contents

List of Abbreviations

ADIA	Abu Dhabi Investment Authority
AHAB	Ahmed Hamad Al-Gosaibi & Brothers
AMF	Arab Monetary Fund
APR	Annual Percentage Rate
Aramco	Arabian American Oil Company (also Saudi Aramco)
ASEAN	Association of Southeast Asian Nations
BBC	Basket Band and Crawl
BIS	Bank for International Settlements
BP	British Petroleum
BSDA	Bankers' Security Deposit Account
CAR	Capital Adequacy Ratio
Casoc	California Arabian Standard Oil Company
CDO	Collateralized Debt Obligation
CEDA	Council of Economic and Development Affairs
CMA	Capital Market Authority
DMO	Debt Management Office
D-SIB	Domestic Systemically Important Bank
EC	European Commission
ECB	European Central Bank
EDF	Economic Development Fund
EME	Emerging Market Economy
EMI	European Monetary Institute
ESB	End of Service Benefits
EU	European Union
FDI	Foreign Direct Investment
FRED	Federal Reserve Economic Database
FRN	Floating Rate Note
GCC	Gulf Cooperation Council

GDB	Government Development Bond
GDP	Gross Domestic Product
GIPS	Global Investment Performance Standards
GMC	Gulf Monetary Council
GMU	Gulf Monetary Union
GOSI	General Organization for Social Insurance
GPFG	Government Pension Fund (General) of Norway
ICM	Investment Committee Meeting
IDB	Islamic Development Bank
IEA	International Energy Agency
IFI	International Financial Institution
IFS	International Financial Statistics
IFSB	Islamic Financial Services Board
IMF	International Monetary Fund
IOB	Institute of Banking
IOF	Institute of Finance
IP	Investment Portfolio
IPO	Initial Public Offering
LCR	Liquidity Coverage Ratio
LDR	Loan to Deposit Ratio
LIBOR	London Interbank Offered Rate
LOLR	Lender of Last Resort
LTV	Loan to Value ratio
MAS	Monetary Authority of Singapore
NCB	National Commercial Bank
NCCI	National Company for Cooperative Insurance
NEER	Nominal Effective Exchange Rate
NPL	Non Performing Loan
OBU	Offshore Banking Unit
OPEC	Organization of the Petroleum Exporting Countries
PIF	Public Investment Fund
PPA	Public Pension Agency
PSIA	Profit Sharing Investment Account
REDF	Real Estate Development Fund
REER	Real Effective Exchange Rate
RMB	Renminbi
RP	Reserves Portfolio
SAA	Strategic Asset Allocation
SABIC	Saudi Arabian Basic Industries Corporation
SAGIA	Saudi Arabian General Investment Authority
SAIBOR	Saudi Arabia Interbank Offered Rate
SAMA	Saudi Arabian Monetary Agency/Authority

SAMBA	Saudi American Bank
S&P	Standard and Poor's
SAR	Saudi Arabian Riyal
SARIE	Saudi Arabian Riyal Interbank Express
SCI	Specialized Credit Institutions
SDF	Saudi Fund for Development
SDR	Special Drawing Right
SME	Small and Medium-sized Enterprises
SIDF	Saudi Industrial Development Fund
SIMAH	Saudi Credit Bureau
Socal	Standard Oil Company of California
SPV	Special Purpose Vehicle
SWF	Sovereign Wealth Fund
TAA	Tactical Asset Allocation
Tadawul	Saudi Stock Exchange
TASI	Tadawul All-Share Index
TIBC	The International Banking Corporation
TIPS	Treasury Inflation-Protected Security
UAE	United Arab Emirates
WB	White Weld and Baring Brothers advisory team

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CHAPTER 1

Background to Saudi Arabia's Financial Challenges

1 Oil

Try to imagine a world that has run out of oil. This is what happened in the 1979 film *Mad Max* made during the second oil crisis. Motorcycle gangs roar through the Australian outback fighting each other for the remaining stocks of fuel and terrorizing the local population. But we need not follow Mel Gibson into this dystopian future to realize how important oil is to us. The automobiles we use to get around in, the trucks that deliver the food we need, the ships that move our goods across the oceans, the planes that crisscross the skies and the power stations that help provide our electricity – without this compact, easily transported form of energy all of them would come to a silent halt. Oil is the most important stuff in the modern world – arguably, it *is* the modern world. Its use is ubiquitous. The plastics made from it are present around and within us, from our mobile phones, to our clothing, to our body implants. Fertilizers made from oil help feed the planet. A world without oil is hard to imagine – it would certainly be a lot harder to live in.

The kingdom of Saudi Arabia is the most important supplier of this precious commodity, with a reputation as a stabilizing force that makes a significant contribution to the maintenance of world order. Yet Saudi Arabia sits in one of the most volatile regions of the world and did not even exist as a nation state until 1932. By any standards the story of the

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emergence of Saudi Arabia into one of the powerhouses of the modern world would be a remarkable one. What made the Saudi kingdom quite extraordinary and of particularly crucial importance was its historical role of the world's oil 'swing producer' with sufficient spare capacity to enable it to step in and prevent any disruption of supply elsewhere from disturbing the world economy. When other countries were unable to supply oil – such as Iran in 1979, Kuwait in 1990, Iraq in the 1990s and Libya after 2011 – until recently it was Saudi Arabia that filled the gap. Likewise, when too much oil was available and the price crashed, the kingdom was ready to limit production. For many years, it stated that its role was to stabilize the world oil price, in the same way that a central bank stabilizes the price level of an economy.

But since 2014 the kingdom's oil policy has changed. Under the impact of American shale oil, it decided to let the price be determined by supply and demand in the market while maintaining its output. The plan in 2014, as prices plummeted, was that a period of cheap oil would discourage shale producers so that a balance of supply and demand could be established at a higher price. At the end of 2016, there were signs that this strategy would have to change to include output cuts.

The 'central bank of oil'[1] is the Saudi Arabian Monetary Authority, renamed in 2016 (SAMA), headquartered for the past 30 years on King Saud Street in the capital city, Riyadh. But just as the kingdom rarely received the credit for its stabilizing role in the oil market, so there is little understanding of how crucially important SAMA is. SAMA has three main jobs: looking after the foreign reserves, managing monetary and currency policy, and supervising the banks and much of the rest of the financial sector. Founded by Ibn Saud in 1952, it is the most successful central bank of any oil exporter – and one of the most successful of any developing country – largely because it understands the particularity of an economy in which virtually all government spending is made possible only by the money earned through oil exports.

2 Paradoxes of an Oil Economy

SAMA's expertise in managing and working within the Saudi economy has been gained over many years. At the core of the Saudi economy lie four financial paradoxes explained below. SAMA plays a key role in managing all of them.

Diversification versus dependency

This paradox underpins the others. Without oil the economy would suffer severely, yet no matter how much Saudi Arabia diversifies it has not substantially reduced its dependence on oil. Up until 1980, the kingdom's oil was viewed as a temporary source of wealth that would be used to fund a diversified modern economy. But things have not turned out the way the theorists expected.

This is not due to a lack of money. Since the first development plan in 1970, the government has spent well over a trillion dollars on modernizing and updating all aspects of the economy and society – infrastructure, education, industrial projects and so forth – with the purpose of diversifying away from oil. But success has been limited. The inescapable reality is that the kingdom has few natural competitive advantages. Saudi Arabia is a hot, barren country with no running water or lakes and almost no resources apart from oil and other minerals, so that most goods and services are either imported or, if produced locally, made by the foreign residents who send most of their earnings home. Education has not helped much to fit Saudis for the jobs in the economy and productivity is low. The majority of Saudi men work in the public sector, while most women work in the home.

The national account statistics present a puzzling picture. They seem to say that, on the contrary, diversification has been a success, and that the oil sector has become less important. But this is misleading, for there is a difference between diversifying the local economy so that the oil sector's contribution to Gross Domestic Product (GDP) becomes smaller and reducing dependence on oil. The kingdom has certainly achieved the first aim. The non-oil economy has grown, as has the scale and diversity of goods and services provided. But the country remains dependent on the flow of income from oil that comes in through public spending – the essential importance of oil has in no way diminished. In 2016, the Saudi government launched Vision 2030, a strategy aimed at breaking the kingdom's dependence on oil revenue by selling off part of the state energy company, Aramco, and building a sovereign wealth fund to create revenue from global investments. Whether it will be successful remains to be seen.

Up to now, despite its best efforts, the Saudi government has had limited success in finding other sources of revenue. Diversifying government revenues by increasing the share of non-oil revenues has been a constant theme of its plans over nearly half a century. But local taxes and charges come nowhere near providing enough revenue for the budget. The simple truth is that most government spending relies on oil money. This was true from the earliest days of oil in the kingdom.

The obvious question is this: how can the non-oil sector have become more important if the government's outlays are fueled by oil revenues, and oil is by far the most important export? The answer lies in the effect of public spending, and the national account statistics are misleading in this respect also. The best way to visualize this is as a version of the Keynesian multiplier. We understand how budget deficits can keep an economy from falling into depression. The story for Saudi Arabia is directly analogous. The government's budget injects demand into the economy. It is similar to a budget deficit in the way it operates because there is almost no local taxation. The spending is paid for by oil money. That demand push is spent on a variety of goods and services and each round of spending results in more GDP being generated (as outlined in the multiplier). Quite quickly demand leaks away into savings and imports of goods or into remittances by expatriate workers (the equivalent to imports of services). As the economy becomes more diversified internally, so the number of rounds of spending increases, thus elevating the size of the non-oil sector.

But if oil money stopped coming in – if, say, there was a breakthrough in the cost of renewables so the oil price collapsed and stayed down – the government would have nothing to spend unless it raised money from local taxes, something that would decimate the economy. Meanwhile, there would be no exports to pay for import needs. To put numbers on the size of the problem, in 2015 oil revenues were $119 billion and GDP was $646 billion, meaning that without oil the government would have needed to find the equivalent of almost a fifth of GDP from a combination of local taxation and borrowing.[2] In practice, the percentage would have been far higher since GDP would itself have slumped. The Saudi economy still depends on oil.

The 90 percent solution

The second paradox is: government spending financed by oil is matched by an equivalent loss of foreign exchange reserves. This conclusion is a sobering one because it means that – unless something changes radically – the kingdom will continue to depend on the deployment of oil revenues that rapidly leak away into imports of goods and services. This is where

SAMA comes into the picture because imports are paid for in the last analysis by the official foreign exchange reserves held at the central bank. As early as the 1970s, SAMA realized that there was a very close relationship between government spending that was not financed by local taxes and outflows of foreign exchange. The team of US and UK advisors working at SAMA called it 'the 90 percent solution.' It was publicly identified by Banafe in 1993.[3] Over a long period of time, the statistical fit is good.

Looking at the long run of decades allows one to understand the most important fact of Saudi government finances. Some government spending is made directly in dollars and that, by definition, is matched by an equal loss of dollars from the reserves. A comparatively small part of local spending is financed by local revenues. The rest consists of a pure injection of purchasing power into the economy. As Fig. 1.1 shows, over the past 30+ years[4] a comparison of public spending in riyals (that part that is not financed by local revenues) and the balance of payments deficit from the private sector in the same year produces a 'best fit/least squares' statistic of 0.893. That is to say, the two series are closely correlated. The line has a slope of 0.894 meaning that on average nearly 90 percent of public spending in the country that is financed by oil revenues flows out as a deficit on the balance of payments in the same year, which is financed by a loss of foreign reserves from SAMA. The relationship varies from year to year. Taking the most recent three years, it was only 76 percent in 2013, meaning not as much government spending financed from oil flowed out of the economy as usual, then 106 percent in 2014 and there was a catch-up in 2015 when the relationship was 121 percent, meaning more money flowed out than the government was spending from oil income. But over the 2013–2015 period as a whole the difference between the two numbers is only $5 billion. It is remarkable and chilling that the 90 percent solution, identified so long ago, continues to work.

The connection between spending and the level of reserves is simply that when the Finance Ministry runs a budget surplus, the oil income flowing into the SAMA reserves as dollars exceeds the outflow from the economy resulting from public spending and the foreign reserves go up. When there is a budget deficit, then outflows are greater than imports of dollars, and the reserves go down. So, the critical variable to watch in the Saudi economy is the amount of foreign reserves at SAMA. Government sales of domestic debt after 1988 to cover the budget deficits were partly an accounting exercise[5] – as is the case in the latest round of issuing debt. The key factor

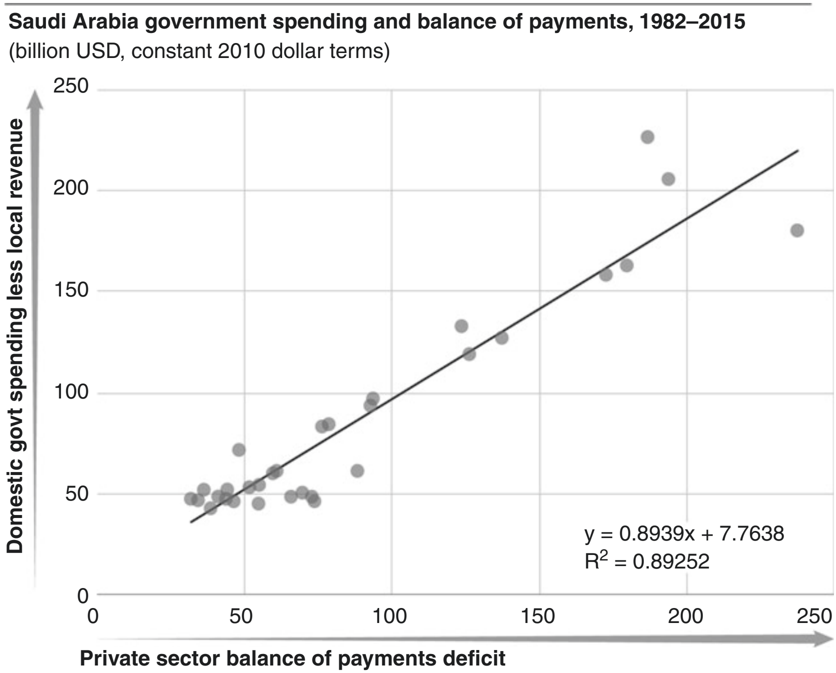

Fig. 1.1 Saudi Arabia government spending and balance of payments, 1982–2015

Source: SAMA Annual Reports

determining how much money the government can spend – and therefore keep the economy going – is the level of foreign reserves.

The roulette wheel of oil

The first two paradoxes would be easier to manage if oil income was reliable. The third paradox is that the oil revenue on which the government depends for stability is unpredictable. The 90 percent solution was tough for Saudi policy-makers to accept. Even harder was the fact that the future oil price is completely unknown. In technical terms, it is a random walk and exhibits no duration dependency: no matter how long oil prices have been in a slump or a boom, the likelihood that they will reverse direction remains constant. It is just like the odds in a game of roulette. The fact that the ball has just landed in a red slot 2, 5 or 20 times in a row does not alter the chance of it landing in a black slot next time.[6] That probability remains constant. The future does not depend on the past. It is unpredictable. As we will see in this book, oil prices can stay low for 20 years as they did in the 1980s and 1990s; or double, halve, then double again in four years, as happened between 2007 and 2011. SAMA has seen it all. It means that forward budget planning is very difficult and the consequence is that fiscal spending has to be highly flexible. The annual budget is largely indicative and the actual amounts have to be adjusted depending on income.

The bottom line for the Ministry of Finance is the size of SAMA's foreign reserves and whether they can cover the budget deficits. This was tested almost to destruction in the late 1990s during the Asian crisis, the most dangerous time for Saudi finances in 40 years. The drop in the oil price meant reserves fell to a level where it seemed that the country was close to running out of the money to pay for imports.

Shariah-compliant versus conventional banking

The fourth paradox of Saudi finances is a comforting one. The population's religious beliefs mean that Western-style banks operating in the kingdom are actually safer and more profitable than elsewhere. It is a core Islamic belief that paying or receiving interest on money is wrong. As a result, the banks have always had access to a pool of deposits on which they have to pay nothing.

The ruling Shariah code is based on the Holy Quran and interpreted by judges who are not bound by precedent. It is essentially a set of prohibitions that have to be followed rather than a clear path of correctness, which is why it is more accurate to talk about 'Shariah-compliant' rather than 'Islamic' banking. The Quran states that God prohibits usury known as *riba*.[7] This prohibition is not unique to Saudi Arabia – many US states have anti-usury laws, but the main difference is that in America usury is usually defined as charging more than a reasonable interest rate. For instance, the Californian state constitution prohibits personal loans that charge 10 percent or more. But the Quranic statement on *riba* can be interpreted to mean that *any* payment of interest is not allowed.

In practice, the question of interest is a gray area. For example, while SAMA's 1957 Charter states it should not pay or receive interest as an agent for the government, the central bank receives interest on the government's foreign exchange reserves. Again, it has to be made clear by SAMA that the rates paid on Saudi government debt are linked to the rates of return on unspecified government development projects – hence the name, 'Government Development Bonds.' This avoids the charge that the government is paying interest.

The paradox is most marked in commercial banking. SAMA's supervision of banks is based on a Western model, but any business that involves interest can be regarded as *riba* under Islamic law. Mortgages are one example. The banks historically found it difficult to seize the homes of defaulting borrowers because judges would not support them and notaries would not register mortgages. Yet a large number of Saudis have unsecured personal loans on which they pay interest. Many of the same people who borrow money refuse to receive interest, so about a third of bank deposits pay no interest, which keeps the banks profitable. Meanwhile, transactions under Shariah law that are theoretically based on sharing risk often have the same economic effect as loans that pay interest.

3 Managing the Foreign Reserves

Given the paradoxes, it is urgently necessary to find some other way to maintain foreign reserves and provide government revenue. Local taxes can meet a small part of the need, but currently the only other major source is the return on foreign investments. SAMA needs to earn a predictable rate of return on its reserve assets. But this has become more difficult.

Expensive oil drove the world economy into recession in 1973–1974 and 1979–1980. This meant that SAMA's cash flow from oil revenues grew just at the time when asset prices fell, so the central bank could buy up assets at cheap prices. This was understood at SAMA in 1980 and it drove the investment of the profits from the second oil shock. The opposite situation applied when the oil price was low. Then the cost of energy was also low and this fueled faster growth – especially in the developing economies – so that asset prices rose. SAMA's cash flow was negative at these times so it was a potential seller of assets at the time when the prices were high. There was, therefore, a natural equilibrating mix in having both oil revenue and financial assets. But since the global financial crisis the relationship between cheap oil and faster growth has not operated clearly: the oil price drop after 2013 has not led to a buoyant global economy.

Yet even under the old paradigm, the kingdom found it hard to set aside a big enough reserve fund to use those cycles and serve future generations.

This is because of rapid population growth in the past, which drove government spending on an ever-higher trajectory. Forty years ago the local Saudi population was less than 7 million: now it is over 20 million. Saudi Arabia has a high population growth rate and depends heavily on migrant labor. If many of the ten million foreign workers could be replaced by Saudis the situation would be a lot better. The large local population also uses vast volumes of oil that could be sold abroad. This demographic situation is probably the country's biggest challenge.

4 Monetary and Banking Policy

In addition to guarding foreign reserves, SAMA has two other tasks – monetary policy and bank supervision. Because the private sector depends on public spending the whole economy is driven by fiscal policy, which is highly stimulatory. The role of monetary policy is to provide the backdrop through the currency peg. It is worth explaining the way oil money is treated in government accounts and how monetary and fiscal policy work together. The relationship is as follows: oil income from Aramco is deposited with SAMA. In return, the government gets a credit entry in riyals in its account with the central bank, and the dollars go into the foreign exchange reserves. The credit in the local currency, the Saudi riyal, is treated as oil income in the annual budget. As the government spends money, SAMA debits the government's account to pay the bills. The private sector accumulates the riyals that the government is spending and since goods and services are imported – directly or indirectly – there is a demand on the banks to provide dollars to pay for imports or foreign workers' remittances home. The banks then come to SAMA to buy the dollars, which are made available at a fixed exchange rate of 3.75 riyals to the dollar (Fig. 1.2).

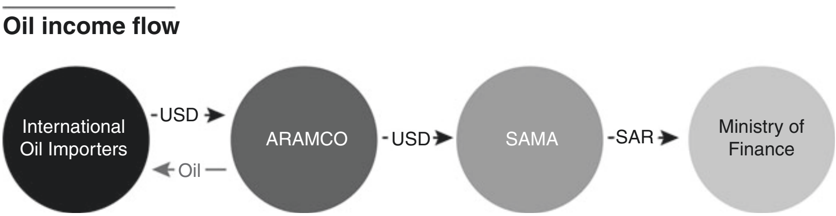

Fig. 1.2 Oil income flow

Source: Authors

The 3.75 exchange rate peg has been in place for a generation and was being seriously debated as reserves fell in 2016. The argument for the peg was that since oil is paid for in dollars and the kingdom could not secure any advantage through selling more oil by making its currency cheaper; there would be no boost to exports from devaluation. Meanwhile, there is limited scope to substitute local products for imports, especially of foodstuffs.

In supervising the banking system, SAMA has always adopted a tough and conservative regulatory policy, stemming from a deeply interventionist culture, combined with official skepticism about what the banks are doing. Until the 2008–2009 financial crisis, this approach stood in marked contrast to the lighter touch adopted by Western regulators. SAMA's public reaction to that crisis zeroed in on weak regulation as the main cause. This approach has its roots in the part the banks played in the chaotic financial conditions of the 1950s when they exploited their connections to abuse capital controls – an abortive experiment that SAMA has never repeated.

One consequence of the first paradox described above is that the banks are tied into a system, whereby the state distributes wealth to the population through its spending. Saudi businesses depend for a lot of their work on state contracts and on preferential treatment. Provided the fiscal situation is kept stable the banks as a whole will not run into trouble. Shariah-compliant banks in Saudi Arabia only engage in business which does not contravene the Saudi interpretation of religious law. For the first half of SAMA's existence, it had nothing to do with this part of the banking system, despite having on its doorstep the biggest Shariah-compliant bank in the world, Al-Rajhi. SAMA has always treated Shariah-compliant banks in the same way as conventional banks but this approach has not been without difficulties.

5 A Note on Numbers

The Saudi riyal is not a currency familiar to most readers. Throughout this book, therefore, riyal numbers have been translated into dollar ones. Since a fixed exchange rate has been in place since 1986 this poses no problems. For the pre-1986 period, the riyal was in effect shadowing the dollar and the shifts amounted to only a few percentage points year by year. No attempt has been made to adjust for inflation, save in this chapter.[8]

We also need to set out why we have tried to avoid analyzing financial numbers by referencing them to Saudi GDP. It is normal practice for asset values such as government debt or flows like budget deficits to be expressed as a percentage of GDP. The reason is that GDP represents the output of a variety of activities that are viable in themselves – such as Silicon Valley in California in the case of the United States – and can be taxed. But the GDP data for Saudi Arabia, while technically correct, becomes problematic when used as a point of comparison with other variables. Saudi government debt went from nil in 1988 to over a hundred percent a decade later. But the critical variable had nothing to do with GDP. It was the foreign reserves that were available to pay for the imports. Most of the private sector would not be viable without the government spending that comes from oil. Put simply, Saudi GDP is not robust in the way that American GDP is.

We have broken our rule on not adjusting for inflation by providing two charts that show how much has changed in SAMA's role in the world since the 1970s. Figure 1.3 shows how SAMA's foreign reserves,

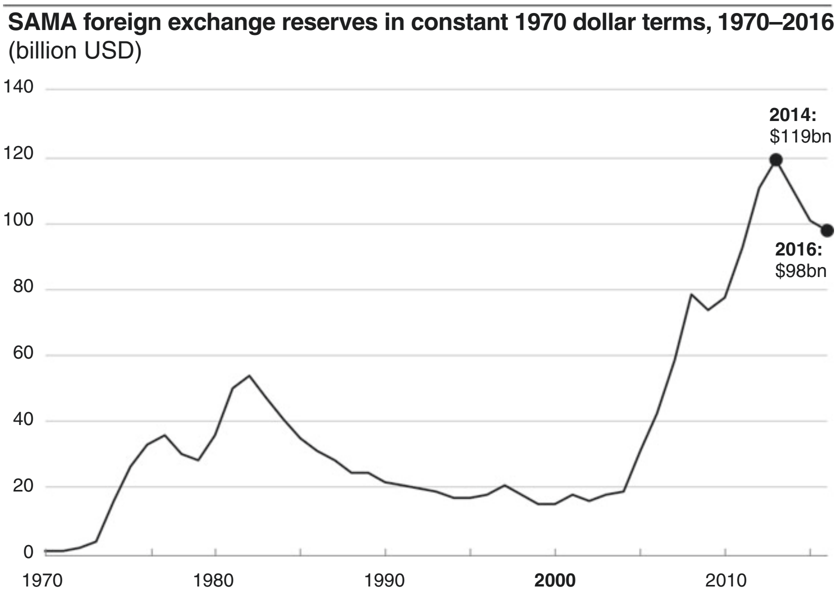

Fig. 1.3 SAMA foreign exchange reserves in constant 1970 dollar terms, 1970–2016

Sources: SAMA Annual Reports, IMF IFS database, FRED database

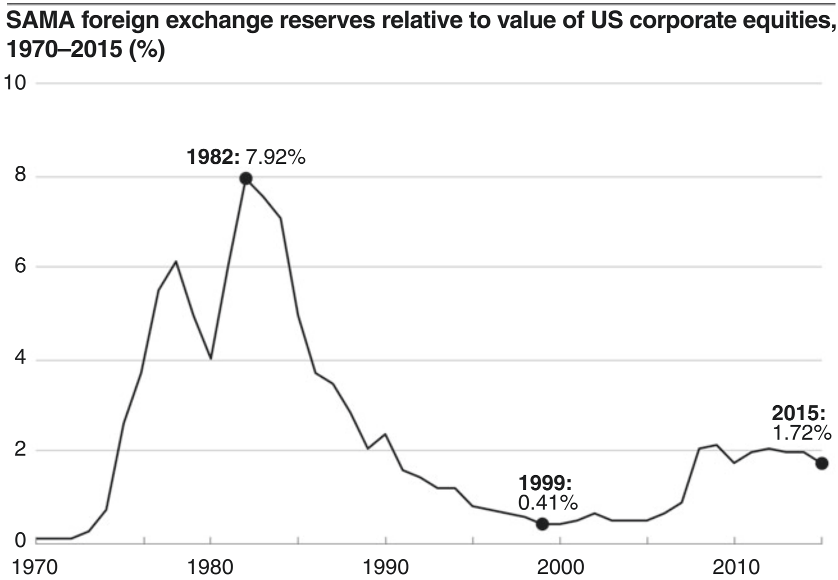

Fig. 1.4 SAMA foreign exchange reserves relative to value of US corporate equities, 1970–2015

Source: SAMA Annual Reports, IMF IFS database, FRED database, Financial Accounts of the US

the key variable in the Saudi system, have moved in terms of dollars measured at the US price level of 1970.[9] This gives a rough idea of how their purchasing power has changed. Note that it took 30 years for the reserves to get back to their purchasing power value of the early 1980s.

Figure 1.4 plots the same story, but this time in terms of the ability of the Saudi government to buy assets in the United States, which was a big concern of American public opinion. In 1982, SAMA's foreign reserves amounted to eight percent of the value of US corporate equities. But this state of affairs was short-lived as the kingdom's income fell victim to the roulette wheel nature of the oil price.

Notes

1. This phrase was first used by then-Vice-Governor Muhammad Al-Jasser in the early 1990s to describe the kingdom's role in stabilizing world output of oil. We borrow it from him and apply it to the central bank with acknowledgments to the former Governor.
2. In 2013 before the oil price fall the revenue from oil was over a third of GDP. And there was no borrowing.
3. Ahmed Banafe, *Saudi Arabian Financial Markets* (Riyadh: Ayyoubi Printers, 1993), 71–75.
4. SAMA has presented this table for 34 years, using balance of payments data for imports and exports of goods and services (excluding Aramco and other entities in the public sector which means it excludes oil and oil products exports so it measures the non-oil economy's imports and exports). To capture other payments for services and for financial transactions, and also for individuals and private companies this is supplemented with numbers for the amount of foreign currency SAMA sells to the banks. Data for the years 1992–1994 following the Kuwait conflict is not available so there is a gap in the series. Over the entire period 1982–2015 the difference between the two series is \$54 billion, or between \$1 and \$2 billion per year. In layman's terms, the numbers are practically identical. The numbers in Fig. 1.1 have been adjusted by the US consumer price inflation index to show them in 2010 dollars, but showing them in nominal dollars would make little difference (the r^2 goes up to 0.93).
5. Chapter 5 tells this story and explains the '90 percent solution' in more depth.
6. The subject of the randomness of the oil price is examined in detail in Chapter 6.
7. The clearest statement in the Quran (2:275) is normally rendered into English as: 'Allah has permitted trade and forbidden interest.'
8. Here we have used US inflation data, because the Saudi inflation numbers are unreliable until recent times.
9. Annual data on foreign reserves and on the breakdown between deposits and investments, and on the foreign assets of quasi-government organizations, comes from SAMA's consolidated balance sheet in the Annual Report. Monthly data on foreign reserves from 2000 is in the Monthly Statistics Bulletin Table 7a, http://www.sama.gov.sa/en-US/Pages/default.aspx (accessed October 25, 2016).

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PART I

History

CHAPTER 2

Discovery of Oil and the Founding of the Saudi Arabian Monetary Agency, 1902–1952

1 King Abdul Aziz

The best place to start the history of money in Saudi Arabia is not in the capital, Riyadh, nor the arid heartland of Nejd, but in Jeddah on the Red Sea coast, in the old part of the city. Here, in contrast to the modern thoroughfares elsewhere in Jeddah, the streets are too narrow for cars and curve round in a comforting way to lead you toward the bazaars. There are no tourists with guidebooks and cameras in old Jeddah, only business visitors and foreign residents who peer into the tiny shops. The old town houses are made of coral blocks from the seashore, tied together with dark wood. Most striking are the bow windows and wooden balconies with their lattice work. The buildings overhang the road and funnel the cool sea breeze up from the street through their windows. At the edge of the old city is a two-storey concrete building – the *souq al-dhahab* or gold market of Jeddah.

A policeman wearing khaki and a black beret is seated on a chair by the double glass doors to the gold market, a glass of sweet tea on a table beside him. Tea boys run with their metal trays from door to door of the little shops, weaving in and out of the crowd. Shops are on the ground floor, offices on the first. Gold is everywhere. Trays heaped with gold ornaments and coins litter the glass counters. Business always seems to be brisk.

A. Banafe, R. Macleod, *The Saudi Arabian Monetary Agency, 1952–2016*, Financial Institutions, Reforms, and Policies in Muslim Countries, DOI 10.1007/978-3-319-55218-7_2

Gold is more than an ornament – it is an investment. Saudi ladies keep their personal wealth in gold jewelry. Muslims are not allowed to earn interest on their investments, but there is nothing wrong in making a profit when the gold price goes up. At the gold *souq* you are allowed a glimpse into Arabia's monetary past, when gold and silver were the main stores of value and the currency of trade. The story of SAMA begins here with the discs of gold on the counter.

Islam has always had a specific set of views about the role of money in the economy. The Quran forbids paying or taking interest. In simple terms, Islam sees all economic activity as an expression of the community. Those who have money must invest it in business, taking risks alongside the entrepreneur and losing their investment when he does. The idea of receiving interest without taking any risk is not part of a pure Islamic system of economics because it allows the rentier to distance himself from the affairs of the community. The single tax in the Islamic economy – *zakat* – is a capital tax that will take away an annual portion of any assets, so they must be productively used to earn at least a return greater than the annual tax. Banks on the other hand – at least in the Western model – are founded on the idea of paying and receiving interest, a principle which underpins the entire Western banking system. So, the establishment of a central bank in the birthplace of Islam was, to put it mildly, a great challenge, one that involved navigating a joint, mutually beneficial path, not just between two fundamentally different and apparently incompatible banking systems, but between what were essentially two distinct philosophical and religious codes. Not surprisingly then, it took the combined efforts of three people to accomplish it. King Abdul Aziz Al Saud (commonly known as Ibn Saud, the founder of the kingdom) made sure that its creation conformed to his understanding of Islam; Abdullah Al-Suleyman – his Finance Minister – pushed it through in the face of opposition; and an American financial diplomat, Arthur Young, came up with the detailed plan. But at the beginning of the twentieth century in the desert of central Arabia money played an unusual and surprising role.

The Ottoman Empire loosely controlled part of the eastern fringe of the peninsula on the Gulf coast, as well as Hejaz, which, bordering on the Red Sea was historically the richest area and home to the holy cities of Mecca (Makkah) and Medina, the centers of Muslim prayer and pilgrimage. The popular image of the Arabian Peninsula is of a vast area of desert over which tribes fought sporadically until the creation of Saudi Arabia and it was only the discovery of vast oil reserves that forced it to connect with the outside world. But this is far from the case.

At the start of the twentieth century, Jeddah teemed with the life of the entire Arab and Islamic worlds. Tall Sudanese mingled with thick-set Egyptians, Somalis with Moroccans and slaves from Ethiopia, while among them slipped the small bird-like Bedouins from the high desert. Each year the town was engulfed by the great pilgrimage, or *Hajj,* to Makkah, bringing together Egyptians, Indians, Malayans and Javanese, multiplying the population for a few weeks and inflating prices until the boats and caravans departed. The merchants of Jeddah, in this way, had connections across the whole Muslim world. Many currencies were in circulation. As Hejaz was under Ottoman rule, the official currency was the Ottoman silver riyal; but much of the trading went on beyond the reach of Turkish jurisdiction and was paid for in whatever came to hand. Gold and silver coins were minted locally and valued by the weight of their precious metal. A Louis d'or coin, for example, would probably never have been closer to Paris than the forge of the local goldsmith.

But the country beyond the Hejaz was so poor that gold and silver coins were a rarity. Most of Abdul Aziz's desert subjects were Bedouin who were not troubled by questions of money. There was no local taxation except for the *zakat,* or levy, fixed at one sheep in a flock of forty, and one goat for every five camels. In the villages, trade was conducted through barter. Dates were swapped for wheat, goats for blankets and camels for guns.

King Abdul Aziz, who died in 1953 at the age of 77, was the most significant and influential figure to emerge from the Arabian Peninsula in modern times. By uniting most of the peninsula, he created a new kingdom in 1932 and crucially succeeded in establishing an enduring system of governance that gave the country a bedrock of stability. A key element of this was a solid monetary system that would be rigorous enough to handle the phenomenal revenue generated from the largest oil reserves on the planet. Saudi Arabia today has arguably the best regulated financial sector in the Middle East. But things might have been very different. Abdul Aziz's rise to prominence began at the age of 25: in 1902, he and a small band of companions retook Riyadh, in those days just a small and remote desert settlement, but capital of the territory controlled by Al Saud – the Saudi clan. One of Abdul Aziz's financial assistants would subsequently recall that during these years the royal treasury was so small that he was able to carry it all in his purse. Stealth and daring – not money – were his tools. The king was personally a frugal man and was used to doing without cash. In 1913, the revenues did not exceed half a million dollars and ten years later, when he was on the verge of conquering Hejaz, they totaled only a million.[1]

World War I forced the peninsula brutally into the modern age. The local Hashemite rulers of Hejaz threw off their allegiance to Ottoman rule, liberating Hejaz with the help of the British under T. E. Lawrence (Lawrence of Arabia). In the meantime, Abdul Aziz Al Saud was moving west with the aim of capturing Jeddah, the capital of Hejaz, now effectively in the hands of the Hashemites. His army was spearheaded by a band of fearless, austere warriors, the Ikhwan. By the start of 1925, Abdul Aziz was closing in on Jeddah and it was obvious that nobody stood between the Saudis and control of Hejaz. Abdul Aziz took Makkah, putting the mint out of commission, and Jeddah finally fell to his troops at the end of 1925. The sultanate of Nejd (central Arabia) and the western region of Hejaz were joined, and in January 1926 Abdul Aziz became the King of Hejaz and Nejd, giving him mastery over the whole peninsula. The new leadership in Jeddah stamped existing coins with the word 'Al-Hejaz' and these continued to circulate. Taxes went up.

Abdul Aziz's right-hand man during this rapid military advance across the Hejaz would play a key role in the history of Saudi financial and monetary policy, straddling an era when money was not essential for the kingdom's governance to one in which Saudi Arabia began acquiring vast revenues from the export of oil.

2 Abdullah Al-Suleyman: The First Saudi Finance Minister

Photographs show a small man with a dark face and piercing eyes, a powerful hooked nose and a black beard in the style of the king; moustache, beard on the chin and a tuft under the lower lip. He was born in Nejd, but he had moved to Mumbai in India as a boy where he became a servant in the house of one of the leading Saudi merchants in the city. He then returned to Saudi Arabia and worked as a clerk in Abdul Aziz's court. The king recognized Al-Suleyman's value and put him in charge of the finances. Al-Suleyman remained in this role throughout Abdul Aziz's reign. During this period and until SAMA was founded in 1952, the history of financial, monetary and foreign exchange policies were intertwined because all the threads ended up in the hands of Abdullah Al-Suleyman.

Al-Suleyman was the consummate insider, a workaholic who needed little sleep and regularly put in 18-hour days. Although he was self-effacing in the court, his relationship with Abdul Aziz was very close because the king trusted him absolutely – to the extent that he headed the agency that in 1944 became the Defense Ministry. In contrast to the warmth he showed in his relationship with King Abdul Aziz, Al-Suleyman was reserved and aloof with others. He did not believe in delegating power and refused to promote talented people for fear they would supplant him in the king's favor. He was also unpopular with the rest of the royal family because he often refused to pay them out of the Treasury. On one occasion Prince Saud, heir to the kingdom, became so angry that Al-Suleyman had to flee from Makkah to Jeddah to escape him. Saud followed Al-Suleyman's car in his own and the royal advisor only escaped by taking refuge on board one of the foreign ships in the port.

The downside of Al-Suleyman's dominance was that his accounting was primitive: he had received no formal training in accountancy and did not use double-entry bookkeeping. For many years, this did not matter; but when the oil money began to flow in the late 1940s Al-Suleyman's system was unable to cope. Finally, after Abdul Aziz's death, King Saud and Prince Faysal eased him out of office. Al-Suleyman had been involved with the affairs of a German construction company that had won a number of contracts and they seized the chance to investigate him.

Al-Suleyman's office was called the General Finance Agency and it combined the roles of a Finance Ministry and a central bank. There were three jobs for a central banker in those days: acting as banker to the government by supplying it with money; looking after the government's foreign currency reserves; and supervising the banks. Since there was only a single bank in the country (the Netherlands Trading Company in Jeddah), this left the office with two jobs: supplying the government with cash and looking after foreign currency reserves. Saudi Arabia had no paper currency so gold and silver served both purposes. Al-Suleyman kept control of the gold and silver and disbursed cash, often personally.

In 1928, the government took an important step which committed it to a currency standard. Two years after taking over Hejaz, Abdul Aziz's revenues had risen to over $7 million and he decided to issue his own silver coinage, declaring that other silver coins were no longer legal tender.

Al-Suleyman organized the changeover. The Saudi riyal was exactly the same as the old Ottoman riyal it replaced. A Royal Decree was issued at the same time marking the birth of Saudi monetary policy. Article 4 of the decree declared that the silver currency was linked to gold at the rate of ten riyals equaling one gold sovereign. The British connection was emphasized by the fact that the silver riyals were made in England.

Silver was fine for everyday transactions, but government taxes were paid in gold because silver was too bulky. By fixing the rate of exchange at ten silver riyals to one British sovereign, the new country placed itself on a bimetallic standard which depended upon a fixed relationship of the local silver coin to gold. But if the world values of gold and silver moved away from the one sovereign equaling ten silver riyals ratio, the relationship of gold and silver inside the kingdom would break down as well. If the silver price rose faster than gold, the riyals would be worth more as melted-down bullion than as coins, and it would pay people to swap gold coins for silver. If gold outpaced silver, then gold coins would be in demand and would likewise disappear, probably abroad. Al-Suleyman believed that there was no alternative to gold and a silver currency. Gold was needed for large payments, while silver was in ordinary use, and the people would not accept coins that were worth less when melted down into metal. Most countries on the gold standard, rather than adopting this approach, followed Britain in having a 'silver' coinage that was actually an alloy of lower value than the face value of the coin. For the next 30 years, the Saudi government stuck to a bimetallic standard despite episodes of coin forgery, melting-down and disappearances, until it eventually collapsed in the crisis of the 1950s.

The Depression of the 1930s hit government finances hard because the money levied on pilgrims fell as their numbers shrank. Merchants supplying the royal court went unpaid as their bills mounted. Al-Suleyman mortgaged his future revenues by selling them exemptions from customs duties. He also borrowed from rich merchants, and by 1933 the king owed them $1.4 million. Al-Suleyman had another problem: his gold and silver currency was disappearing. After the United States left the gold standard that same year, the world price of gold and silver in dollar terms went up and the coins were worth more as bullion outside the country. Eventually, oil came to his rescue when a group of Americans became interested in prospecting after the discovery of oil in neighboring Bahrain. The geological survey of the formations around Dhahran that followed indicated substantial reserves. Abdul Aziz initially offered the concession to the British but they turned him down and in early 1933 a team from the Standard Oil Company of California (Socal) arrived in Jeddah. Al-Suleyman was of course in charge of the negotiations from the Saudi side and he wanted payment in gold.[2]

By April the Socal delegation had made its final offer to Al-Suleyman: an advance loan of $150,000 in gold to Saudi Arabia, with a second gold loan of $100,000 18 months later. The first annual rental payment of $25,000 would be in gold, but all subsequent ones, including the royalties on oil produced, would be made in dollars in New York. The negotiation was on the verge of being completed when the US Treasury announced a prohibition on exporting gold without a license. The gold eventually arrived from the Morgan Guaranty Trust in London on board a P&O steamer and the sovereigns were counted out under Al-Suleyman's own eyes at the offices of the Netherlands Trading Company in Jeddah.

It was not only money that started to flow into Saudi Arabia with the discovery of oil but modern technological products as well. Socal secured an agreement to use aircraft in its exploration campaign. Two pilots were recruited by the newly formed California Arabian Standard Oil Company (Casoc) and a twin-engine plane was flown out from the United States. After weeks of aerial reconnaissance, the prospectors discovered what became known as the Dammam Dome. Drilling began in the summer of 1934. Finally, after a series of disappointments, oil was discovered in 1938. King Abdul Aziz and Saudi Arabia were on the road to fortune. The relationship between Al-Suleyman and the Americans, who had become his main source of money, grew closer. Already in 1936 Casoc (renamed the Arabian American Oil Company or Aramco in 1944) was hiring an increasing number of Saudis, but there were problems in paying them because nobody would accept the Indian rupee notes that were all the Americans had as paper money, and the problems with the silver price meant that there was a shortage of riyals. So, Casoc lent Al-Suleyman the dollars for more silver coins to be made so they could pay their own workers.

When World War II broke out in 1939 the flow of pilgrims almost stopped and the Jeddah merchants hid their gold. Britain helped Al-Suleyman out with some aid, but more importantly, in 1943 President Roosevelt declared that Saudi Arabia was vital for the defense of the United States, thus qualifying the kingdom for assistance under the Lend-Lease Act. The Americans immediately supplied silver riyals followed by gold bars and gold discs equivalent in weight and fineness to an English sovereign. Aramco also advanced more money.

By the war's end, the Ras Tanurah refinery on the Gulf coast was producing over 50,000 barrels per day (b/d) of petroleum products. Payments by Aramco soared. Al-Suleyman received $10 million in 1945 and over $100 million in 1951, figures that were to transform Saudi finances – given that the king's total income in 1950 was only $113 million. But spending rose as well and the aged Al-Suleyman began to lose his grip on finances. The Saudis had been drawn by their oil into the Cold War world, with Saudi Arabia regarded as an important ally in the West's campaign to stop the spread of communism. Aramco was running large development projects on Abdul Aziz's behalf and the Americans persuaded Al-Suleyman to accept a mission to help him improve the management of the nation's finances – especially by the establishment of a central bank.

3 Arthur Young: The Plan for a Central Bank

A Douglas DC-3 Dakota twin-engine propeller plane used to stand in front of the old airport of Jeddah. It had been the first one to operate a regular service from Cairo, reaching Jeddah in five hours. The steamer service had taken three days. After take-off from Cairo, the plane would head south over the pyramids at Giza, following the blue ribbon of the Nile. Then it would turn east and fly down the Red Sea to Jeddah while the passengers lunched on sandwiches and coffee from a thermos jug. The landing strip was to the east of the old town of Jeddah, which was changing fast under the impact of oil money. Abdullah Al-Suleyman had ordered the old city walls to be demolished, but the outlines remained clearly visible from the air. Bechtel Corporation used the coral blocks taken from the walls and the old houses nearby to make breakwaters for the expanding port. At one end of the lagoon the Medina gate and the caravanserai, or resting place for travelers, could be made out. But now new buildings of imported concrete were springing up outside the city walls.

On a steaming July afternoon in 1951 (it was almost 50°C in the shade), US Ambassador Raymond Hare waited with a small group of fellow-Americans in the shade of the terminal building while the aircraft from Cairo taxied to a stop. Down the gangway came a short man with crinkly gray hair and a broad open smile. Hare shook hands with the new arrival before whisking the visitor and his team off to a brand new villa in the Bechtel compound. Arthur Young and the financial mission to Saudi Arabia had arrived. Others had been there before them, including a Dutch financier who attempted to explain double-entry book-keeping to Al-Suleyman. But the Young mission was ground-breaking.

Arthur Young was 56 years old, about the same age as Abdullah Al-Suleyman. But their backgrounds were very unlike. Young had spent his life doing different jobs for the State Department in expanding America's sphere of influence. Born to middle-class parents in Los Angeles, he took his undergraduate degree at Occidental College where his father was President. After World War I, he joined the State Department as an economic adviser. When countries asked the United States for technical assistance to sort out their finances, Young was one of the people who would be dispatched. He would typically move into the Finance Ministry and supervise the implementation of a program of spending cuts. Then he would set up a central control of spending so that the budget was balanced. As America's global reach grew, Washington sent Young as financial adviser to Chiang Kai-Shek in China. He stayed there for 18 years, only to see his work count for nothing when the Communists took over.

Young was a careful man by nature and his time in China had taught him that matters could not always be hurried along. This was useful because anti-American feeling was on the rise. Three years before Young's arrival in Saudi Arabia, the face of the Middle East had been changed by the establishment of the state of Israel. The anger felt by Arabs in Saudi Arabia at this development was as great as that expressed elsewhere in the region. There was also dismay in the kingdom at the degree of American support for Israel, leading to a period of cool relations with Washington.

Young flew to Aramco's headquarters in Dammam where he found a kingdom within a kingdom. There was modern sanitation, running water, electricity, shops, hospitals, schools and colleges. Over 14,000 Saudis worked for the oil company and were provided with all they needed. Aramco expatriates lived on compounds where everything was shipped in. The problem, Aramco's financial experts told Young, was that within the kingdom silver and gold was substitutes for one another at a fixed rate. This was what the bimetallic standard meant. But the real exchange ratio was determined on the world market. In particular, if the silver bullion price rose the silver coinage would disappear from circulation as it was sent abroad to be sold. Most of the kingdom's revenue was in US dollars so a dollar link was logical but Al-Suleyman did not like paper currency and sold the dollars in return for gold and silver riyals.

As Young listened, the core of the problem became clear to him. The large amount of oil money meant that Al-Suleyman had to accept payment in dollars in New York. When he exchanged the dollars for gold and silver in Jeddah, he was bidding up the bullion price against himself. If the banks were short of gold coins or silver riyals, the exchange rate would rise and the amount Al-Suleyman received for his dollars would be less than he expected. As oil revenues rose, the banks were finding it more difficult to come up with the gold and silver currency. Aramco suspected that the banks were manipulating the exchange rate against Al-Suleyman. When Al-Suleyman instead used dollars to get more gold and silver coins minted abroad, the exchange rate was affected by the extra supply of money. His problem was also Aramco's because it constantly needed local coinage to pay its workers. As Aramco expanded, the banks became the only place from which it could acquire coins, and the shortage of coins meant the company paid more dollars for the same amount of riyals. Aramco wanted the riyal linked to the dollar, a paper currency and a central bank where they could buy all the riyals they needed at the fixed exchange rate rather than going through the local banks.

Young flew back to Jeddah for a meeting with Al-Suleyman and told him he would work on setting up a central bank, but meanwhile the exchange rate had to be fixed against the dollar. Al-Suleyman agreed to hold back enough riyals at the Finance Ministry so that he could intervene in the Jeddah bullion market. When the silver riyal was falling against the dollar Al-Suleyman bought riyals in and paid for them in gold. When the riyal was rising he did the reverse, issuing extra riyals onto the market and receiving gold sovereigns in return. The plan worked well except during the *Hajj*, or pilgrimage, season. The influx of visitors buying riyals traditionally strengthened the currency and the stabilization plan was suspended until they left. By the summer of 1952, the silver riyal was stable against the dollar. But how long would that last?

Young was not the first financial expert to try to come to terms with the kingdom's finances – there had been British, Dutch and American advisers before – but he was the most successful. In many ways, his experience was typical. It was months after his arrival before he could get anything done. He thought he would stay for only a short time but spent over a year in the country. While he found the desert beautiful, he lived isolated from Saudis in a prefabricated house on an American compound. But he differed from some of his successors in one important respect: he won the trust of the king and his Finance Minister.

After a deterioration in relations due to American support for the newly founded state of Israel, Washington successfully recalibrated its relationship with a state that was fast becoming a giant producer of oil. The Cold War was intensifying and the Americans wanted to make sure that Saudi Arabia stayed in the Western camp. They were happy to keep foreign aid flowing in the belief that in the long term the increasing flow of oil would solve the kingdom's financial problems. And the oil was flowing faster. Aramco expanded the refining capacity of Ras Tanurah to 150,000 b/d. By 1950, the first cross-country pipeline was built to Sidon on Lebanon's Mediterranean coast, so oil could be easily delivered to a Europe that was threatened by the Soviet Union. The kingdom's oil had become a major strategic asset, as promised in 1943 when Washington was told by Aramco that the oil reserves were as big as those of the United States. Al-Suleyman's problem was that spending was rising faster than income. Abdul Aziz was admittedly generous in his allowances to the royal family. But the speed with which the kingdom was now changing also brought with it unprecedented levels of expense. The population of Riyadh grew from 47,000 in 1940 to 83,000 a decade later, and the city became a large construction site, with new palaces, government buildings and other facilities springing up. Al-Suleyman tried and failed to cut spending. And by the end of 1949, the government was forced to postpone some development projects. By the time Abdul Aziz died four years later, the government was $200 million in debt.

Against this background, Young sent a report to the US State Department on setting up a central bank. The idea had first been put forward 20 years before by the British, and by 1951 it was clear that some type of national body was needed. Each of the foreign powers who felt they had a stake in Saudi Arabia's development had their own ideas about how to proceed. The British had proposed a Currency Board with 100 percent bullion backing of a paper currency. The French plan included a 'Stabilization Administration,' while the Dutch had suggested that the Netherlands Trading Society set up a national bank which would initially be majority Dutch-owned. Some in the royal family led by Crown Prince Saud were attracted by the idea of a national bank that could print money and solve the government's budget deficit.

The core of Young's November 1951 plan was that a Saudi Arabian National Bank should be set up with capital from the government. It would have four objectives. First, it should be responsible for operating a stable monetary system. Second, it should regulate the local banks. Third, it should be the bank to the Saudi government, receiving its income and making payments to it. Fourth, it should advise the government on financial problems. When SAMA was set up, it followed Young's report almost to the letter.

Two months after submitting his proposals, at the start of 1952, Young contacted the Finance Ministry to find out what was happening. Al-Suleyman's deputy replied that his boss was in Riyadh with King Abdul Aziz. They were talking about the central bank idea. Young told him that George Blowers, who ran the central bank of Ethiopia, was in town and was ready to accept the post of Governor. The minister told them to come up to the capital right away.

Young packed his bags and he and Blowers flew to Riyadh. But the central bank project was stalled. Although the king was in favor, his advisers were telling him he could not approve setting up a bank which was against the Islamic principle of *riba*. Over lunch the next day, Crown Prince Saud told them that he was personally in favor of the plan, but the new organization should be headquartered in Riyadh, not Jeddah. Young suspected this was because the prince wanted the bank to be more subject to political influence than he thought desirable, so he hedged. After the meal, Young received a summons to meet the king. Young had deliberately avoided referring to the new institution as a bank, but his suggested name of 'Saudi Arabian Financial Institute' still implied that interest would be payable. If he talked to the king perhaps a compromise would emerge. But the meeting was put off and Young returned to Jeddah with nothing settled.

Young had experienced long periods when nothing happened, punctuated by short and unheralded periods of frenetic activity. Now he wondered whether another long delay was likely. It was not. Two days after his return, the telephone by his bed rang at eight o'clock in the morning. He picked it up. Al-Suleyman was on the other end. He was back in Jeddah and needed the decree setting up the central bank drafted by the afternoon, as well as a charter setting out its aims and purposes. After seven months in Saudi Arabia, Young had half a day to write out SAMA's birth certificate.

In Young's own words:

> I got in my secretary and wrote a charter of twelve articles for this central bank, or monetary agency. I'd write an article and it would be translated into Arabic. I produced the whole thing by the middle of the afternoon, a complete charter for the central bank; and really a surprise because ordinarily

> when you set up a central bank you have hearings and expert studies and all that sort of thing, and it takes months or years. Fortunately I had prepared a report in which I explained just what I thought the institution should be and what its powers should be, and what it should do and not do. I had that right before me, and so it was just a question of drafting, of getting these things in the form of a simple charter of about twelve articles.[3]

The charter Young drafted is a curious document. In some ways, SAMA was set up like a normal central bank. For instance, all government revenues had to be paid into SAMA and all funds were to be disbursed from it (although in practice this has not always been the case). Since most revenue was in dollars the receiving account would ultimately have to be in New York. SAMA would pay riyals in exchange into the government's local account. It would also set up a foreign exchange reserve account to provide for reserves if oil income should be insufficient to pay the import bill and have the authority to regulate the local banks. But the list of things that SAMA could not do was in some ways as vital. It could not pay or receive interest – a stricture that would complicate relations with the local banks. Nor could SAMA lend money to the government. This was a standard prohibition the Americans had used before in advising developing nations.

The last prohibition was the most important, and fortunately it did not last. SAMA was forbidden to print currency notes. Gold and silver coins were to remain the sole legal tender of Saudi Arabia. Furthermore, all the bullion currency in circulation had to be backed 100 percent by setting aside the same value of gold and dollars in SAMA's vaults. This was Young's idea. He was looking ahead to the time when paper money would come in, and he wanted a limit on its issuance. The commercial bankers of Jeddah were allies of the big spenders among the princes. The old Netherlands Trading Company had even suggested to Al-Suleyman that it should be given the right to issue paper money itself and act as the central bank. Many powerful people besides Crown Prince Saud wanted a 'soft money' policy, with liquidity issued by SAMA. Young was a 'hard money' man. He believed that if the government could print money without any backing through the reserves, then spending would be unrestrained. The backing rule meant that an inflationary course of printing money would be hard to begin.

The following day Young went to Al-Suleyman's office and they worked on the details: Young's initial suggestion of 'Financial Institute' was rejected, so he proposed 'Currency Institute' instead. They eventually decided on 'Saudi Arabian Monetary Agency' or SAMA for short. SAMA would be located in the commercial capital, Jeddah, despite the objections of some of King Abdul Aziz's advisers led by Crown Prince Saud who still wanted it to be headquartered in Riyadh. As for paying and receiving interest, this would only apply on overseas accounts. Al-Suleyman assured Young that there was no difficulty in this respect with the religious authorities as they were only concerned about what happened inside Saudi Arabia. Two months passed with nothing happening until in April 1952 Young was summoned to Riyadh again to meet the king. He later recalled what happened next:

> …we went to see King Ibn Saud. An audience with him was like a scene from the Arabian nights. He sat on a throne in the corner of a large hall, with rich oriental rugs, and with retainers with pistols and daggers lining the walls. He motioned me to a seat at his side, while the Finance Minister and interpreter knelt before him. He was a large man, well over six feet and strongly built –but lamed by more than twenty wounds from his desert wars. Of course, in that country the king takes the initiative. The minister had said, 'The king may bring up this question of the bank; if he doesn't bring it up, you bring it up.' And so we went on and I exchanged a few pleasantries with the king and the king didn't bring it up.
>
> And then I opened the subject and I said, 'Your Majesty, we've been working on a project for setting up a financial institution to help with the currency and finances.'
>
> The king said, 'That's fine.'
>
> I said, 'I think we might call it a monetary agency.'
>
> He smiled and he said, 'Yes, I think that's good.'
>
> And then I said, 'I suggest that we should also put in the charter a clause to the effect that nothing should be done contrary to the Islamic law.'
>
> And then the king – this was all done through the interpreter kneeling in front of him – said that he agreed. He spoke for five or ten minutes on the Islamic law, showing that he was very deeply religious, and that he really believed in the Islamic law.[4]

King Abdul Aziz played a critical role in SAMA's establishment because without his approval nothing could have happened. By indicating originally that he wanted the project to go forward he put the burden on its opponents to stop it. When they objected to the use of the term 'bank' (which was really about the problem of *riba*) he simply delayed for a while.

When at last Young was granted his audience, the king's silence forced Young to make the running and to put forward his own ideas. King Abdul Aziz had been careful not to suggest anything himself. For him, the key thing was that nothing should be done contrary to Islamic law.

While SAMA had been created, it had no Governor, no building, no staff and – critically – no control over the money supply. It had been a long birth and the young institution had some teething troubles ahead. The security and stability of what was fast becoming a major oil exporting state with rising revenue would depend on these problems being resolved quickly and decisively.

Notes

1. Alexei Vassiliev, The History of Saudi Arabia. (London: Saqi Books, 1998), 305. This is a good general history of the country from a non-Western (Russian) viewpoint. The Soviet Union was the first foreign power to recognize the Kingdom of Hejaz and the Nejd (as Saudi Arabia was then known) in 1926, seven years before the United States. The number of Moslems inside the Communist domain meant that understanding Saudi Arabia and what would happen to the Hajj under the new desert ruler was important to Moscow. Abdul Aziz broke off diplomatic relations in 1938 and they were only re-established after the fall of the Soviet Union.
2. Mohammed Almana, Arabia Unified: A Portrait of Ibn Saud. (London: Hutchinson Benham, 1980), 226. This is an excellent memoir of the period, written in old age. Almana was from Zubair and like Abdullah Al-Suleyman was educated in Bombay where his father dealt in Arab horses for the Indian Army. He was the king's constant companion as his English-language translator from 1926 to 1935 and his is an insider's account of Abdul Aziz's personality, his style of government and financial problems.
3. Truman Library oral history project 1974, Arthur Young interview, http://www.trumanlibrary.org/oralhist/young.htm (accessed October 21, 2016)
4. Truman Library oral history project 1974. The sequence of meetings is different in Young's later written account.

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CHAPTER 3

Financial Development before the First Oil Crisis, 1953–1974

1 Date Palms and Sheep

The new central bank needed a home. Arthur Young toured buildings in Jeddah, finally choosing an old, solidly built villa near the airport, constructed from three-foot thick coral blocks. Next to the villa was a patch of waste ground where the vault to house SAMA's bullion and dollars could be built. Crown Prince Saud (soon to succeed his father) formally opened the central bank, arriving with his half-brother Prince Faysal in a royal limousine with uniformed retainers standing on the running boards and accompanied by armed guards in red jeeps. The official court poet recited a poem for the occasion and the princes inspected the offices, which had been furnished for the day with rugs and gilt chairs borrowed from a nearby palace, before the royal guests drove off through the cheering crowd.

From its very first days SAMA faced a number of issues that still remain intractable today, over 60 years later. Most are connected in some way with fluctuations in the revenue from oil (Fig. 3.1).

It is difficult by citing numbers to grasp the magnitude of the change that oil brought about in Saudi Arabia in the 20 years after SAMA was set up. One good way is to think of the government spending from oil as an object that the country's inhabitants in those days would have recognized

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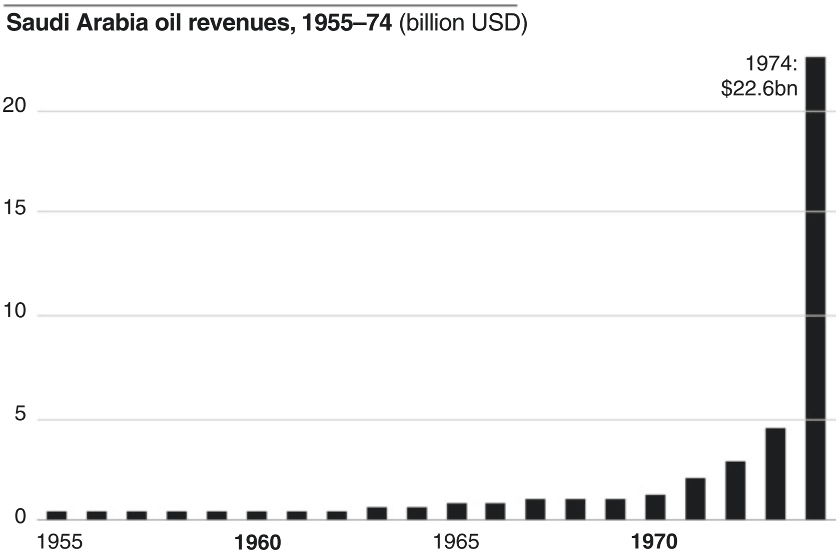

Fig. 3.1 Saudi Arabia oil revenues, 1955–1974

Source: SAMA annual reports

as having value, such as a grove of date palms. In King Abdul Aziz's last years, the budget was about $300 million. Imagine that as a single palm sapling only a meter high sitting in the grove. Ten years later, in 1963, the budget was about $490 million, so beside the first sapling there is now a bigger palm sapling nearly two meters high. The 1971 addition to the imaginary palm grove measures 8 meters, while the one in 1974 is more than 90 meters tall. It towers above its neighbors and is far bigger than any real tree, an image which is appropriate because nothing like this scale of revenue growth has been seen in the modern world.

The problems of managing the upsurge of revenue were immense: not just disbursing it but handling the amounts that could not be disbursed. The earliest data we have for the reserves is 1958 when Anwar Ali became Governor. They amounted to two million dollars. Continuing the analogy of items that the Saudis valued, such as a herd of sheep, let us think of these reserves as representing a single sheep. By the time Ali died in 1974, the size of the reserves had grown to the point where his flock was nearly 11,000 strong. But first there was a crisis.

2 Crisis: 1953–1957

The years after SAMA was founded were very difficult. Oil income stagnated, the Finance Ministry took away the new authority's independence, there was a major geopolitical crisis in the Middle East in the form of the Suez affair, and the central bank introduced exchange controls which were a failure. The currency depreciated, inflation rose and the intervention of Prince Faysal in 1957–1958 was needed to restore stability. As all this was taking place, SAMA was dealing with the failure of the bimetallic standard for the riyal and the introduction of paper currency.[1]

The core of the problem was the fact that oil revenues, which had grown rapidly in King Abdul Aziz's last years, flattened out and the kingdom's debts rose in a disorganized manner. The government borrowed from Jeddah merchants and did not pay its bills on time. Externally it borrowed from Western banks. The background to this was the first major Middle Eastern crisis of the modern era, the Suez crisis of 1956–1957, in which Britain and France connived with Israel to create a pretext to attack Egypt in the hope of retaking the recently nationalized Suez Canal. It resulted in the blocking, for several months, of the waterway through which most Saudi imports and exports passed. Stockpiling of imports by Jeddah merchants during the crisis was one reason for the introduction of capital controls.

Arthur Young left Jeddah for good a few weeks after the opening ceremony and was succeeded by George Blowers. After a year in office his working environment changed dramatically. King Abdul Aziz died in November 1953 and the new monarch and Abdullah Al-Suleyman – who did not last long – found themselves at loggerheads over spending. But first there was a problem with the bimetallic monetary system that Young had not been able to touch. The world price for gold fell after the end of the Korean War, so the face value of a Saudi gold coin in terms of buying silver became worth more than its value in metal. This was an opportunity for arbitrage – illegal arbitrage to produce counterfeit gold coins in order to buy up the silver coins then turn them into dollars for a profit.

That summer, SAMA heard rumors that counterfeit gold coins were circulating. A gang was arrested in Beirut, but the inflows continued. The coins were replicas containing the full stipulated weight of gold, making it virtually impossible to distinguish between them and in December 1953 SAMA announced that, for a fixed period of a month, it would redeem the counterfeits at their face value and exchange them for silver coins. After that, all counterfeits would be confiscated without any compensation. But nobody wanted to be left holding a gold coin which might be a counterfeit. By the year end, SAMA's vaults were almost empty of silver as most gold coins were handed in. From this point on gold coins effectively stopped circulating and Saudi Arabia operated on a silver standard.

It was silver's turn next. By early 1956, the international price of silver had risen by far enough that the metal in the coins was worth more as bullion, so smugglers took silver riyals out of Saudi Arabia and melted them down into ingots to sell for dollars. In May 1956, the government withdrew the silver riyal from circulation and replaced it with a paper currency, based on the pilgrim receipt, which had been invented as the answer to the monetary problem posed by the influx of pilgrims each year for the *Hajj*. Their demand for riyals pushed up the exchange rate and then brought it back down again when they left. In 1953 Blowers printed temporary money – the pilgrim receipts – and sent it to the central banks of Egypt, Pakistan and Iran, where many of the pilgrims came from. These banks transferred dollars to SAMA's account in New York and sold the paper money to pilgrims before they set off. The receipts increased the money supply in Saudi Arabia during the *Hajj* season. In theory they should have been withdrawn afterwards but SAMA let them circulate and soon they were widely accepted. The introduction of paper money was a success but SAMA's credibility was threatened by a budget crisis.

King Saud reacted to the stall in oil revenue by giving the new Finance Minister – he had fired Abdullah Al-Suleyman – effective control of printing money. Saud was already heavily in debt to the merchants in Jeddah, as well as to domestic and foreign banks, and he had just announced a hugely costly rebuilding of the Great Mosque at Makkah. The spending was financed by printing money. The government pretended the currency notes were covered by gold and silver in SAMA's vaults, but by 1957 this amounted to barely 14 percent of the value of the paper currency issued. Access to foreign currency at the official rate was restricted, a black market in foreign exchange developed, the riyal fell in value against the dollar and inflation took off.

Blowers resigned in protest after two years in office. His successor Ralph Standish was a commercial banker, incapable of imposing his authority. The Finance Minister was already President of SAMA's Board of Directors but now he exercised full executive authority – including approving new banks. Foreign reserves were close to zero when, in April 1955, exchange controls were imposed for the first (and only) time in Saudi Arabia. In November 1956, following the Suez crisis and the withdrawal of British and French forces under US pressure, a system of licensing was introduced to control access to foreign exchange for imports. The main effect was to enrich people who had access to dollars at the official rate via either SAMA or the commercial banks.

SAMA was unable to control the usage of scarce dollars at a time when the black market rate for the dollar was well above the official rate. The theory was that by checking on the banks' use of foreign exchange, controls could be implemented effectively. But the system was open to abuse. People who could acquire an import license had access to dollars from the banks at the lower, official, exchange rate. Best of all was to set up your own bank by getting the Finance Minister's personal approval. Banks, meanwhile, had every incentive to buy as many dollars as they could from SAMA, so they could sell them to customers on the black market.

Because the system relied entirely on the honesty of the banks there were no effective checks on their activities and SAMA was unable to take action against any of them. When it requested extra resources from the Finance Ministry it was told not to interfere with the private running of the banks. After the Suez Crisis, Saudi Arabia provided funds to help the Egyptian government recover. There was also a short oil boycott of the UK and France. Both factors made the budget deficit bigger.

The black market value of the dollar rose, and the Saudi riyal (SAR) fell until it stood at the exchange rate of SAR 6.25 to one dollar by 1958, compared to the official rate of SAR 3.75. More and more of the scarce foreign revenue was diverted to pay the interest on Saudi government borrowings abroad. Members of the royal family, led by Prince Faysal, supported a reform program. In April 1957, a committee of senior ministers was set up to regularize the way the government used foreign exchange and three months later SAMA's Foreign Exchange Department was established to monitor the banks.

When Prince Faysal visited SAMA in early 1958 he specifically asked to see the physical gold and silver in the vaults. He found only 300 silver riyals that had not been pledged against foreign loans.

3 Recovery: 1958–1962

In March 1958, Prince Faysal became Prime Minister and Chairman of the Council of Ministers. He also became Finance Minister for two years. Faysal was determined to balance the budget and get back to a single exchange rate with no capital controls. The recommendations of the International Monetary Fund (IMF) mission headed by Anwar Ali, which had arrived in Jeddah in 1957, included currency reform with a new dinar to replace the devalued riyal; but Faysal took no action on this. However, he appointed Ali to succeed the ineffective Standish and amended SAMA's Charter so that Ali became head of the Board of Directors, bringing the Finance Minister's direct control of SAMA's affairs to an end.

Prince Faysal succeeded his half-brother as king in 1964 after Saud was deemed by leading members of the ruling family to be unfit to rule. Saud resisted the first attempt to oust him, and Faysal was reluctant to see him forced off the throne. But in the end Saud agreed that it was in the kingdom's best interests for this to happen. Faysal ruled with a firm hand, and his restoration of financial stability was based on the principle that austerity should be led from the top. He lived in a modest villa and disliked ostentatious behavior. In 1958, before becoming king, he had introduced a freeze on government spending and a decree that no spending or borrowing of money was allowed without prior approval of the Council of Ministers, which he chaired. Imports of private cars were forbidden and old retainers of the Court were retired. Those who had fought alongside King Abdul Aziz and their descendants received fixed pensions. But the amount of money made available to members of the royal family as a whole was cut back.

Faysal's technique was to trust a small band of public servants and to back them up against all others, including members of the royal family. The foremost of these servants was Anwar Ali, a Pakistani, who remained SAMA's Governor until his death in 1974. Ali was an economist who had worked in the Indian Finance Ministry before independence. When Pakistan was created he did the same job there and then was appointed to the board of the central bank before moving to the IMF. He was the first SAMA Governor to have a background in the public finance of both developing and developed economies and this made him invaluable to Faysal.

The twin aims of their program – to cut spending and stabilize the riyal – were closely linked. Only with a balanced budget and debts repaid could order be restored in the foreign exchange market. At the peak of its indebtedness, the government owed more than one year's revenues, both at home and abroad. As Faysal cut spending, Ali introduced a two-tier exchange rate system to gradually squeeze out the black market. The official rate of exchange of SAR 3.75 applied only to essential consumer and capital goods. In practice, this amounted to subsidizing the basic living costs of the poorer citizens. SAMA recognized the free market in foreign exchange and supplied it with dollars to encourage the market rate to rise. Profits from the sale of dollars were directed into public revenues. The free market value of the riyal rose steadily to SAR 4.75. Finally, in 1959, Ali pegged the riyal at SAR 4.50 to the dollar and the black market vanished. By 1962, the country owed no money abroad or at home. Saudi Arabia joined a small club of countries whose currencies were freely convertible and where there were no capital controls.

Rising oil revenues due to rising output were a critical component of the recovery. Oil income jumped from the $300 million level in the late 1950s to over $400 million in 1962. As the budget moved into surplus SAMA's foreign exchange holdings also rose. The links between domestic budget surpluses, economic growth and the accompanying propensity to import on one side, and gains in foreign exchange reserves on the other, were noted by Ali in the first and second SAMA Annual Reports:

'With a narrow productive base, the domestic expenditures of the Government largely reflect themselves into a demand for foreign exchange for imports and other purposes.'

'Government spending in Saudi Arabia (based itself largely on oil revenues) is the primary determinant of the general level of incomes in the country. The level of budgetary expenditures influences the volume of economic activity directly as well as indirectly i.e. through the multiplier process.'[2]

This was a public recognition of the first two paradoxes of Saudi finances: government spending from oil money is the main determinant of economic growth, and the budgetary stance largely determines the size of the foreign exchange reserves. Well into the 1960s the fact that the government could not spend all its revenues led to a rise in foreign exchange reserves.

4 Long Boom: 1963–1974

The long boom period can be divided into two roughly equal parts: up to 1970 and after. In the 1960s, growth was steady and the problem was budgetary: getting the infrastructure projects completed. After 1970, the picture changed because oil income exploded: an 18-fold increase in five years. SAMA became the repository of quantities of dollars that were the envy of the rest of the world.

Saudi finances in the 1960s were dominated by a strong and steady increase in oil revenues. In contrast to the early 1970s, the rise was reasonably predictable, since the bulk of it was due to export growth (Fig. 3.2)

Output continued to increase, even as reserves estimates rose, so there was no sense of a drawdown of oil but rather of limitless growth. From around a million barrels per day (b/d) in the late 1950s, output crossed the one and a half million b/d barrier in 1962, exceeding two million b/d in 1965 and three million b/d in 1968. But the export revenue from a barrel of oil did not rise during this period: in fact, it fell slightly, despite the country's efforts to add value by investing in facilities that could export refined products. This price slump was the reason for the establishment of the Organization of Oil-Exporting Countries (OPEC) in 1960. The final factor was positive: the share that the government received of the total oil revenue rose, as Aramco handed over more of its income, before being gradually nationalized.

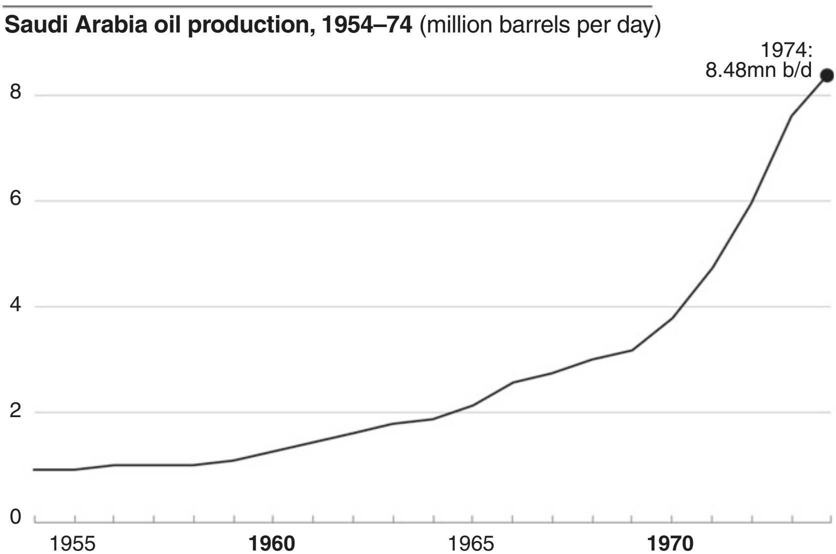

Fig. 3.2 Saudi Arabia oil production, 1954–1974

Source: SAMA anual reports

Faysal spent the money wisely on the essential infrastructure of a modern state, exemplified by a big road-building program. In terms of their importance in unifying the country, his highways can be compared to the first trans-continental railway line in the United States. When he took power there were only 900 miles of surfaced road in Saudi Arabia. His first great project was the Red Sea-Gulf highway running nearly a thousand miles from east to west, which was finished in 1967. This was followed by a north–south road stretching from Tabuk province on the northern Red Sea to Jizan on the Yemen border. He also provided drinking water by drilling wells, and expanded the land under irrigation. He built new hospitals and schools and trained more doctors and teachers.

The king controlled spending by allocating money to different funds, so that he could halt projects in one area while accelerating them in another. In this he used the central bank as a partner. Faysal allocated money to an Economic Development Fund (EDF) maintained by SAMA. In 1964, for instance, total government spending was 50 percent more than the official budget because of the use of the EDF. For development projects outside the oil sector, foreign investors were exempted from taxes for five years. This was in complete contrast to the state-directed plans of other developing economies at this period.

The steady growth of the 1960s received a setback with the Arab–Israeli war of June 1967, when the Arab armies and air forces were defeated in just six days. Growth in oil production slowed and the closure of the Suez Canal damaged both imports and exports, as had happened during the Suez crisis of 1956. Faysal helped the two countries that had suffered most in the war with Israel, sending \$35 million immediately as assistance to Egypt and Jordan, followed by nearly \$300 million over the next two years. This was the beginning of the policy under which Saudi Arabia has supported its neighbors at times of financial trouble. These early assistance programs were financed domestically by borrowing from the banks, but the dollar advances were paid by SAMA out of the foreign exchange reserves.

This resulted in the central bank losing about \$110 million in reserves between October 1967 and the end of the year, but despite this the reserves still stood at a substantial \$740 million. Domestically, the war was a shock to sentiment and bank loans declined temporarily. No capital flight occurred, however, and SAMA did not need to take action to stabilize the banking system. In 1969, oil revenues continued to rise, but by less than the government expected. In 1970, they surged.

During the 1960s, SAMA was able to diversify its growing foreign assets. In 1958, these had amounted to only a couple of million dollars. But the situation improved so rapidly that in 1963, SAMA began to acquire investments in short-dated US Treasury bonds, to supplement its holdings of gold and foreign exchange (mostly in dollars with banks in New York). The US Treasury holdings were worth more than $180 million by 1968.

In those years, Anwar Ali also began to design the modern Saudi banking system. During the 1950s, banking supervision had been almost non-existent. Many people and businesses had no bank accounts, as they either dealt in cash or used the Shariah-compliant system of moneychangers which stayed outside SAMA's control for many years. While formally all banks had to be licensed with adequate capital and they had to send SAMA information about their balance sheets and activities, SAMA did little with the data the banks supplied. Furthermore, there was no domestic monetary toolkit other than the printing of banknotes. No reserve ratios were applied to the banks, which were free to expand their balance sheets without limit.

One of the first things Anwar Ali did was to put in place the classic instrument of control that central banks exercise over commercial ones by instituting a reserves ratio. He directed the banks to hold the equivalent of 15 percent of their overall deposits with SAMA (cut to 10 percent in 1962 and adjusted thereafter in line with monetary conditions). The idea was that the reserves ratio would restrict the banks' lending activities and also give SAMA access to their assets to repay depositors in case of a bank failure. At this time, there were two types of banks operating. First, banks that had been established for some years primarily to finance imports, of which the most important was the Netherlands Trading Company (later Saudi Hollandi Bank), which began operations in 1926. [3] Second were the new banks. National Commercial Bank (NCB), founded in 1953, was the biggest and grew out of the Bin Mahfouz money-changing business.[4] The new banks had sprung up during the crisis years when capital controls were in place. In those circumstances, it had been immensely attractive to set up a bank with the right to buy dollars from SAMA at the official rate.

The new banks had done well out of the black market and what amounted to theft of official supplies of foreign currency, but under the sterner rule of Faysal they ran into problems. Riyad Bank in particular came under great pressure and in 1959, together with another small bank, Bank Alwatani, it turned to SAMA for help. SAMA merged the two under Riyad Bank's name, injected liquidity in its role of lender of last resort and nominated three board members. As a result, SAMA discovered that senior officials and board directors were receiving large personal loans. It instructed the bank to stop this practice and place extra deposits with SAMA. The following year the Finance Minister fired the bank's chairman and put his own nominee in charge. There was chaos among the senior management as they accused one another of fraud (a former managing director was found dead in his hotel room). Finally, the government took a 38 percent equity stake. Riyad Bank survived and eventually prospered.

The 1966 Banking Control Law imposed extra prudential standards on the banks. Deposits were limited to 15 times capital and reserves, and if this was exceeded SAMA had the power to force a bank either to increase its capital or to deposit extra funds with SAMA. The statutory reserve ratio was formalized and the law endorsed actions that Anwar Ali had already taken to adjust the amount of credit in the economy. Other provisions limited the size of loans to any single person and prevented the bank's directors from making unsecured loans to themselves (one of the problems with Riyad Bank).

Banking in the 1960s was dominated by the three indigenous Saudi banks: NCB, Riyad Bank and the small Ibrahim I. Zahran. In 1966, these accounted for about two-thirds of deposits and of loans and advances. SAMA's policy was that the banking sector should be Saudi-owned. Locally owned banks were easier to monitor than branches of overseas ones, which could both receive and pay away funds to their parents abroad. Also, SAMA had more influence over local management in implementing its guidelines. Finally, Saudi banks were more likely to participate in SAMA's plans to spread banking throughout the country by opening local branches in rural areas. SAMA helped by opening offices across the kingdom. Ali's bank Saudization program took its first step in 1968 when the local branch of the National Bank of Pakistan became a company with majority Saudi ownership.

Payments throughout Saudi Arabia were still mostly made by cash. The introduction of paper money made this easier than it had been previously when heavy silver coins had to be used, but clearly the next step was the use of bank checks. Checks between banks were cleared through bilateral arrangements. SAMA pushed for check clearing houses to be opened in the big cities so that payments between banks could be settled speedily. A clearing house was opened in Jeddah in 1967, followed by Riyadh and Dammam. The other major banking development of the 1960s was the birth of the Specialized Credit Institutions to help areas of the economy that the banks were neglecting. These were government agencies like the Agricultural Bank, but closely connected with SAMA, which looked after their banking needs.

Saudi Arabia had become the world's largest exporter of crude oil and, as we have seen, it was unable to spend the export revenues immediately because of the problems of development. But by the late 1960s, the country was absorbing higher government spending. Meanwhile, the global picture for oil was changing. Negotiations between oil producers and the major oil companies started with Libya's demand for a greater share of revenues. Following that, the oil majors accepted higher sovereign taxes on their oil revenues and kept up their profits by raising the price of oil. The 1971 Tehran Agreement between the Gulf oil producers and the majors increased Saudi oil revenue by a half. But just as importantly, both sides agreed to an annual escalation in the price of oil.

The pace of price rises became helter-skelter. The dollar fell after the United States went off the gold exchange standard at the end of 1971, and since oil was priced in dollars, this became another reason to raise the price. In 1971, Saudi Arabia agreed with Aramco to raise the posted price of its oil from $2.37 (already a rise from 1970) to $3.18 per barrel. Prices went up throughout 1972 and 1973. In October 1973, after the Arab–Israeli war, OPEC raised the price to $5.12 while Saudi Arabia announced it was cutting output by 10 percent and banning sales to the United States and other countries that had supported Israel (more on this in the following chapter). The political impact of the oil embargo was not great because the ban on sales was brief. Even so, the oil price shot up again to almost $12 per barrel. Meanwhile, the staggered nationalization of Aramco, which had been a hundred percent American owned, began.[5] The state acquired a 25 percent holding immediately after the 1973 war, 60 percent in 1974 and took full control of the company in 1980, giving the government access to still more oil revenues. A few brief boom years followed: output continued to climb, the oil price went up and the country's share of oil income also rose as Aramco was nationalized. All three engines driving government revenues were operating at full throttle. In 1974, oil revenue reached a staggering $22.6 billion.

SAMA faced two challenges in the early 1970s: fighting inflation and handling the explosion in foreign exchange reserves. The tremendous increase in government spending resulted in price rises in all areas of the economy, but SAMA did not have either the evidence or the tools to respond effectively. Housing costs in particular were difficult to estimate as rents rose by leaps and bounds. As an illustration, the one-time head of the Investment Department at SAMA used to tell a story that as a junior civil servant in the 1970s in Riyadh, the rent on the house he shared with other colleagues rose by a factor of ten in three years, until it swallowed up most of his salary.

SAMA tried to slow down imported price rises by continuing the link with gold so that the riyal appreciated against the dollar. Saudi Arabia revalued the riyal against the dollar by 22 percent in 1971–1973 after 12 years of a stable exchange rate. In 1975, there was a 2 percent revaluation to 3.475 and the riyal was formally pegged to the Special Drawing Right (SDR) of the IMF. To back up the anti-inflation drive, the government lowered or abolished a range of local taxes and customs duties.

The consumer price basket rose by 5 percent in 1971 and by marginally less than that in 1972. But in 1973 measured consumer inflation (a very approximate measure of what was happening to the economy) rose to 16 percent, peaking at 35 percent in 1975. The real problem was housing where the supply was inelastic in the short run. Housing costs rose by 390 percent between 1970 and 1977 compared with 204 percent for the overall cost of living index and 154 percent for foodstuffs.

What the inflationary episode of the early and mid-1970s showed was that SAMA could not control inflation in an economy where cash was being injected far in excess of the capacity to absorb it. The only action open to SAMA would have been to raise reserve requirements at the banks. But the banks found demands for loans slack, so raising reserve requirements would have been pushing on a string as inflation was not driven by credit demand, but was due to a cash boom. The currency appreciation dealt with imported inflation to some extent but not with housing. The Saudi response reflected the importance of fiscal policy, because it was a fiscal one through subsidies – the Real Estate Development Fund was set up in 1974 to give interest-free loans for private housing.

Once again, the oil money outpaced the ability of the government to spend it. Local currency was credited to the government's account with SAMA in exchange for the dollars which went to boost the foreign reserves, and much of it stayed there. In the budget announcements, revenue was always balanced by spending plans but cash spending lagged. In 1974, for instance, a mere 42 percent of budgeted spending actually took place. Not

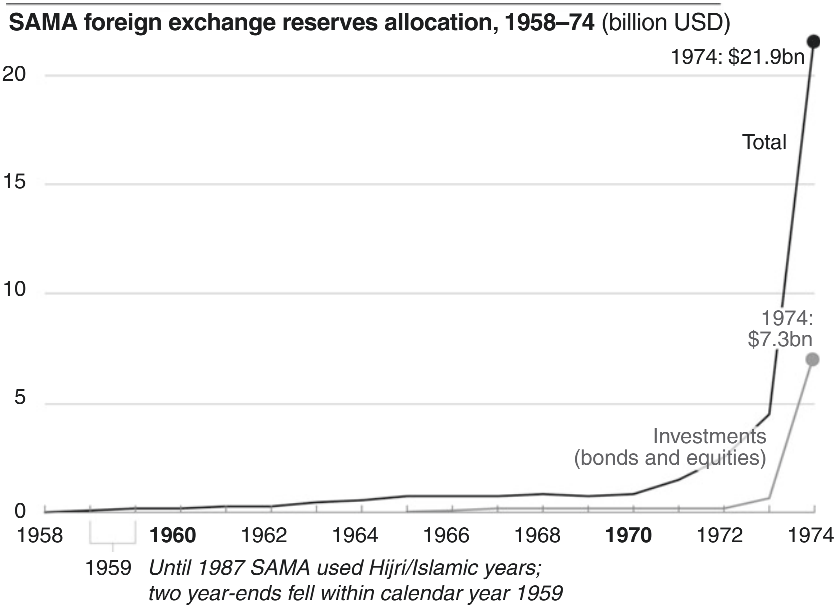

Fig. 3.3 SAMA foreign exchange reserves allocation, 1958–1974

Source: SAMA anual reports

Note: Invesments include currency cover

only did SAMA's foreign exchange reserves rise (see Fig. 3.3), but the commercial banks began to accumulate foreign assets. Banks spread their branch networks across the country to take advantage of the tremendous positive cash flow. But they had to compete with the rising numbers of government agencies which were also available to meet credit needs. The Public Investment Fund established in 1972, financed government investments, while the Saudi Industrial Development Fund was started in early 1974 to provide interest-free financing for private businesses in an effort to diversify the economy. The result was that the banks, unable to profitably expand their loan books, piled up cash and accumulated foreign assets.

In the early 1960s, SAMA had taken the first steps to diversify its holdings away from gold (a core holding and one that it did not add to) and bank deposits into bond markets. Its initial investment was in short-dated US Treasury bonds and bills, which could be managed by SAMA's correspondent banks in New York, where most of the deposits were held. The Treasury holdings did not offer a yield pickup over bank deposits for similar maturity, but they meant that SAMA could invest the money for longer periods. This relieved SAMA of some of the burden of rolling over bank deposits. The US Federal Reserve and SAMA cooperated closely, and Treasuries quickly built up to comprise about 20 percent of the total reserves by 1970. But the diversification program stalled because SAMA staff were fully occupied in coping with dollar inflows and placing them on deposit, and had no time to do more than roll over the maturing Treasuries. By late 1972, Treasury holdings had dropped to barely 6 percent of assets. SAMA urgently needed trained people who could invest longer term instead of merely rolling over deposits.

The shock to the developed economies from the rise in oil prices was profound. In effect, it was a transfer of purchasing power to the oil producers, but the problem was that they could not spend enough, so the overall result for the world economy was a drop in demand as well as a shift to Western external deficits and Middle Eastern external surpluses. There was great pressure on Saudi Arabia to recycle the oil revenues it could not spend so as to help oil importing economies finance their balance of payment deficits. In practice, this was inevitable as SAMA's foreign exchange reserves built up, and they had to be placed somewhere: mainly in US bank deposits. But Saudi Arabia also willingly saw its IMF allocation rise so that funds were placed at the disposal of the Fund in SDRs.

In 1974–1975, the direction of the country's financial affairs changed. Anwar Ali died of a heart attack while attending the annual meetings of the World Bank and the IMF in Washington DC in November 1974. He had been Governor for 16 years and had suffered two attacks already before he admitted himself to a Washington hospital – but too late. It was fitting that he should die in harness, for he was a tireless administrator and guiding spirit. Under him, SAMA had grown enormously in stature and in scope of operations.

The shock of his death was followed by a far larger one: the assassination of King Faysal in 1975. Faysal had managed the country's finances with the aid of Anwar Ali and his uncle, Finance Minister Musaed bin Abdul Rahman. Prince Musaed's resignation in the wake of the king's death meant that the directing spirits of the country's financial and monetary development had all gone in the space of a few short months. SAMA faced big challenges under its fourth Governor, who, after two Americans and a Pakistani, was the first Saudi national to hold the office.

Notes

1. A. Mcleod, Bimetallism in Saudi Arabia (Typescript, SAMA Library in Riyadh, probably 1958) is the invaluable (but unpublished) source for the failure of the bimetallic standard and the introduction of paper currency. Mcleod was part of Arthur Young's team and stayed on as first head of SAMA's Research Department. So he was present throughout the twisted story.
2. SAMA Annual Report 1380 AH (1961) p. 2 and 1381–82 AH (1962) p. 6, http://www.sama.gov.sa/en-US/EconomicReports/Pages/AnnualReport.aspx (Accessed October 21, 2016). The Annual Reports provided for many years the only English language commentary on economic developments and they are still the most authoritative.
3. It originally traded as the Netherlands Trading Company. The others were: in 1947 Banque de L'Indochine (later Saudi French Bank), Arab Bank of Jordan (1949) and in 1950 the National Bank of Pakistan (later Al-Jazira) and British Bank of the Middle East (later SABB).
4. The other new banks principally comprised: Saudi Cairo Bank (1954) Lebanon and Overseas Bank (1955), First National City Bank (later Citibank) also in 1955, Riyad Bank (1957) and the National Bank (1958).
5. Profits had been shared equally with the government since 1950 when Abdul Aziz threatened nationalization. But since the American management and their accountants decided on what the declared profits were, the Saudis could never be sure they got their fair share.

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CHAPTER 4

Petrodollar Recycling and Saudization of the Banking System, 1975–1982

1 The First Oil Boom

In the winter of 1974, America was feeling the icy fallout of war in the Middle East. In October of the previous year, Egypt broke through Israeli defenses along the Suez Canal triggering a conflict which eventually ended in the defeat of the Arabs, following American support for the Jewish state. In response Saudi Arabia led a boycott of Arab oil exports to those nations which had supported Israel. As Americans waited in line to fill their cars with expensive gasoline or shivered in under-heated office buildings, calls for the United States to invade the Eastern Province and provide cheap oil grew loud, and neither the Nixon administration, nor the subsequent Ford government, attempted to silence them. Indeed, Secretary of State Henry Kissinger saw the threat as a means of putting pressure on the Saudis. In a magazine interview, he discussed the circumstances in which the United States would need to take military action to secure its supplies of oil. President Ford backed Kissinger up while his Secretary of Defense claimed that occupying the Gulf was 'practicable.'[1]

There were precedents for the West using force in the Middle East. In 1956, Britain and France had attacked Egypt to, among other things, stop President Gamal Abdul Nasser closing the Suez Canal to oil tankers. Two years later, American marines went ashore in Lebanon to safeguard Western interests in the region and in 1961 Britain moved forces into

A. Banafe, R. Macleod, *The Saudi Arabian Monetary Agency, 1952–2016*, Financial Institutions, Reforms, and Policies in Muslim Countries, DOI 10.1007/978-3-319-55218-7_4

Kuwait in response to Iraqi threats. Reacting to the American rhetoric, King Faysal ordered Prince (later King) Abdullah to reinforce the National Guard's protection of the oilfields with orders to destroy the facilities if the Americans attacked.

Faysal's actions in the wake of the 1973 Middle East war had led to a massive price increase and the transfer of a significant volume of the developed world's foreign exchange reserves into SAMA's vaults. In 1972, the United States paid $4 billion for its oil imports; in 1974 it paid $24 billion. For the developing economies, the oil price hike was even worse news. But from the Saudi perspective matters also looked dire. Faysal was uneasy about the scale of the oil income. The country could not spend all of it immediately on development, so it had to be turned into monetary assets. But if it was invested in US banks, it could be held hostage by an assets freeze. There were intemperate voices on both sides. Some in Riyadh wanted to cut oil exports back to reduce revenue to a level that the economy could absorb. In June 1975, Farouk Akhdar, a senior planning official, caused a furor in Washington when he said that Saudi Arabia would stop investing in the United States unless it could be assured of friendly relations.

But Faysal knew that any attempt to restrict oil production would send oil prices soaring and make the recession in the West worse, thus increasing the risk of military intervention in the Gulf. This kinetic route to resolve the oil price crisis was probably never considered seriously in Washington. It would have been foolhardy in the extreme because Saudi oil production would have fallen as the Gulf became a combat zone. The alternative to occupying the Saudi oilfields was for both sides to manage the consequences of the high oil price. Saudi Arabia took the initiative in responding to the crisis and SAMA played a pivotal role in shaping that response. Saudi policy was revealed to Treasury Secretary William E. Simon by Anwar Ali in September 1974. The foreign reserves would be recycled back to the West, and particularly to New York, for investment in US banks and Treasury bonds to help finance the balance of payments deficits that the oil price hike had produced. This phenomenon became known as petrodollar recycling and the dollar was the key currency since the oil surpluses were being accumulated in dollars.

But this was just a temporary answer. The second part of the plan was for Saudi Arabia (together with other oil nations) to spend enough money on its own development to revive countries around the world by enabling them to sell goods and services to the kingdom. The vehicle for this was the Second Five-Year Plan, designed by the Saudis with American help. The First Plan (1970–1975) had been designed to spend $9 billion. The Second Plan (1975–1980) eventually earmarked $180 billion.

This deal reflected the fact that Saudi Arabia had no effective alternative economic partner. The oil revenues came from Aramco in dollars, and it was neither sensible nor practical to convert them into other currencies. Most of the country's imports were priced and paid for in dollars and the scale of the foreign exchange transactions to turn the dollars into other currencies was beyond SAMA's capabilities (and probably beyond that of any commercial banking system at that time). The recycling of the surpluses to finance the corresponding deficits of Western and importantly other developing economies was mostly being done in dollars. Finally, the dollar bond and equity markets were then, as they are today, the most liquid in the world. In effect, Americans and Saudis were locked in an uncomfortable but necessary embrace.

2 Working Out Petrodollar Recycling

In early 1974, Anwar Ali had been in a difficult position. He had historically placed the bulk of the oil income with commercial banks, mainly in the Eurodollar market. This was unsatisfactory now because it meant placing and then rolling over short-term deposits in huge amounts, stretching SAMA's investment team and its telex operators to the limit. It also meant that SAMA was exposed to Western banking risk. For the recession unleashed by the oil price rise had caused a banking crisis. Ali already had an arrangement to buy small amounts of US Treasuries but needed royal approval to do so in the size necessary to diversify his exposure in a meaningful way. Meanwhile, he pursued two other routes – setting up a development bank and lending money to the International Monetary Fund (IMF).

The foreign reserves held with commercial banks abroad were loaned by them to developing economies which were heavy oil importers. But this did not help other less developed economies which were not perceived as good credit risks. Over the next couple of years, SAMA helped create the Islamic Development Bank (IDB) to partially open up a second route to nations that fell into this category. But this took up less than a billion dollars of SAMA's ever-growing reserves.

For Ali, the former IMF official, using the IMF was an obvious route but it foundered on two obstacles. The Fund would not pay him a commercial rate of interest for buying assets, known as Special Drawing Rights (SDRs). If Saudi Arabia could have been given an executive directorship, there might have been a compromise on the rate of interest. But the Americans were concerned about losing their veto over the Fund as their share of the quota declined, and they stalled consideration of this idea, so SAMA made no big SDR purchase. In 1981, a Saudi national finally joined representatives of the G-5 nations – the United States, Britain, Japan, France and Germany – as a permanent member of the board. It was the first time in post-war history that a developing economy had achieved this sort of position.

Ali was left with the government-to-government route. This was the most promising option as the biggest world economies – America and Japan – needed Saudi money. By buying large amounts of government bonds direct, SAMA would help these countries finance their current accounts and –more importantly – fund their budget deficits. The United States was Ali's focus. As the situation developed rapidly in 1974, he knew that only Treasuries offered SAMA the right combination of a market rate of interest and a risk-free investment. Direct purchases would give SAMA the size it needed. But there were political complications as it would tie SAMA investment policy to the US government. He referred the decision upwards to the king and waited for a response.

In practice, SAMA could not have bought Treasuries on the scale it needed without official help. The US banking sector held $110 billion of federal and agency securities at this time, so SAMA's cash flow could have bought up the banks' entire holdings of government assets from them in a few years.[2] Purchases on this scale were impractical and Ali needed the cooperation of the US government to sell him bonds direct if he was going to buy enough to engineer a major shift out of bank deposits.

For its part, the Treasury was keen to sell him debt as its borrowing needs were unprecedented in peacetime. Talks began in April 1974 when two US Federal Reserve (Fed) officials visited Jeddah. A meeting three months later between Treasury Secretary William E. Simon and Ali resulted in no immediate action. But by September Ali had the top-level clearance he needed to go ahead with direct purchases. As a US government account of a meeting with Simon recorded: '…it was made clear by Governor Ali that SAG (Saudi Arabian Government) has made a political decision at the highest level to make substantial investments in US securities. Governor

Ali said these investments would be made for the most part through the Federal Reserve Bank of New York . . . It was anticipated that these investments would be made both in marketables and, when it suited the governments of both parties, in Treasury Specials. The Governor forcefully emphasized, however, the importance to SAG of complete confidentiality of the amounts and types of the instruments.'[3]

The best picture available of the reserves at this time is from Ali's September discussions with the Americans (Fig. 4.1). He told them total assets were $12.7 billion. In January 1974, reserves had been $4.5 billion so the picture was changing fast and it seems likely that what he handed over was not up to date. Over three-quarters were in short-term bank deposits, mostly in the Eurodollar market, and the amount with international financial institutions such as the IMF was small.

The new SAMA Governor, Abdulaziz Al-Quraishi, was an experienced civil servant who had been educated abroad and had a reputation as a problem-solver. Kissinger was told that: 'Although he has no experience of

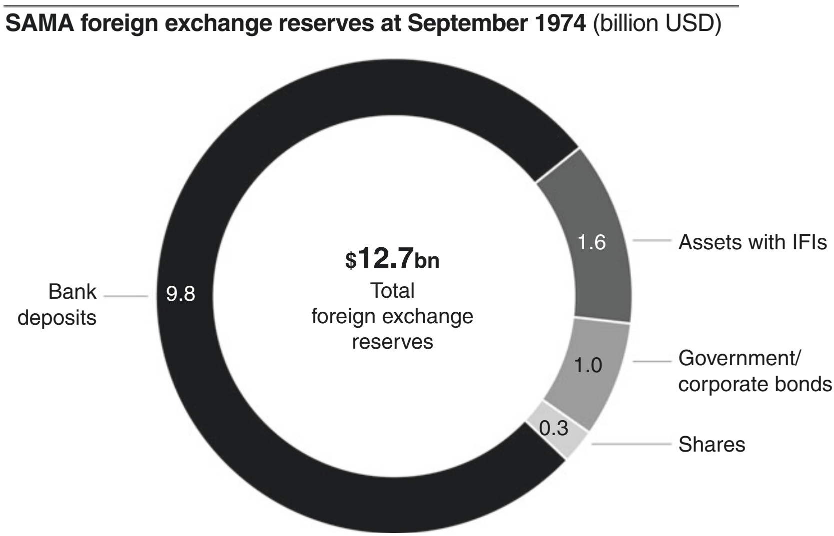

Fig. 4.1 SAMA foreign exchange reserves at September 1974

Source: US Embassy Jeddah report, September 1974

Note: IFI assets mostly with IMF/World Bank finance, he is a quick learner.'[4] Before year-end 1974, Al-Quraishi made SAMA's first direct or 'Treasury Special' purchase by acquiring an additional amount of a Treasury issue direct from the US Treasury at the average price at which they auctioned it publicly to banks. If the Saudis wanted to sell before maturity they would notify the Fed which would buy them back.

With the direct investment program supplementing SAMA's bank deposits, and decisions taken about the size of investment in the IDB and the Fund, Al-Quraishi's position was further bolstered by the arrival of a group of top US and British advisers who joined the investment team of Ahmed Abdullatif, the director general of the foreign department.

3 The WB Team: 1975–1978

At the start of 1975, an American and an English banker walked together into the dealing room of SAMA's investment department on Old Airport Road in Jeddah, next to the Kandara Palace Hotel, and took their seats for the first time. David Mulford from White Weld & Co. and Englishman Leonard Ingrams from Baring Brothers shared intellectual brilliance and an Oxford education, but were very different personalities. Mulford, at 37, was powerfully built with a commanding presence: he had steely gray eyes, a moustache and a mop of dark hair that was turning gray. The Saudis quickly appreciated him for his understanding of their problems and ability to cut rapidly to the core of any issue. Ingrams, four years Mulford's junior, was a quieter man, an excellent viola player who later promoted a chamber music ensemble in Jeddah. He was related to the Baring family on his mother's side and followed his father to become a director of the bank. But behind the eyes that twinkled through glasses perched askew on his nose, and the nervous laughter that gave the impression of a man giggling at his own witticisms, lay a fine brain that could wield a rapier wit.

The arrival of Mulford and Ingrams, together with their assistants, was the result of Ali's decision to bring in foreign specialists who could help manage the money.[5] They would give him professional advice and take on some of the work of placing deposits. Ali had been under a lot of pressure in his final months: there was talk elsewhere in the government of setting up a sovereign wealth fund along the lines of the long-standing Kuwait Investment Authority. This was the route that would be followed by the Abu Dhabi Investment Authority in 1976, but Ali wanted SAMA to handle things in-house rather than see the investment management division split off.

Ali sought advice from the heads of the Fed and the Bank of England. They suggested two relatively small, conservative and discreet private institutions. The Fed recommended White Weld, and the Bank of England chose Baring Brothers.[6] Ali's sudden death did not interrupt negotiations: less than a month later White Weld and Barings met the Finance Minister who agreed that three Americans and two British bankers would join SAMA as advisers. They were contracted on a cost-plus basis for a six-month trial period. In the event, the relationship continued until 1989 and the banks became known collectively as WB because investment recommendations were initialed as coming from White Weld (W) and Barings (B).

By January 1975, the team members had settled in and were beginning to work out what assets SAMA held. Ingrams characteristically ordered his team to wear jackets at their desks even when the air-conditioning broke down. In his entertaining memoirs Mulford described his first morning:

> We found our way to a small office on the third floor of our very dilapidated building. In the room six desks were pushed closely together and six Saudis sat cross-legged on the desktops drinking sweet tea, their sandals lined up neatly on the floor. Before them lay large, leather-bound, open ledger books in which line after line of entries had been made by hand in beautiful Arabic script . . .
> 'How do you look at your whole portfolio of deposits say, for example, by their maturity structure?' we asked. They didn't keep a spreadsheet record of maturities, they explained. But they did have, on a single piece of lined paper, a long handwritten list showing the total deposits of SAMA with all eighteen depository banks. My eye went to the bottom of the column, which was in Arabic numbers. The total at the bottom read something like $20,160. As an investment banker I automatically added three zeros. 'Over 20 million dollars,' I observed. 'No,' they corrected me, 'here you have to add six zeros.'[7]

Gradually, they transcribed the office ledgers into English and made out lists of bonds and deposits held, putting them onto file cards for easy reference. A picture slowly emerged of how exactly the assets were invested, with which banks and in which currencies. But WB did not get a handle on how much oil income to expect, nor when dollar outflows to the banks or from government would occur.

Once a month the chief dealer in the Investment Department would take a crucial phone call – one that would determine SAMA's investment strategy over the next four weeks. It came from the Aramco headquarters in Dhahran, giving three figures in dollars which, totaled together, made up the monthly oil income payment. The man from Aramco was always polite but never gave SAMA's chief dealer any more information. This fundamental uncertainty about next month's income underpinned the monthly strategy meeting that followed. The oil money went into SAMA's bank accounts with Morgan Guaranty and Chase Manhattan in New York. If nothing was done, or as often happened in 1975 it took some time for the telex operators to get the funds transferred out, the money would sit there.

Just as SAMA had no idea ahead of time what size the dollar inflows from Aramco would be, it also had to make dollar payments out without warning. Government outlay affected SAMA's assets in two ways. Most government departments did not have their own foreign currency accounts (the Defense Ministry was an exception). So if the Health Ministry wished to buy medical scanners from Texas for a new hospital it would ask the Finance Ministry to issue a foreign currency check (usually in dollars) drawn on SAMA, and SAMA would in turn issue a check on one of its own bank accounts with a New York bank to pay the Texans. Because the central bank never knew beforehand what bank checks had been signed, it was necessary to maintain very large call accounts in dollars. For instance, in 1978 the overall balance on call accounts was never allowed to fall below $2.5 billion.

The bulk of spending was, however, made by way of riyal bank checks drawn on local banks. If the Health Ministry needed to pay for materials locally, such as steel girders to build a new hospital, it would write a riyal check on its account with a local bank. This was riyal spending. But since steel was all imported, the contractor would ask his bank to make a foreign currency transfer to pay the foreign supplier. As the bank naturally wanted to balance its foreign exchange position, it would go to SAMA to sell riyals for dollars. The result was again a reduction in the level of dollars on SAMA's call account. Either way – dollar or riyal spending – the money came out of the reserves. This was WB's first acquaintance with what they called 'the 90 percent solution.'

By the middle of 1975, Mulford and Ingrams were ready to make a comprehensive report to Al-Quraishi's investment committee, which habitually convened in his office after the monthly oil figures were received. They found most of the reserves were invested in developed markets. The Saudis were supposed to be passing huge sums of money to other Arab states, but at the end of 1975 WB calculated that less than $2 billion was being lent in this way. Each time a fresh loan was negotiated, WB would be consulted about the right interest rate for the credit, but the decision as to whether to lend the money was out of their hands.

Reportedly, from time to time SAMA was instructed as agent for the Saudi government to provide support to neighboring states in other ways that did not appear on the balance sheet, namely by guaranteeing loans. SAMA's cooperation with the local subsidiary of Algmene Bank Nederland (later to become the Saudi Hollandi Bank) provides a good example of how this worked. Abdullah Al-Suleyman always had a close relationship with this bank, which had been established in Jeddah to service the needs of pilgrims from the Dutch East Indies (later Indonesia). SAMA would make a fiduciary deposit with the Dutch bank which would then lend the same amount of money to a government nominated by SAMA.

Early on, WB advised Al-Quraishi that there was no need to make major changes to his investment plan. Cautious purchases of American equities should be continued, but they did not advocate big buying of stocks. WB's economic outlook was that bonds, particularly shorter dated government bonds, were a better and safer home for the cash. The emphasis should be on accumulating government bonds in the United States and also in Europe and Japan. Public sector bonds guaranteed by the government were an acceptable substitute. Government-to-government deals were the quickest way to invest the cash and another simple method was through private placements where SAMA would buy an entire bond issue. The implication was that as the issue would not be liquid in the market, SAMA would then hold it to maturity.

Figure 4.2 shows that this was a good investment plan for the next three years: in 1975 both one-month Eurodollar deposits and five-year Treasuries were yielding less than US inflation (so-called negative real yields) but yields on bonds were higher than deposits, so it made sense to switch cash to bonds that would not be needed for several years.

The Japanese were particularly keen to sell bonds direct and in early 1976 an agreement was signed with the Bank of Japan, which was very similar to the one made with the United States. By 1981, with another oil

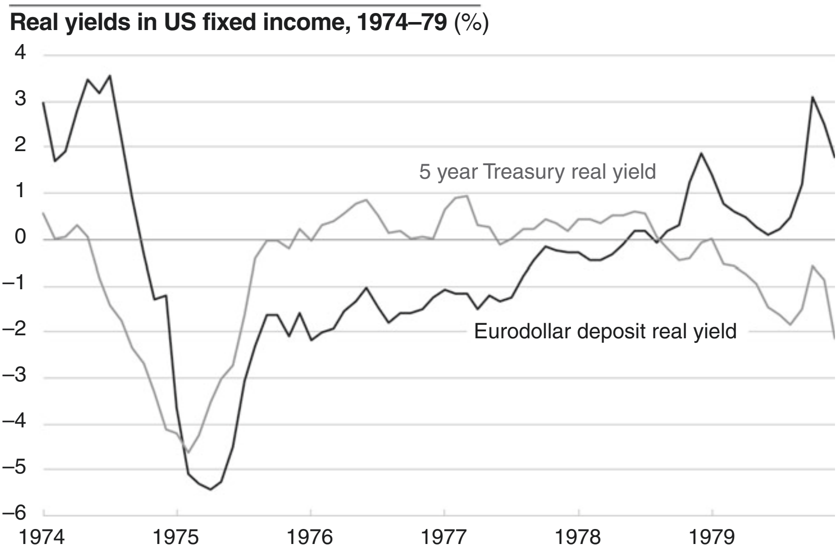

Fig. 4.2 Real yields in US fixed income, 1974–1979

Source: Federal Reserve Economic Data

price boom under way, 7 percent of SAMA's assets were in the yen. The Europeans had no common policy toward Saudi Arabia. The Italians were very keen to sell bonds to Riyadh, but the Swiss were not. The British and French governments both put forward government agencies instead of doing so directly. This suited SAMA very well as the interest rate negotiated was always at a higher level than the government bond yield, with the additional advantage of a government guarantee underpinning the loan. The Germans, like the Swiss, were not interested because they were weathering the oil price hike better than their competitors. They had no objection to SAMA giving business to German banks and buying Bunds in the market but that was as far as they would go.

Conditions in boom-time Jeddah were not what the visitors down to sell bonds were used to. If the bankers were lucky they could pay up to share a room in a rundown hotel with cockroaches and bedbugs, while the unlucky had to rent a cab for the night and sleep in the back seat. In the morning the bankers freshened up as best they could and lined in the narrow corridor outside Al-Quraishi's office, sometimes sitting on the stairs and sipping sweet tea, before they were ushered in to see Ahmed Abdullatif with a WB team member. If the proposal was interesting, the WB team would go into conference with the bankers until terms were agreed. Then the lawyers in London or New York would be telexed to draw up the documentation. Mulford later recollected that during a lengthy meeting with one particular delegation he worked out that he was already $300 million behind on the money he needed to invest that day. It was an exhausting schedule for the junior members of the WB team who had to stay in the office late to roll over the deposits on the telex after the close of business in Jeddah.

The list of borrowers who queued up to pass through SAMA's doors in December 1975 alone shows what a press of business there was. The London consortium bank Orion wanted Deutschmarks on behalf of the government of Ireland, also dollars for a Swedish shipbuilder. The Belgian Société Générale de Banque was asking for a loan on behalf of a nationalized industry. Credit Suisse wanted Deutschmarks for a borrower in Germany. So did Norddeutsche Landesbank, on behalf of the state of Lower Saxony. But these demands were put in the shade by the list of corporate borrowers paraded by the New York banks. Morgan Guaranty was representing Standard Oil of Indiana and Royal Dutch Shell (to no avail because SAMA would not lend money to oil companies that could compete with Aramco). White Weld in New York, who were clearly not bothered about conflicts of interest, had the Illinois Bell telephone company looking for dollars, Smith Barney represented a second Swedish shipbuilder and Kidder Peabody had a mandate from Michelin.

SAMA moved into a five-storey office building across the road from its first headquarters on Old Airport Road in 1957, but as far as communications with the outside world were concerned the infrastructure remained primitive. The key daily task facing the WB team continued to be to get the money out of the call accounts and earning more on fixed-term deposits, and then to roll those deposits over on the best terms. SAMA had a single ticker tape machine which printed out financial prices. The only way for WB to stay in touch with what was happening in the financial markets was for someone to stand by the machine and read the printout as it scrolled out. Every morning this tape, with prices of bonds and exchange rates, was supposed to be gathered up and placed on Mulford's desk, so he could see what had been happening overnight and price the day's private placements and the best maturity for the fixed deposits with foreign banks. But if nobody remembered to put a new reel in the machine before they left the office the night before it would run out of tape. When that happened (which it did regularly), WB were ignorant of the latest developments on Wall Street. They were flying blind

Every transaction with the international financial markets, whether it was a bond purchase, a foreign currency switch, or placing a fixed deposit with a bank, had to be done via a telex machine. This made life hard for Tony Hawes and Jonny Minter, the two British money dealers from Barings. Some of their Saudi colleagues would not use the machines because the 'send' button on the keyboard was identified by the sign of a Maltese cross. Jonny Minter solved this by taping the mark of a crescent moon on top so that SAMA employees could touch an Islamic rather than a Christian sign.

In 1975, there were only 700 telex lines in the entire country and access to them was highly prized. Operators at the Jeddah telephone exchange occupied a uniquely influential position during those boom years. SAMA had no direct telex lines and everything had to be done through the telephone exchange. Some hotels such as the Kandara Palace next door, which was where the bankers who were attempting to do business with SAMA invariably stayed, had their own telex machine to make reservations for clients and pass their messages back to head office. The Kandara Palace had a cozy relationship with somebody at the Jeddah telephone exchange who gave priority to its telexes. A banker who had met with SAMA that day and wanted to report back on developments need only pay a fee to the hotel telex operator to ensure that his message was given priority over those of his rivals. This clogged up the telex lines. Neither WB nor Ahmed Abdullatif could stop the practice. SAMA dealers trying in vain to dial through to New York or London would get the engaged signal from the Jeddah exchange. Meanwhile, the telex operator at the Kandara Palace would be promptly sending out bankers' reports on the state of negotiations with SAMA.[8]

The result was that dealers had no alternative but to stay at their posts endlessly dialing and re-dialing through the night until a line became available. Meanwhile, the oil money would sit on the New York call accounts. When WB finally did manage to contact a bank, it might alone get the whole of the day's business because WB dealers could not be sure of being able to make contact with other banks to take alternative quotes. But at least Hawes and Minter had the consolation of being able to work in shirtsleeves late at night when Ingrams had gone home.

4 Currency Diversification and the Exchange Rate: 1975–1982

There was never a conspiracy to keep SAMA's assets in dollars. We have already grasped one of the two reasons why Ali and his successor Al-Quraishi left most of the foreign reserves in dollars in the 1970s – the problem of investing large amounts of money outside the dollar markets. The other reason is that they believed the rise in foreign reserves would soon be spent on the massive program of development. The Second Five-Year Plan projected annual spending of over $30 billion. Reserves at the start of 1975 were $22 billion, so in theory they could run out in less than a year. The future flow of oil revenues depended on the oil price, which was uncertain. It made perfect sense to manage the bulk of the assets conservatively by keeping them in dollars on short-dated maturities or in short-dated Treasuries.[9] But in the mid-1970s the idea took root that the reserves would not quickly run down.

SAMA's foreign exchange reserves had risen to nearly $60 billion by the end of 1977. Most were in dollars. Only about 6 percent were in Deutschmarks and yen. The idea of making some reserves available for longer term investment outside the dollar now made sense and there were dollar sales, which went into Deutschmarks (via commercial banks) and yen through the direct government bond arrangement with the Bank of Japan. Some was invested in equities. By 1977, there were two Swiss equity accounts and one German with external fund managers, and a small British share portfolio. SAMA would have liked to buy more German assets, but was hampered by the informal agreement with the Bundesbank, which restricted sales of German assets to SAMA by German commercial banks.

There was no connection between diversifying the foreign exchange reserves and exchange rate policy. In particular, there was no idea of matching currency composition to the mix of currencies in which imports were billed. SAMA linked the riyal to the SDR rather than the dollar in the mid-1970s, but this was in part a political gesture. The SDR link sat well with SAMA's public commitment to international cooperation. The imperative was to try to quell domestic inflation. Measures such as cutting or eliminating taxes, raising civil service pay and subsidizing foodstuffs loosened fiscal policy and so fueled price rises. They did not tackle the root problem, which was the pace of government spending. Currency policy did have some limited effect and SAMA was able to defuse calls for larger changes in parity by making small moves at the decade's end.

The riyal was officially linked to the SDR in 1975 but in current parlance, exchange rate policy up to 1986 would be described as a 'dollar-shadowing' or 'dirty floating' era. The aim was to avoid importing inflation through a weakening dollar, and when the dollar strengthened, to follow it up, but only partially. The rise in the riyal was engineered when the dollar was weak in the late seventies and as the dollar rose after 1980 SAMA informally broke with the SDR link in 1981 and adjusted the dollar peg down until the rate was set at 3.75 riyals to the dollar in June 1986 (Fig. 4.3).

SAMA did not have the technical resources to do more. WB were too busy investing the money – they did not try to adjust the rate on a daily basis, nor to destabilize speculators' positions by interventions or unexpected moves. In fact, the central bank struggled to avoid being predictable in its currency adjustments: this became a source of occasional profit for local banks and offshore banking units (OBUs) in Bahrain who could see changes in the exchange rate coming. There was probably some limited effect on the inflation rate as a result, but

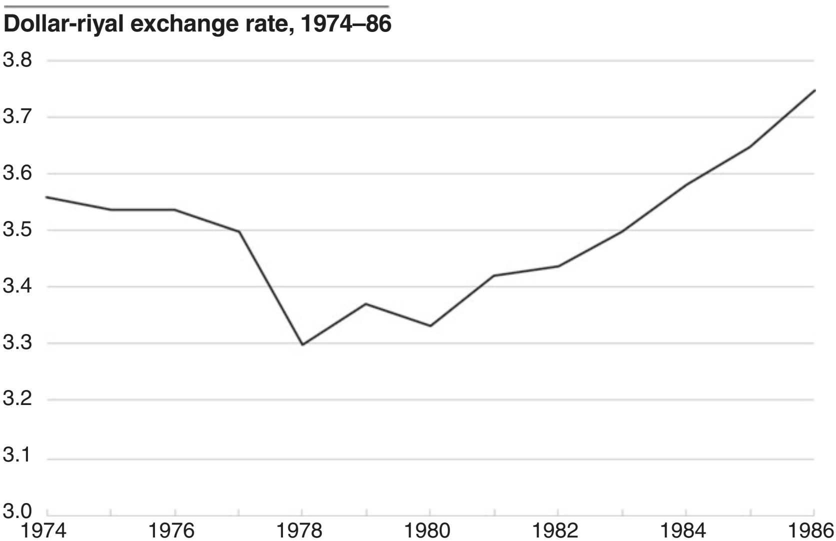

Fig. 4.3 Dollar-riyal exchange rate, 1974–1986

Source: SAMA Annual Reports

after 1982 as the oil price fell, inflation ceased to be a problem. And in practice, the riyal followed the dollar reasonably closely.

5 The Second Oil Boom: 1979–1982

In October 1978, SAMA moved from its home in Jeddah over six hundred miles east to the desert capital of Riyadh. The WB team packed all their office files into Land Rovers and drove up Route 40 in a convoy. The journey from Jeddah took two days and the team camped under the stars on the way up. Finally, they reached a line of cliffs on their left, the Tuwaiq escarpment a couple of hundred meters high, which guards the Western approach to the Saudi capital. Running for over a hundred miles it was a tough obstacle which generations of travelers to Riyadh had to navigate until King Faysal built the modern road. When the Egyptian army marched up from the coast to destroy Diraiyah, the old Saudi capital, in 1818 they had to dismantle their cannon and pack them onto the backs of camels for the trek over the escarpment. Travelling rather more comfortably along Faysal's road, the WB team could see the old track as they slowed down to approach the police checkpoint at the summit. From there, it was only a brief drive to the outskirts of the city.

Beyond the escarpment, there were a few farms clustered around a natural spring – Riyadh means in Arabic 'green places' – but everywhere else the land was baked brown. There was no sudden panoramic view of Riyadh. Gradually the houses became more frequent and then the convoy was running between continuous lines of habitations. Behind the houses, there were glimpses of palm trees, a reminder that Riyadh was a collection of small oases when it became the Saudi capital in the late nineteenth century. Before long, the road widened, the traffic thickened and soon there were shops and stalls, and they were on the edge of the main market or *souq*.

The convoy passed over the scrimmage on a series of flyovers called 'the seven bridges' and came to the palace area of the royal family in the north of the town. All around were concrete mixers, dump trucks, piles of steel girders, great holes dug into the Nejd rock for foundations, half-finished buildings and, above all, dust that whirled about in the desert wind. By day a dust cloud hung over the city and at night it partly blocked out the stars. Only on Friday, the day of rest and prayer, did the Riyadh air become wholly clear. The place of greenery had become the world's largest building site. A quarter of a century after Crown Prince Saud suggested to Arthur Young that SAMA should come to Riyadh, the Land Rovers pulled into a parking lot on Riyadh's Airport Road. The central bank was housed in temporary quarters in the Pension Fund building, in front of the Ministry of Defense, until its new head office next to the Riyadh Intercontinental Hotel was completed. The Investment Department's home in the temporary building was an open-plan office on the sixth floor of the north tower. There was a large trading room and enough direct telephone and telex lines to ensure uninterrupted connections with the markets. Soon a Reuter's terminal was installed, which gave 24-hour coverage of financial news. The ticker tape machine had been left behind in Jeddah.

Despite the plush new offices and the progress WB had made in diversifying the foreign exchange reserves, they were still operating in conditions that were far from easy. Spending continued to rise and foreign reserves fell during the course of 1978. WB started to reverse some of the asset diversification, and ran down bond holdings as they matured by transferring them to the cash accounts to maintain a balance between bonds and deposits. It seemed likely that assets would continue to fall as oil income stabilized while government spending continued to rise. The Third Five-Year Plan for 1980–1985 was even more ambitious than the second and envisaged spending nearly $50 billion each year. Then something happened which aided SAMA's financial position immeasurably. In November 1978, workers in the Iranian oil fields went on strike and production collapsed. After months of rioting, which disrupted oil production, the Shah of Iran fell from power in January 1979, opening the way for the creation of the Islamic Republic of Iran, thenceforth the advocate of Shia Islamic expansion in the Middle East. Iran's oil production did not rebound.

This was the first – but not the last – occasion when the kingdom gained financially from its political stability in a volatile region. Crude oil prices had hovered around the $11–12 level since 1975. By April 1979, the price was close to $16 per barrel and over the next 12 months, as the Islamic revolution in Iran gathered pace and Iraq made ready to attack its Ayatollah Khomeini-led Shia neighbor, the price shot up to $40 (this was a peak that would not be reached again for another 30 years, adjusted for inflation). The first oil boom in Saudi Arabia had been the result of a combination of three factors: rising production, a larger share of the profits and a higher oil price. The second oil boom was due mainly to higher prices and its effect on the country would be just as big. But first there was a terrible price to pay.

Muharram, the first month of the *Hijri* year 1400 (according to the Islamic calendar that began in 622 AD), corresponding to November–December 1979, was extraordinary in showing almost no government disbursements. It was as though the country had come to a halt. In fact, this was close to the truth. On the first day of the new century, the great mosque at Makkah, site of the Kaaba and the holiest shrine of Islam, was seized by extremists. It took a week for the mosque to be recovered and fighting continued in the catacombs beneath the shrine for another week. Zaki Yamani, Saudi Arabia's long-serving Minister of Oil, attributed the oil price rise in part to panic buying at this spectacle of internal dissension within the kingdom, and he increased production, honoring long-standing contracts at prices well below spot rates.

The windfall from the second oil price hike was far greater than from the first, and Table 4.1 shows how the consequent burden was once again mainly shouldered by the developing economies.

The first oil price hike had sent the industrial economies into recession and their recovery had been weak, dogged by persistent inflation. The response of economic theorists such as Milton Friedman was to declare that high inflation could be tackled by controlling the money supply. Right-wing politicians such as Ronald Reagan and Margaret Thatcher saw inflation, not recession, as the enemy, and high interest rates as the weapon to slay the giant. This would also strike at the power of organized labor. These monetarist ideas now held sway in the United States, and Mulford and Ingrams believed that there were great investment opportunities for SAMA as the developed economies slipped into recession in 1979–1980.

Table 4.1 Global cumulative current account surplus/deficit, 1973–1975 and 1979–1981 (billion USD)

Global data	*1973–1975*	*1979–1981*
Oil exporters	98	255
Industrial countries	19	-58
Less developed economies	–107	–244
Residual	10	–47

Total amounts for each set of three years. 'Residual' is statistical errors
Source: IMF Annual Reports 1976, 1982

Back in 1975, WB had recommended that most of the reserves should either stay in deposits or go into short-dated bonds. This was against the background of high inflation and American real yields that hovered around zero. But the second oil price hike panned out differently. President Jimmy Carter nominated Paul Volcker as chairman of the Fed in August 1979 and two months later in the 'Saturday Night Massacre,' Volcker raised the Fed funds rate by 1 percent to 12 percent over the weekend with a package of measures targeting inflation. Bond yields soared in real inflation-adjusted terms. By late 1980, Treasury yields were well into double-digit territory. As inflation fell, holders of long-dated bonds did well (Fig. 4.4).

This time round demand for oil dropped sharply, inflation peaked quickly and interest rates became sharply positive in inflation-adjusted terms. The WB team understood that high interest rates would make the recession worse, which in turn would bring down Western demand for oil. They had already seen this play out after 1975 when oil prices stagnated. But here was an opportunity to grow the foreign reserves through smart

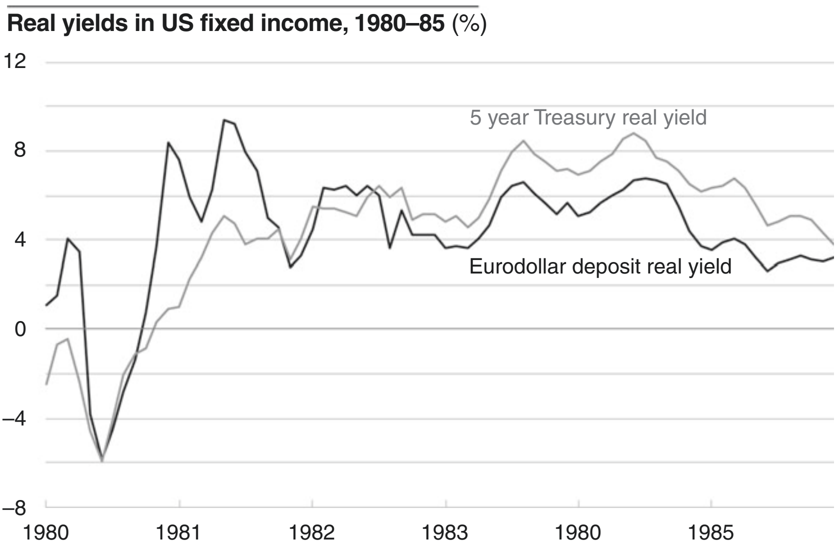

Fig. 4.4 Real yields in US fixed income, 1980–1985

Source: Federal Reserve Economic Data

investment decisions and they presented Al-Quraishi with a three-point plan to maximize investment return from the second oil boom. First, buy longer dated bonds. Second, cut exposure to bank deposits. Third, accelerate the move out of the dollar.

SAMA was in the position to make a huge financial killing as it uniquely had the money to buy bonds when they were cheap. Even though government spending was rising, the assets were growing faster. Oil income in 1981 was close to $100 billion, four times the level of 1974. But Al-Quraishi limited bond buying to maturities of ten years. In any event, the possibility of buying long-maturity bonds through the direct arrangement with the Treasury was not broached. The Treasury portfolio became much larger but its average maturity rose only from 12 months to 3 years. The Investment Department and outside managers bought bonds in the market, but the opportunity to acquire large amounts of longer dated Treasuries when yields were at post-1945 highs was missed.

On the question of reducing exposure to the commercial banking system, the Governor agreed. Al-Quraishi was convinced by the argument that the rise in American interest rates would bring about a crisis in those developing countries which borrowed in dollars, at the same time as the US recession reduced demand for their exports. Well before the Mexican crisis of 1982 alerted markets to the dangers of lending to developing economies, SAMA had reduced its bank exposure as a percentage of assets. Cutting the absolute amount on deposits was hard to do as oil income came into the call accounts (Fig. 4.5).

WB's last – and most critical – recommendation was a big push to reduce the dollar exposure. The team felt it was unwise to have so large an exposure in a single currency, especially one that had begun to use financial weapons for political goals. In November 1979, the Americans froze all Iranian assets in dollars after the revolutionaries seized hostages at the US Embassy in Tehran. Dollar assets held by US banks in the Eurodollar markets, mostly in London, were included in the freeze. There was no prospect of this happening to SAMA's assets but it would have been unwise to completely ignore the possibility in the future that dollars held anywhere in the world could be frozen. Al-Quraishi agreed that half of each month's oil income should be sold for Deutschmarks and yen.

Currency diversification on this scale was possible because the international monetary system had developed since the first oil crisis, and crucially the Germans wanted to cooperate with SAMA. Germany had run a current

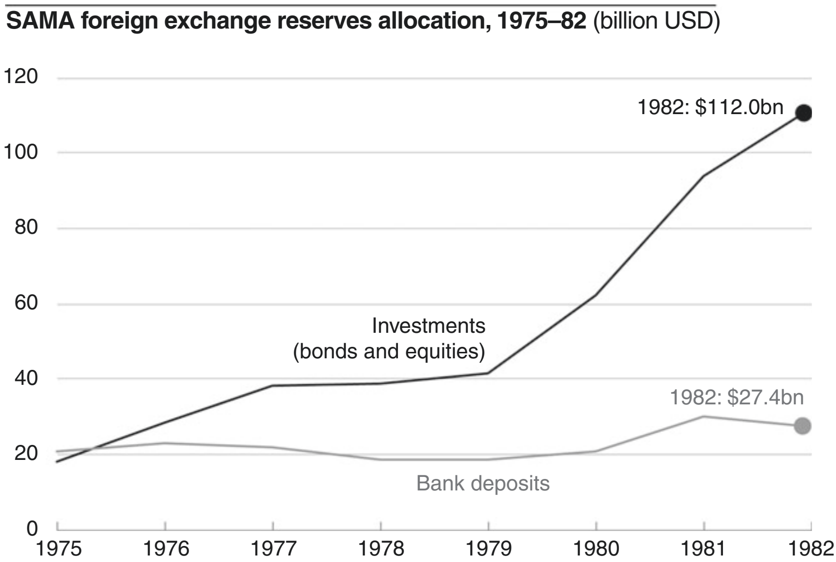

Fig. 4.5 SAMA foreign exchange reserves allocation, 1975–1982

Source: SAMA Annual Reports

Note: Investments include currency cover

account surplus during the first oil price hike of 1973–1975, but by the time of the second oil price boom, this had switched around to a deficit (Table 4.2).

Now the German government changed its mind about talking to SAMA. In March 1980, a team from their Finance Ministry flew to Riyadh. Bonn would follow Washington and Tokyo by offering SAMA direct purchases of Bunds.

The currency diversification was a big success. By 1986, under a third of SAMA's reserves were in dollars, less than was held in the German and Japanese currencies. For some years, as the dollar continued to strengthen, it looked as if acquiring holdings in non-dollar assets had been a mistake, but after the 1985 Plaza Accord, which weakened the dollar, the gains ran into tens of billions of dollars. This was certainly the most important investment call made by WB and endorsed by Al-Quraishi, and was largely the reason why the country was able to sustain budget and external deficits later in the 1980s.

Table 4.2 Cumulative current account surplus/deficit for industrial countries,1973–1975 and 1979–1981 (billion USD)

	1973–1975	*1979–1981*
UK	–12	32
Canada	–11	–10
Italy	–7	–11
France	–4	–10
Germany	28	–10
Others [mainly EU]	6	–56
Japan	–1	–11
US	19	18
Total	**19**	**–58**

Total amounts for each set of three years. 'Others' are other European Countries, Australia and New Zealand. Totals may not add up due to rounding.
Source: IMF Annual Reports 1976, 1982

6 Saudizing the Domestic Banks

'In theory, theory and practice are the same thing, but in practice they are different.' This aphorism is especially apt in the field of economic development. In theory, banks have a crucial role to play. They are supposed to encourage savings by offering checking and deposit accounts, and to mobilize credit to industrial projects in an efficient way. But in practice, banks often do not behave as the theoretical model tells them they should. They can be shut out of the development process by government policy and find it easier to focus on import financing and currency speculation. They can also act as conduits to send deposits offshore. In other words, they can be as much a problem as an opportunity for the central bank. In Saudi Arabia, banks were also hampered because a large section of Saudi society was uncomfortable with Western-style banking for religious reasons.

After 1974, SAMA struggled to develop the commercial banking system in ways that would help the domestic economy. Western-style banks were always going to find it challenging to operate within the constraints imposed by the Saudi legal system and Islamic culture. Many had particular difficulty in making loans at interest and in taking a legal charge over collateral which they could seize if a borrower defaulted. According to a theory popular at the time, as the economy developed this problem would go away, but in practice Islamic banking strengthened as an alternative to the Western-style banks. SAMA's regulation of money changers, after a local scandal in 1981 caused by the worldwide collapse of silver and gold prices as monetarism got a grip on inflation, led to the biggest money changer of all, Al Rajhi, becoming a bank later in the decade.

During the oil boom, the kingdom's banks could not lend out the money fast enough to match the flow of domestic cash. The ratio of loans made to deposits taken in dropped from 97 percent in 1971 to 60 percent by 1975. In 1978, only one riyal in three on deposit with the banks was used to make a domestic loan. Subsidized loans from the government's coffers were one reason. For example, the Saudi Industrial Development Fund (SIDF) offered interest-free medium and long-term financing for up to half the cost of a project. Why should any borrower visit a bank when a deal like this was on offer? In 1976, the loans and investments of commercial banks were larger than Specialized Credit Institutions' (SCIs) lending, but five years later loans from the SCIs were more than twice those made by the commercial banks.

The second issue with the banks revolved around SAMA's determination that the riyal should not become an international currency. The riyal lacked the fundamentals of an international currency. It did not have a conspicuous role in international trade and finance. The domestic financial markets were not well developed and were not accessible to foreign investors, and the riyal was not used as a reserve currency for foreign exchange holdings and investments. But the banks, finding it hard to make money at home, moved their assets heavily outside the country, and by 1986 one riyal in every two was held offshore, either as riyals in Bahrain or as foreign currency. Some of the offshore loans went back into Saudi Arabia, some were converted to dollars to fund the foreign contractors working on the country's development program and some were loaned outside the economy. An offshore market in the riyal made the implementation of monetary policy more difficult, especially as SAMA lacked some basic instruments necessary to do the job, such as the ability to provide the banks with safe and attractive assets onshore by selling them government or central bank bonds (Fig. 4.6).

Saudi banks became involved in the offshore business through deposit placements and foreign currency swaps to employ their excess riyal liquidity, simply because their deposits exceeded loans inside the country. The OBUs typically had an unhedged exposure to the riyal. The challenge for SAMA came in the form of unpredictable bouts of accumulating or losing foreign exchange as the OBUs adjusted their currency positions, which were accompanied by disruptions to domestic liquidity.

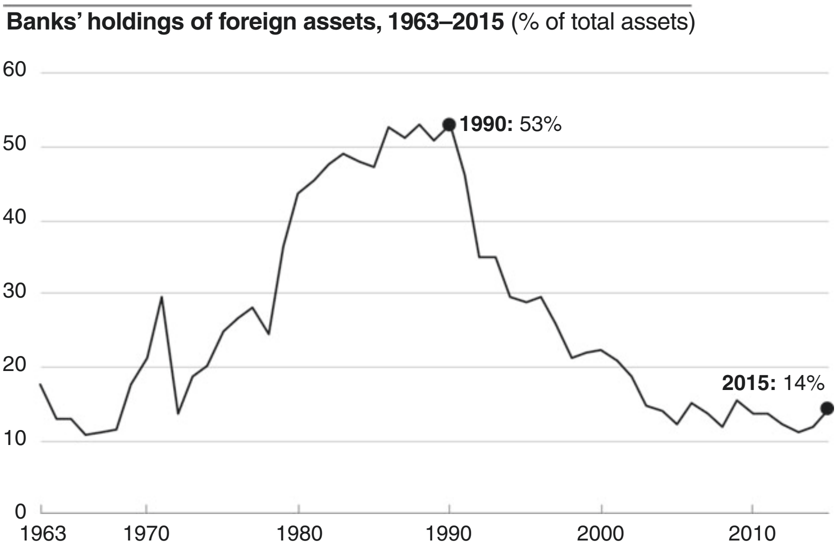

Fig. 4.6 Banks' holdings of foreign assets, 1963–2015

Source: SAMA Annual Reports

SAMA tried to get a handle on the situation without setting up capital controls, mindful of the disastrous episode in the 1950s. For instance, in 1983 domestic banks were ordered to consult SAMA before bringing OBUs into syndicated riyal loans and were prohibited from participating in riyal loans arranged outside the country. This had some effect, but the offshore business continued to be a headache.

The final issue where SAMA and some of the banks found themselves at odds was Saudization. When he took office, Al-Quraishi wanted to bring the country's banks firmly under SAMA's control by ending foreign ownership. By way of incentives, he held out the prospect of a Saudized bank receiving government deposits and a five-year tax holiday. But he did, in fact, have a stick ready: if banks did not Saudize they would be prevented from raising fresh capital, thus making their future expansion impossible.

In 1974, there were 12 foreign-owned banks in the country besides the two local ones, Riyad Bank and National Commercial Bank (NCB). Most went along with the idea and discussions revolved around technicalities. At issue were how much the foreign holders could retain, the price at which the local investors would buy shares and how widely the shares would be dispersed to prevent any single group taking control. The foreign owners kept 40 percent of the shares and were able to protect their investment by being granted a ten-year management contract. The local branch of Algmene Bank Nederland became Saudi Hollandi Bank and the British Bank of the Middle East became Saudi British Bank respectively. Saudi French Bank and Saudi Cairo Bank followed. SAMA made sure that government deposits were placed in the Saudized banks. This both helped to ensure their stability and gave SAMA an additional lever of control as the banks did not want to see the deposits withdrawn. The big winners were two banks which did not have to go through the process because they had always been locally owned: NCB and Riyad Bank. By now, they ran 70 percent of bank branches across the country.

But Citibank, which operated as the First National City Bank (FNCB), still held out. Walter Wriston, Citibank's aggressive chairman, gave in only when SAMA refused to allow US bank examiners to study FNCB's books and closed his branch office in Dhahran. Finally, Wriston capitulated and in 1980 Saudi American Bank was formed from FNCB with 60 percent local ownership. When United Saudi Commercial Bank was set up in 1981 by the amalgamation of three small banks, the Saudization process was complete.

As the second oil boom was quelled by Volcker with high real interest rates in the West, traditional inflation hedges fell sharply in price and bullion speculators went to the wall. There were losers in Saudi Arabia as well. Saudi Cairo Bank's top officials had been secretly trading in bullion and the bank was technically insolvent. They hid the losses as fictitious deposits with foreign banks. SAMA placed soft deposits with Saudi Cairo on concessionary terms in order to keep it afloat and later the Public Investment Fund took a 50 percent stake. In 1981, a smaller but arguably more significant event brought the Shariah-compliant side of the financial system under the central bank's wing. A Dammam-based money changer went bust, losing his clients' money on bullion speculation. Money changers were not considered by SAMA to be carrying on banking business but in reality they made up a sizeable part of the banking system. In response to public pressure, mainly from those who had lost money, SAMA began to regulate the money changers, including Al Rajhi which was the third largest deposit taker in the country. The end result was that, in 1987, Al Rajhi decided to become a bank and Shariah-compliant banking became a major factor in the domestic financial system.

7 Avoiding Mistakes

Ian Fleming's hero, the British spy James Bond, was an icon of the 1960s with his gas-guzzling sports cars. In Fleming's 1961 novel, *Thunderball,* the villain Blofeld, head of Special Executive for Counter-Intelligence, Terrorism, Revenge and Extortion or SPECTRE, decides to put the organization's funds in the safest currencies in the world: Swiss francs and Venezuelan bolivars, for reasons of prudence. At the time, Venezuela was ranked alongside Switzerland as a safe-haven currency because of its vast oil reserves, something that seems absurd today after decades of economic mismanagement by successive Venezuelan governments. Saudi Arabia, by contrast, did not mismanage its oil wealth. SAMA may not have got everything right, but crucially for success it did not make any big mistakes.

The subtlety of SAMA's approach can be shown by the fact that at the same time as Al-Quraishi was showing Walter Wriston a stick and stopped the Comptroller of the Currency from auditing his bank, he was simultaneously working with the Fed to improve banking supervision. Fed officials provided a draft manual and trained staff in Jeddah as well as supporting SAMA's banking training program for private sector bankers and its own staff. In 1980, SAMA was able to roll out a much more detailed bank inspection regime which allowed it, among other things, to analyze the banks' foreign assets as between riyals and foreign currencies

The petrodollar recycling years of 1975–1982 were a peaceful win–win solution. The international banks got Saudi deposits which they recycled to developing economies, and direct government-to-government arrangements meant developed economies received the funds necessary to finance their current account and budget deficits. Meanwhile, SAMA was able to manage its transition into the first major proxy sovereign wealth fund (since the assets were held in the account of the central bank instead of a special fund). Kuwait had possessed a small fund for some years and Abu Dhabi started one in 1976, but the giant of the decade was SAMA. It was a success story. The imported WB team blended well with the permanent SAMA staff and the big calls they made were all successful ones: to move from bank deposits into shorter dated government bonds in the 1970s and then to extend maturity in the early 1980s when real yields were extremely attractive. There were problems along the way, as with the telex machines, but SAMA developed a reputation as a sensible partner with whom to do business.

Currency policy was a secondary matter for Al-Quraishi: effectively SAMA was doing a 'dirty float' against the dollar and quelling public concern about inflation. Al-Quraishi's dealings with the banks were long and complex and he did not stop a sizeable offshore business developing. But he did prevent the riyal becoming an international currency and he achieved local ownership of the Saudi branches of the foreign commercial banks. It could be argued that SAMA should have taken a looser approach by allowing internationalization and free entry for foreign banks. But this would probably have brought more banking problems in when the oil market collapsed soon afterwards. Other countries in the Gulf had banking crises at this time. The United Arab Emirates, for example, had to rescue a number of banks in 1977, while the Souk Al-Manakh affair in Kuwait in 1982 left every bank save one technically insolvent. But SAMA, supervising a far larger banking system than Kuwait and the Emirates, had to deal only with the Saudi Cairo fraud. This may be counted a relative success. Probably SAMA did about the best job it could under the circumstances.

By the end of 1982, the second oil boom was fast shading into a slump. The oil price, which had finished 1980 at nearly $40, dropped to $30 as monetarist policies in the West deflated their economies and demand for oil fell. More importantly, Saudi production had been cut to try to stabilize prices. Lower prices plus lower pumping was a toxic combination for the flow of cash into SAMA's reserves, especially as government spending continued to rise. SAMA's foreign exchange holdings peaked in October 1982 at $140 billion. By September 1985, they were down under $100 billion. The boom era was well and truly over and the central bank faced a new set of challenges.

Notes

1. Vassiliev, *History of Saudi Arabia,* 413.
2. Federal Reserve Board, 'Flow of Funds Accounts of the US 1965–74,' Tables L210 and L211 give bank holdings of Treasuries, https://www.federalreserve.gov/releases/z1/current/annuals/a1965-1974.pdf (Accessed October 23, 2016). In any event, the banks needed some Treasuries to meet regulatory standards. Foreign holdings of Treasuries had risen substantially from 1970 but were only $58 billion in 1974.
3. Wikileaks, 'Kissinger cables.' Declassified cable: Kissinger to Jeddah Embassy December 3 1974, https://wikileaks.org/plusd/cables/1974STATE265655_b.html (Accessed October 23, 2016). Kissinger finally

brought the diplomats in Jeddah up to speed on a conversation that had taken place in Washington DC three months earlier before Ali's death.

4. Wikileaks. Declassified cable from Jeddah Embassy to State Department December 8 1974, https://wikileaks.org/plusd/cables/1974JIDDA07206_b.html (Accessed October 23, 2016)
5. Previously, Ali used a group of senior international bankers who flew down to Jeddah regularly to advise him.
6. It may not have been a coincidence that a member of the Barings dynasty, Evelyn Baring, Lord Cromer, who had been a managing director at the family bank, served as Governor of the Bank of England in the 1960s when he would undoubtedly have met Anwar Ali, and in 1974 was British Ambassador in Washington DC.
7. David Mulford, Packing for India (Washington DC: Potomac Books, 2014), 110. That is to say, the list amounted to over $20 billion. Most of the money was on current account and earned no interest. But the banks did not tell SAMA this.
8. Forty years on, the hotel is still open for business. It now has an internet connection and has retired the old telex machine.
9. The events of the period are grist to the mill of conspiracy theorists. David Spiro, The Hidden Hand of American Hegemony: Petrodollar Recycling and International Markets (Ithaca: Cornell University Press, 1999) is useful when he stops seeing American exploitation of its dominant position behind what was simply the most practical way of resolving the problem. Coming from a later intellectual generation, Rachel Bronson, Thicker Than Oil: America's Uneasy Partnership with Saudi Arabia (Oxford: Oxford University Press, 2006) is a straightforward account.

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The report on SAMA's foreign reserves in September 1974: Canonical ID: 1974jidda05097_b.

Jeddah Embassy's report of the WB discussions in August 1974: Canonical ID: 1974jidda04784_b.

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Jeddah Embassy's considered assessment of Quraishi in December 1974: Canonical ID: 1974jidda07206_b.

Saudization of the foreign-owned banks: Canonical ID: 1975jidda01928_b.

CHAPTER 5

Declining Foreign Exchange Reserves and Iraq's Invasion of Kuwait, 1983–1993

1 War for Oil

The war for oil that people in the Middle East had been expecting for so long finally erupted in 1991 in a small town called Khafji. This otherwise unremarkable place lies on the shores of the Gulf, in the Eastern Province of Saudi Arabia, 12 miles south of Kuwait City in the middle of the Khafji oilfield. On the night of January 29, three battalions, two armored and one mechanized, crossed the border from Kuwait, which had been occupied by Iraqi forces in August 1990, heading for Khafji. The Iraqi tanks were spearheading three armored divisions and 60,000 soldiers, a column of men and machinery so vast it stretched back ten miles from the border, nearly as far as Kuwait City. Yet despite the scale of the resources being put into the attack, it was largely unsuccessful. A fleet of fast Iraqi patrol boats intended to support the land invasion was driven back to Kuwait by British helicopters. The mechanized battalion was forced to retreat northwards, while one armored unit ran into American forces and was hit by helicopter gunships and close-attack aircraft. In the darkness, however, one armored battalion with supporting infantry managed to brush aside the Saudi defenders and reach Khafji. Once there, it deployed among the apartment blocks and wide boulevards of the town.

The task of liberating Khafji fell to the commander of the Saudi armed forces, Major-General Khalid bin Sultan, son of the then-Defense Minister and nephew of King Fahd who had succeeded King Khalid in 1982.

A. Banafe, R. Macleod, *The Saudi Arabian Monetary Agency, 1952–2016*, Financial Institutions, Reforms, and Policies in Muslim Countries, DOI 10.1007/978-3-319-55218-7_5

He sent in one of his subordinates, Lieutenant-Colonel Hamid Matar, with the second National Guard Brigade's seventh Battalion, composed of Saudi infantry in V-150 American personnel carriers and two Qatari tank companies using the French AMX-30 battle tank. The Americans agreed to provide supporting fire. Matar waited for darkness to fall. At dusk on January 30 the battle to retake the town began. Matar's first attack failed and he was reinforced by another National Guard battalion and a ministry of defense brigade before the fight was renewed the next morning, January 31. The battle continued throughout the day. An Iraqi relief column trying to get through to the town was caught and destroyed, and the Saudis then attacked from the north and the south. The besieged Iraqi general pleaded for permission to withdraw, but President Saddam Hussein refused and left his soldiers to their fate. Saudi troops eventually liberated Khafji after two days of street fighting. Hundreds of Iraqis were killed, wounded or taken prisoner. Combined Saudi and Qatari losses, by contrast were just 19, with 36 wounded. Khafji was the first land battle of the war to liberate Kuwait, a war that was to end less than a month later with victory for Saudi Arabia and its allies.

The battle for Khafji was just one incident in the Kuwait war, which was about control of the Gulf's oil. Mismanagement of oil money was at the root of Iraq's problems. Saddam's forces invaded Kuwait in August 1990 because he needed more oil. He had ruined Iraq's economy and believed he could become solvent again by absorbing Kuwait. Ten years earlier Iraq's economy had been healthy with annual oil income of $26 billion and another $35 billion as foreign reserves. But Saddam made the mistake of taking on Iran, having become convinced that he could inflict a swift defeat. It was a move which triggered a protracted and expensive war lasting eight years, which left the Iraqi economy in ruins. The foreign reserves were gone and the country owed $80 billion to overseas lenders, money it had no hope of repaying. In the year before the invasion of Kuwait, Iraq had earned just $13 billion in oil revenues. Its expenses were double that. International bankers would not lend Iraq any more money while the oil price stayed low. Raising the price was impossible unless other oil exporters also agreed to cut production. Or Iraq could invade Kuwait and appropriate its oil reserves – or use the threat of invasion to extort money from the Kuwaitis.

During the first months of 1990 Saddam started to put pressure on his Gulf neighbors. In June, he demanded that Kuwait lend him $10 billion. By July 27, eight Iraqi divisions were massed on the Kuwaiti border.

The Kuwaitis still thought Saddam was bluffing. But the Iraqi leader had already decided that the time for negotiations was over. On the morning of August 2, he invaded Kuwait, a move which led to a six-month crisis ending in war and defeat. [1]

Oil was at the center of Iraq's decision to begin the Gulf war. Saddam grossly mismanaged his country's oil revenue, but his biggest blunder had been to alienate the country's private sector and the banks. They had billions in foreign assets that could have financed the regime for years, but he could not persuade them to repatriate their dollars. Nobody in Iraq trusted Saddam with their money.

Why did Saudi Arabia have so much more financial success than Iraq during the 1980s? After all, the kingdom certainly shared one of Saddam's three financial problems – an inability to balance revenue and spending. However, the Saudi Arabian Monetary Agency's (SAMA's) skillful handing of its reserves generated good returns, which bolstered assets and meant that the long-feared crisis was perpetually deferred. But the biggest difference between the Saudi and Iraqi experiences lies in two words: profligacy and trust. Saddam squandered state funds in reckless foreign adventures – the war with Iran and the invasion of Kuwait. As for trust, the story of how SAMA invented the debt market illustrates the trust that the Saudi private sector had in its government. After debt was first issued, SAMA was able, paradoxically, to increase its foreign assets because the private sector brought back dollars and took up government bonds. Any analysis of SAMA's achievements during this period also needs to include a critique of the monetary policy aspect of the Kuwait war and its aftermath, the financial panic of August 1990 and SAMA's textbook response to restore confidence.

2 Economic and Fiscal Background: The 90 Percent Solution

Working out how much foreign exchange was likely to flow out every month was a major task for the White Weld and Baring Brothers team (WB) but they eventually got a handle on it. Overall leakage of government spending into losses of foreign exchange in the late 1970s and early 1980s averaged about 90 percent of total spending. If $100 million came into the reserves as oil revenue and the government's account was credited with the equivalent in riyals, about $90 million would quickly flow out again. To calculate foreign exchange outflows, you just needed to know how much the government was spending and subtract 10 percent from the number. Hence the phrase 'the 90 percent solution.' Now that the budget had gone into deficit and 'the 90 percent solution' remained unchanged it became a subject of intense and pressing interest.

Public spending was cut and development projects effectively came to a halt. But Fig. 5.1 shows that even as spending fell, the budget deficit rose. By 1989, the gap between spending and revenue was $11 billion and tougher years lay ahead.

The non-oil economy should have been coming on stream to develop export markets and make substitutes for imports; but it did not. Nobody had expected this outcome. The giant Five-Year Plan of 1975–1980 had been intended to relieve dependence on oil. Although the country had few natural advantages besides oil, economists predicted that local labor would replace foreign workers and the cost of assembling materials would be less than importing finished products. Modern industries were being built around oil. This was at least the theory. What evidence was there by the mid-eighties that the Plan had been successful?

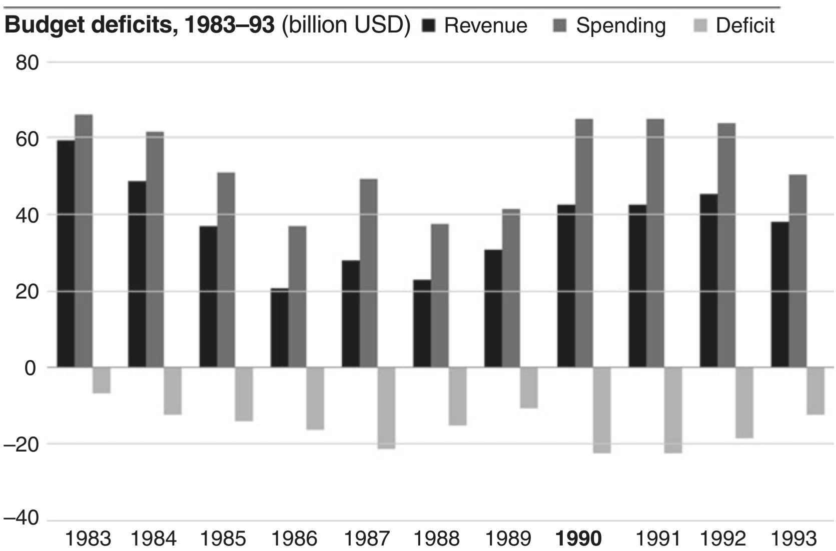

Fig. 5.1 Budget deficits, 1983–1993

Source: SAMA Statistical Bulletin

Note: Data for 1990 and 1991 was reported as combined: the chart divides it into two equal parts

Tax revenues should in theory have gone up as the local economy grew. But the evidence was not there – in 1993, for example, non-oil revenue amounted to only about one-fifth of the budget. And a large part of this consisted of transfers from SAMA and state corporations (Fig. 5.2).

Figures for import substitution and exports also showed a depressing picture. Non-oil exports were less than one-tenth of oil exports and far below the planners' target. Diversifying the economy away from oil was still a distant dream.

Although public spending kept the economy from collapsing, the country nonetheless went through a major depression with output and prices falling. In inflation-adjusted terms, Gross Domestic Product dropped by 24 percent between 1982 and 1987. The oil sector fell by over half but the decline in the non-oil private sector was held to 10 percent because it was supported by government outlays. The picture improved as the oil price recovered at the decade's end; but on the eve of the Kuwait war the Saudi economy was still smaller than it had been eight years earlier. Without countercyclical deficit spending, the situation would have been far worse (Fig. 5.3).

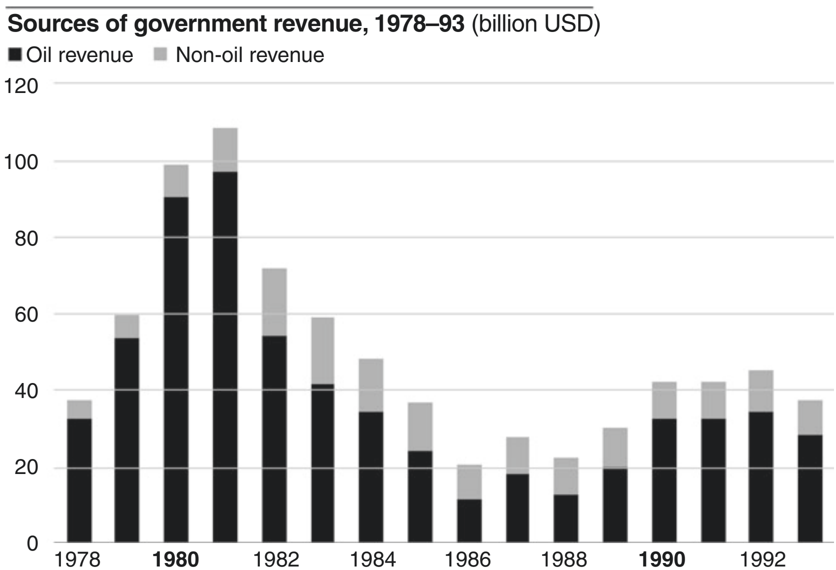

Fig. 5.2 Sources of government revenue, 1978–1993

Source: SAMA Statistical Bulletin

Note: Data for 1990 and 1991 was reported as combined: the chart divides it into two equal parts

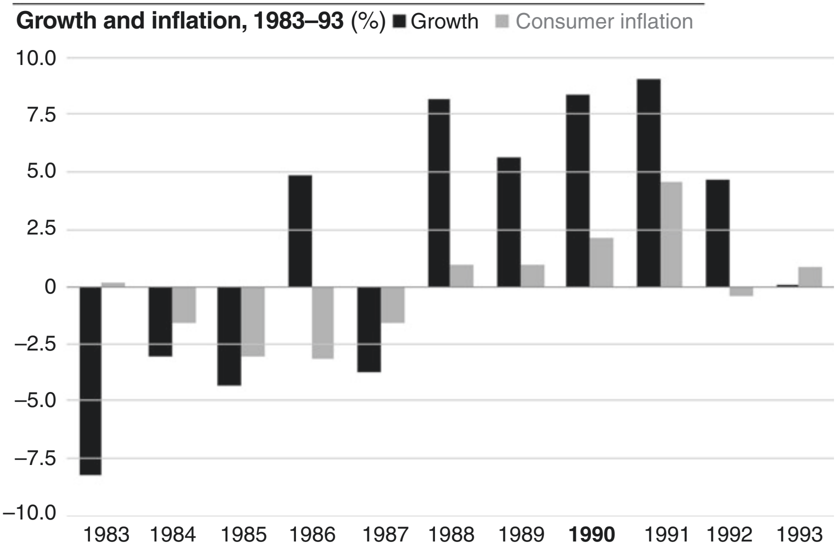

Fig. 5.3 Growth and inflation, 1983–1993

Source: International Monetary Fund, World Economic Outlook Database, October 2016

As long as SAMA's reserves held up public spending could continue, even with big budget deficits. Why did the reserves matter so much? The best way to look at this is to compare Saudi Arabia's finances with those of the United States. The crucial difference between the two is that the US government taxes the private sector and spends the money. When the budget is balanced and revenue equals spending – which admittedly rarely happens – the result is mostly just a shift in demand from one part of the economy to another. But a big spending impulse only happens if government spends more than it taxes. This is the multiplier effect as the public spending ripples through the economy

In Saudi Arabia, most of the government's spending comes from oil money. This is a pure injection of purchasing power because the money is earned from countries that use Saudi oil. Saudis themselves pay almost no taxes. Any spending over and above local tax revenue is, therefore, the equivalent of a budget deficit in America, since almost none of the government's revenue comes from taxing Saudis. What effect does government spending have on the Saudi economy? The difference with the US is that Saudi Arabia has a very large import sector. Food, clothing, automobiles, in fact most goods, are imported. Labor is also imported in the form of millions of foreign workers, who send their earnings home. The huge leakage into imports in each spending round in the non-oil economy consequently weakens the multiplier effect. Countries exporting goods and labor to Saudi Arabia benefit from the demand instead. India, for example, pays for oil but gains from the remittances of Indian expatriate workers. As money is spent on imports, SAMA's reserves fall. The 90 percent solution explains the key dynamic underpinning the economy.

The 90 percent solution worked most of the time as a predictor of foreign exchange outflows, but it was still something of an over-simplification. In practice, there were other factors at work, all related to the financial sector. The banks were naturally seeking to make secure investments with parts of the assets they were not lending out. In a developed economy, government debt is bought by the banks to fill the gap, but this was not issued until the late 1980s in Saudi Arabia, with the result that the banks tended to accumulate foreign assets as they turned their riyals into dollars and invested them abroad. The rest of the private sector (businesses and wealthy individuals) did much the same thing, diversifying their assets by investing outside the country. The actions of this second group of investors were virtually impossible to predict. However, what *could* be predicted was that if private sector banks, businesses and wealthy individuals all expected riyal devaluation, they would be likely to take the prudent step of moving their money into the dollar. This would be dollarization of the economy – a worry always present in the minds of policy-makers – and really drain down SAMA's reserves. After 1986 when a credible dollar peg was established, this stopped being a source of anxiety until the Iraqi invasion of Kuwait which, not surprisingly, caused financial panic, albeit short lived.

All these elements played important roles at various times during the eighties and they gradually began to work to SAMA's benefit. The establishment of the government debt market in 1988 provided the banks with a secure long-term investment home for their riyals, enabling them to run down their net foreign assets. As the domestic economy was squeezed, some of the private assets came back from abroad, especially after the threat from Iraq had been removed by the 1991 Gulf War. Finally, the credibility of the dollar peg after 1986 reduced the likelihood of speculative attacks on the riyal.

For SAMA, the 90 percent solution meant that the question of financial solvency was simple. The only variables to consider in estimating how fast the foreign exchange reserves would drop were oil income and government spending. If government spending (after allowing for local revenues) was greater than oil income then reserves would fall. In other words, running budget deficits meant the reserves would fall. And the budget deficits which started in 1983 ran on until 2003 when the average price paid for Saudi oil rose above $25 per barrel for the first time in 20 years.

3 Changing Personalities

The Wadi Hanifah (*wadi* means valley in English) is one of the great riverbeds of central Arabia. Most of the year it is dry until the first winter storm of the year when the skies open and several inches of rain fall in an hour. The water runs off the dry desert floor into the *wadi* and a flash flood, a wall of water several feet high, rushes downstream carrying with it uprooted trees, stones and small drowned animals. The *wadi* runs between Riyadh and the Tuwaiq escarpment, and its course on the outskirts of the city is marked by market gardens and small clusters of date palms. A few miles north of the city a dam has been built to prevent the flash floods from washing through the farms. The historic walled town of Diraiyah lies just below the dam on the edge of the *wadi*.

Nowadays, Diraiyah is preserved as a national monument and some of the old palaces have been rebuilt, making it a popular place for family outings and picnics. Across the road from the main entrance to the walled town sits a palatial house surrounded by a garden of date palms. It has been built in traditional style with long cool rooms with high ceilings running into one another. Meals are prepared in a kitchen separated from the living quarters by the courtyard. It is the perfect place for entertaining guests. After SAMA moved to Riyadh the house was rented by David Mulford until his departure in 1983 to become Assistant Secretary of the US Treasury, soon followed by his right-hand man at SAMA, the charming and urbane Tom Berger who became Deputy Assistant Secretary. The Diraiyah palace was inherited by Mulford's successor. Steve Wilberding was a distinguished Vietnam veteran in his forties, a wiry figure with dark brown wavy hair and a powerful nicotine habit. He was an old Wall Street hand with Merrill Lynch, which had taken over the American side of the WB contract when they acquired White Weld. His approach was technocratic and analytical. An early user of the personal computer, he was frequently found in front of a screen, a cup of sweet tea to hand and cigarette burning in the ashtray beside him, as he doggedly analyzed the financial problems that SAMA faced.

Personalities changed on the Saudi side as well. Ahmed Abdullatif was promoted from running the Investment Department to become Deputy Governor for Foreign Investment, moving up from the second to the fifth floor. He was succeeded by Ahmed Al-Malik who came from the defense ministry.

In April 1983, news came of a much bigger change. At the opening of the Saudi Investment Bank's offices in Riyadh, Governor Al-Quraishi toured the vaults and was given a gold key to box No.1 as a gift. He jokingly asked whether this was meant for him, or for the governor of SAMA. Nobody knew what he meant until three days later the news was made public that he had chosen to step down. The 43-year-old Vice-Governor Hamad Al-Sayari, who had moved to the job from one of the quasi-government organizations, was appointed to stand in as acting governor and was confirmed in office two years later. SAMA's move to Riyadh had been a sign that the days of the bank's staffing by the Hejazis from the western region on the Red Sea coast were drawing to an end. As they left, their places were filled by Saudis from the central province of Riyadh, formerly Nejd.

These were pleasant years for the WB team members. The new head office was ready in 1985. It was a vast improvement on the old one. For one thing, it had an excellent staff canteen. Investment issues were often discussed by WB members and Saudi staff around a lunch table. Leonard Ingrams was appointed to the post of special adviser to Governor Al-Sayari. His successor was another holder of an Oxford doctorate, Jeremy Fairbrother, a scientist by training, who had the ability to arrive intuitively at the same decisions that Wilberding reached through step-by-step validation. His wife Linda became a star on Saudi Arabia's English-language TV channel, hosting a weekly magazine program called 'In Focus.' Good hotels, restaurants, bookshops and record stores were opening. The Fairbrothers threw themselves into the vibrant cultural life of the desert city. A classical orchestra rehearsed in the basement of a hotel and the British team members of WB staged amateur dramatics with a troupe that had its own playhouse at an out-of-town compound. Western musicians flew in to give concerts to diplomats and their guests. It was a happy time when modern facilities were up and running, and before the effects of the fall in wealth and spending were felt.

4 Selling Assets: 1983–1990

The monthly 'carve-ups' as they were known at SAMA changed their character in 1983. The investment meetings had acquired this nickname because historically the work had consisted of estimating the cash flow for the month and recommending to the governor where it should be placed: into equities, bonds and deposits, and how much should go into non-dollar currencies. Now the discussions turned to which assets to sell. Equity holdings were not affected: in fact, as we will see below, SAMA was actively buying equities. But bonds and deposits were a different matter. Increasingly, they were not reinvested, but put into the call accounts where they disappeared to meet the demand for dollars (Fig. 5.4).

At the start of 1984, Wilberding and Fairbrother told Al-Sayari that if things did not change over the next year, there would not be enough positive cash flow to meet the daily outflows of dollars. SAMA would have to enter the bond markets as an active seller and raise cash, and the team wanted to train up for this by trading the bond portfolios to improve the return.

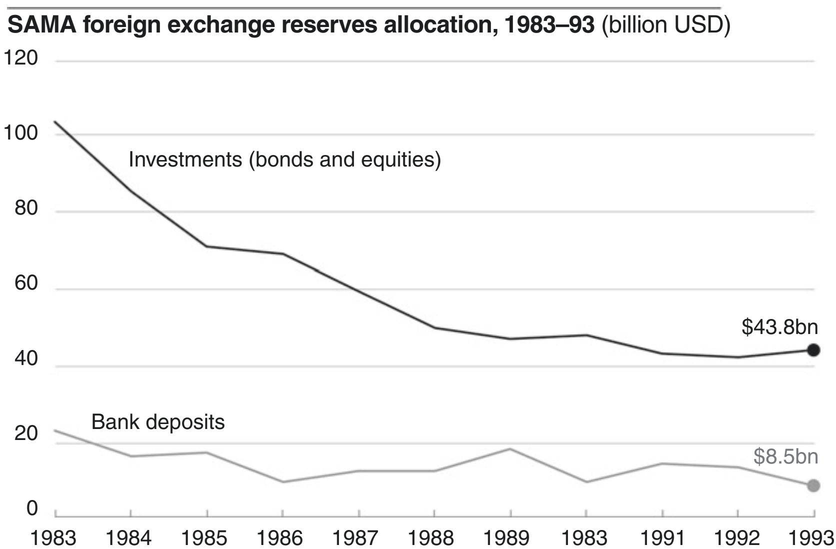

Fig. 5.4 SAMA foreign exchange reserves allocation, 1983–1993

Source: SAMA Annual Reports

The bonds held under the three deals with America, Germany and Japan could mature naturally, but they would not provide enough money. Some bonds would have to be sold. The WB team argued that the US currency was fundamentally over-valued and that dollar bonds should be sold first. Only if the dollar cash flow was inadequate to keep feeding the call accounts should other currencies be sold. This plan paid off the following year when the dollar fell sharply as a result of the 1985 Plaza Accord reached by the G-5 nations on currency intervention. Al-Sayari also agreed that the International Monetary Fund (IMF) would receive no more big allocations of money. Finally, some of the external bond managers were dismissed, especially those who were not achieving their target index return – the first time that the central bank took this step.

The department's workload increased dramatically after the governor gave the go-ahead for it to start trading the internal bond holdings. The Saudi members of the team already handled most of the routine work of placing deposits and executing the monthly bond-buying program. Now they started swapping bonds of the same duration (duration is related to maturity and measures the change in a bond's price when its yield to maturity moves). This practice had the advantage of getting the markets accustomed to SAMA's activities. Opportunities to do duration-neutral switching and increase yield (typically by buying high coupon issues that traded well above par) abounded 30 years ago when bond markets were less efficient than today.

The routine was that each day the dealers talked to the major investment banks around the world. The banks were given the criteria for a bond swap and asked to put up proposals for swaps in cases where there was a gain in yield to maturity but no change in duration or credit rating. Their proposals increased the yield on SAMA's portfolio and the dealers learned a lot from the work. By 1986, bond sales were being inserted into the swapping plan. Typically, a SAMA dealer would phone up an investment bank and pretend he was executing a bond swap using two counterparties, as had happened so many times before, but he would only do one leg of the swap. The dealer concerned only saw the bond sale and not the subsequent purchase with the other counterparty (which never happened) so he did not grasp that the trade was an outright bond sale. Meanwhile, the moment was approaching when, unless government spending was matched by income, non-dollar bonds would have to be sold in markets that were less liquid than New York.

Spending was not the core of the financial problem: it was income, and particularly oil income. In the first oil boom, the kingdom had benefited from higher production and higher prices, but after 1982 the situation was reversed. Both the quantities produced and the money received for each barrel sold dropped sharply. The price fell from an average $34 in 1982 to $13 four years later and was at the same level in 1989 just before Iraq invaded Kuwait (Fig. 5.5).

As world demand weakened, Saudi Arabia cut output to try to stabilize the price. The monthly oil payment made by Aramco to SAMA was over $8 billion in 1982. Over the next three years, it dropped to just $2 billion. The crisis year was 1986, when oil revenue fell by half from the previous year. Yamani, the long-standing Minister of Petroleum, had been using the kingdom as a 'swing producer' within OPEC. He hoped that by restricting its output, Saudi Arabia could force a recovery in oil prices. By July 1986, the spot price was under $10 and SAMA received only $700 million from Aramco that month, less than a tenth of the 1982 number (Fig. 5.6).

In September, Yamani publicly suggested that the oil price would stay under $18 for another four years. If this happened, the consequences

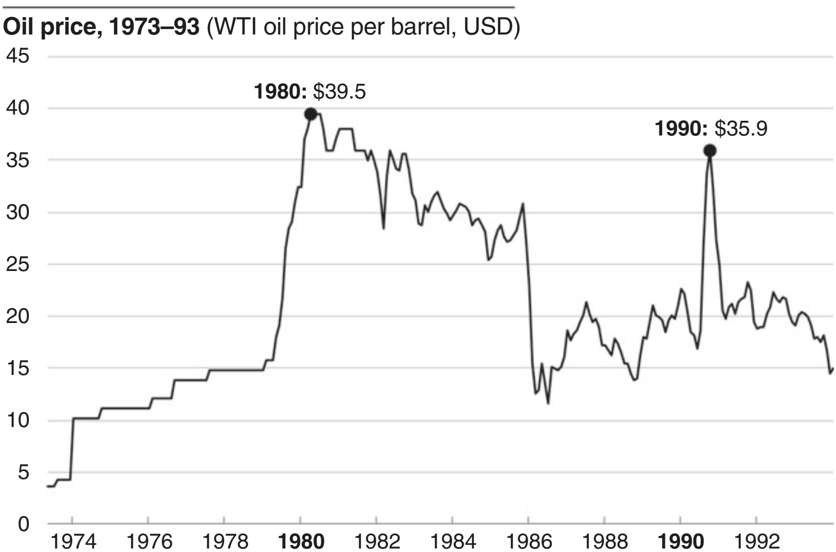

Fig. 5.5 Oil price, 1973–1993

Source: Federal Reserve Economic Data

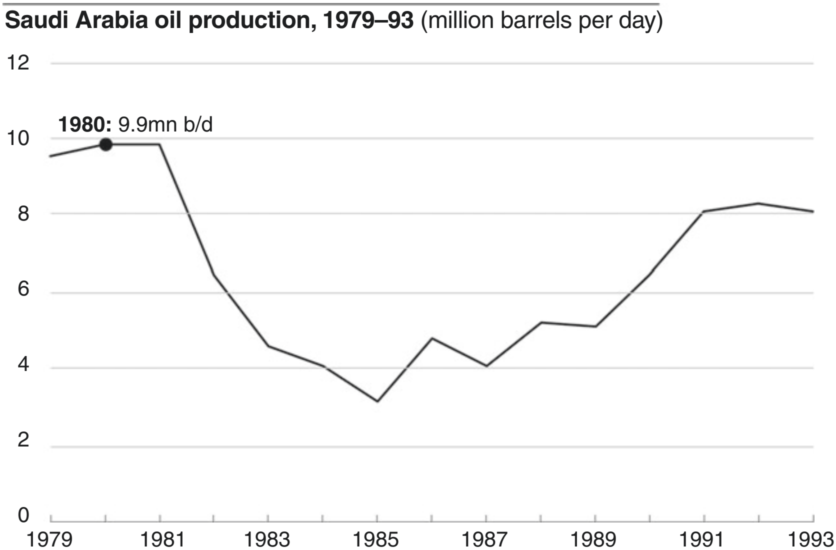

Fig. 5.6 Saudi Arabia oil production, 1979–1993

Source: SAMA Statistical Bulletin

would be devastating. Al-Sayari called Wilberding into his fifth floor office; with him came Bill Black, who had replaced Fairbrother as the third and final head of the Barings team the previous year. A tall quiet Scot in his fifties with a fine ruddy profile and a mane of gray hair, Black was an accomplished linguist, a lover of foreign places and skilled at business relationships. He gained the Saudis' confidence and stayed on as an independent adviser for eight years after the rest of the WB team left in 1989.

Al-Sayari wanted to know what would happen to the foreign reserves if Yamani was correct. The WB team told him how serious it could get. Assets would fall to $25 billion by 1990, which they considered a dangerously low level. The 90 percent solution dictated that the only thing to do was cut spending and raise taxes to rein back the spending (and therefore importing) power of the ordinary Saudi. Finance Minister Mohammed Abalkhail felt he had no choice but to freeze public spending, with the result that the Saudi government stopped paying some of its bills. Civil servants received their monthly pay, but foreign contractors got nothing. Within a few weeks, there was gridlock. Since foreign firms were not being paid, they stopped paying their bills in turn and suppliers of all sorts quickly ran out of cash. Soon nobody was paying bills.

Around New Year 1987, the cutbacks hit SAMA. Office perks, such as trips abroad, were restricted and projects put on hold. Working days were filled by the mechanics of selling dollar bonds. Oil income had lifted at year-end but fell again in early 1987. The gap between inflows and outflows was running at $40 million per day. In February, the Fed raised rates and, as interest rates rose in the United States, bonds around the world began to sell off (Fig. 5.7). SAMA needed to sell non-dollar bonds now if their price was falling. Japan was the first target. One spring morning, the WB team and the department's dealers came into work very early. They wanted to catch the Tokyo bond market in the middle of the afternoon in Japan, before liquidity drained away near the market close. Al-Sayari believed that selling Japanese bonds was bound to attract publicity, but it had to be done. SAMA was the biggest holder of government bonds outside Japan. The WB team members parceled out the sell orders. Each man sat at his own telex machine and started typing out the sales orders to different Tokyo banks. When the messages were ready to go the chief dealer gave the signal. The 'send' buttons were pressed simultaneously and the biggest sale order in the history of the Japanese bond market arrived at half a dozen Tokyo banks.

The team sipped tea and waited. Before long, the responses came chattering back down the telex lines. Every bond that SAMA wanted to sell was taken up at the right price. The team anticipated the gossip and rumors that would feed back to Riyadh via the world's financial markets. The sales continued for weeks. By the middle of 1987, it was the turn of the Frankfurt bond market and by year-end nearly $15 billion equivalent of non-dollar bonds had been sold, and the proceeds turned into dollars to feed the foreign exchange outflow.

The weak oil price produced strains elsewhere in the government. For example, the Defense Ministry ran out of money to make payments on the fleet of Tornado attack aircraft they had bought from the British. The original deal involved an exchange of oil for planes, but the oil price had fallen so steeply that the country was behind in its payments. One day in July 1987, half a billion dollars was taken from SAMA's dollar call accounts without any notice. It was credited to Saudi International Bank in London and sat there out of SAMA's reach. The interest was used to pay British Aerospace. This was a common practice at the time, when money was tight and interest rates were high. A pool of public money would be segregated for a number of months and then SAMA

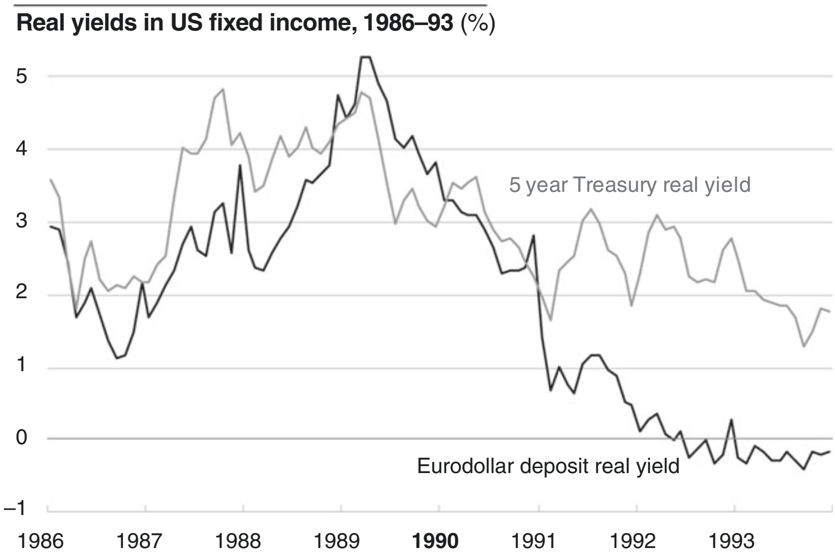

Fig. 5.7 Real yields in US fixed income, 1986–1993

Source: Federal Reverse Economic Data

would be reimbursed. But although the capital sum was intact the interest that had been earned had gone elsewhere.

SAMA's assets were of course not the only foreign ones held by the government. For instance, Yamani handed over $4 billion from the oil ministry before he was replaced in 1986. When nearly $20 billion in other assets owned by the government but managed by SAMA are included, the total, two months before Iraq's attack on Kuwait in 1990, was just under $80 billion, as Figure 5.8 shows. The picture was better than WB had expected several years before – on the assets side the non-dollar holdings had gone up in value as the greenback fell, and the inflow of cash was rising as the daily production of oil went up from 3 million barrels in 1986 to 5 million in 1990.

5 Buying Equities: 1983–1990

From the outside, SAMA's head office looks like a giant cardboard box. As you walk toward the huge bronze doors, you see the box has thin fluted columns running from ground to roof, and you hear a loud humming noise

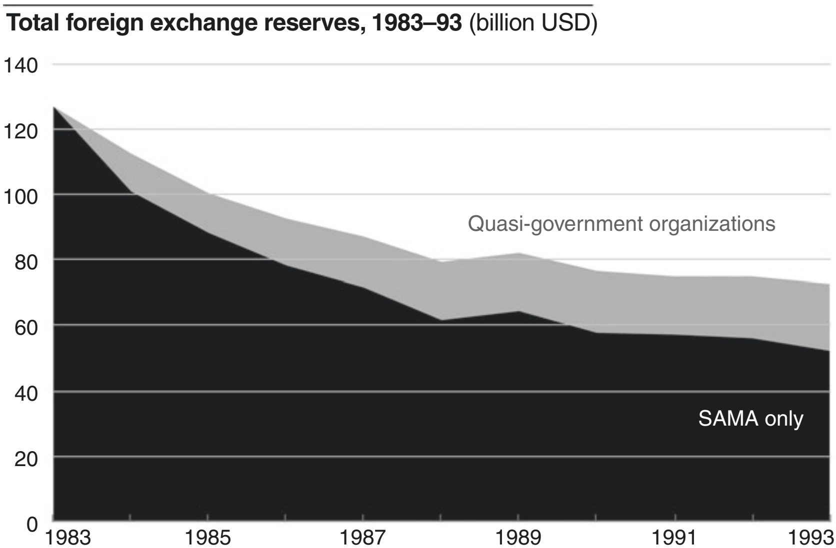

Fig. 5.8 Total foreign exchange reserves, 1983–1993

Source: SAMA Annual Reports

like a helicopter flying past. That is the sound of the air conditioning units set inside the flat roof. The bronze doors are part of the panoply of a central bank. When SAMA was still in Jeddah, Ahmed Abdullatif, the head of the Investment Department, used to joke that the authenticity of a central bank could be judged by the size of its doors, and promised vast ones when the new headquarters were finished. Abdullatif got his wish in 1985 when SAMA occupied the building. SAMA's doors are twice the size of those of the Bank of England. Inside, a massive hall greets the visitor. You almost feel that you are outdoors again. All the offices and meeting rooms fit into the sides of the box and the central space dominates everything.

The sheer size of the atrium gives the illusion of not really being inside a building at all. The eye moves easily to the furthest corners, upwards to the glass dome in the roof and down again to the sheen of light it casts on the slightly slippery marble floor and the reflecting pool. The five-storey building (the Investment Department is on the second floor at the back) conveys an air of tranquility that would be envied by any central bank. At intervals throughout the day employees leave their desks momentarily to lean on the balcony and look at the view.

On the morning of Tuesday October 20, 1987, anybody looking across the atrium would have seen the two heads of the WB team walking toward the elevator leading to the governor's office on the fifth floor. Carrying piles of green folders, along with pocket calculators and Reuter's printouts, Steve Wilberding and Bill Black were in no mood to appreciate the view. They felt sick. The US stock market had fallen nearly a quarter the previous day and SAMA's equity holdings at nearly 20 percent of foreign exchange reserves had suffered significant damage – as had their own private portfolios.

Nobody in the WB team was a specialist in investing in equities. Nonetheless, SAMA had become a major player in world stock markets, using outside managers who were given mandates in a particular market. When the WB team first arrived in Jeddah, they found several equity portfolios; but by 1981, equities were still only 4 percent of the total. That changed as a result of the second oil boom. Al-Quraishi reasoned that if the WB team was right and bond yields were set to fall as a result of monetarist anti-inflation policies in the developed economies, then sooner or later stock markets would look cheap.

Al-Quraishi's target was 10 percent in equities, which meant a much bigger portion of the cash flow. But he wanted this done through the anonymity of third-party portfolio managers. While the Kuwaitis were building up big holdings of shares, SAMA did not get involved in buying direct. The timing of the move was good: at the end of 1982 equity markets began to rise, led by Wall Street. But oil revenue continued to fall to a level below that of spending. This had the long-predicted consequence of a drop in foreign exchange reserves, creating a dilemma: should SAMA continue to invest in volatile equities? The potentially high returns seemed to be part of the answer to the problem and the investment program continued until the equity holdings were double Al-Quraishi's target (Fig. 5.9).

Managing the equity portfolios was much more complex than looking after direct holdings of bonds and cash. By the end of 1985, there were 20 fund managers running 26 portfolios with thousands of individual stock investments. Equity managers regularly came to Riyadh to update SAMA on their activities. All members of the Investment Department could attend these meetings, and a WB member sat in and took notes. At the end of the quarterly round, WB gave a presentation to Al-Quraishi (and later to his successor Al-Sayari). More and more time at the monthly investment meetings was spent discussing the

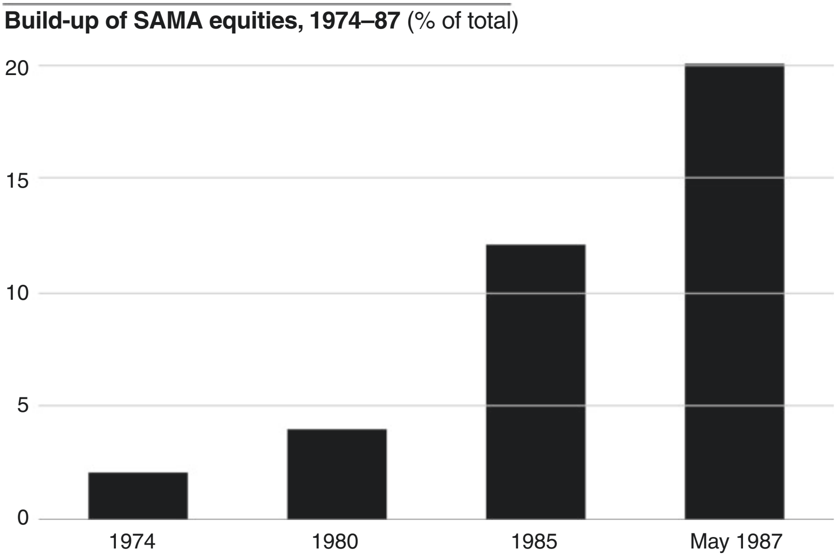

Fig. 5.9 Build-up of SAMA equities, 1974–1987
Source: Authors

equity markets and the performance of fund managers. Which equity market would be the next one to take off? From being a small part of the portfolio, equities had assumed great significance and managing them took a lot of time.

This was partly because of the difficulty (many would say impossibility) of linking a manager's historic performance to his ability to do well in the future, or to predict where equity markets were going. These were matters that the WB team could help with by analyzing the data. But it was also an administrative burden. SAMA had a small Investment Department: no more than 30 people in total, in an age when data processing was in its infancy.

One issue that swallowed up a lot of time was the Arab League boycott of companies doing business with Israel – an attempt to isolate the Jewish state economically, a measure that has not consistently been enforced. The list of companies arrived from Damascus, the headquarters of the boycott campaign, in the form of a vast computer printout. The American members of the WB team would have nothing to do with the list, saying that they would be breaking their laws by doing so. So the job was left to the

British. The banned companies had to be checked manually against each manager's holdings, a process made more laborious by the fact that each name had to be traced up to the top listed company. Then the fund managers would be told to sell the offending holding. SAMA's senior management was also determined not to invest in alcohol, pork products, gambling or pornography, all aspects of Western business of which they disapproved. Fund managers were given the general guidelines and told to get on with it.

Al-Quraishi's insight into the potential of the stock markets in the early 1980s was correct and the value of SAMA's equities soared. By May 1987, they accounted for nearly 20 percent of foreign exchange reserves; but the team missed the warning signs that the bull run was coming to an end. WB's last review before the October crash concluded that equities were not likely to fall soon. A month later Wall Street crashed. At the October 20th meeting the governor agreed that trying to sell equities would only spread further panic around the world. Wilberding and Black told him that there would be a bounce in prices even if the crash was the harbinger of an extended bear market, and SAMA still had some months during which it could get better prices than the current ones. They suggested setting a target for the stock market: when the index went up past the target, then SAMA would sell stock. The Governor agreed and the idea was applied to the Tokyo market as well. By summer 1988, $4 billion of equities had been liquidated. The episode was a warning to Al-Sayari that his timescale for investments was now shorter. He had been unable to sit out the crash because the fall in foreign exchange reserves had forced him to liquidate assets, even equities.

SAMA's first major venture into the equity markets had taught the bank a couple of expensive lessons. Nobody had warned of what would happen. None of the equity managers predicted the crash or advised SAMA to take its profits and move the money it had invested with them. In the aftermath, the governor became far more skeptical about what his managers said. He learned from bitter experience about the principal-agent problem in fund management, namely that SAMA as principal and the managers as agents did not share the same goal. Managers wanted to maximize their fees, even if that was at the expense of their clients' returns. Looking back, SAMA's decisions had been the wrong ones for the time frame within which it needed to operate. It had bought into a rising market when it could not bear the pain of losses. Equities were a long-term investment and SAMA would return to them later.

But another pressing problem in the New Year of 1988 was domestic, not external, and concerned debt, rather than equities. Abalkhail at the Finance Ministry had entrusted Al-Sayari with issuing Saudi government debt.

6 Issuing Government Debt: 1984–1993

It is a reflection of SAMA's solid reputation that it was asked to assume the role of a debt management office, rather than allocating this responsibility to the Finance Ministry or another department.[2] From this point onwards, Saudi Arabia's central bank managed the government's debt, as well as looking after its foreign currency assets.

Leonard Ingrams, the English side of the original WB team, was the architect of Saudi Arabia's first domestic money market instrument. He had enjoyed his time in Jeddah and Riyadh and disliked being back in London, so he left Barings and worked for SAMA as an independent adviser, in particular on the growth of offshore banking units (OBUs) in Bahrain. From time to time, the chairmen of the commercial banks would be invited into SAMA for coffee and a chat, part of Al-Sayari's efforts to ascertain why they had so much money offshore. The bankers sipped their coffee and complained that the kingdom had no government debt to buy, so they did not have the option of earning a return at home by investing in the highest quality domestic instrument, that is to say, one issued by the government.

Ingrams suggested to Al-Sayari that he should create a domestic money market instrument that would be desirable to the commercial banks because it offered a good return. SAMA would benefit because it would gain direct influence on domestic liquidity through its ability to vary the amount of the instrument issued. The new vehicle would also attract some of the commercial banks' money back into the country. The OBUs operated in dollars as well as riyals, so discouraging their businesses in Bahrain was a way of slowing down the banks' demand for dollars from SAMA.

A name was needed for Ingrams's new instrument. In an echo of the debate some 30 years before about a name for SAMA, it could not be called anything that smacked of paying interest. Al-Sayari accepted the suggestion of Bankers Security Deposit Account (BSDA). These were designed to compete with dollar deposits because the Saudi interbank rate was closely related to the dollar interest rate. The essential idea behind the BSDAs was simple and profound: the banks should actively want to buy them. The first BSDA was issued in February 1984 for a three-month term. It was a zero coupon instrument which was issued at less than 100 and repaid at par. This was in line with Treasury and central bank bills around the world and also circumvented the clause in SAMA's founding charter that prohibited it from paying interest.

Over time BSDAs became an instrument for regulating domestic liquidity. The banks could temporarily sell overnight up to 75 percent of the value of a BSDA back to SAMA (technically this is known as a repurchase or repo facility) whenever they wanted. By making a repo more or less attractive, SAMA could drain or inject liquidity into the money market.

BSDAs had some success in deterring the banks from building up their offshore assets but the program stayed small. In the early 1980s, 30–40 percent of banks' offshore assets were in riyals, either invested back into the country or lent outside. By 1990, only some 10 percent of offshore assets were held in the riyal. The peak in the use of riyals coincided with the introduction of BSDAs. But the fall was probably more due to other factors, since the liquidity shortage in the domestic economy meant that conditions were changing for the banks, which no longer had so much surplus liquidity in riyals to put into the OBUs.

The significant fact was that SAMA and the banks had engaged in a mutually profitable enterprise. SAMA had learned how to price, issue and use a government obligation and the banks had embraced it as a liquidity tool. The repurchase or repo facility incorporated into the mechanism was particularly important. A bank could repo a BSDA with the central bank to meet a liquidity shortage, a procedure SAMA made more flexible over time, allowing the possibility of multiple repos during a given day. These lessons proved invaluable when the Finance Ministry turned to SAMA for help in financing its 1988 budget deficit.

On December 31, 1987, the budget announcement for the following year sparked a crisis due to the inclusion of big cuts in subsidies and hikes in taxes. Foreign workers were shocked by the imposition of income tax on them[3] and they protested in the only way they could – by stopping work. The national airline, Saudia, was grounded because engineers would not service the planes, while hospitals refused to accept new patients. The stoppages ended only when it was announced that King Fahd had cancelled the income tax plans.

In all the uproar, most people missed the couple of sentences in the speech about borrowing. The king announced that the equivalent of \$8 billion would be raised by selling government debt. He did not refer to interest payments, but spoke instead about payment at an 'earnings rate' equivalent to the return on selected government projects. Wilberding and Black were not surprised at the announcement since their team had spent months working on the details of how to design and market the debt.

There had been budget deficits for several years without government debt being issued. This year was different because of a bookkeeping issue. The previous deficits had been met by running down the government's general reserve account at SAMA. Whenever SAMA received foreign currency from the oil ministry, it credited the general reserve account with the riyal equivalent. But by the end of 1987, the government had run its way through the general reserve, leaving only the equivalent of \$2 billion available. In other words, the riyal equivalent of all the money ever earned from oil over the years had gone. If the government wanted to spend more than it earned during the course of 1988 without raising taxes, it would have to find some way of filling up the general reserve account at SAMA.

The previous October, Al-Sayari had asked the WB team what difference it would make if government debt were used to bridge the budget gap. One possible answer was to issue foreign currency debt externally and credit the government's account with the riyal equivalent. This would give SAMA sufficient foreign currency to pay for the outflow when the government spent the money. But the plan would draw attention to the kingdom's economic plight, and within government questions would be asked about why it was necessary to borrow abroad if the foreign currency reserves were so large.

The alternative was to issue riyal debt within the country, just as had been done with the BSDAs for the previous three years. Wilberding and Black believed that both the banks and the public sector could be potential buyers. Simply moving riyals from them to the government account would have little effect on the economy because the 90 percent solution was what mattered. The government's need to issue riyal debt was an accounting issue. But the BSDA program showed that the banks' allocation of assets could be altered if debt was priced attractively, and under the right circumstances they could be encouraged to sell their foreign currency assets and buy riyal debt, which, in turn, would boost the foreign reserves.

There was a more immediate target: the billions in foreign assets held by other government agencies. They could be encouraged to use their dollars to buy the debt. Three quasi-government funds were in the frame: the General Organization for Social Insurance (GOSI), the Public Pension Agency (PPA), which invested for the armed forces' and civil servants' retirement, and the Saudi Fund for Development (SDF), which made concessionary loans abroad. They controlled as many assets as all private savings held by the commercial banks combined. And their finances were already deeply intertwined with SAMA because the central bank looked after the money. Part of their deposits were in riyals and part was placed into foreign currencies by SAMA's Investment Department, which invested it in bonds and deposits just as it did for the foreign currency reserves. GOSI and the PPA in particular needed long-term assets to match their liability to pay out for pensions in the future. Government debt in riyals would be an ideal asset for them to hold. It seemed that swapping dollars for government debt should suit both sides.

Abalkhail and Al-Sayari were both waiting to see how the religious authorities would react to the king's announcement about a rate of return linked to the profits on selected government projects. But when the short Riyadh spring turned into summer, the Finance Minister was ready to push the button. One morning in June Al-Sayari called Wilberding and Black upstairs to his office and told them he wanted the terms and conditions for the bonds by lunchtime. Saudi government business often proceeded in this way. Long periods in which nothing happened were succeeded by days of frantic hurry, as was the case with Arthur Young and the SAMA charter.

Abalkhail agreed that the document should be sent by special messenger to the chairmen of the banks. A covering letter warned them to keep it under lock and key. This was because the paper contained a potentially explosive admission. Although it repeated the king's December 31 statement about an 'earnings rate' reflecting the returns on government projects, the details made clear that the bonds would pay a market rate of interest that was linked to US Treasury yields. SAMA would decide on what constituted an attractive market rate and the banks would be offered the bonds at a fixed price. In New York, new issues of Treasuries are sold by auction but SAMA would not go that far.

While the banks considered the proposition, SAMA talked to the heads of GOSI, the PPA and SDF. The funds were reluctant, but went along with the plan and took government bonds as private placements from SAMA. The accounting effect was that they swapped some of the assets they had deposited with SAMA for bonds and the general reserve account was credited with their investment. To begin with, they paid in the riyals they held at SAMA and there was no benefit to foreign exchange reserves; but within a short time they were dragooned into paying in dollars as the Finance Ministry had wanted.

By the end of the year, there were bonds at maturities of one to five years, paying interest at a rate closely related to US Treasury yields at the same maturity. The banks were not buying many bonds and when they did, they were using the same tactics as the funds: not repatriating any dollars but simply switching their money out of BSDAs. Meanwhile, the general reserve account became overdrawn and SAMA plugged the gap by buying bonds for its own account and crediting the government.

Al-Sayari told the banks they could sell the bonds on to the public in blocks of a million riyals (about $270,000), on the principle that only high net worth individuals, who presumably already had interest-bearing assets, would buy them. The religious authorities did not react and by 1993 there was almost no restriction on who could buy a government bond. The banks also sold the debt to the public through mutual funds.

Things improved the following year, 1989. The riyal interest curve became very steep, generating greater appetite for longer term debt among the banks, which could pick up extra yield by moving from deposits to government bonds. In setting the coupon on each bond issue, SAMA made sure that government bonds stayed attractive as an investment. As time went by, the banks began to repatriate their foreign assets to buy government debt. Their holdings of foreign assets peaked in 1990 at $32 billion. By 1993, they held the equivalent of over $11 billion in Saudi government bonds.

Looking back, the debt program had both an easy target and a difficult one. The quasi-government funds were the easy target. The Finance Ministry could persuade them to switch out of dollars into riyal debt. The banks were far more difficult to deal with. Here, progress in selling bonds was slow, but SAMA's insistence on working with the banks to provide attractive interest rates above Treasuries was eventually vindicated. Gradually, the banks were persuaded to voluntarily bring their money home (Fig. 5.10).

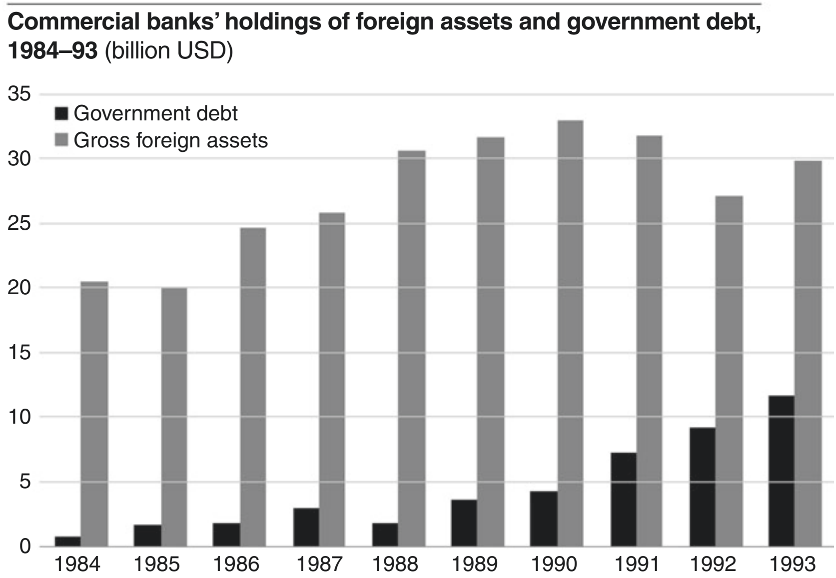

Fig. 5.10 Commercial banks' holdings of foreign assets and government debt, 1984–1993

Source: SAMA statistical bulletin and annual reports

Note: Government debt includes SAMA BSDAs from 1984

7 A Line in the Sand: 1990–1993

The oil price reacted immediately as the news spread across the world on the morning of August 2, 1990, that Saddam Hussein's army had invaded Kuwait. In the weeks that followed Saddam was isolated politically. Economic sanctions were imposed on Iraq and oil prices jumped from $15 in July to $32 in October as an embargo took effect.

The Kuwait war proved to be a short-term boost for Saudi finances. The combination of rising prices and higher output led to an echo of the two previous oil booms. Before the war, daily output was in the 5 million area: afterwards it rarely fell below 8 million. The year 1989 marked the end of the era of low production and heralded a change for the better in the kingdom's fortunes, initially at the expense of Kuwait and Iraq. When the fighting stopped at the end of February 1991, it became clear that several years would be needed to fully repair the damage to the Kuwaiti oil wells which had been set ablaze and that during this time Iraq would remain shut out of world oil markets. Saudi production, meanwhile, pushed through the 8 million barrels level during the same year and government oil revenues jumped by 60 percent to $34 billion in 1991 and 1992. Even when the oil price fell back to $15 as Kuwaiti production resumed, oil revenue in 1993 remained 40 percent higher than its pre-war level.

In January 1989, the WB team had left Riyadh. Al-Malik, now the Vice Governor, had successfully pushed for Saudization and team members spent the final months before their departure training up their Saudi replacements. In some areas – such as bond trading – the local staff coped well; but asset allocation and relations with the IMF were more complex subjects. Al-Malik's task was greatly eased when Bill Black, head of the Barings team, agreed to stay on as an independent adviser.

So the Saudi staff at SAMA were very much on their own when Riyadh awoke on a stifling August day in 1990 to hear the news that Kuwait had been invaded. The financial markets panicked, understandably, as Iraqi tanks could, in theory, cover the 300-odd miles to the Saudi capital in a couple of days. The Kuwaiti dinar collapsed overnight. Kuwaitis in hotels all over the world were stranded as their credit cards stopped working. Many Saudis decided to take their money out of the bank. Physical banknotes seemed safer. Others bought dollars from the banks in case the Saudi exchange rate went the way of the Kuwaiti dinar. In any financial panic, there is a flight to liquidity and safety and this is what happened on invasion day and the succeeding days.

Like Iraq's neighbors, who had thought Saddam was bluffing, Saudi bankers were caught by surprise. It was high summer and many senior executives were on leave. Banks had tied up funds in longer term operations as they had been expecting interest rates to fall, so they were short of cash. Also, they had exposure to Kuwaiti banks amounting to about 10 percent of their capital and reserves, which could not be repaid and looked like being a complete write-off. Riyal interest rates soared while the banks struggled to find liquidity. The deposit and forward markets effectively vanished overnight. For a couple of days even the central bank seemed to be stunned, but it recovered quickly.

The first half of August saw big withdrawals from the banking system – perhaps up to the equivalent of $4 billion in dollars and riyals. The overall exodus of money during that month was later estimated at $5 billion. Some left the kingdom. The rest became part of families' survival kits, consisting of food, water, gasoline, tape to seal a room from poison gas, plus a few thousand dollars and riyals. With cash in their hands, families could drive to Jeddah and buy a ticket to leave the country if Iraqi forces invaded. The banks themselves did not have enough dollars and organized airlifts of currency into the kingdom to meet demand.

The first week was the worst. On August 8, President George HW Bush told the American people that a line had been drawn in the sand as he sent combat forces to Saudi Arabia. But the Saudi central bank had drawn its own line in the sand several days earlier. The first and most important act was to stress that it was business as usual in the currency market, by continuing to meet all demands for dollars, which mostly stemmed from retail customers looking for safety. This statement of confidence produced a positive effect and demand for dollars dropped off. SAMA also provided funds directly to banks that needed them by placing large deposits and increasing the volume of bank notes in circulation. Banks were allowed to overdraw their accounts with SAMA overnight if they needed the funds. Finally, SAMA increased the use of the repo facility to supply cash to the market and send a signal on interest rates. SAMA charged interest at 10 percent, effectively putting a cap on riyal interest rates, since the banks knew they could always access central bank money at this level. SAMA was also busy making sure that those members of the Kuwait royal family who had sought refuge in Riyadh, along with their followers, were adequately supplied with funds. It was a classic example of the central bank acting as lender of last resort and supplying liquidity to the system.

As the pre-eminent central bank in the Gulf, it fell to SAMA to assume an international leadership role during the crisis and Al-Sayari worked with the heads of the other GCC central banks in devising a coordinated approach. The central banks concentrated on calming currency markets rather than worrying about the Gulf equity markets. The Iraqi military had annexed Kuwait and abolished its currency. The international community responded by freezing all Kuwaiti accounts in order to stop any attempts by Saddam to loot them. Throughout this difficult period, when the Kuwaiti government was in exile, Al-Sayari and his colleagues kept the country's finances afloat by supporting the Kuwaiti dinar. In London, the Kuwait Investment Office set up a $7 billion facility to meet unsettled transactions, and the London branch of the National Bank of Kuwait became the official clearing house.

At home, SAMA asked its domestic banks to support these efforts and to keep rolling over their exposure to Kuwaiti banks. Together with the other Gulf central banks, SAMA provided a market for the Kuwaiti dinar at the pre-invasion exchange rate, as well as for other Gulf currencies that might be affected by the crisis. The domestic banks accepted all Gulf currencies at the official rate (most were in any case pegged against the dollar, like the Saudi riyal). But not everybody was happy about this. When rumors reached Al-Sayari that some moneylenders were buying Kuwaiti dinars at a discount, he immediately informed all Shariah-compliant financial institutions that failure to support the official currency rates would lead to their licenses being withdrawn. To back this up, the central banks in the GCC agreed to accept all Gulf currencies at the official rate and provide dollars in exchange. In effect, this amounted to providing open-ended swap lines to each other. It was an example of unprecedented monetary cooperation in the region and all the Gulf currencies remained stable during the six months of the crisis, a remarkable achievement.

Although SAMA had taken the lead in calming the Gulf markets, the world's financial system took steps to reduce its exposure to the region. This made matters more difficult because it isolated Gulf banks from sources of funding. Some countries behaved better than others. The Japanese banks were the worst offenders in rejecting deposits from the region outright. Other foreign banks suspended or reduced foreign exchange and money market facilities for Gulf countries. Some Canadian banks cancelled their credit lines to the Gulf, and US banks reportedly asked for margin deposits before they would deal in foreign exchange with Saudi ones. Al-Sayari decided to make a gesture of Arab solidarity in response and SAMA stopped doing business with the banks in question. This constituted a significant break with SAMA's previous commercial position, which had always been based entirely on objective criteria, specifically credit ratings, in deciding which banks to deal with and how much business to give them.

In January, the war against Saddam's forces in Kuwait intensified when the American-led coalition launched missile attacks against Iraq. The Iraqis retaliated by firing crude Soviet-era missiles at Riyadh and Dammam, as well as invading the Eastern Province at Khafji. Several civilians were killed by the rockets, but despite this the population remained calm. The allied air attacks were followed by a land offensive which liberated Kuwait. In the end, in large part due to the solidarity displayed by the region's financial institutions, with SAMA at the helm, the war ended without any repetition of the initial financial panic. A tense six months had passed and there had been tough moments at SAMA, particularly at the start; but Saudi Arabia's financial system had suffered only minimal damage. The key to SAMA's policy had been to maintain a voluntary market and work through persuasion rather than coercion, even though it was under great pressure itself during this period.

All wars are both destructive and expensive. One estimate is that this one inflicted physical damage of $160 billion on Kuwait and $350 billion on Iraq. The losses suffered by other countries were only financial. The military expenses of the foreign allies were paid by Gulf nations, and these gross costs have been estimated at about $84 billion. In addition, Saudi Arabia bore all the local costs of the hundreds of thousands of troops stationed in the Eastern Province. Yet Saudi Arabia was in the fortunate position of being able to offset the costs of the war against higher oil income. The country also gained from the termination of Saudi aid to Iraq, which had averaged over $3 billion per year throughout most of the 1980s. This saving helped to offset the substantial payments of $3–5 billion made to Egypt and Syria for their support during the war.

How much did the war cost Saudi Arabia on balance, taking into account the higher oil income? The budget deficits in 1990 and 1991 were about $12 billion higher than the previous trend, putting the war cost at about $24 billion. Government transfers and contributions abroad ran at $10–15 billion in the late 1980s, but jumped to nearly $50 billion for 1991 and 1992 taken together (Fig. 5.11). At face value, this would indicate a foreign exchange cost of around $20–30 billion.

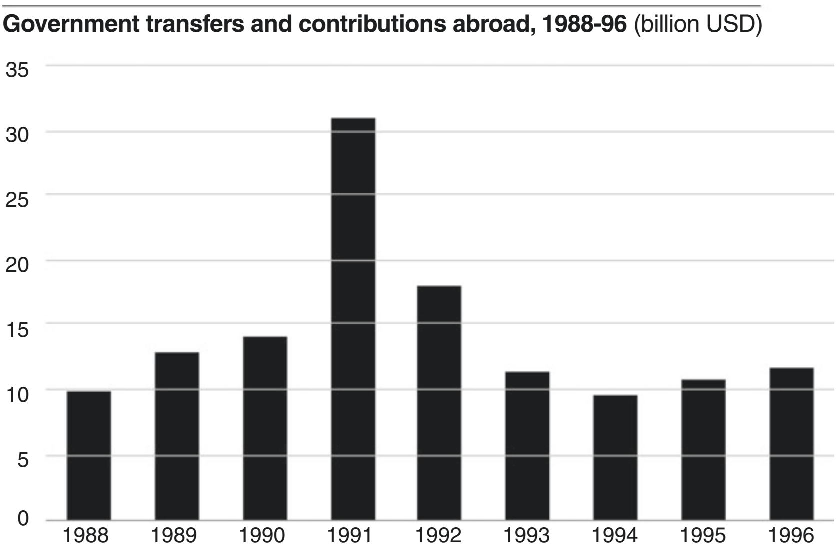

Fig. 5.11 Government transfers and contributions abroad, 1988–1996

Source: SAMA Annual Reports

However, the impact on foreign exchange reserves appears to have been negligible. SAMA's balance sheet shows that reserves fell by only $2 billion between June 1990 and June 1992. These figures should be treated with caution because they use book costs rather than market values, so take no account of moves in equity markets and currencies. It is also likely that dollars in other government accounts were being shifted into the reserves to enable SAMA to make the war payments. But even allowing for these factors, the difference between the numbers is striking.

The key fact is that SAMA's reserves hardly suffered from the war. But with peace, the same old problem returned. Budget deficits led to a drain on the reserves. Despite pumping at a daily rate of over 8 million barrels, earnings from oil fell from war-time highs. It was becoming clear that as the country modernized, domestic oil demand would lead to a reduction in exports, further threatening oil revenues. This problem was compounded by the fact that rapid population growth was driving spending ever higher. Disaster had been averted, but the slow bleeding away of reserves was continuing. How long could SAMA hold out? The rest of the decade was to hold some surprising answers.

8 A Question of Trust

Margaret Thatcher reportedly said, when asked about her decision to launch the Falklands War, that sometimes it is good not to know what the future holds. It is an observation that is particularly relevant to this decade of SAMA's history. If either Al-Quraishi or Al-Sayari had been given the unwelcome gift of foresight in 1983 they might well have cleared their desks and resigned. They would have been appalled to learn that over the next few years the oil price would fall to under $10 per barrel, the diversification program would be revealed to be in large part a failure, the government would run budget deficits every year, the 90 percent solution linking spending with loss of dollars would continue to apply, and Iraq would occupy Kuwait and attack the Eastern Province. Yet, while all of these things did happen, the government still managed to keep the economy going, and the foreign reserves ultimately financed the spending. SAMA deserves much of the credit for this.

Part of the reason for Saudi Arabia's finances holding up so well is simply that the reserves were extremely large to begin with. Saddam had a war chest of $35 billion in the central bank in 1980 when he attacked Iran, while Saudi Arabia had more than twice as much a decade later. But another, equally important, part of the answer is that those reserves were managed prudently. A third factor is that SAMA sought to work cooperatively with the commercial banks, just as a central bank in a developed economy would have done, turning problems like offshore banking into win–win situations. SAMA combined commercial incentives with a degree of moral persuasion. It did not apply capital controls, but as the financial situation deteriorated it incentivized the banks to buy government assets. When war came, the partnership with the banks made it far easier to collaborate successfully to survive the crisis. The last piece of the jigsaw must be the rise in the oil price caused by the war itself. Between 1982 and 1987 the Saudi economy had shrunk by a staggering 25 percent. As late as 1990, the economy was still smaller than it had been eight years earlier, but that year did at least mark a return to economic growth.

One last point is worth making. The Saudi state operated on the basis of consensus. It was often slow to make decisions because it placed great emphasis on keeping the trust of the governed. Banafe has argued that the Saudi government bond market '. . . even helped Saudi banks weather the panic withdrawals seen in August 1990 by allowing the central bank to provide them temporarily with liquid funds through a bond repurchase facility.'[4] Events showed that when a government issues debt it actually strengthens its position during a crisis, provided the buyers of the debt are reasonably confident that they will be repaid.

The first historical example of this process in operation can be seen in the collapse of the Jacobite rebellion in England in 1745. British governments, which had issued debt to finance the wars against France, found that the national debt had the effect of reinforcing political support for the Crown. Those who bought Britain's debt also acquired, at the same time, a vested concern in supporting the regime that owed them money. When England was invaded by Bonnie Prince Charlie, the Jacobite pretender to the throne, his troops were repelled. The English middle class stood by King George II. They trusted that the German-born monarch, uncharismatic though he was, would honor his debts and believed that the gallant 'Bonnie Prince' would not. It was ultimately a matter of self-interest – or perhaps a matter of paying interest.

In the case of Saudi Arabia, from the first issue of BSDAs the banks willingly bought central bank bills and then government debt. This was attractive to them for a variety of reasons. The bonds gave them a yield pick-up over Treasuries, they were denominated not in dollars but in riyals which was the banks' base currency, they were issued by the government so they were by definition 'riskless' (in the sense that government could always pay because of its control over the riyal), they helped the banks' prudential ratios and, crucially, through the repurchase arrangement with the central bank, the bonds could be turned into cash instantly. This long-standing collaboration with the banks made it much easier for SAMA to deal with the crisis of August 1990.

Notes

1. Lawrence Freeman and Efraim Karsh, *The Gulf Conflict 1990–91: Diplomacy and War in the New World Order*. (London: Faber & Faber, 1994): 42–63. This is a good near-contemporary account of the Khafji battle and the Kuwait conflict.
2. In 2016, a debt management office was set up inside the Finance Ministry.
3. Technically, an earlier tax was reimposed.
4. Banafe, *Saudi Arabian Financial Markets*, 187–201. This is the most comprehensive account of the Kuwait episode from a financial point of view.

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CHAPTER 6

Low Oil Prices, Rising Government Debt and External Crises, 1994–2004

1 Betting on Red

One evening in August 1913, some of the richest in the world were gathered around one particular roulette table at the famous Monte Carlo casino. There was a buzz of excited chatter and anticipation, which grew stronger with every spin of the wheel. Instead of randomly generating red or black winning colors, this particular wheel came up with black time after time. The gamblers started placing large bets. Not on black – no, they reasoned, that would be illogical. The big bets were on red. Surely red would come up on the next spin of the wheel? But the ball kept bouncing into a black slot. Bigger and bigger wagers were made as it seemed impossible that black should keep repeating. But it did – 15, 20, 25 times. Finally, on the 27th roll of the wheel, red came up. Some rich people had suddenly become poor people by doubling up their bets on red. But in fact the odds on red had remained exactly the same at each spin.

Ten years later the American economist Irving Fisher compared the Monte Carlo effect to the length of the business cycle. Under the label 'random walk' his hypothesis has been used to examine stock market cycles, investment expertise and other puzzling sequences of events. Another term for random walk is 'no duration dependence'. This can be

A. Banafe, R. Macleod, *The Saudi Arabian Monetary Agency, 1952–2016*, Financial Institutions, Reforms, and Policies in Muslim Countries, DOI 10.1007/978-3-319-55218-7_6

explained by the example of ordering a cup of coffee. As you wait for it you experience different types of duration dependence. First, you believe that there is a rising probability your coffee will appear on the counter. This is positive duration dependence. When five minutes have gone by, and no coffee appears, you experience negative duration dependence as your hopes fall of ever seeing the coffee. In a situation of no duration dependence – and this can be quite hard to imagine – mathematically your chance of getting the coffee is exactly the same after two minutes, two hours or two days. The probability of an event happening is independent of time. This is not an intuitive idea and needs to be learned.

The concept of no duration dependence applies to the prices of commodities. In 1999, oil prices had been below $20 per barrel for longer than anybody expected. Saudi Arabia's finances were in poor shape. As people wondered how long the slump could last, International Monetary Fund economists looked at the price history of oil and 35 other commodities between 1957 and 1999 and found no evidence of either positive or negative duration dependence. Their paper concluded with a warning to the future:

> There is an important implication of our finding of no duration dependence in commodity prices. If market participants do not take account of the nonexistence of duration dependence, there is a danger that they may misperceive the nature of commodity price cycles. On the one hand, they may mistakenly believe that because prices have been in a boom period for a long time, that there is a new paradigm, so that cycles are no longer relevant. Conversely, they may also mistakenly believe that the longer adverse movements in prices continue, the more likely it is that this period of falling prices is about to end. However, we show that the probability a boom or a slump will end actually remains constant.[1]

Despite this – and more recent work confirming their conclusion – people still speak of commodity cycles as though Cashin and his colleagues had never examined the data. It is natural to find randomness difficult to accept. For instance, in 2008 a government task force in the United States reported on what had been driving oil up over the previous five years. The report concluded that speculators had played no role. Their answer as to why the oil price had shot up was a description not an explanation: 'As it is very difficult to substitute for oil in the short-term, very large price increases have occurred as the market balances supply and demand.'[2] They offered no explanation of how this occurred, because they had no answer. Of course, there are times when an event causes a change in oil prices – Iraq's invasion of Kuwait is an obvious example – but the concept of no duration dependence tells us that we cannot predict where oil prices are going.

The team in Saudi Arabian Monetary Agency's (SAMA's) Investment Department waas not surprised to read this article on commodity prices. It told them what they already knew – that the oil income that came to SAMA every month was not predictable. The trick was to combine political and financial skills. The Finance Ministry balanced public expectations about spending against their lack of knowledge about what oil income would be. The central bank needed sufficient foreign exchange reserves to enable deficit spending to continue until the oil price went up – that is, when the next spin of the wheel placed the roulette ball in a red slot rather than a black one.

2 Economic and Fiscal Background

Public spending was still the overwhelming force in the economy since the non-oil sector was unable to develop independently. The Finance Ministry tried to keep growth positive by using deficit spending. This meant heavy domestic borrowing and falling foreign reserves. Overall, they succeeded. The most difficult period was the Asian financial crisis of 1997–1998, but even then growth was negative only in 1999 and inflation stayed around zero because the riyal was pegged to the dollar (Fig. 6.1).

In 1981, the government spent a record \$84 billion, but a decade later in 1991 spending was around half to three-quarters of that level. In the same year, nevertheless, the kingdom did not see a return to the depression and deflation that characterized the hangover from the second oil boom, because current spending was kept up and there was a multiplier effect as civil servants spent their salaries in the local economy. Capital spending had never provided this. In 1981 \$50 billion had gone on capital projects but most were dollar contracts paid to construction firms using foreign labor who lived in work camps, so almost all the money went outside the country immediately as fees, imports of machinery and remittances from workers.

Deficit spending kept the economy growing and the Finance Ministry controlled spending in a pragmatic way rather than using a formula. The budget was constructed each year around an implied oil price and likely export volumes. The key to any planning was the oil price and this fluctuated widely (Fig. 6.2).

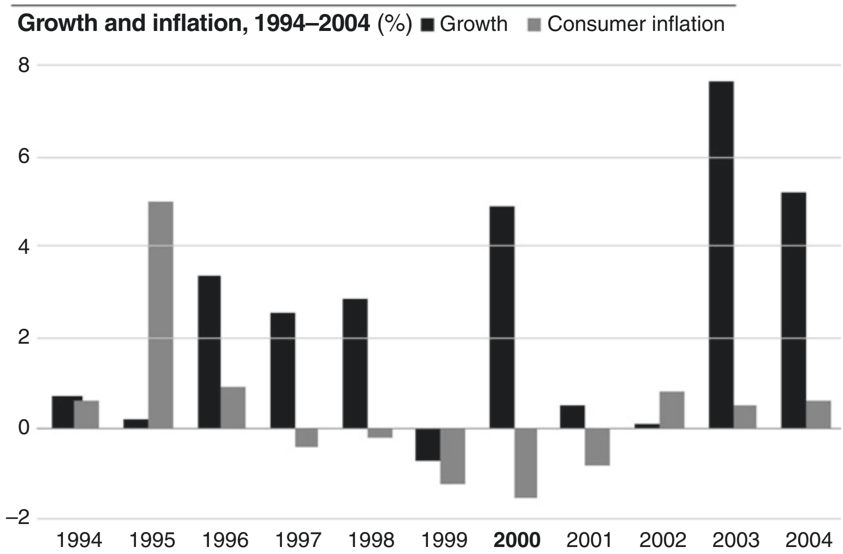

Fig. 6.1 Growth and inflation, 1994–2004

Source: International Monetary Fund, World Economic Outlook Database, October 2015

The Saudi population meanwhile had grown fast. From 8 million citizens in 1980, it doubled to 16 million by 2004. The demand for health services, schooling and housing kept rising. Public spending had to rise faster than the population's growth rate of 2.5 percent per year or real incomes per head would fall.

The rising number of unemployed men presented another problem. The policy of diversification was introduced with the aim of creating more jobs for young Saudis, and many jobs in the private sector could, indeed, have gone to Saudis, but for the fact that businesses preferred to employ foreigners. In 1974, there had been fewer than a million foreign workers in the kingdom. According to SAMA's annual reports for the relevant years, there were 4.8 million in 1994 and 6.1 million a decade later. In the crisis year of 1998, the country earned $21 billion in oil income, but foreign workers sent home $14 billion, or two-thirds of that. A significant share of the remittances were – and still are – actually the earnings of small, informal expatriate business owners (*tasattur*) who rent business licenses from Saudi nationals, so that not just wages but also profits flow out of the country.

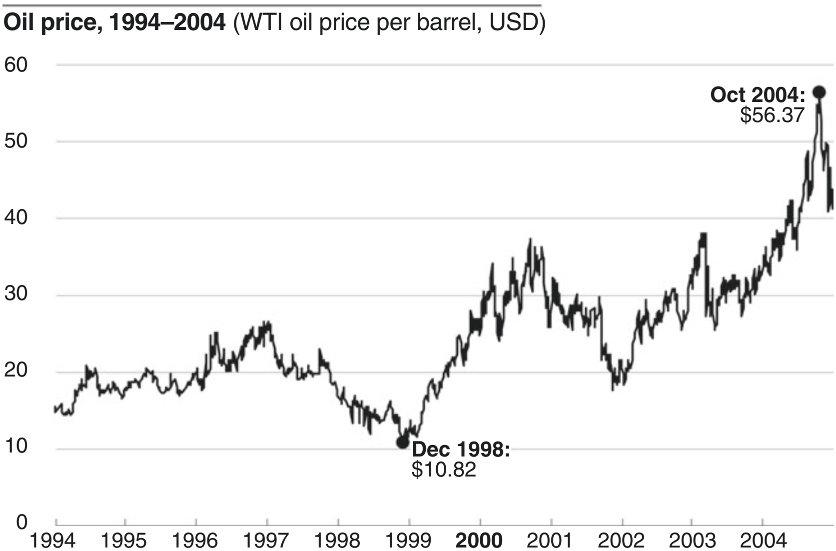

Fig. 6.2 Oil price, 1994–2004

Source: Federal Reserve Economic Data

Oil revenues also depended upon how much could be exported and domestic demand threatened this. The financial crisis of the 1980s occurred when both price and output slumped. Saudi Arabia had cut exports and the oil price had fallen. Oil production in the 1990s was stable at around 8 million barrels per day, but there was a growing gap between the volumes produced and those exported. Fuel was used wastefully at home because of the low price and as consumption went up exports had to fall. Ministry of Petroleum figures – which are on the conservative side – show that direct usage of refined oil by the public rose by a third from 0.98 million barrels per day in 1994 to 1.3 million barrels per day in 2004. This oil was not available to earn dollars abroad.

After the Finance Ministry decided what oil income to expect, it negotiated with other ministries to arrive at a spending total to be used in the king's budget speech. Actual spending was different from the budget and these negotiations continued throughout the year depending on oil revenues. In the mid-1990s, this process worked well because revenue was higher than predicted and the government spent the extra revenue so the

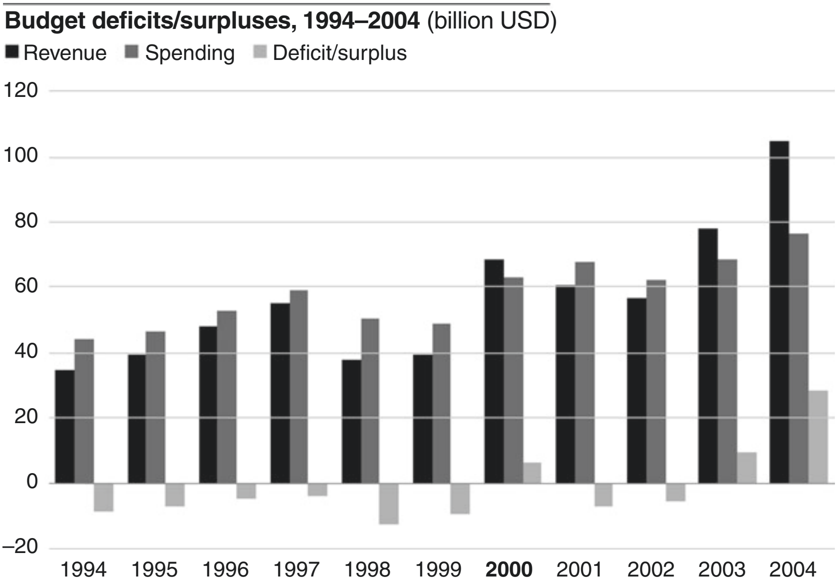

Fig. 6.3 Budget deficits/surpluses, 1994–2004

Source: SAMA Statistical Bulletin

deficit stayed the same. But there was a 'ratchet effect' so that when the oil price was stronger government spending moved up more than it fell in bad years. In 1997–1998 – the years of the Asian crisis – it took a couple of years for the Finance Ministry to rein back spending (Fig. 6.3).

For 1994–2004 as a whole, actual budget deficits were $60 billion less than the amount budgeted because the Finance Ministry stopped all the extra unexpected oil revenues from being spent. It is worth noting how the Finance Ministry kept a lid on spending considering the uncertainty around oil revenue and the pressure it faced to spend more.

3 Managing Government Debt

SAMA's key role was to make sure foreign exchange reserves were large enough to pay for the imports on which the fiscal deficits were ultimately spent. Failure to provide for imports would mean an economic and social crisis. Managing government debt was less vital but it carried a big risk. Failure to sell government bonds at home could lead to a drop in confidence in the currency and – with an unlimited promise from SAMA to sell dollars at the pegged rate – capital flight out of the riyal. Then a vicious circle would develop when the falling reserves would make it logical for more Saudis to switch from riyals to dollars. This had started to happen in the panic of August 1990.

The debt strategy continued to be managed by the Investment Department. During this period, SAMA stuck to the idea that the banks should actively want to buy Government Development Bonds (GDBs). One way of ensuring this was to offer them at reasonable spreads over Treasuries and the floating rate dollar London Interbank Offered Rate (LIBOR), and others were to provide the types of bonds and maturities that the banks wanted, and to include GDBs in SAMA's liquidity ratio calculations. In short, the banks were given what they wanted.

In 1996, SAMA widened its range to include three- and five-year maturity riyal Floating Rate Notes (FRNs). The FRNs were priced over three-month Saudi Interbank Offered Rate (SAIBOR). Since SAIBOR was very closely linked to dollar LIBOR, this followed the model where fixed rate GDBs were priced against Treasuries. They were attractive to the commercial banks because the bonds matched their short-term funding. But the banks paid no interest on a third of bank deposits so their average cost of funds was well below SAIBOR. In practice, SAMA was guaranteeing the banks a profit in order to incentivize them to buy more bonds. Ten-year bonds replaced three-year bonds in 1998 when the banks said they wanted longer dated securities. Finally, as the budget slipped back into deficit in 2001, SAMA was compelled to widen the spread over Treasuries. It disguised this by saying it was now pricing bonds from US dollar swap rates which paid more than Treasuries. SAMA increased the spread over Treasuries in order to make the bonds more attractive.

Not surprisingly, given the guaranteed profit margin offered to them, the commercial banks voluntarily bought many bonds. Their holdings peaked at $40 billion equivalent in 2003. This was partly funded by running down their foreign assets, to the benefit of SAMA's foreign currency reserves. In effect, the banks took fewer dollars from the central bank to fund their activities as they moved into GDBs, thus helping SAMA to hold onto its reserves.

SAMA also continued to sell to the quasi-government funds[3] on the same basis as the banks. These funds were pressured into buying by the Finance Ministry. As a result, their appetite for dollars to buy assets overseas was reduced, thus helping SAMA's reserves still further. Nevertheless, the total of what the commercial banks, the quasi-government funds and all

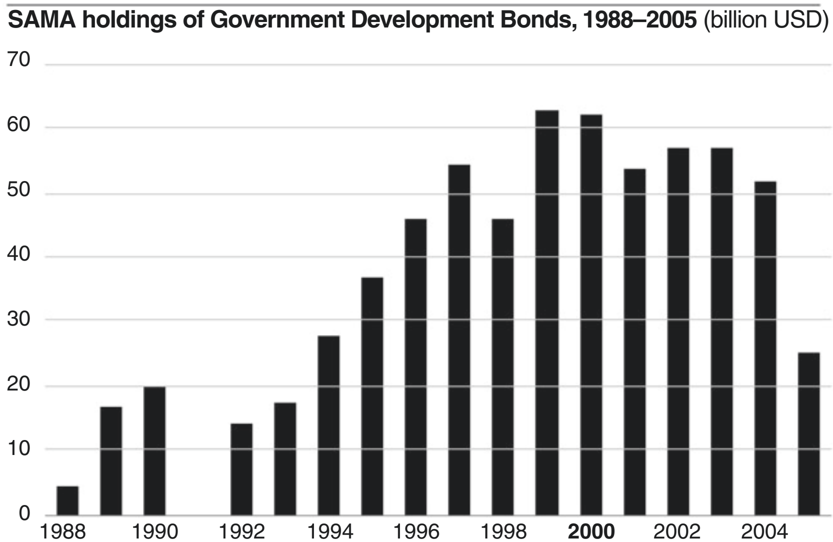

Fig. 6.4 SAMA holdings of Government Development Bonds, 1988–2005

Source: SAMA Annual Reports (SAMA Balance Sheet)

Note: No data available for 1991

other sources of assets within government were able to buy still fell short of what was needed to fund the deficits. The only answer was for SAMA to buy government bonds itself and pay the money into the government account. The practice took off after 1993 (Fig. 6.4). The quasi-government funds had reached their limits and for the next four years it was SAMA's intervention fund that financed most of the budget deficit.

This practice reached its height in 1999 when the government settled its unpaid arrears by paying contractors with bonds instead of cash. The banks bought up these bonds cheaply with the result that they had little appetite to take more. A huge funding gap meant SAMA had to step in, pushing its holdings up by over $16 billion. This was the peak. From 2002, higher oil revenues led to disappearing deficits, with SAMA on the list of those repaid as the bonds matured.

Who bought most of the debt? The major holders were in fact SAMA and the quasi-government funds. Only 30 percent was held by the private sector, mostly by the banks (Fig. 6.5).

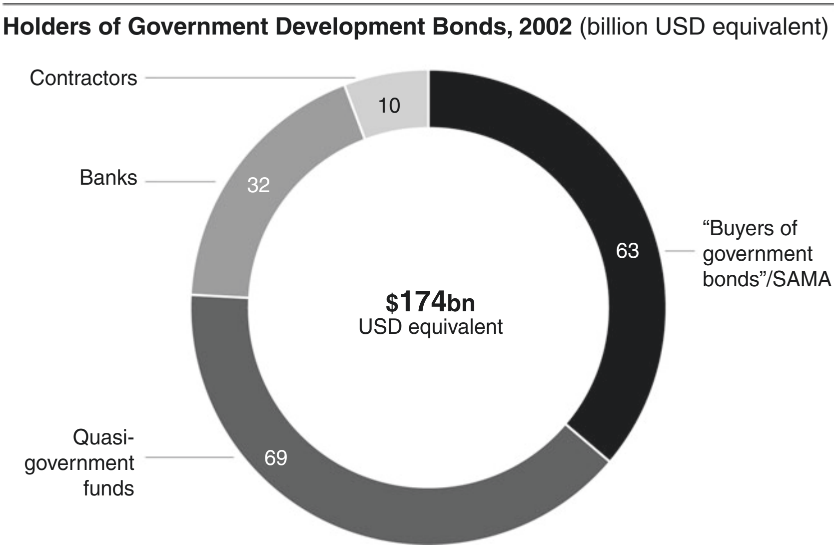

Fig. 6.5 Holders of Government Development Bonds, 2002

Source: Authors

SAMA had given the Saudi banks everything they asked for but the private sector had not even come close to funding the budget deficits. On the plus side, there is no evidence that the debt program damaged investment. The banks limited their lending to the government, while private sector lending grew rapidly. Without budget deficits funded by debt, there is no doubt that the economy and investment would have fared far worse. SAMA's bond purchases were what came to be described later as quantitative easing. It amounted to printing money, but it had no effect on inflation, which remained subdued: indeed, the level of consumer prices in 2002 was slightly lower than six years earlier. And after the emergency was over SAMA made sure it was repaid quickly.

4 Looking after the Foreign Exchange Reserves

Like the Roaring Twenties, this was a golden age for stock markets, especially in the United States where SAMA had the bulk of its assets. The Asian economies provided cheap goods at low cost for the West which paid for them by borrowing more. Capital barriers tumbled, so the West could offer the developing world financing to increase investment and exports. Inexpensive raw materials also helped, especially the vital ingredient of global capitalism – cheap oil.

In the Roaring Nineties corporate profits were strong and inflation stayed low. All the major US stock indices returned between 9 and 11 percent per year.[4] A high-grade US corporate bond index rose by over 7 percent per year. Investors did very well.[5] As Figure 6.6 shows, real yields on bonds and deposits stayed high until 2001, so SAMA could reinvest its income at attractive rates.

The foreign exchange reserves were the final line of defense for the kingdom's finances and how much money SAMA could make became a matter of national importance (Fig. 6.7). By now, the 90 percent solution that related domestic government spending to foreign exchange outflow in a close to one-to-one ratio was fully understood. The money came in as oil revenues and went out again as imports. The Investment Department knew that the 90 percent solution was only a guide taking one year with another, but had found that it worked well when applied over a period of several years.

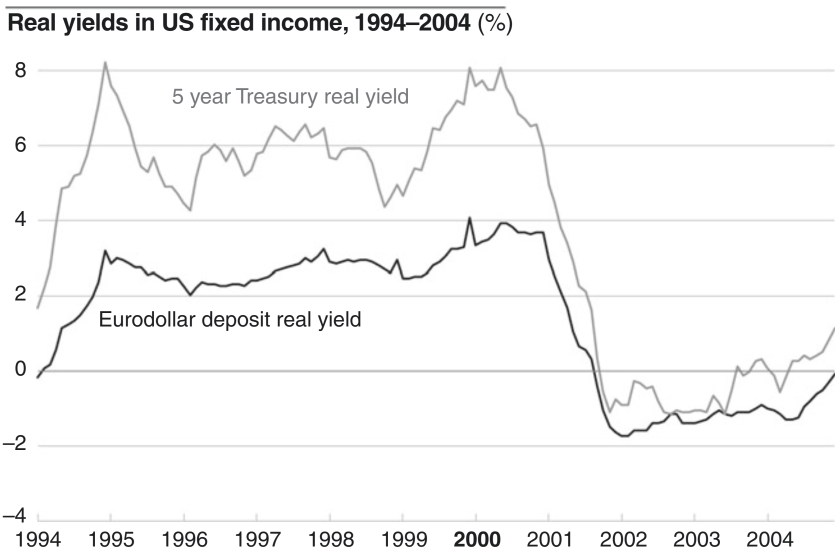

Fig. 6.6 Real yields in US fixed income, 1994–2004

Source: Federal Reserve Economic Data

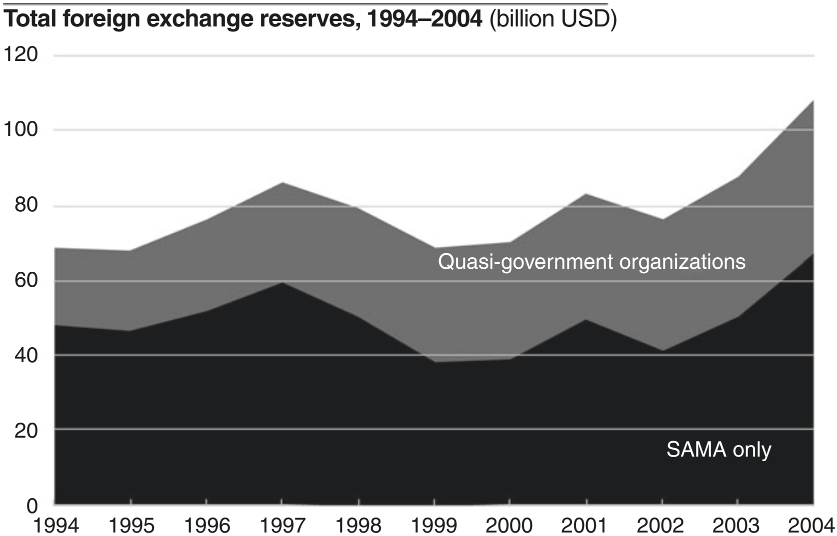

Fig. 6.7 Total foreign exchange reserves, 1994–2004

Source: SAMA Annual Reports

However, the central bank did not run out of reserves as – on the face of it and taking the published numbers for government spending from oil revenues and the private sector's balance of payments deficit – the 90 percent solution predicted it would. The reserves stood at $48 billion in mid-1994. The 90 percent solution should have meant that the net outflow from foreign reserves would be very similar to the cumulative budget deficits, which totaled $55 billion between 1994 and 2002. In theory, SAMA should have run out of money. But this did not happen despite the fact that the data shows the 90 percent solution kept working and external deficits matched budget deficits. The worst year was 1999 when reserves fell to $38 billion. But by 2002 they were back over $40 billion, only $8 billion less than they had been eight years earlier. How did this happen? Part of the answer is that realized market gains helped to keep the reserves steady. The bulk of the reserves were in bonds and equities. For these assets the published reserves numbers represent historical book costs, which did not reflect the rise in the markets (Fig. 6.8).

US securities markets doubled during these eight years and non-dollar markets also did well. SAMA's assets would have recorded big gains. The rest

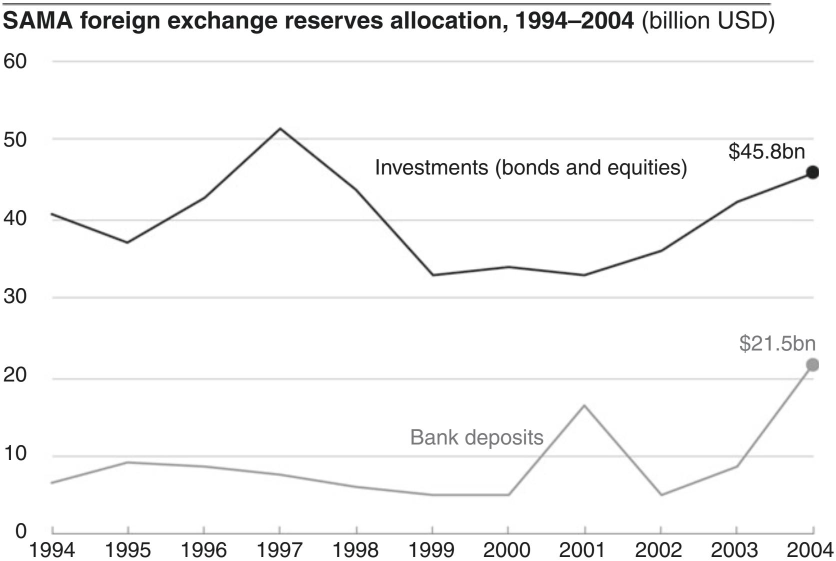

Fig. 6.8 SAMA foreign exchange reserves allocation, 1994–2004

Source: SAMA Annual Reports

Notes: Invesments include currency cover

of the answer is that the reserves gained dollars from other sources as the banks and quasi-government funds sold dollars and bought GDBs.

Investing in the way SAMA did was not a riskless strategy. The years from 1997 to 1999 were a time of real stress due to the Asian crisis. The reserves fell by more than $21 billion. But the Investment Department stuck to its guns, arguing that it was justified in taking risks because low oil prices were good for the world economy and financial markets

Dangerous moments did occur. The big risk was a loss of confidence in the exchange rate peg. A minimum level of reserves was needed to stop the market worrying that the riyal would be devalued. Once that happened the banks would inevitably buy dollars and sell riyals to SAMA in order to protect themselves. Nobody knew what that requisite reserves minimum was – until it was reached. The signal would be an increase in the spread between riyal and dollar interest rates, something that would also show up as a discount in the forward rate for the currency, implying a loss of value for the riyal.

It never happened. The spread in interest rates was at its highest in 1998 and 1999, the years of the Asian crisis, when oil revenues plummeted. At times, the spread was a lot higher than the averages shown in Figure 6.9 and the forward exchange rate weakened. In 1999, the reserves were at their lowest point and SAMA's holdings of government bonds peaked at just over 100 percent of Gross Domestic Product. But an average spread of less than 1 percent between Saudi and American interest rates was certainly not evidence of a full-blown crisis. SAMA responded by making several small interventions through the forward foreign exchange market to underpin the riyal against the dollar.

The other dangerous time came between the 9/11 terrorist attacks in the United States on September 11, 2001 and the invasion of Iraq in March 2003. Public debate in the West contained the idea that the involvement of Saudis in the attacks might result in retribution of some kind, but the Americans decided instead to vent their anger on Iraq. When the coalition forces encountered problems in the wake of the invasion,

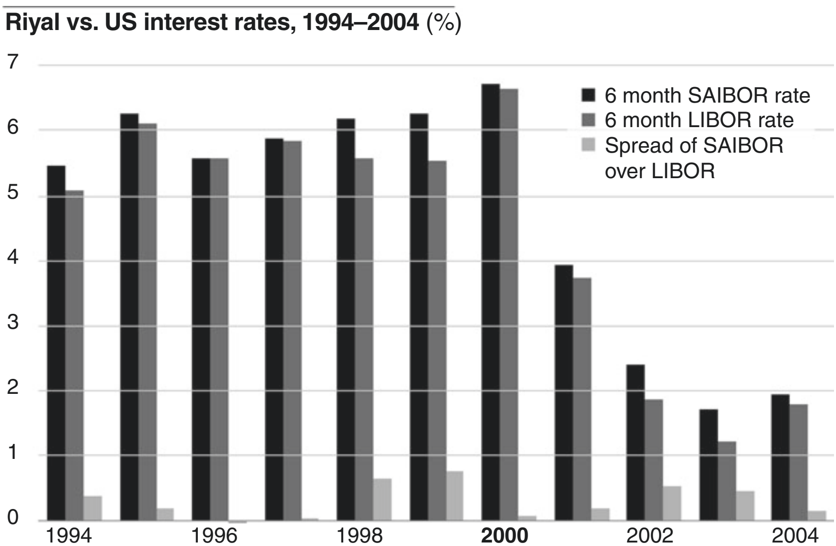

Fig. 6.9 Riyal vs. US interest rates, 1994–2004

Source: SAMA Annual Reports

Note: Annual averages

it became clear that the kingdom itself would not come under pressure from Washington. The interest rate spread shrank and by 2004 the average difference was only a fifth of a percent.

5 Challenges in the Banking System

The biggest challenge for a central bank is a financial crisis, but the kingdom's banks remained mostly profitable, despite problems in the economy. The exception was National Commercial Bank, where the government (via the Public Investment Fund) acquired a 50 percent interest in 1999 when the bank – still controlled by the bin Mahfouz family – revealed some problem loans. SAMA had modest success in persuading the banks to play their part by buying debt while helping the non-oil economy. Meanwhile, Shariah-compliant banking was growing as Saudis increasingly linked their financial practices to their own personal piety and adherence to religious tradition. This led to a rise in non-interest bearing deposits, which boosted bank profits further.[6]

The story of the struggle to develop the insurance market ties these elements together – the role of the banks, the drive to develop capital markets and the challenges presented by Islamic law. Insurance products had been sold by foreign insurance companies operating in the kingdom for at least 40 years before 1986, when the National Company for Cooperative Insurance (NCCI) was set up. The banks sold insurance products through their branches, but the cash from insurance premiums had always left the country. NCCI was meant to change this. It was owned by the government and Islamic jurists approved it as an equitable way of sharing risk; yet despite this it grew very slowly.

In the late 1990s, SAMA wanted to see competitors to NCCI as part of its push to develop local capital markets. The prize was to create a big life insurance industry. The banks were keen to sell life policies and this was a way for individuals to invest in long-term savings funds and help the stock and bond markets, including buying more GDBs. In 2004, a new insurance law placed SAMA in control of all aspects of the insurance business, which was to work on a cooperative basis. Soon after, NCCI was part-privatized and floated on the Saudi Stock Exchange (Tadawul). But there was a serious obstacle: Islamic scholars could not agree on whether life insurance interfered with a person's destined fate. Some took the view that insuring a life was not allowed, because paying out on a death meant gambling on that person's life.

The central bank was careful to stay within Islamic guidelines as it drew up the new insurance rules – which took four years. It organized conferences to educate companies in how these rules would work but it took until 2008 before it began approving competitors to NCCI. Within a few years there were dozens of insurance companies competing in the market, including joint ventures with local banks, all in line with the Islamic principles that SAMA had codified. Yet even so, life insurance was only a small part of the industry.

Despite failures to develop its range of products, at the turn of the millennium the banking system was one of the soundest in the world. Comfortably within the regulatory regime's prudential ratios, the banks had made loan loss provisions that seemed adequate to cover bad loans. The system was profitable as well as highly liquid, with loans less than customer deposits, which made it more secure against a run on deposits.[7] Furthermore, SAMA had succeeded in keeping most banking business in the riyal. The specter of a dollarized economy, which always haunts a commodity exporter, was absent, as evidenced by the low ratio of foreign currency loans and deposits within the system.[8]

Loans were kept considerably lower than deposits to guarantee that the banking system could stay liquid in the event of a crisis. SAMA set a ceiling of loan to deposit ratio at 65 percent (later raised to 85 percent and then to 90 percent in 2016), with a requirement that the banks keep at least 20 percent of their deposits in assets that could be converted to cash in 30 days.

But although it was resilient, liquid and profitable, the Saudi banking system remained small compared with those of developed economies. More than ever, the commercial banks dominated the system, having by now become much more important players than the state-owned specialized credit institutions and possessing more assets than the quasi-government funds. Their links with the government were strong and this made SAMA's task of supervising them easier. The public sector had equity stakes in several, and bank shares were actively traded on the stock market. Since Al-Quraishi's Saudization program 20 years before, foreign equity stakes had been limited, but six banks still had foreign investors. Saudi British Bank, for instance, was 40 percent owned by HSBC, which ran the operation through a service contract.

The big question was whether to let the global banks into the Saudi market. This would improve competition but it would also increase the risk that the Saudi banking system would be affected by any global financial crisis, prompting the big banks to pull their lending lines. This, indeed, was to happen in 2008. But by then the central bank had awarded licences to three global banks: Deutsche Bank, JP Morgan and BNP Paribas.

SAMA hoped that by doing so it would bring trading expertise in market-making and develop the bond market. Its own attempts to encourage the local banks to trade in GDBs had largely failed because they put the bonds they bought into their investment accounts rather than their trading accounts. This meant they could hold them at cost and did not have to mark them to market. For regulatory purposes, SAMA classified them as liquid investments (this was the one way that SAMA had adjusted the rules in order to make GDBs more attractive).

At the time, there were worries that GDBs would lead to a reduction in lending to the private sector. There must inevitably be some 'crowding out' of the private sector in percentage terms when a government borrows from its banks. However, the banks continued to extend credit and in fact private sector lending grew from under \$20 billion to over \$80 billion during the period of GDB issuance, which started in 1988 and began to run down after 2004 (Fig. 6.10).

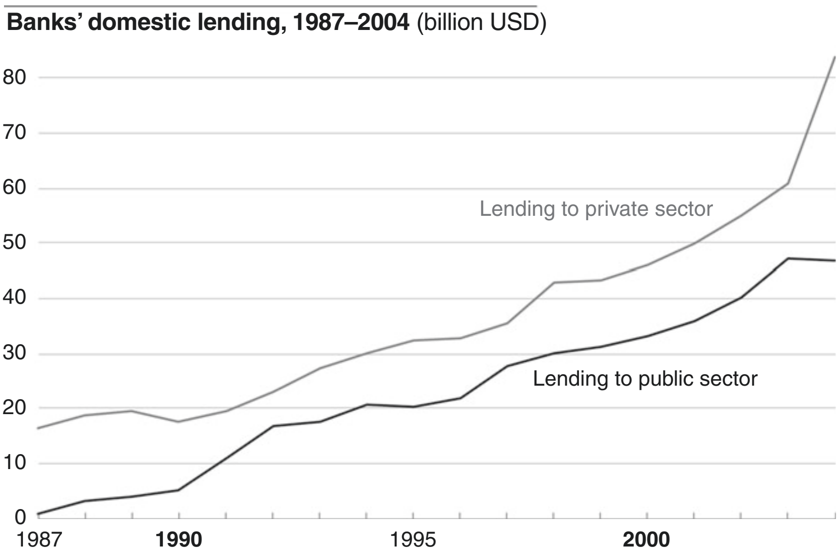

Fig. 6.10 Banks' domestic lending, 1987–2004

Source: SAMA Annual Reports

One new product appeared in these years, and the banks profited as consumer loans proliferated like rabbits across the Saudi financial landscape when an improved electronic payments system was introduced. The country had set up a central payment network as early as 1980, but SAMA's goal was to achieve something that few countries outside Europe and North America possessed: a real-time gross settlements system like the US Fedwire, the UK CHAPS or the ECB Target 2. Until all bank claims on each other could be settled simultaneously, the banks continued to use old-fashioned methods, including telex, bank checks, phone calls and fax messages. The Saudi Arabian Riyal Interbank Express (SARIE, meaning 'fast' in Arabic) came into being in 1997 and enabled banks to make and receive payments directly from their accounts with the central bank and to credit beneficiaries with funds that were good value the same day. Banks operated more efficiently now they could monitor their financial positions on a continuous basis. An unexpected result of SARIE, however, was that the banks were able to make many more consumer loans.

Under Shariah law, as we will see below, it was virtually impossible for banks to pursue borrowers who did not keep up with interest payments, so they needed a way of making sure interest was paid on time. SARIE provided them with this mechanism, because under the new system salaries paid by the civil service and public sector employers were credited directly to the employees' bank accounts. The banks simply made the borrower set up a direct debit for the repayments, which SARIE also made practicable. By 2004, consumer and credit card loans – rare in the days before SARIE – accounted for two-fifths of all bank credit to the private sector. Indeed, the rapid growth of bank lending in these years was entirely due to consumers borrowing for unspecified purposes against the security of their salary. The news was unwelcome to SAMA, which wanted the banks to make loans to local businesses rather than consumers and worried that these loans would be used to speculate on the stock market.

Up to this point, the central bank had supervised Western-style banks. Shariah-compliant banking was brought fully under SAMA's control as a result of a sensational set of fraud cases. In 1991, a tough time for the Saudi economy, a new type of business sprang up which inveigled unwary customers into handing over their money by promising unrealistic returns. Many were in effect fraudulent Ponzi schemes and the police closed nearly 30 under various ordinances. These doubtful operations called themselves 'Islamic' and took nearly \$2 billion from credulous investors. By comparison, deposits at the licensed Shariah-compliant bank, Al Rajhi, the biggest in the country, had risen to $13 billion. Al Rajhi was growing fast because it operated as a fully Shariah-compliant bank (SAMA did not allow it to describe itself as Islamic).

Shariah-compliant finance is a simple idea that eliminates the rentier (who lends money and takes no risk). There are two sides to it: receiving deposits and making investments. The bank pays no interest on deposits. If the customer wants a return he must buy an Islamic investment product that involves sharing the risk of the borrower. A complete definition of what is Islamic is not possible as Islam has no codified legal system and depends upon a judge's interpretation of Shariah law.

Meanwhile, the conventional banks were operating in a gray area, neither within nor outside the law in some of what they did. While it could be perfectly legal to offer high returns on certain Islamic investments, not all Western-style banking business was supported by Shariah. At the same time, the banks profited from the rise of Shariah-compliant banking because they too took in deposits on which no interest was paid. Al Rajhi was an exceptional case, but by 2000 about a third of deposits at conventional banks did not pay interest either. This was one reason why the return on assets and bank equities stayed high. By 2004, Saudi banks were among the most profitable in the world.

Against this benefit, the banks had a problem in making loans: there was no legally secure mechanism for them to pursue defaulters. The law of the land was Shariah and the courts did not recognize a conventional loan as legally enforceable. In the late 1980s, SAMA had set up an arbitration committee to try to resolve some of the banks' non-performing loans, and banks in many cases were able to recover money. But it was tougher to recover money from personal borrowers. This made the banks reluctant to go about the basic banking business of lending against good collateral – such as real estate. If, for instance, a bank made a personal loan to help buy a house and the borrower later defaulted on that loan, the bank could not repossess the house and sell it to settle the debt, as is common practice in the United States. The Islamic courts invariably sided with the borrower.

The other major aspect of banking business, making investments, was also in theory hard to do. Under Islamic law, there can be no investment without risk sharing. But in practice, the main Shariah-compliant vehicles operated in similar fashion to Western-style loans. In its simplest form, a Shariah-compliant bank can buy an asset from a customer in return for giving him cash, and he will agree at the same time to buy the asset back in the future at a higher price. The asset remains in the possession of the borrower. This can qualify as a Shariah-compliant transaction on which no interest is payable, but its economic effect is identical to making a loan at interest. Similarly, trading in the forward currency markets can be acceptable, even though the difference between the spot and forward prices of the riyal against the dollar incorporates the interest rate differential between the two currencies.

SAMA did not distinguish between Shariah-compliant and conventional banks in carrying out its supervision and it continued to treat both areas equally even after Islamic banking came under its umbrella. This allowed it to obscure the distinction between Islamic and non-Islamic banking and the fact that lending money at interest could be seen as against the law.

SAMA supported Shariah-compliant banking provided it stayed away from dubious practices. The difficulty was that banking supervision had been designed for a Western-style system. SAMA adopted a two-pronged approach to the problem, but its success was limited. The first involved regulating lending practices. A lot of the money consumers borrowed went into the local stock market, where Shariah-compliant arrangements were available to small investors so they could leverage their money and take more risks. A cap on consumer lending would prevent these businesses from flourishing. So SAMA clamped down on the amount a bank could lend to an individual on an uncollateralized basis by limiting instalment payments to 30 percent of salary. The central bank also sponsored a Saudi Credit Bureau (SIMAH) so that borrowers with poor credit records who were using Shariah-compliant vehicles could be identified by the banks.

The second tactic was to liberalize the rules under which the Shariah-compliant banks worked by cooperating with other countries. Since the kingdom had stricter Islamic law than most others, applying international rules could systematize Shariah-compliant products and make the rulings of the courts more predictable. With this aim in mind, SAMA joined the Kuala Lumpur-based Islamic Financial Services Board (IFSB) when it started up in 2002. The IFSB tried to regulate Shariah-compliant finance across the world, just as the Basel regulatory regime's rules unified Western banking practice. The problem for the IFSB was that, as in Saudi Arabia, the central banks found it impractical to incorporate its standards into their own regulations because local jurists had their own judgments based on their personal interpretation of Shariah. So standardization made only limited progress.

SAMA struggled with this problem. The central bank needed to find a better way to regulate Shariah-compliant lending within a single supervisory code and to make sure that banks fully provisioned against the risks they were taking. The problem was growing because Shariah-compliant products nearly doubled between 1999 and 2003, when they amounted to nearly a quarter of bank investments. In some areas Shariah-compliant practices dominated: for instance, 75 percent of the mutual funds that banks (including the Western-style banks) sold to retail customers were advertised as being Shariah-compliant. But nobody in the banks or SAMA could say exactly what this meant.

6 The Stock Market Comes of Age

The Middle East is a turbulent region and in every chapter of this book we have seen how political events bring financial consequences in their wake – usually unexpected ones. For example, the Middle East war of 1973 precipitated the first oil shock, which transformed the Gulf, and the overthrow of the Shah of Iran in 1979 caused the second one. In the last chapter, we saw how the 1990 Kuwait war and subsequent sanctions against Iraq boosted the price of oil and helped Saudi finances at a difficult moment. In the same way, the attacks of 9/11 in the United States contributed to a boom in the local stock market, which was followed by a huge crash in 2006.

The 9/11 terrorist attacks put the kingdom in the spotlight of media attention. Lawsuits were filed against Saudi Arabia by the relatives of those killed, and this highlighted how the relationship between the kingdom and the United States had changed. Private Saudi wealth began to leave the United States amid concerns it might become tied up in similar lawsuits. Some of this money came home, in search of high returns, and the Saudi stock market or Tadawul was the beneficiary.

Share trading in Saudi Arabia dates back at least to 1935 and the Tadawul had been under SAMA direction since the mid-1980s. Yet it had never matched the kingdom's growth because most big businesses – with the exception of the banks – were owned by the state or wealthy families, and neither needed to raise equity capital. This also explained why bank lending had not played a bigger role in the country's development. The stock market was small, inefficient and illiquid. As late as 2003, only 69 companies were listed on the Tadawul. Buying and selling shares was costly as the banks typically charged 1 percent commission, and trading was mostly in the partly privatized petrochemicals corporation Saudi Arabian Basic Industries Corporation (SABIC) and commercial banks such as Al Rajhi. For its part, SAMA had concerns that short-term 'hot money' flows from outside the country would lead to volatility and damage the confidence of local investors. So, the central bank restricted share ownership to Saudis and excluded foreigners.

In the late nineties, the picture started to change. SAMA opened the market up slowly. The first step came in 1997 when it approved a closed-end fund, which raised $250 million in London. The next year the International Finance Corporation (IFC) – a World Bank affiliate – included Saudi Arabia in its index of emerging markets. Two years later, foreigners were allowed to buy the same mutual funds that the banks sold to their Saudi retail customers. SAMA also improved the infrastructure of the market. Shares started being settled on a same-day basis and banks cut their commission rates. The Tadawul All-Share Index was launched as the market benchmark. But liquidity was still poor.

The Saudi economy was the biggest in the Arab world and the stock market now began to reflect this. In 2001, it accounted for nearly half the total stock market capitalization of all Arab equity markets. Foreigners represented only a tiny fraction of the market, which was worth $300 billion by 2004, while a significant proportion of the Saudi population, over 220,000 citizens, had invested money in mutual funds. But there were only a limited number of big shares to trade in, and many of them were banks.

Before the market could play a significant role in the economy investors had to be able to buy shares in state-owned companies. The government had never been wedded to state control of the economy, save in the oil sector, and as early as the 1980s 30 percent of SABIC had been sold to the public. Over the following decade, there was pressure to privatize in order to help fund the budget deficit. After the Supreme Economic Council was set up in 1999, with SAMA's Governor as a member, it drew up a list of companies for sale. Attitudes to the stock market changed dramatically after 9/11. The private sector held far more money abroad than the government. Estimates of private Saudi wealth overseas in 2003 ranged from $480 to $900 billion, compared to government funds of $87 billion. The stock market was the obvious place for the wealth that came home.

With so much money returning to the kingdom, the potential existed to create a virtuous circle if the Tadawul were expanded. Companies in the non-oil sector could raise capital for expansion. Savings from pensions and insurance products would find a natural home. This would in turn encourage domestic savings and capital formation, and reduce the impact of the 90 percent solution, whereby most government spending flowed out of the riyal. Stock market financing would furthermore reduce the role of specialized government credit agencies, which had promoted a culture of reliance on the state through soft loans and government contracts. The first state flotation after 9/11 was in 2002, when shares were sold in the nationwide utility, Saudi Electric, followed the next year by the sale of 30 percent of Saudi Telecom, which raised the equivalent of $4 billion. But the state still retained majority stakes in both companies.

In a sign of the times, in July 2003 the new Capital Market Authority (CMA) became the independent market regulator of the Tadawul. The hand of SAMA was clear in the appointments. The CMA's first chairman was ex-SAMA Deputy Governor Jammaz Al-Suhaimi, and another of its five members was Mohammed Al-Shumrani who had succeeded Mohammed Omar Al-Khatib as head of the Investment Department. The CMA and SAMA both wanted to see the stock market grow in a gradual and sustainable way. They were worried by its rapid expansion, believing that there was too much cash chasing a limited number of companies.

The value of the Tadawul rose fourfold in the three years following 9/11 (Fig. 6.11), and in 2005 the market doubled in value again.

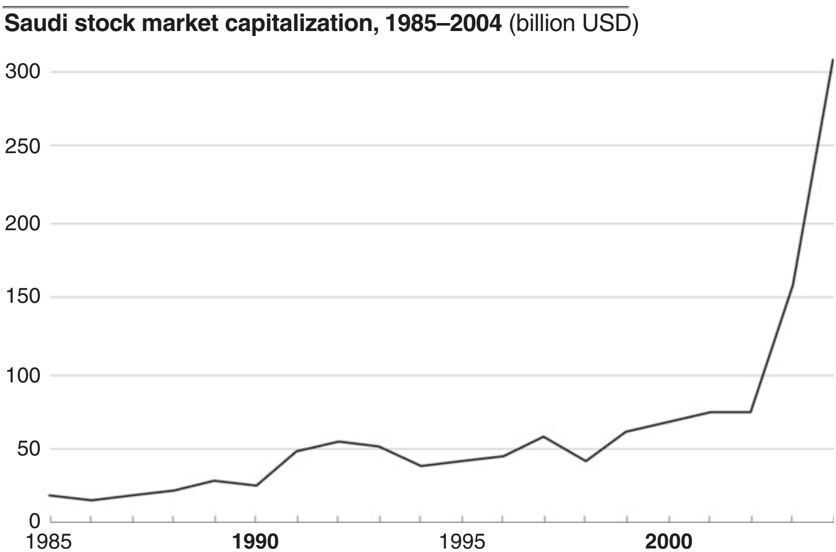

Fig. 6.11 Saudi stock market capitalization, 1985–2004

Source: SAMA Annual Reports

Turnover soared – by 2004 it was over 150 percent, with many small investors day-trading and using leverage to increase their exposure to shares at higher and higher valuations. The authorities would have a major problem when the stock market faltered, because hundreds of thousands of small investors would exert pressure on the government to support the market. These fears were well founded because in early 2006 this is exactly what came to pass. That story is told in the next chapter.

7 A Global Player

The charming Swiss town of Basel on the river Rhine is home to the Bank for International Settlements (BIS). It was founded here in the wake of World War I to sort out the debts of combatant nations. From being the last relic of the Treaty of Versailles, the BIS has gradually evolved into 'the central bank for central bankers' because it offers banking services only to central banks. The BIS is one of the most important and little-known organizations in the world of finance.

Membership of this club is not handed out lightly, but in 1996, nine central banks from developing economies were invited to join. They came from Brazil, Russia, India, China (and Hong Kong), Korea, Singapore, Mexico – and Saudi Arabia. Three years before the kingdom joined the G-20 group of major economies, SAMA was accepted as one of the top central banks, reflecting the kingdom's importance in the global economy.

Common to all the new members of the BIS was their credibility. In the case of SAMA, this was especially hard-earned because of the close linkage between fiscal and monetary policy in Saudi Arabia. The fact that fiscal policy was outside its control and the currency peg was fixed, placed a huge burden on SAMA to manage the reserves and meet demand for dollars. Foreign exchange inflow came from oil income, which nobody could predict, and the 90 percent solution showed no sign of changing. SAMA had to manage the reserves prudently and make sure that there was no loss of confidence, which would have led to individuals and banks cashing in riyals and buying dollars.

The Saudi government never lost faith in the idea of voluntary financial markets, although it must have been tempting at times to SAMA and the Finance Ministry to act through coercion by forcing the banks to buy more bonds than they wanted. This openness was applied to foreign direct investment (FDI) as well. In 2000, the Saudi Arabian General Investment Authority (SAGIA) was set up to increase FDI into the country. The new law gave foreign investors the same level of incentives, benefits, and guarantees as Saudis themselves received, apart from the tax rate on profits.

Only a year after the controversial idea of taxing expatriates was raised again, SAMA felt confident enough to ask the US ratings agency Standard & Poor's (S&P) to assign ratings to Saudi Arabia for the first time. In July 2003, four months after the invasion of Iraq, the American analysts reported that they had found a broadly positive picture. The economy had remained stable despite severe shocks, due to the counter-cyclical policy of running budget deficits ultimately financed by SAMA's reserves. S&P applauded the pegged exchange rate, the history of low inflation and a sound banking system. SAMA's reserves could finance about ten months of imports and were more than equal to the repayment of short-term debt incurred abroad by the banks. Above all, the government had almost no external debt and owned sizeable domestic assets in the form of state-controlled companies.[9] Riyal government debt was a burden, but most was in medium and long-term bonds, and the biggest holders were the government's own funds.

S&P went into the realm of political governance when they criticised the country as possessing a slowly developing socio-political system. Their argument was that Western countries found it easier to raise taxes and cut spending when necessary. While tax hikes and spending cuts are tough to implement anywhere, a good case can be made that this is especially hard in a middle-income country like Saudi Arabia where the major employer is the government. The public sector payroll had grown every year for half a century. Attempts to impose direct taxation – including the 1987 decision to re-impose a tax on expatriate workers – had not been successful. What S&P failed to grasp was that the Saudi political system had been stable and effective precisely *because* it developed slowly. The decade ahead would see this approach tested with the collapse of the stock market at home, the biggest international financial crisis in 70 years, and the regionwide impact of the Arab Spring.

Notes

1. Paul Cashin, John McDermott and Alasdair Scott. 'Booms and Slumps in World Commodity Prices,' International Monetary Fund Working Papers G99/8 (1999):11. They reached this conclusion after looking at 36 commodity price series for a 42-year period from 1957 to 1999. This is the key paper finding a random walk (no duration dependence) in the historical data series of most commodity markets. Random walk applies to other financial markets as well, such as stock and bond markets which have so far proved

impossible to predict systematically. More recent work has not seriously dented the random walk conclusion about oil prices. See for instance Ron Alquist, Lutz Kilian and Robert Vigfusson. 2011. 'Forecasting the Price of Oil,' Federal Reserve System International Finance Discussion Papers Number 1022.

2. Commodity Futures Trading Commission, 'Interim Report on Crude Oil – July 22, 2008.' Interagency Task Force on Commodities Markets, http://www.cftc.gov/idc/groups/public/@newsroom/documents/file/itfinterimreportoncrudeoil0708.pdf (Accessed October 24, 2016). It does not appear that the Task Force submitted a further, final report.
3. The main ones were the Public Pension Agency (PPA) and the General Organization for Social Insurance (GOSI).
4. MeasuringWorth.com, Stock markets database, http://www.measuringworth.com/DJIA_SP_NASDAQ/ (Accessed October 25, 2016). Accurate data including dividends reinvested is not easily available. In practice SAMA's managers in developed stock markets tended to under-perform the indices.
5. Federal Reserve Bank of St. Louis, Economic Research, FRED database, https://research.stlouisfed.org/fred2/series/BAMLCC0A2AATRIV/ (Accessed October 24, 2016). This provides credit market returns on the Bank of America Merrill Lynch US Corporates AA Total Return Index. This gives an approximate return from the sort of bonds in which SAMA's external fund managers in the United States would have been investing.
6. International Monetary Fund, 'Financial Sector Assessment Program (FSAP) 2006,' http://www.imf.org/external/index.htm (Accessed October 24, 2016). This program involved officials from the Fund coming to Riyadh for several weeks and discussing with SAMA what they saw as the current issues in the banking system. The IMF published the reports with a delay (typically a year or so). The first report was published in June 2006 but the work was done in November 2004 and included banking data from 1999 and 2003.
7. Return on equity was 9 percent in 1999 after the Asian crisis, and 23 percent in 2003.
8. Foreign currency loans and deposits were each 18 percent of the total in 2003, having fallen from a 1999 level of 20 percent for deposits and 29 percent for loans.
9. There was a 2004 estimate of gross external debt of $18 billion from the IMF's regular Financial Stability Assessment Program (FSAP) visits, some of it owed by quasi-government organizations.

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CHAPTER 7

Impact of the Global Financial Crisis and Its Aftermath, 2005–2016

1 The Stock Market Crashes: 2006

In December 2005, former President George HW Bush ('Bush 41') flew to Kuwait to deliver the keynote speech at a conference on the future of capital markets in the Gulf. The man who had rid the country of Saddam Hussein's army 15 years earlier was a welcome guest. Bush spoke at the end of a year when local stock markets had soared and the Tadawul had doubled in value. The Dubai market had risen even further. The Gulf markets, he said, were no longer local and illiquid. They were capitalized at nearly a trillion dollars and the region was booming on the back of a higher oil price. But his vision was that the fortunes of the Gulf were not just linked to oil. Governments were privatizing, and non-oil growth would drive the future. His message was clear: these stock markets can only go up.

The climate was assuredly one of great optimism. The oil price was going up. The Saudi Arabian General Investment Authority (SAGIA) aimed to attract a trillion dollars of direct investment over the next 20 years. International banks were lining up to lend money for mega-projects. The International Monetary Fund (IMF) was praising the privatization program that fueled the stock market's rise. What could go wrong?

What indeed? Discussions at the Kuwaiti conference should have focused more on the technical factors driving the Saudi Stock Exchange (Tadawul). The index was dominated by state-owned companies and

A. Banafe, R. Macleod, *The Saudi Arabian Monetary Agency, 1952–2016*, Financial Institutions, Reforms, and Policies in Muslim Countries, DOI 10.1007/978-3-319-55218-7_7

banks. Private investors (only Saudis and other Gulf Cooperation Council citizens could buy shares direct) believed the state would protect them from losses. Buying shares in privatized companies became a sure-fire way of making a profit because they were sold at below their real value to distribute wealth to the people. Small investors were favored and big investors were forced to drive the price higher as they built up stakes. At the same time, new technologies made it easy to become a speculator. Investors could subscribe to the new companies at bank branches, over the telephone and on the internet. In August 2005, public sector pay rose when King Abdullah succeeded King Fahd and ascended to the throne, and civil servants were able to increase their borrowing against the security of the regular salaries credited to their bank accounts. Saudis with access to the internet could trade during working hours. Office chat was full of stories about money made day-trading on shares. By 2006, over half a million Saudis were trading online.

A week after Bush's speech, at the end of December share subscriptions closed for Yansab, a start-up operation to produce ethylene – a feedstock for the manufacture of plastics – from oil. Yansab was not an obviously attractive concern. It was being sold by the state petrochemical company SABIC and needed to raise nearly $5 billion, mainly through project finance deals, which were yet to be arranged and it would produce no ethylene for two years. Nevertheless, small investors hurried to subscribe and by the subscription deadline nearly nine million individual bids to buy shares had been received (many were multiple applications by speculators). This was the standard pattern for a public offering. Small shareholders were heavily favored in the hand-out with nobody getting more than five thousand shares. Those small investors who had bought on credit prepared to sell to the bigger investors. Trading on the first day was chaotic as the Yansab price soared.

In January 2006, the market in bank stocks received a boost when Standard and Poor's raised the credit ratings of the biggest Saudi banks. But by Saturday 25 February 2006, the bubble in share prices was ready to burst. The market index stood at 20,966, having risen by a quarter since the start of the year. The valuations were extraordinarily stretched: by this time the average price earnings ratio was 46 times, and prices 11 times book values. The stock market was worth three times Saudi Arabia's Gross Domestic Product (GDP). Wall Street, by comparison, was worth less than twice American GDP at the height of the dotcom bubble in 2000. The following day prices began to fall and within a week the index had dropped 15 percent. The Saudi market was not alone – other Gulf economies were also affected: by early March stock prices had dropped by a third in Dubai. The Capital Market Authority (CMA) suspended two share dealers and imposed daily limits to stop the market falling, but the selling continued.

By May, the market was down by half and members of the government's advisory council, the Shura, called for the state to support shares by official buying. But, unlike in the United Arab Emirates (UAE), the government did not intervene to support the market. Instead, gasoline prices at the pump were cut to compensate small investors for their losses. In the summer, the pace of the fall slowed. By October, the market was 60 percent below its peak.

The Tadawul crash occurred just over a year before the sub-prime collapse in New York. And there was a link. The Riyadh bubble in stocks was directly connected to the activities of global banks, and SAMA – like other central banks – found itself buffeted by events beyond its power to control. The crisis in developed markets in 2008–2009 infected the Saudi banks and briefly threatened the Saudi riyal (SAR) itself. And yet the Saudi economy continued to do well. The price of oil, as always, was the single most important factor in the economic well-being of the kingdom, and fortunately this remained high until 2014. Then it fell and another crisis loomed.

2 Economic and Fiscal Background

Fueled by rising oil prices, the Saudi economy roared ahead. Between 2005 and 2014, income from oil was nearly one trillion dollars over budget. But in 2014–2016, matters became much more somber as the oil price dropped and stayed low.[1]

Budget surpluses replaced deficits up to 2014 – save in 2009, when the global financial crisis meant that demand for Saudi oil slowed – and SAMA's foreign reserves soared, peaking in 2014. As in the 1970s, a proportion of the revenues were used to facilitate economic diversification. The government also encouraged state-controlled companies (such as Yansab) to borrow from the global banks. High public spending fueled inflation and SAMA had to grapple with the problem, which led to pressure to revalue the riyal in the run-up to the financial crisis. Inflation was close to 10 percent in the months before Lehman Brothers collapsed in the autumn of 2008. The following year saw the start of the world

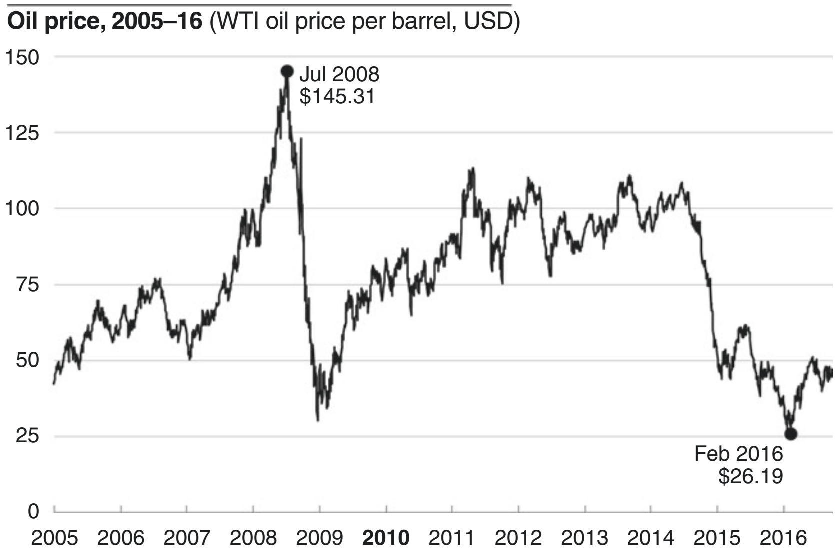

Fig. 7.1 Oil price, 2005–2016

Source: Federal Reserve Economic Data

recession, and growth turned negative. Oil prices fell and the government kept the economy going by running a deficit – a pattern repeated after 2014 (Fig. 7.1).

On the downside, the non-oil economy continued to disappoint. It was still driven by government activity, whether directly through social spending or indirectly by state-controlled businesses. The government's revenue base also remained firmly anchored to oil. No sales or income taxes were imposed. Although non-oil income doubled from 2005 to 2013, it still accounted for a smaller proportion of revenue than 20 years earlier.

The extra public spending mostly went on projects: schools, universities, hospitals and communications, such as the planned Riyadh metro and the high-speed Haramain rail project linking Makkah and Madina in the west of the kingdom. Demand for steel and cement far outstripped local capacity and their prices rose sharply, contributing to the inflation (Fig. 7.2).

Revenue and spending were strong when the oil price was strong, and weak or negative when the price was weak (Fig. 7.3). The economy

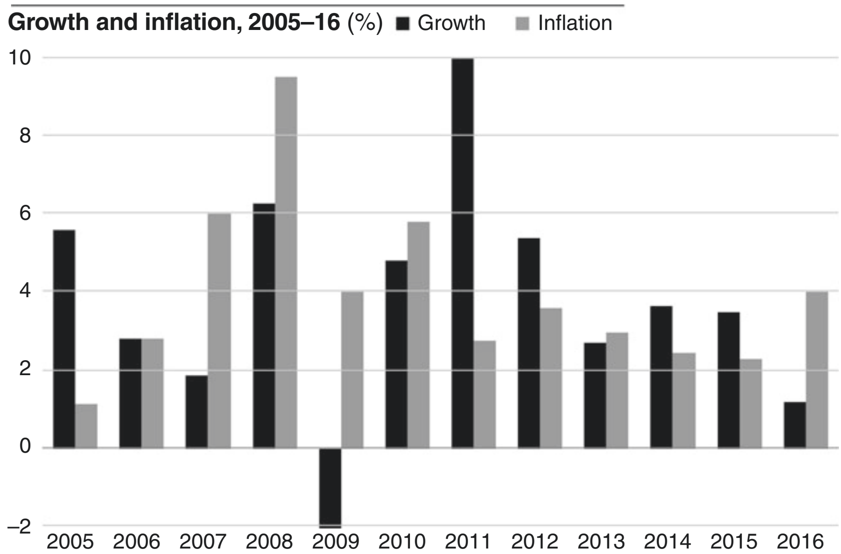

Fig. 7.2 Growth and inflation, 2005–2016

Source: IMF WEO Database, October 2016

continued to be tethered to the price of oil. At the end of the period, the government made its most determined attempt to diversify away from oil.

3 Replacing Government Debt

When the oil price rose in the new millennium there was a difference of opinion between SAMA and the Finance Ministry, which was determined to use its surpluses to repay government debt. This had peaked in 1999 at 100 percent of GDP. The argument was simple: debt cost money to service so there was no reason to keep it unless you needed to borrow. Moreover, the IMF and the rating agencies pointed to the fall in debt as a prime reason for upgrading the country's ratings.

The central bank, on the other hand, wanted to keep some government bonds in issue as a way to develop the bond market. Saudi finances had seen rapid and unpredictable swings and SAMA believed that developing a secondary market in debt was the key to keeping costs down in the long run. This could not be done if it was all repaid. The central bank also

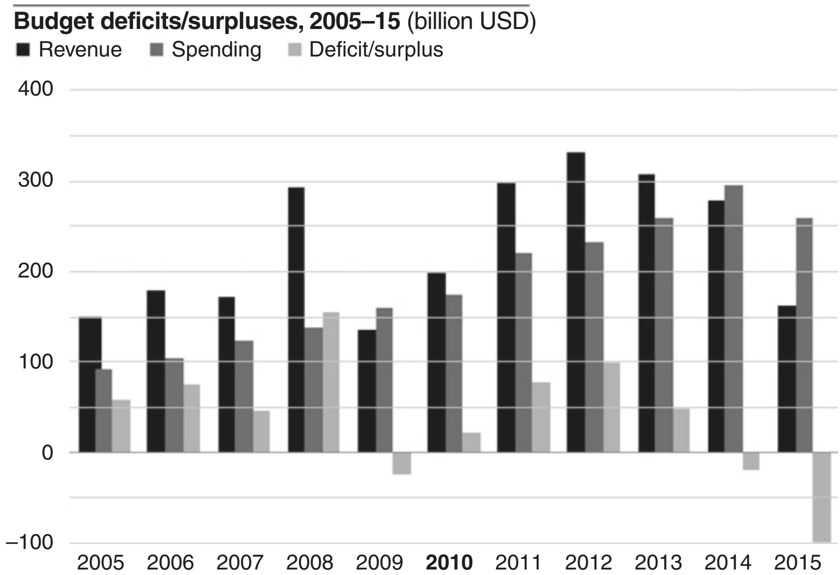

Fig. 7.3 Budget deficits/surpluses, 2005–2015

Source: SAMA Annual Reports, Ministry of Finance

worried that if Gulf Monetary Union happened without liquid bond markets in Riyadh, Saudi Arabia might find itself hosting the new central bank, but Dubai would keep the prize of being the Gulf's financial hub – as in the euro area, where the central bank was in Frankfurt, but London remained the financial center.

The Finance Ministry won the argument and debt was repaid as it matured. Without government debt, SAMA found itself back in the era of the early 1980s when it had to bear the costs of issuing its own debt. SAMA's tools for fine-tuning liquidity were the repo and the reverse repo facility, but to make them work the banks had to hold government bonds which they could repo. So when government debt issuance ended in 2007, new SAMA bills were issued for monetary control purposes, as well as murabaha instruments (the Shariah-compliant equivalent of bills for Shariah-compliant banks such as Al Rajhi). SAMA bill issuance increased after Lehman failed in September 2008 and Saudi banks tightened their credit standards. For the next two years, lending to the private sector stalled and the banking system was flush with cash. SAMA stepped in with more bills to drain liquidity.

At the same time, the new Basel III requirements – designed, among other things, to prevent another liquidity shortage like the one that had occurred during the global financial crisis – meant that SAMA had to introduce a new Liquidity Coverage Ratio (LCR) to ensure that likely cash outflows were matched by liquid assets, either cash or government bonds. The banks were caught out by both sides of the Basel equation. They were highly dependent on short-term deposits, which meant that at any given time cash outflows could be large, and they did not have enough liquid assets to match this potential liability. This meant that if SAMA did not provide more bills, the risk was the banks would put money abroad. SAMA started issuing more bills from 2010 ahead of phasing in the LCR in the following years.

By early 2015, central bank bills amounted to more than 6 percent of GDP, the level SAMA had believed back in 2007 was the minimum required to keep a government debt market alive. It had substituted its own debt, paid out of its income, to give the banks the assets they needed. Effectively, the original Bankers Security Deposit Accounts were back, only their name had changed. But the SAMA bills were not traded between the banks and the opportunity to build a yield curve had been missed.

Then, later in the year, the government returned to the domestic debt market. Interestingly, this time it was a matter of choice. The earlier episode of debt issuance that began in 1988 happened because the government's account with SAMA was out of riyals, but in 2015 this was not the case. The government wanted to act by issuing debt on good terms before it was forced to do so. This pro-active approach was part of the new style of running the country introduced under King Salman who succeeded his brother in that year. In the following year, a debt management office was set up within the Finance Ministry and the prospect was raised that SAMA's role of managing government debt would diminish, even if it remained the paying agent, and continued to issue its own SAMA bills.

4 The Battle for the Riyal: 2007–2008

Food and housing prices soared in the pre-financial crisis years as government spending grew rapidly. The question of how to tackle inflation without revaluing the currency dominated SAMA's agenda until it was pushed off the table by the global financial crisis. The debate was conducted in public and the Shura Council became involved.[2] All Gulf countries had the same problems, so attention was paid to what Saudi Arabia's neighbors were doing. When Kuwait left its dollar link in May 2007, SAMA came under sustained pressure to follow suit.

Rising food prices are a politically emotive issue in Saudi Arabia. But SAMA can do little about them. Broadly speaking, there are two different kinds of inflation, one caused by increased demand and the other by higher costs. While a central bank can try to dampen demand by raising interest rates, little can be done about higher costs if they originate from abroad – other than to revalue the currency. The rise in food prices was cost-push inflation, and it was not confined to Saudi Arabia. The world price of raw foodstuffs measured in dollars soared by more than a half in the four years up to 2008. The main reasons were a weak dollar in the run-up to the financial crisis and the rise in oil prices, which drove up a range of agricultural costs. The very resource that lay behind the country's renewed prosperity was also responsible for dearer food.

A rise in the price of food posed a direct threat to the nutrition of poorer Saudis. Their diet was already dependent on the basic foodstuffs like lentils and rice, so they could not react to higher food prices in the way an American consumer would do, by eating out less or buying a cheaper cut of meat. Housing costs were the second-biggest element of the price index. After several years of stable prices, from 2006 onwards they jumped strongly, largely driven by the renewed influx of foreign workers. Expatriate numbers rose by more than 2.5 million in the four years up to 2009, and all of the newcomers needed accommodation. New housing put extra pressure on the supply of building materials, which drove inflation up further, while competition for existing housing also caused rental rates to rise.

SAMA could do little about food and housing inflation without changing the currency peg. If inflation had been due to a rise in bank credit, then it could have acted. But this inflation was not due to the banks lending aggressively. As in earlier oil booms, liquidity was abundant at the banks but it was not feeding through into the rest of the economy. If the riyal had been a floating currency, it would have naturally risen on the back of the current account surplus caused by high oil revenues and this would have made imports cheaper; but the dollar link stopped this happening. Likewise, the central bank had only limited freedom of action domestically to hike rates, due to the exchange rate link, which meant that Saudi interest rates had to keep fairly close to dollar ones.

Looking around the Gulf in early 2007, it was clear that inflation was related to how fast each economy was growing. Qatar was booming and had inflation in double-digit territory, followed by the UAE. Kuwait and Saudi Arabia, by contrast, were both at the lower end. So it was a surprise when in May Kuwait announced that it was shifting from a dollar link back to the pre-2003 currency basket. Kuwait's inflation rate, like Saudi Arabia's, had been rising, but the move was not so much a revaluation as a pre-emptive strike against further dollar weakness. Financial markets around the Gulf were electrified by the news, which represented an unexpected hurdle in the path toward a common Gulf currency.

If the link to the dollar could be broken in Kuwait, then the same could happen in Riyadh. Critics argued that inflation could never be conquered as long as the dollar exchange rate stayed at SAR3.75. Instead, they favored either revaluing or following the Kuwaiti plan of moving to a basket. Policy-makers within the central bank debated whether the peg was sustainable. Should the currency move to a wider band, or simply be fixed at a higher level?

The obstacles to widening the currency band were daunting. The economy had no experience of managing foreign exchange risk. For example, many firms lacked computer systems that could cope with a changing dollar-riyal rate. The private sector borrowed in riyals and dollars interchangeably. SAMA would need to adopt an inflation target. Yet the central bank was ill-equipped to do this. SAMA decided to continue to adhere to the policy of the fixed peg.

The second question was whether a revaluation made sense. On the plus side, it would not hurt exports because the price for oil and petrochemicals was determined in dollars on the world market, so a more expensive riyal would make no difference. But for over 20 years, the fixed rate had provided an anchor for monetary stability in tough times as well as good ones. Any shift would change the rules of the game for the banking system.

SAMA's conclusion was that the level of the peg was not perfect, but the implications of changing it were worse and so it made sense to play for time. The dollar would not stay weak forever. Meanwhile, sustaining the peg was a technical matter of sterilizing the speculative inflow of dollars being turned into riyals by those hoping to make a profit from an uplift in the exchange rate, and draining the liquidity these extra riyals created.

As 2007 advanced and the dollar weakened, pressure to revalue the currency intensified. The crisis in the American banking system was obvious, and the Fed's rate cuts in response widened the gap between interest rates in riyals and those in dollars, and made it cheaper for speculators to borrow dollars and take out long riyal positions, betting on a revaluation. SAMA did not follow the Fed because it needed to appear to be tough in its fight against inflation. This was a battle that the central bank was also fighting through the media; by repeatedly issuing public denials that it had plans to revalue, with Governor Al-Sayari saying in September that it was not necessary to follow a cut in American interest rates, especially when there was high liquidity, and emphasizing rate stability and transparency were important for investors. Vice Governor Al-Jasser worked hard to explain SAMA's stance and its argument for defending the peg in speeches and seminars. But the Saudi media were convinced that revaluation was imminent.

The toughest day for the riyal was November 26, 2007 when it rose to SAR3.70 against the dollar. The one-year forward rate implied a 5 percent revaluation and SAMA had to intervene repeatedly by buying spot dollars from the banks. The pressure eased as SAMA started lowering policy rates. At the same time, SAMA drained liquidity from the banks by raising the cash reserve ratio.

Meanwhile, political pressures had increased. In the second half of the year, the price index rose at an annual rate of more than 10 percent and food prices went up by nearly double that. There were calls for an increase in food subsidies, but although the government cut the costs of numerous government services, there was little action on foodstuffs. The authorities also increased civil service salaries and welfare benefits. This was the standard fiscal policy response, but raising the incomes of ordinary Saudis boosted purchasing power and offset the disinflationary effect of the subsidies.

Critical comments were made online, in newspaper columns and even in the Shura Council, and in July 2008 a Shura committee publicly called for a 20 percent currency revaluation to stop inflation. But economists in the commercial banks endorsed SAMA's policy of keeping the peg at its current rate. A small revaluation, they believed, would merely encourage speculators to push the riyal up further.

SAMA took comfort from their backing and stuck to its strategy of buying time. This slowly began to work. The scale of attacks on the currency peg decreased as 2008 progressed and there was a slow improvement in the inflation picture. The dollar stopped falling and then rallied strongly as the financial crisis spread outside America. Global banks scrambled to acquire dollars, and speculators sold riyals to get the dollars they needed and eased off the pressure – they had their hands full merely trying to survive the crisis. Kuwait's example also became less compelling when it became clear that its action had not led to a drop in inflation, which stayed close to that in Saudi Arabia.

The dollar's stability after April 2008 slowed the food price hikes. SAMA's actions to mop up liquidity in the banking system became more effective. The banks could park their excess funds with the central bank by buying SAMA bills.

By the autumn of 2008, SAMA was winning its year-and-a-half battle to defend the peg. Food prices hardly moved after July and, with the biggest item in the index stabilizing, inflationary pressure gradually fell to 8 percent in the second half of the year. Meanwhile, the worst recession in 70 years had hit the world in 2009. Now the central bank turned from defending the currency to defending the financial system itself.

5 SAMA and the Global Financial Crisis: 2005–2010

The prelude to the crisis for Saudi Arabia was the project finance boom from 2005 onwards, when Saudi Arabia became a target for international banks eager to fund the country's development program with wafer-thin margins. This encouraged the sale of shares in some state-controlled enterprises, which fueled the stock market. The Tadawul crash of 2006 was a sign of how finance had changed. When the international banks withdrew during the global crisis the following year they caused problems for Saudi banks and drew SAMA into the heart of the tempest.

The third oil boom in the first decade of the new millennium differed significantly from the others because of the involvement of international finance. In the 1970s and 1980s the growth in Saudi financial assets had resulted from SAMA's policy of 'recycling.' The global banks did not get involved in the kingdom's investment plans, which were financed internally by oil revenues. So Saudi Arabia remained semi-detached from global finance. It deposited money with banks in major financial centers, but they did not invest the money back in the kingdom.

A generation later the financial world had become fully globalized and foreign banks saw profitable opportunities in the country's development. SAMA could not rein back their activity, although it did ensure that the commercial banks it supervised remained well capitalized. Nor did it have a full set of tools to cool down the Saudi stock market. The currency peg limited its influence on interest rates, and supervising the Tadawul had been hived off to the independent CMA. Finally, the central bank was influenced by the financial orthodoxy of this period, which was that the markets knew best.

Financial bubbles always have a rationale, and a compelling logic underpinned the project finance boom. By 2005, Saudi Arabia needed large investment to update its infrastructure and diversify the economy. A backlog of projects to improve electricity and water supplies had built up during the years of cheap oil. Black-outs and brown-outs plagued the kingdom in the summer months. Water desalination plants were needed to replace the fast-dropping natural aquifers, the only source of water in a country without a single river system. Diversification from oil would, it was planned, be based on industrial cities along the coasts clustered around giant state-sponsored petrochemical plants, which would also generate electricity and desalinate water.

As the inward investment agency, the role of SAGIA was to estimate the country's needs and help obtain the funding. The numbers it released were startling. In early 2006, SAGIA announced that it aimed to attract $1 trillion in foreign direct investment (FDI) over the following 20 years, or $50 billion annually. To put this in context, China had FDI flows of $61 billion in 2004. The International Energy Agency in Paris, for its part, had concluded the Gulf nations would have to spend up to $0.5 trillion by 2030 in order to satisfy global energy demand. These numbers were so large as to be meaningless but they fueled the giddy optimism of the boom.

Who should finance this massive spending? Confidence in the power of the markets to do the work had taken the place of the old idea that the state should play a leading role. It was also the prevailing belief that leverage was the key to good investment returns: that is, a high level of debt finance and a low level of equity. The government's plan was that Tadawul flotations of the state-owned companies that were managing the projects would provide them with equity funds and the banks would come up with the debt. But project finance requirements over the subsequent three years alone were greater than all Saudi bank deposits in 2006, so money would have to come from banks outside the kingdom. The structure of Saudi finance also complicated the launch of project financing. A Saudi bank buying into project finance faced a complete mismatch of assets and liabilities. The banks depended on deposits, which could be withdrawn on demand, while the projects required a commitment of funds for up to 20 years.

Saudi banks did become involved to a small extent, but they faced stiff competition from the global banks which attempted to gain market advantage by moving into domestic banking. The World Trade Organization agreements, which the kingdom had signed in 2005, allowed for majority foreign ownership of a bank. This could have been disastrous in the over-heated atmosphere of the years just before the financial crash. Fortunately, SAMA became more cautious and stopped awarding new bank licenses.

Investment banking, on the other hand, was changing under the control of the CMA. It wanted to see more competition and, by mid-2007, just as the sub-prime problem became public knowledge, it had licensed 64 investment banks, including a joint venture between Morgan Stanley and a local investment bank, The Capital Group, which was headed by Fahd Almubarak, who a few years later would run the central bank. Morgan Stanley Saudi Arabia proved to be just the sort of innovative player the stock market needed. In 2008, Almubarak's bank gave foreign investors direct access to the Tadawul for the first time through a swap transaction, and followed up by launching Saudi Arabia's first open-ended mutual fund open to foreigners.

By early 2006, as the Tadawul was at its dizzying peak, evidence was emerging of financial excesses elsewhere. The massive refining and petrochemicals project at Rabigh on the Red Sea is a good example. In September 2005, the cost of this project had been estimated at $8.6 billion, but five months later, before work had even started, it became clear that it would require at least another billion dollars. Under the terms of their contracts the banks had to come up with the extra money. The banks financing Rabigh included names that would soon become casualties of the financial crisis, including BNP Paribas, Credit Agricole, Westdeutsche Landesbank and Citibank. They had committed to a long-term, multi-year project in an area where they had little previous experience and where the margins would only be profitable many years later – assuming that the project was not refinanced at lower terms in the interim.

The collapse of the Tadawul was a warning of the coming financial crisis. For a while afterwards a sense of unreality persisted, with international lenders aggressively competing for projects and agreeing to increasingly harsh terms. They still accepted 'open book' deals like Rabigh, where they committed to pay for any escalation in costs, while the projects had higher and higher leverage and longer loan periods. Many ventures lacked the long-term supply contracts necessary to lock in revenue in order to repay the loans. But by October 2007, any international interest in projects had disappeared. Although Rabigh went ahead, it was later estimated that a quarter of Saudi energy projects had been shelved.

The world financial crisis began in 2007 and a decade later we are still living in its aftermath. History has been aptly described as an argument without end, and historians will long argue about what caused the financial crisis. Any list would include the build-up of global imbalances, income inequality, transfer of investment to east Asia and a rise in consumer debt in the United States, a failure of regulators and excessive confidence that markets would always stabilize.[3] What is clear is that the crisis began with rising defaults in the American mortgage markets and that by June 2007 other credit spreads had started to widen. SAMA was aware of the problems because of its regular meetings with its American fund managers. It grasped that the financial system was in trouble, but felt it could watch the crisis from afar, since the problems centered on the rating of collateralized debt obligations (CDOs) that consisted of subprime mortgages for borrowers who had poor credit records, an area that SAMA had steered clear of investing in.

The central bank also felt relaxed because it knew that Saudi banks were not involved. At the onset of the crisis, banks' exposure to mortgage-backed securities and other securitized assets amounted to only 3 percent of total assets, figures which flew in the face of conventional wisdom of the time, which held that securitizing assets and using derivatives was desirable as a way of reducing risks by spreading them around the financial system. When the American fund managers returned to Riyadh in the late summer of 2007 they brought worse news with them. A vicious downward spiral had gripped the United States. As the rating agencies downgraded CDOs, investors tried to reduce their exposure. They found no buyers and so prices plunged. The problem was made worse because investment banks like Bear Stearns and Lehman were in over their heads with exposure to problems in the CDO market that was larger than they were aware of, or could handle. By August, the whole American financial system was in trouble. The most serious consequence was that banks stopped lending to one another because of fears about solvency. Some banks found it difficult to raise funds.

Like a cancer metastasizing, the panic generated by the problems of the United States infected the global financial system. The next stage in the crisis was a flight to quality: risk-free bond prices soared while corporate bonds sold off and structured products were shunned by investors. Equity and real estate prices fell, with financial institutions badly hit.

In SAMA's view, the central banks reacted appropriately. Both the US Federal Reserve Board (Fed) and the European Central Bank (ECB) injected very large amounts of cash into the money markets to bring rates back down and allow banks to postpone the liquidation of CDOs. The Fed also cut the discount rate. By September 2007, some degree of stability had returned to the markets but it was only a lull. Interbank rates remained at elevated levels, and it soon became apparent that many investors needed to sell their holdings of structured products, but were unable to find buyers at an acceptable price. Until they could be sold, the crisis was still far from over. It was about to move instead into a more serious and unprecedented phase.

SAMA knew from its fund managers that this would not be a small adjustment. The banking supervision team was also watching with concern because the Saudi banks were creditors of Wall Street and the City of London, and would be damaged by any failures there. SAMA had many years' experience of bank regulation in a financial system dominated by commercial banks, and it knew that a regulator has failed in his job if he does not understand the banks' business. But in New York and London, the regulators faced a new challenge because the problems had arisen outside the traditional banks, in the shadow banking system.

So far, in contrast to the Asian crisis of 1998, the emerging markets were largely unscathed. It seemed to SAMA at the end of 2007 that the most likely way in which Saudi Arabia could be affected was not through the financial markets but through a slowdown in the world economy, which would lead to a drop in the oil price. As the higher cost of borrowing continued, this seemed an increasingly plausible scenario. The Saudi banking system weathered the first year of the storm in good shape. Customer deposits and loans both rose. Critically, liquidity remained abundant.

Lehman's failure in September 2008 is the iconic event of the crisis. As the problems spread SAMA stuck to fighting inflation by withdrawing liquidity just as central banks in the United States and Europe were injecting it. The central bank was aware of this anomaly but did not feel it could afford to switch course until the autumn. The trigger came from New York. Specifically, it came from 745 Seventh Avenue in midtown Manhattan, the headquarters of Lehman Brothers. When Lehman filed for bankruptcy on September 15, capital markets all over the world went into meltdown and the dollar–riyal interest rate gap widened sharply as Saudi banks were shut out of international funding. This was followed by problems in the Dubai real estate market and a sharp drop in the oil price. Panic gripped the Gulf and there was a scramble to withdraw funds from the banks. Four weeks after the Lehman collapse, on October 11, the UAE central bank guaranteed all bank deposits. SAMA the following day made the first cut in its repo rate in four years and reduced bank reserve requirements. On October 16, in an effort to avoid a bank run, the Supreme Economic Council announced it would ensure the safety of bank deposits, while indicating that this was not an open-ended guarantee.[4]

The announcement prevented a panic of the sort that had marked the start of the Kuwait crisis 20 years earlier. But the banks still had an immediate funding problem. SAMA responded by placing dollar and riyal deposits with the banks. This restored confidence that the banks would have sufficient funds and six months later the deposits had been repaid.

Action shifted from the banks to the currency and the stock market. Rumors abounded that a flight of money out of the riyal into the safety of the dollar would force a devaluation. The Saudi riyal weakened to SAR3.77 from its official peg level of SAR3.75 but it quickly recovered. The stock market rose in response to the announcement of October 16, and the risk premium between riyal and dollar rates narrowed. Between October 2008 and January 2009, SAMA reduced its repo rate from 5 to 2 percent. The Saudi financial system weathered the storm with almost no damage because it was underpinned by continued growth in the economy, driven by public spending. Over the year as a whole, bank profits barely fell, while credit to the private sector and banks' capital and reserves increased sharply. Most importantly, the system remained liquid at levels Western banks looked at with envy.

Muhammad Al-Jasser, SAMA's Vice Governor at this time, deserves much of the credit for the kingdom's rapid response to the Lehman crisis. Born in 1955, he had been second in command of the central bank for 14 years after a spell as the Saudi representative at the IMF in Washington, and he replaced Al-Sayari as Governor in February 2009. Al-Jasser was an outstanding example of the new generation of Saudi technocrats who had grown up in the oil boom. He had excellent public speaking skills as well as an incisive approach to problems. His aim was to lead Saudi Arabia out of the crisis by instilling confidence in the banking system, which was still under pressure. By April 2009, bank lending to businesses was falling and growth in deposits had stopped. Al-Jasser felt that further interest rate cuts might endanger the currency peg, so he lowered the banks' reserve requirements instead to give them access to more cash. This was a prescient move because two months later a major corporate bankruptcy exploded on the financial scene. Up to this point, the problems had originated outside the kingdom. But this was a wholly domestic story.

In June 2009, a spectacular corporate default occurred in the Eastern Province that shone a searchlight into how Saudi businessmen financed themselves. The banks were shocked because about half of all corporate credit in the kingdom was to private businesses. The company involved was the conglomerate Ahmed Hamad Al-Gosaibi & Brothers (AHAB), one of the Gulf's most respected companies, which ran a construction business, controlled the kingdom's Pepsi bottling plants and – critically – had for decades run a money exchange for Aramco's foreign workers to help them send their earnings home. This developed into the financing arm of AHAB, raising funds for the group and servicing its debts. AHAB also established a finance subsidiary in the offshore center of Bahrain.

Almost none of the banks which lent to AHAB wanted security on their loans. They simply handed over money on what was labeled in the Gulf a 'name basis'; taking comfort not from collateral but from the apparent success of AHAB's overall business and its tightly knit family structure. For many years, the financing business had been run by Maan Al-Sanea, a Kuwaiti member of the Gosaibi family through marriage who ran his own business, the Saad Group.

The sequence of events that toppled Saad and AHAB began with the attack on the World Trade Center eight years earlier. After 9/11, SAMA clamped down on unregulated money exchanges, fearing they could be used as a source of financing for terrorists. In response, AHAB moved its funding offshore and formed a new bank, The International Banking Corporation (TIBC), in Bahrain, outside SAMA's regulatory remit. The banks loaning money to TIBC believed they were funding the AHAB group. But AHAB has alleged that Al-Sanea took much of it for himself. There was, moreover, a mismatch between TIBC's long-term assets and short-term liabilities, which made it vulnerable to liquidity drying up. In May 2009, it missed a payment to the Dubai-based Mashreqbank, which in response filed a suit against TIBC. This triggered a rush among the banks that had lent to AHAB and Saad to call in their loans. The credit agencies downgraded them and within a couple of weeks both groups were massively in default.

No Saudi bank has disclosed publicly its exposure to AHAB and Saad, but probably over $6 billion in loans was involved. Exposure was even bigger outside the kingdom. The biggest lender to Saad was reportedly a French bank, BNP Paribas, and estimates of the total amount of failed borrowings by the two groups ranged up to $22 billion. The good news for SAMA was that no bank ran into serious difficulties as a result, despite the fact that most of the borrowings were not backed by collateral.

The Saudi authorities acted quickly to warn other large Saudi borrowers that they would be dealt with harshly by freezing Al-Sanea's assets, followed by AHAB's. AHAB maintained that TIBC was controlled by Al-Sanea and that he had embezzled $9 billion from them. In September 2009, SAMA extended the freezing order to cover the assets of other Al-Gosaibi family members. A travel ban prevented Al-Sanea and the Al-Gosaibis from leaving the country until the matter was resolved. AHAB and Al-Sanea went to court around the world, counter-suing each other. Eventually AHAB offered a global settlement with its creditors. In mid-2017 litigation was going on in the Cayman Islands.

SAMA took prompt action but its power to resolve the affair was limited, and hampered by the absence of a bankruptcy code and rock-solid legal power to collect money from debtors. It assessed Saudi banks' exposure to other private companies, and established a working group with representatives of the banks to review what had gone wrong. Here the Saad-AHAB affair has a parallel with the collapse of Lehman. Whoever was actually in charge of TIBC, both AHAB and Al-Sanea were taking advantage of regulatory arbitrage by locating a bank offshore in Bahrain, just as Lehman used different financing rules in London and New York to disguise its liabilities. Although SAMA talked regularly with Bahrain's central bank, there was no formal arrangement for cooperation. Regulators like SAMA around the world relied – as they still largely do – on informal arrangements.

Critics argued the central bank took too long to resolve the issue and favored Saudi creditors over foreign ones, thus hurting the international creditworthiness of Saudi businesses. But the actual resolution of AHAB's debts was not the responsibility of the central bank. It was a legal matter which went to a committee chaired by the country's chief legal official with representatives of SAMA, the CMA and the Interior, Commerce and Justice Ministries. In the absence of a comprehensive bankruptcy code – like that in the United States – the situation could not be resolved quickly. SAMA did not favor Saudi banks, which used their set-off privileges, as did the foreign banks, to collect some of their debts. Foreign banks were given the right to file their suits with SAMA's Banking Dispute Settlement Committee but most of them had lent to AHAB outside SAMA's jurisdiction. Astonishingly, the Saad-AHAB affair did not mark the end of 'name lending' by the banks. In theory, SAMA can stop it by requiring the banks to obtain and scrutinize financial statements before making loans. But in line with central banking practice worldwide and its own commitment to market-based solutions, SAMA feels that responsibility lies with the banks and the auditors.

The Saad-AHAB affair was followed in October 2009 by the suspension of payments by the UAE conglomerate Dubai World (although Saudi banks' exposure was small). After that, the banks were through the worst of the storm. The Dubai collapse effectively marked the end of the financial crisis in the Gulf. Credit growth resumed in 2010. By the following year, most of the damage to the banks had been repaired. In November 2011, Standard & Poor's concluded that the Saudi banking system was one of the safest in the world, ranking alongside Germany, France, Norway, Sweden and Singapore. The American credit agency cited SAMA's close and conservative supervision as a key factor.

When the IMF came to Riyadh in 2010 to carry out its first assessment of the financial system for six years, the structure of Saudi banking had changed little since the last review and was arguably much the same as in 1980. Although SAMA licensed two new banks, and foreign banks now had branches within the kingdom, the system remained highly concentrated with the three largest (state-owned National Commercial Bank, the Shariah-compliant Al-Rajhi and the old Citibank operation trading as Saudi American Bank) holding nearly half of Saudi assets and deposits. The banks continued to benefit from not paying interest on a large part of their deposits. It was clear to the IMF that the global crisis had damaged the banks far less than the long period of cheap oil before 1999. Banks' return on equity was down from pre-crisis highs but was still well into double digits, with a loan-to-deposit ratio that was highly conservative.

While the structure of the financial system had not changed much, its penetration of the economy had. The most useful way of looking at this is through bank credit relative to the size of the non-oil sector. This picture was one of steady expansion: from 20 percent of the size of the non-oil economy in the 1970s to 100 percent at the start of the financial crisis. The banks had largely taken over from the government's own lenders, such as the Saudi Industrial Development Fund (SIDF), which had traditionally provided low-cost lending to the industrial sector. This sort of credit fell from about 70 percent of non-oil GDP in the 1970s to 20 percent by 2009.

Looking at the trends in bank credit, the commercial bond market was still in its infancy. SABIC had led the way with the country's first corporate bond, structured as a Shariah-compliant (sukuk) bond in 2006. This was followed four years later by the first international bond issue by a Saudi corporate, but few other companies went on to follow SABIC's example. Access to banks and credit had also gone up, partly because of the growth in personal credit secured by the banks' access to public sector workers' incomes. By 2014, 65 percent of adult male Saudis had a bank account. The latest figure for women was a lower 58 percent.[5]

Figure 7.4 shows that the global crisis hardly damaged the banks. Primarily this was because they were firmly tethered to the government. Government spending kept the economy growing, made the banks' loans to the private sector safe and cushioned their exposure to project finance. By stepping in to fund them where necessary, SAMA successfully stopped the banks from taking risks. It kept a close watch on foreign exchange positions and the banks' derivative exposures were mostly on behalf of clients and fully hedged. The situation was enormously helped by high government spending and the redemption of government debt, so the banks were continuously receiving liquidity from the fiscal operations of the government. In one sentence, the dominance of government spending from oil in running the Saudi economy had been vital.

SAMA rejected the creed of market fundamentalism that in the pre-crisis years captured the imagination of academics and policy-makers worldwide. For Saudi Arabia, the uncertainty of oil income had stopped policy-makers becoming too relaxed about the inevitable excesses of the market economy. In the advanced economies, a pro-cyclical bias was built into systems of financial regulation so that booms ran on for too long, while governments denied themselves the fiscal room to maneuver by continuing to run budget deficits. When growth was stronger the kingdom ran fiscal surpluses – which were counter-cyclical – but arguably it should have run even bigger ones to cool the economy down. And in 2008–2009 keeping up spending worked counter-cyclically against the risks of the global financial crisis. But the kingdom stored up trouble in the years ahead by continuing to raise spending rather than building up the foreign exchange reserves by running bigger budget surpluses. It was effectively betting the oil price would stay high.

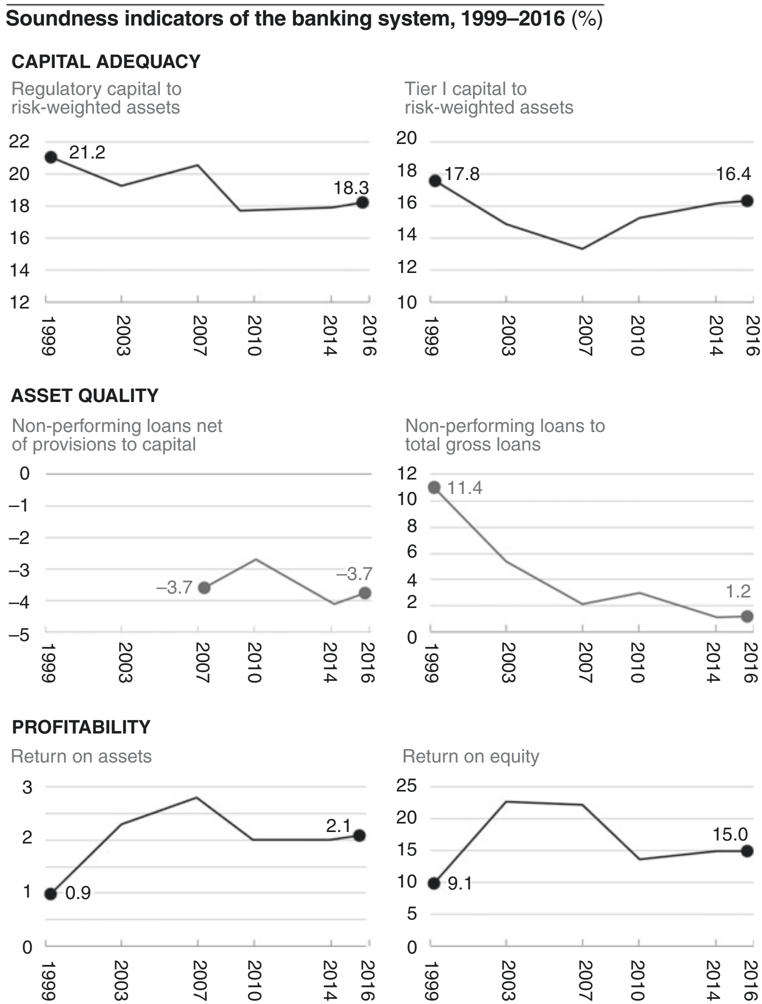

Fig. 7.4 Soundness indicators of the banking system, 1999–2016

Source: IMF Financial Stability Assessment Report 2006, 2011; Irish Stock Exchange: The Kingdom of Saudi Arabia Medium Term Note Program, September 2016

Notes: **Tier I** Common stock + reserves (shareholders' funds)

Non-performing loans: When payments of interest and principal are past due by 90 days or more

Provisions: Money set aside to offset potential losses from loans or other receivables (25% of NPL for 90 days, 50% for 180 days and 100% for 360 days. Additionally there is a general provision of 1% of performing loans; IFRS provisions are different from SAMA caculations).

Assest quality: No data for NPLs net of provisions to capital before 2007

6 The Foreign Exchange Reserves: Coping with Falling Yields

The damaging effect of falling yields on investment returns is little understood. The inflows into SAMA's foreign reserves from 2005 onwards were on a scale that the central bank had never seen before. The challenge was how to invest the cash prudently. On the fixed income side, bond yields were falling and the quality of credits had deteriorated so that the central bank's opportunities for earning a high safe rate of return effectively disappeared, and it had to diversify into lower quality credits. Meanwhile, huge moves in global equity markets were testing SAMA's nerves. At a time when other developing countries were establishing sovereign wealth funds (SWFs) outside their central banks, there was pressure to follow suit. But Saudi Arabia kept to its policy of concentrating foreign assets at the central bank, although in 2016 it signaled that the Public Investment Fund (PIF) would become a fully fledged SWF.

This was a difficult period for investors. From a roaring start, equities followed Wall Street down during the recession before recovering under the stimulus of central bank action to cut interest rates. The major US stock indices returned only between 5 and 8 percent on a compounded annual basis from 2005 to 2015, roughly comparable to bonds. Returns were pedestrian compared to the previous decade, and volatility high – in 2008 American stock indices dropped up to 40 percent.[6] Critically, interest rates for bonds collapsed alongside those for cash.

Everybody understands that when interest rates on bank deposits fall, it is bad news for savers. But it is true for bondholders like SAMA as well. SAMA looks at holding bonds to maturity, and rising bond prices and falling yields destroy long-term returns because the reinvestment rate of the bond coupons goes down. The conventional way of describing the return an investor can look forward to when he buys a bond is through a formula known as 'yield to maturity.' But this is not an assured return for the buyer – in fact it is an assumption that will only come true if the income from the coupons can be reinvested in the same bond at the same yield throughout its life.

When interest rates fall the price of a bond rises, so that investors who reinvest the bond coupon in more of the same bond can afford fewer bonds, which means, in turn, that the income from the additional bonds they buy will be lower. Consequently, their yield to maturity is less than they expected when the bond was bought. Conversely, if interest rates rise, the bond's price falls and the investor will be able to buy more bonds with his coupon. Perhaps surprisingly, rising yields improve long-term returns for investors who hold bonds to maturity.

This is why the fall in bond yields over the past 30 years has been bad news for long-term bond investors such as pension funds, especially since the crisis, when yields have fallen below inflation, so that they are negative in real terms. Figure 7.5 shows what happened to inflation-adjusted yields in dollars, but the picture is roughly the same for the German, Japanese and British markets. Five-year Treasury yields for instance, have been negative in real terms since 2011. The picture for bank deposits is even worse – they have been negative since 2008.

The decade saw massive growth in foreign reserves and because after 2005 SAMA published regular and accurate data it is possible to look at the pattern on a monthly basis.[7] From less than $100 billion at the start of 2005, foreign exchange reserves soared to reach $445 billion in November 2008 before oil prices fell. It took nearly two and a half years for the reserves to regain that peak, and they continued to grow until August 2014 when

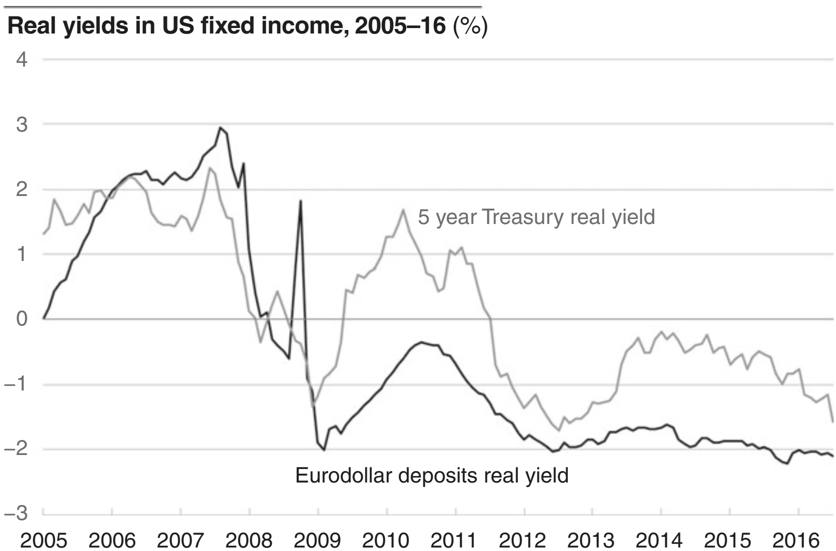

Fig. 7.5 Real yields in US fixed income, 2005–2016

Source: Federal Reserve Economic Data

they reached an all-time high of $731 billion. This figure increases to $851 billion if the funds belonging to the quasi-government organizations are included, as they should be, since SAMA can access them by swapping their foreign currency holdings for government bonds (Fig. 7.6).

With larger reserves SAMA could afford to fix a longer term horizon for its investments and move into assets that promised higher returns but were less liquid. The Investment Department[8] had already developed customized benchmarks for bonds and equities. Now it went a step further by including emerging market assets in its criteria. This meant SAMA had to accept more risk since emerging markets were less liquid and more volatile (there is an old financiers' joke that 'an emerging market is one from which you cannot emerge in an emergency').

The second step was to revise the strategic index, that is, the mix of cash and bank deposits versus bonds, equities and so-called alternative assets. Bonds remained the dominant asset class because they promised a fixed coupon through time that could be relied upon. But alternative assets such as private equity also began to make an appearance. Another question was:

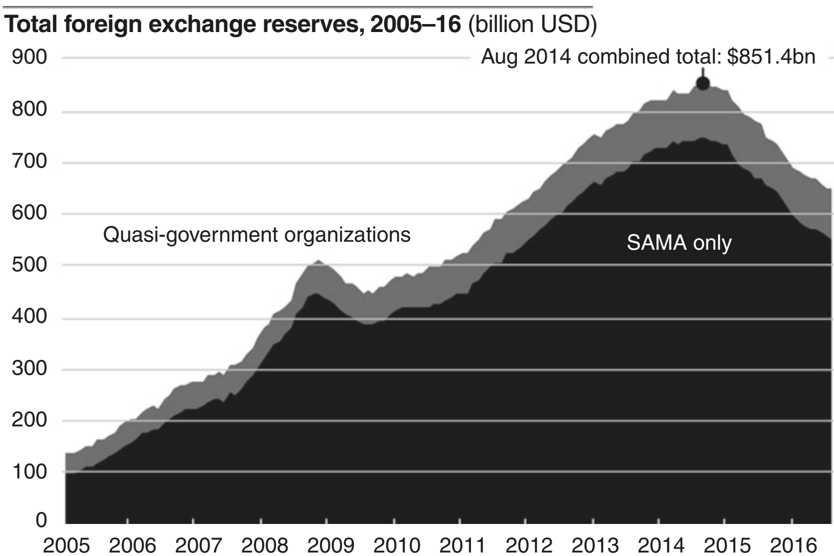

Fig. 7.6 Total foreign exchange reserves, 2005–2016

Source: SAMA Annual Reports, IMF, SAMA Monthy Statistical Bulletin how far should SAMA go to diversify out of dollars? Once the percentages in the strategic index had been fixed, the investment committee, chaired by the Governor, decided what positions to take against the benchmark (so-called tactical allocation) as recommended by the Investment Department.

In the final stages of the bubble that preceded the financial crisis, SAMA switched out of deposits into bonds and equities. By mid-2007, bank deposits made up less than one-tenth of its assets. But during the crisis the central bank built up the deposits steadily until, by 2012, they accounted for nearly a quarter of reserves (Fig. 7.7). SAMA used the rating agencies to grade banks, including Fitch. Its approach was to look for a guarantee for the financial institution from its home government. This could be either an explicit promise – as Ireland guaranteed its banks in September 2008 – or an implied promise based on the fact that the bank was 'too big to fail.'[9] During the crisis, this meant a reduction in the number of banks with which the central bank placed funds.

The result of declining returns in government bonds was that SAMA moved away from Treasuries and other government bond markets into

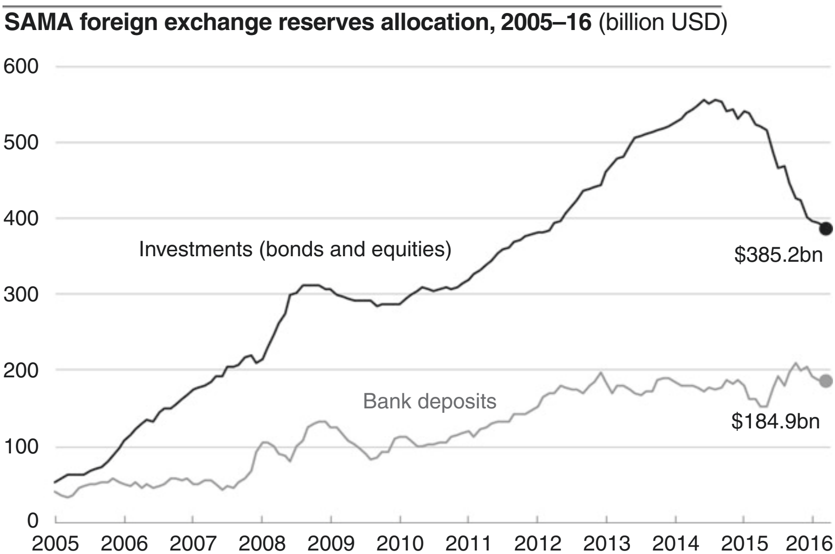

Fig. 7.7 SAMA foreign exchange reserves allocation, 2005–2016

Sources: SAMA Annual Reports, SAMA Monthly Statistical Bulletin corporates and emerging market bonds, starting in Asia with dollar issues followed by local currency ones. Traditionally, it did not buy bonds rated below AA or equivalent, but now it relaxed its credit standards.

7 Bringing Home Mortgages to Saudi Arabia

Housing problems had already contributed to the inflation scare during the financial crisis, but the subject took on a higher political salience as part of the government's response to the 2011 popular uprisings elsewhere often referred to as the Arab Spring. King Abdullah announced funding to build half a million new homes, but the project faced a shortage of available land for development. As long as the price of undeveloped real estate continued to rise year on year, property developers preferred to wait. Four years later, the government announced it would tax undeveloped land in urban areas to force it onto the housing market. But, meanwhile, prices kept rising with the result that many families could not afford their own home.

Most Saudi families have only a single income. The public sector is not a generous employer: the average civil servant earned the equivalent of about \$2,000 per month in 2013. Even though there are no personal taxes, a house in the capital city Riyadh or Jeddah is too costly for people living on this income. Even a basic apartment can be out of the reach of a young couple with children. Forty years earlier, the Real Estate Development Fund (REDF) had been established to provide interest-free loans for housing, but around 30 percent of its loans had defaulted and it could not extend new ones. An estimated 1.7 million families were waiting for an REDF loan in 2012. A new way to fund housing was needed.

SAMA could not help solve the housing shortage as long as the banks were unable to lend money for mortgages. This was a long-standing problem which involved Shariah principles. In Europe and North America, the idea of a mortgage on real estate has been established for centuries, while in Saudi Arabia, mortgages were prohibited by law, and lending to finance the purchase of real estate constituted less than 5 percent of the banks' loan portfolios. The problem dated back to 1981 when the Supreme Council of Judiciary had established three legal rulings that killed the nascent market: taking interest was illegal, any mortgage law would be un-Islamic because banks had no right to seize the property of defaulters and finally Shariah law should be left to the interpretation of individual judges. Ever since then public notaries, responsible for registering rights over properties, had refused to record mortgages. It was not until a generation later in 2008 that the central bank and the Finance Ministry were able to put a set of mortgage laws before the Shura Council and not until 2013 that the regulations went into practice.

The new laws allowed mortgage loans to be made by banks and real estate finance companies – the latter were a new class of financial institution licensed by SAMA. The mortgages would be formally registered with clear title deeds and lenders would have the right to seize property if payments were not kept up. Judges, trained in the new legislation, would back this up in the courts. To avoid the default problems experienced by the REDF, the laws also took a big step toward establishing a comprehensive database of individual credit records. Now, nobody would get a mortgage unless they had a clean record with the Saudi Credit Bureau (SIMAH), which SAMA had established.

The central bank hoped that registering of entitlements to property would encourage the development of Shariah-compliant products that would work as mortgages, but would also not offend against Islamic principles. In a conventional mortgage, the bank only holds the title deeds of the real estate, but in the Islamic equivalent, the bank owns the property while renting it to the borrower, who acquires it at the end of the transaction.

The reforms also shook up the governance of housing finance, a sector that had previously been based on interest-free loans handed out by the state. A new Ministry of Housing was set up and all REDF lending was frozen. The PIF promised to set up a mortgage refinancing arm – the Real Estate Refinancing Corporation – that would operate like the Federal National Mortgage Association (FNMA or Fannie Mae) in the United States to kick-start a secondary market in mortgages.[10]

The central bank also took on a new area of responsibility. At the same time as mortgages were made legal, financial leasing was allowed. By the end of 2014, SAMA had promulgated new regulations and established a new arm to monitor the finance companies. Mindful that over-enthusiastic lending by banks had been behind the collapse of property markets elsewhere in the Gulf, SAMA agreed a loan-to-value ratio that capped lending at 70 percent of the value of the property (later revised upwards when the economy turned down after 2014).

The story of the mortgage laws illustrates the pragmatism of the government, encouraged by SAMA, when faced with legal restrictions on

Western banking practices. However, Saudi judges are independent of the government and are not bound by precedent either, while the government remains unable to challenge the legal rulings of individual judges in repossession cases. This places additional responsibility on the banking system to produce products likely to meet judges' standards. In principle, a judge is still at liberty – despite the mortgage laws – to reject a bank's claim to seize the property of a family which is not paying the interest on its mortgage. But if the product is demonstrably Shariah-compliant he will find this much more difficult to do.

8 Changing of the Guard

For nearly a quarter of a century up to 2009 Hamad Al-Sayari served as SAMA's Governor: in the next seven years the central bank saw three successors. In December 2011, Fahd Almubarak took over from Al-Jasser. The first Saudi Governor without a background in the civil service, he used his four and a half years in office to reform the way SAMA was organized, before he was replaced in May 2016 by Ahmed Alkholifey who had been the Deputy Governor for research and international affairs. Al-Jasser was moved to the job of Economy and Planning Minister. He had been in office for less than three years, but his tenure had spanned a time of storms. The global financial crisis had been unprecedented in terms of its stress and complexity, but the outgoing Governor piloted the central bank through choppy seas to a safe harbor: the banking system had stabilized, the currency peg was secure and foreign assets were once again rising rapidly. But his successors saw the country buffeted by more storms.

The events of the decade had a common theme – the risks of financial globalization. The boom in world stock markets and the frenzy among foreign banks to take part in project financings drove the Tadawul up to unsustainable levels in 2006. The central bank had almost no tools that it could use to slow the rise and smooth the fall in the stock market. In a similar vein, SAMA struggled to control inflation a couple of years later. It tried to prevent the surplus liquidity in the banking system from feeding through into price rises, but it would not change the currency regime.

SAMA officials, led by Al-Jasser, had to recognize that their country's banks had been sucked into the global financial crisis. Despite the decades of sound and conservative regulation there was a brief crisis of confidence when Lehman failed. But Al-Jasser was able to stem the panic by implementing the traditional policy of cutting rates and injecting government deposits. Likewise, the Saad-AHAB affair had an international dimension since it demonstrated that private firms could get funding from foreign banks that knew little about business in the Gulf but were prepared to lend on a 'name basis.'

It was tempting to hope that SAMA might be about to enjoy a quiet period during which Almubarak's reforms could bed down safely, but less than three years after he took office, the oil price halved, the government started running a budget deficit and foreign assets began to fall again. In 2015, the government issued domestic bonds and the following year Almubarak was replaced. Nobody could tell whether this was a rerun of the 20 years of low oil prices that had been the hallmark of the 1980s and 1990s, or whether it would turn out to be an episode lasting only a few months, as was the case during the financial crisis and recession. The rule of no duration dependence in the oil price still held sway. One thing was sure however - SAMA had coped with similar events in the past.

Notes

1. International Monetary Fund reports provide background for this period. Financial Sector Assessment Program (FSAP) reports continue to give information on topics not covered in SAMA's Annual Reports as well as insight into what policy issues were discussed; IMF Article IV reviews with Saudi Arabia from 2001 onwards improve progressively and are more general oversights of the economy.
2. The Majlis-ash-Shura or Consultative Council was revived and strengthened in 2000. It advises the King and can summon public officials for questioning.
3. Howard Davies, The Financial Crisis – Who Is to Blame? (London: Polity Press, 2010) is the best survey of what may have caused the global financial crisis, written by a regulatory practitioner who had been in charge of the Financial Services Authority in the UK before the crash and was involved in what went wrong.
4. The Supreme Economic Council, chaired by the King, looked after domestic policy. It was replaced by the Council on Economic and Development Affairs (CEDA) in 2015.
5. World Bank, 'Global Financial Inclusion Database,' World Bank, http://www.worldbank.org/en/programs/globalfindex (Accessed October 25, 2016). This surveys the degree to which different segments of the population in any country have access to bank accounts. The figures should be treated with caution but since there is nothing more authoritative they have

been used for the latest period available. For comparison, the same database records that 94 percent of American adults have bank accounts.

6. MeasuringWorth.com, Stock markets database, http://www.measuringworth.com/DJIA_SP_NASDAQ/ (Accessed October 25, 2016). Federal Reserve Bank of St. Louis, Economic Research, FRED database, https://research.stlouisfed.org/fred2/series/BAMLCC0A2AATRIV/ (Accessed October 24, 2016)
7. SAMA, Monthly Statistical Bulletin (Table 7(a)) gives monthly data on foreign exchange reserves, Annual data on breakdown between deposits and investments, and on quasi-government organizations, comes from SAMA's consolidated balance sheet in the Annual Report, http://www.sama.gov.sa/en-US/Pages/default.aspx (Accessed October 25, 2016)
8. It became the Investment Deputyship in 2013 headed by a new Deputy Governor.
9. Without the Irish promise SAMA was ready to withdraw all deposits from Irish banks on the following day.
10. At mid-2017 this had still not happened.

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PART II

SAMA in the Modern World

CHAPTER 8

The Future of Gulf Monetary Union

1 Attractions of a Common Currency

When the Iraqis invaded Kuwait in 1990, they tried to abolish the Kuwaiti dinar. Yet despite the fact that the country was in Saddam Hussein's hands, SAMA led the other Gulf Cooperation Council (GCC) central banks in making sure that the Kuwaiti dinar was still exchanged for dollars at its pre-invasion rate. This represented the high-water mark in Gulf monetary unity. In contrast, during a more recent crisis the GCC was found wanting. When Dubai World, a government-linked corporation in the United Arab Emirates (UAE) involved in construction and real estate in the emirate, failed in 2009 to meet its debt obligations, help came from Abu Dhabi, Dubai's richer sister emirate, and from the UAE central bank; but the GCC was absent. Most observers were not surprised, given the limited progress it had made in coordinating the region's monetary affairs.

The reason for the difference in the responses is that Dubai was reluctant to ask for help from its Arab Gulf neighbors. In the Kuwait crisis, by contrast, Saudi Arabia had to involve all other Gulf countries in formulating a common stance. If ever there was a chance for the GCC to show its financial usefulness, Dubai World's default offered it. Yet the coordinating mechanisms that would have made it possible for Dubai to request assistance were not in place although for well over 30 years, the GCC has pursued the project of financial integration and a

A. Banafe, R. Macleod, *The Saudi Arabian Monetary Agency, 1952–2016*, Financial Institutions, Reforms, and Policies in Muslim Countries, DOI 10.1007/978-3-319-55218-7_8

common currency, usually referred to as Gulf Monetary Union (GMU). Why has the theory not translated into practice and will it ever do so?

The attraction of having a common currency is clear. There is no reason why the Gulf economies should not one day enjoy the benefits of having a globally used currency like the euro. By 2020, the GCC population will exceed 50 million and its nominal Gross Domestic Product (GDP) could be around \$2 trillion. That amount of output, and the importance of oil exports, would make the common currency attractive internationally. It could then reap some of the privileges enjoyed by issuers of internationally held currencies, such as the British pound sterling and the Swiss franc. But there are other significant advantages. Gulf economies are likely to borrow more abroad as they diversify away from oil. If this borrowing could be denominated in their own unit, it would free borrowers from exchange rate risk. A single financial market means their securities would offer higher liquidity, which would give them lower borrowing costs.

So much for the theory. Practice is more difficult. Relinquishing national currencies in a broader currency union means losing some of your sovereignty. A member nation will find its banking system answering to a supra-regional regulator. Perhaps most onerous, each member's fiscal policy will be constrained. This is needed to address the 'free-rider problem': if a member's borrowing is repayable in a common currency, it has an incentive to borrow more and let the consequences fall on the group. Given these challenges, in this chapter we try to answer three questions: not only 'How did GMU get as far as it did?' but also 'Why is it a good idea?' and 'What is needed to make it succeed?'

2 Groundwork for a Common Currency: 1975–98

The dream of pan-Arab unity has proved elusive but enduring. The ambition to create a monetary union among the Gulf states dates from the first oil boom and was prompted by a feeling of vulnerability after Britain's military departure from the Gulf. Between 1975 and 1978, Bahrain, Kuwait, Qatar and the UAE attempted to coordinate issuing a Gulf dinar. A form for the new currency was agreed, as well as the exchange rates between the old currencies and the new unit. The agreement, however, failed to get political traction, Saudi Arabia did not become involved and the plan lapsed.

The next attempt was made by the Arab Monetary Fund (AMF), established in 1976 and backed by Gulf money to lend to poorer states. This was a similar idea to the Saudi-financed Islamic Development Bank, set up at about the same time in Jeddah, but it has not led to very much. Today the GCC economies contribute 37 percent of the AMF's paid-up capital, with Saudi Arabia alone contributing 15 percent. The AMF's articles of agreement stipulate that it shall promote economic and monetary integration, leading eventually to monetary union.

Headquartered in Abu Dhabi, the AMF started lending to borrowers mainly in North Africa, Yemen and Jordan. The portfolio has grown to $2.5 billion, but it is not in the league of even the smallest of the regional development banks. For example, it is one-third the size of the African Development Bank, relative to population. The AMF has long forgotten its pan-Arab aim of a single currency and now offers the standard fare of international financial institutions, such as building up technical capacity among member bureaucracies and hosting discussion forums.

In the end, a political event acted as a catalyst, kick-starting the first serious attempt to translate the pan-Arab vision into reality – albeit in the Gulf region alone. The 1979 Revolution in Iran meant that overnight one of the region's biggest and most powerful countries became the first Shia theocracy in the Middle East. The revolutionary regime showed heated ideological antipathy to the Gulf Arabs. Tehran's rhetoric sparked fears that it would seek to undermine Saudi Arabia and other Gulf states in its ambitions to export Shia theocracy throughout the Middle East. Iraq's subsequent attack on Iran in 1980 and the war which followed was the backdrop to the decision by the six Gulf states to move ahead on their own. Taking advantage of Saddam Hussein's preoccupation with the war, Saudi Arabia, the United Arab Emirates, Kuwait, Qatar, Oman and Bahrain signed the GCC Charter in Abu Dhabi in 1981.

The GCC is much like the European Union (EU) in its formal set-up. At the center sits the Supreme Council, with six heads of state. This is like the EU's European Council, which sets the group's broad priorities. Also in the center is the Ministerial Council, similar to the Council of Ministers in Brussels, which comprises the member states' foreign ministers. An executive Secretariat is based in Riyadh, akin to the European Commission (EC) in Brussels. Absent is the equivalent of the European Parliament, the elected assembly that holds the bureaucracy in the EC accountable. The GCC has instead the Advisory Board of the Supreme Council, made up of 30 experts, five each designated by the GCC members. Its recommendations to the Supreme Council are non-binding.

While security was the aim of the GCC's founding fathers, it has achieved more in the economic sphere. Among the most important achievements is a common market, called for in the 1981 GCC economic agreement. This sought a repeal of tariffs among members and a uniform external tariff by the end of the decade. Little happened until ten years after the target date, when in 1999 the GCC revitalized the common market. Once again politics was the explanation. Iraq's 1990 invasion of Kuwait had presented an existential challenge to all the Gulf states. Weak oil prices left Gulf leaderships worrying about the future. The customs union was finally formed in 2003 and a common market came into being in 2008.

The 1981 agreement also called for freedom of labor and capital movement among states as steps toward a common currency, and this is largely in place. The region now enjoys freedom of mobility for its nationals and a high degree of cross-border investment freedom. Each of the GCC members exhibits greater levels of trade and financial openness relative to the world average, particularly when compared to OPEC members outside the GCC. Labor mobility is more complicated because the bulk of workers in the private sector are foreigners on temporary visas, and this needs to be taken into account in any assessment of whether the region is suitable for a common currency.

3 Working Toward Monetary Union: 1999–2016

If the euro is any guide, the lag between saying you want a common currency and its eventual creation is about 30 years. In the euro's case, it began with the 1971 adoption of a ten-year roadmap to unification and ended with the euro's launch in 1999. If this timetable had been applied in the GCC, its peoples would now be handling a Gulf banknote. The GCC Charter was accompanied by an economic agreement in 1981, giving flesh to the founding principles of greater coordination and economic integration. The goal of creating a common currency was ratified in 1982, although the first constructive step did not come until 1987, when members agreed to coordinate their exchange rates. This was put on hold when governments failed to agree a common anchor currency.

Little was done over the following ten years until the coming of the euro in 1999 breathed new life into the GMU project. The creation of the euro was a momentous achievement, prompting speculation that the number of national currencies would drastically shrink. The dream of regional currencies was attractive: for example, the Association of South-East Asian Nations (ASEAN) looked at the idea. The project came to nothing, but generated a swathe of policy papers at the time. As Figure 8.1 shows, by economic and population size, the GCC would sit comfortably among other issuers of global 'hard' currencies – it is not dissimilar to the United Kingdom. An ASEAN common currency, by contrast, would be an outlier in terms of its low income per head.

Three years after the launch of the euro and in tune with the financial wisdom of the day, GCC heads of state agreed a timetable for monetary union. Article 4 of their agreement stated:

> For the purpose of achieving a monetary and economic union between Member States, including currency unification, Member States shall undertake, according to a specified timetable, to achieve the requirements of this union. These include the achievement of a high level of harmonization

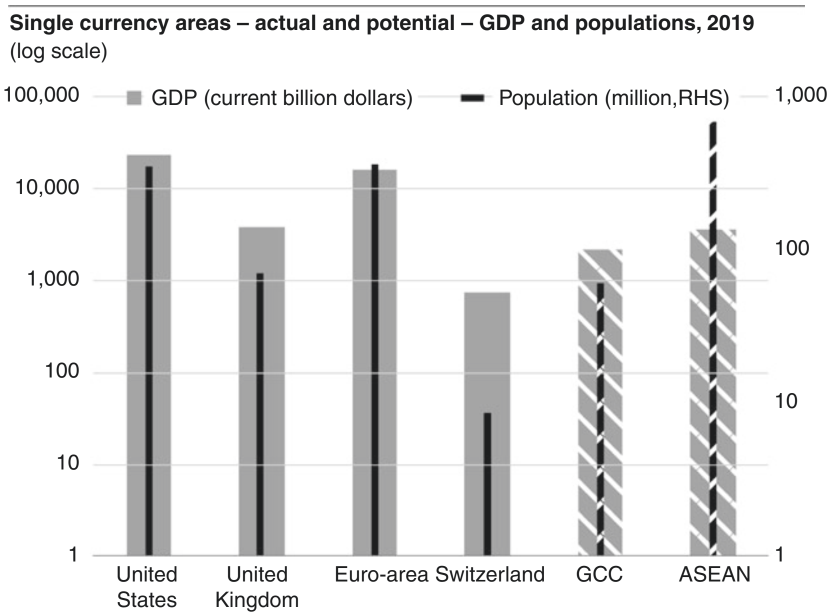

Fig. 8.1 Single currency areas – actual and potential – GDP and populations, 2019

Source: IMF WEO October 2016

> between Member States in all economic policies, especially fiscal and monetary policies, banking legislation, setting criteria to approximate rates of economic performance related to fiscal and monetary stability, such as rates of budgetary deficit, indebtedness, and price levels.[1]

The Muscat Agreement set out an intense convergence process, scheduled to begin in 2005 and culminate in monetary union in 2010. In the early years things went well. The most tangible requirement was for all countries to fix their currencies to the dollar by 2003. As all GCC members apart from Kuwait had been pegged to the dollar for several decades, this amounted to a call for Kuwait to abandon its long-standing peg to a currency basket. Kuwait met the deadline and adopted a dollar peg in November 2003, although it altered the peg level several times (Fig. 8.2).

Unsurprisingly the convergence criteria for each economy were carbon copies of those used during planning for the euro, specifying common debt/GDP levels, along with fiscal deficit and inflation numbers. A committee of central bank and finance ministry officials started working on the actual numbers and again the results looked like those for the euro area:

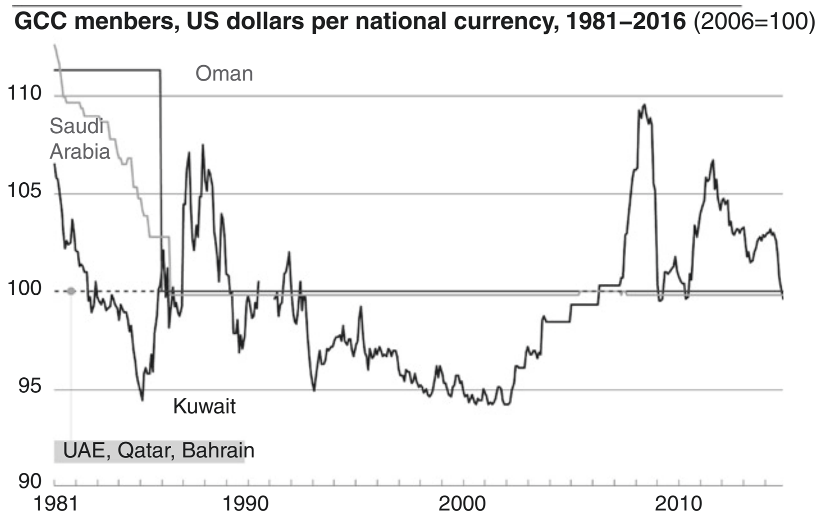

Fig. 8.2 GCC members, US dollars per national currency 1981–2016

Source: IMF

deficits should normally be capped at 3 percent of GDP (although with more flexibility than in the euro's Maastricht criteria), government debt capped at 60 percent of GDP and inflation held at or below 2 percent. The principle that the new central bank for the area should be independent was also in line with the set-up of the European Central Bank (ECB).

The Gulf heads of state agreed to this in 2005 and the following year the GCC signed a memorandum of understanding with the ECB to formalize their cooperation. But the Maastricht criteria do not really make sense. For the European project, the 60 percent debt-to-GDP limit was arbitrary and has never worked. For the Gulf economies, the notion of government debt as a constraining factor is misplaced. GCC economies are all net international creditors. More importantly, government debt is consciously used as a counter-cyclical tool to cope with the roulette wheel of the oil price. While the euro's architects saw debt as a problem to be minimized, GCC policymakers deploy it as a tool to stabilize their economies across the peaks and troughs of the oil price. They ran fiscal surpluses for some years on the back of high oil prices, although this changed after 2014 as the price dropped. For Saudi Arabia, one consequence of cheap oil is a rise in government debt, but this has happened before.

The swings in fiscal position since the millennium caused by the change in the oil price and their impact on debt/GDP ratios for the Gulf states can clearly be seen in Tables 8.1 and 8.2.

The first crack in the project appeared in December 2006, when Oman said it would not be able to meet entry criteria by the deadline. Six months later, Kuwait left the dollar peg and returned to the basket peg. The Kuwaiti dinar then briefly appreciated 8 percent against the dollar but the inflation rate accelerated in 2008 before falling back. Whether this

Table 8.1 GCC members' fiscal positions, 2000–16 (budget surplus/deficit as percent of GDP)

	2000	*2005*	*2010*	*2016*
Bahrain	7.6	2.9	–5.8	–14.7
Kuwait	31.6	43.3	26.0	–3.5
Oman	14.3	13.2	5.7	–13.5
Qatar	4.7	10.5	6.7	–7.6
Saudi Arabia	3.2	18.0	3.6	–13.0
United Arab Emirates	20.3	20.2	2.0	–3.9

Source: International Monetary Fund, World Economic Outlook Database, October 2016

Table 8.2 GCC members' gross government debt positions, 2000–16 (percent of GDP)

	2000	*2005*	*2010*	*2016*
Bahrain	25.7	24.2	29.7	75.2
Kuwait	35.4	14.1	11.3	18.3
Oman	26.1	9.9	5.9	21.8
Qatar	52.5	19.2	41.8	54.9
Saudi Arabia	86.7	37.3	8.4	14.1
United Arab Emirates	3.1	6.6	22.2	19.0

Source: International Monetary Fund, World Economic Outlook Database, October 2016

decline in inflation was due to the basket peg is unclear because the higher inflation, which characterized the years before 2007, was probably due to Kuwait's strong growth. Its economy was among the fastest-growing in the GCC from 2002 to 2007 and has been among the slowest since then. However, the move helped create some room in monetary policy and better aligned the peg with Kuwait's trading and investment pattern. Short-term expediency had triumphed over the common goal.

Kuwait said it still wanted currency unification with the rest of the GCC, but acknowledged that it would either have to re-adopt the dollar peg or convince the other Gulf states to adopt its basket peg. It took a while before the project was formally put on hold. At a gathering of GCC monetary authorities after Kuwait's departure, SAMA Governor Al-Sayari acknowledged that the 2010 deadline looked difficult. But GCC heads of state concluded their December 2007 meeting by affirming the deadline and the following year they approved a draft agreement on monetary union. Away from the international stage, when detailed work by each government was necessary, the process stalled. In March 2009, with a year to go, the GCC secretariat admitted that a new deadline was needed.

To sustain the project's momentum, the GCC established the forerunner of the common currency's central bank, the Gulf Monetary Council (GMC). In considering the parallels with the formation of the euro, it could resemble either of two pre-euro institutions. It has some kinship with the European Monetary Institute (EMI), the 1994–97 forerunner of the ECB. The EMI mainly produced technical documents outlining the various monetary-policy framework options for the ECB. But the GMC's remit is much broader, ranging from coordinating monetary policies to developing statistical systems across the GCC.

It looks in fact more like the body that preceded the EMI: the European Monetary Cooperation Fund set up in 1973 by the European Economic Community to support its efforts to limit exchange rate fluctuation between European currencies (the so-called snake). With the Bank for International Settlements (BIS) in Basel acting as agent, it settled foreign exchange intervention accounts and provided some balance-of-payments support to members. While the GMC will not do this, it does have a role in coordinating exchange rate policy. In the event, the snake was undone by persistent inflation differentials among member states and the euro project stalled for 20 years.

Arguably the act of setting up the GMC further damaged the prospects for monetary union because its location was a pointer to where the federal central bank would be based. With nearly half of the Gulf's GDP produced in Saudi Arabia – and with that country containing nearly 60 percent of the Gulf's population – Riyadh was surely its natural home. Yet this was rejected by the UAE which had wanted to host the central bank itself and it left the table, meaning over a quarter of the Gulf's GDP is outside the plan (Fig. 8.3).

So when the Monetary Union Agreement was finally signed, it carried only the four signatures of Bahrain, Kuwait, Qatar and Saudi Arabia,

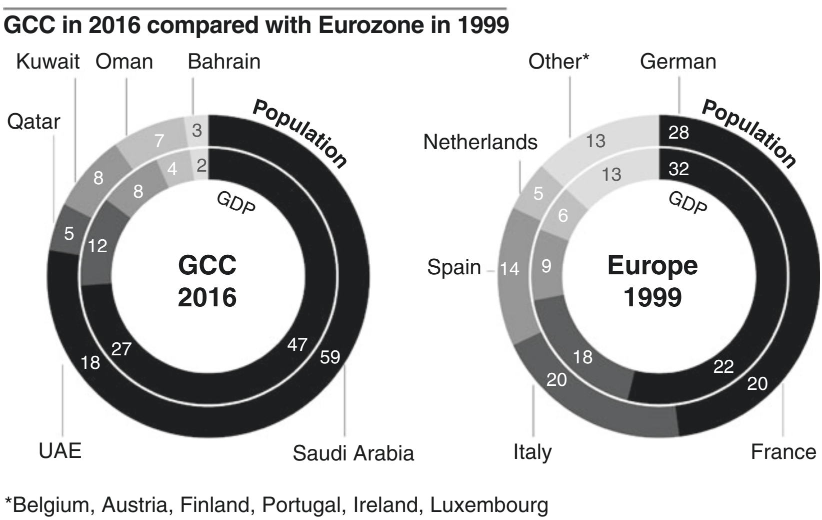

Fig. 8.3 GCC in 2016 compared with Eurozone in 1999

Source: IMF WEO

although it provides for Oman and the UAE to join later. In March 2010, the GMC convened for the first time. The board is composed of Governors of the four national central banks and it has discussed the necessary operational requirements of the monetary union and formed a statistical committee.

Today, the doors of the GMC's headquarters are open but the dream of monetary union remains unfulfilled. That the region's capital markets hardly responded to the UAE's departure spoke volumes about their view on the prospects for currency union. Just as the euro's formation in 1999 encouraged the GCC leaders, so its woes since 2010 have chastened them. The drama of sovereign bailouts and economic distress that followed has led Gulf leaders to reappraise their own plans for monetary union. It is telling that there is now no deadline to work toward.

4 What Is Needed to Make a Common Currency Work?

The theory of optimal currency areas was the 1961 brainchild of Canadian economist Robert Mundell. An optimum currency area is a geographical region, such as North America, where the economic benefits could be expected to exceed costs if the area shared a common currency. At the core of Mundell's analysis is a trade-off between the benefits and costs of sharing a currency. Among the benefits are reduced transaction costs, less uncertainty and network externalities (what might be termed, the 'bigger is better' effect). Some of these have been borne out in the case of the euro area, at least before the 2010 Greek crisis. Empirical estimates suggest that the euro was responsible for an extra 15 percent increase in trade between area members in the first few years of currency union. The birth of the euro certainly led to more cross-border investment, as evidenced by the plummeting borrowing costs of the southern members. However, this is a dubious achievement because the north-south capital flows of the euro area before 2009 lie at the center of its current predicament.

Mundell is often remembered as the cheerleader for common currencies, but he was careful to emphasize the pain involved in sharing a currency, most of which arises from members' varying abilities to adjust to member-specific shocks. Consider the individual members of the successful currency union called the United States of America. Imagine that the state of Oregon suffers a unique shock – for example, timber prices fall. What mechanisms are available to alleviate the pain, considering currency solutions are off the table? Recall that there is no individual Oregon currency, so there is no chance that an Oregon dollar could depreciate to restore the state's prices in relation to the rest of the country. Recall too that the monetary authority for the currency union (the US Federal Reserve) cannot target Oregon with a cut in the interest rate. If the Fed cuts interest rates, it has to do so universally, and so risks stoking excessive growth in the healthier states, which in normal circumstances would be inflationary.

Lacking an immediate monetary response, there is a need for other mechanisms in order to make a common currency area workable. One is labor mobility. In our hypothetical model, Oregon workers can move to California where the economy is more buoyant and can offer better wages. Labor mobility is thus one criterion for judging whether an optimum currency area exists. What else can help? Oregon workers might accept lower wages and Oregon businesses might offer lower prices in response to the fall in income from timber sales. This would attract capital from outside and preserve the state's competitiveness vis-à-vis other states and the rest of the world through an internal devaluation. Wage and price flexibility are two further criteria for determining the existence of an optimum currency area.

However, the picture changes if Oregonians owe money to anybody. A fall in wages and prices means a reduction in income for labor and business. Debts and interest payments do not decline even though wages and prices are falling. In other words, downwardly flexible wages and prices can lead to higher real interest rates, which matters when credit is an important part of the way the economy works. On the other hand, a more open economy also makes recovery easier. What good would it do Oregon to have cheaper wages and prices if its neighbors won't buy its timber? Another criterion is fiscal integration: will fiscal arrangements help offset the shock? Much of the public spending in Oregon does not come from the state capital at Salem but from Washington DC. Such arrangements (sometimes called automatic stabilizers) offset the loss in income from the fall in timber prices.

There is a final criterion for judging whether a common currency would be appropriate, which we can test on our model. Oregon's distress would have been alleviated by synchronized business cycles across the country. Not only would they have reduced the chance of individual states suffering specific shocks in the first place, but they would also solve the difficulties created by a one-size-fits-all monetary policy, something that would be problematic even before a shock and which indeed might be the cause of one. This is illustrated by the example described above, where the Fed can only control one interest rate, and that rate might suit some states but be too loose or too tight for others.

5 Applying Common Currency Theory to the Euro and GMU

Now that we have used the example of Oregon to identify the eight criteria by which a common currency project should be judged – mobility of labor and capital, wage and price flexibility, low debt, an open economy, high fiscal transfers and synchronized business cycles – we can analyze how well the euro and GMU stand up in the light of theory.

The euro is only one example, but it provides a fascinating insight into the mechanics of very large currency unification. Taking the eight criteria in turn, the picture is not encouraging. Labor mobility is low in practice. People do not move around Europe to the extent seen in the United States, partly because of different languages. Capital mobility is high, but it stimulated the wrong sort of growth in the southern euro members. Wage and price flexibility has been disappointing, insofar as the southern euro members' inflation-adjusted, trade-weighted exchange rates have been slow to come down. This problem with the euro is illustrated in Figure 8.4 where price competitiveness of the economy is measured by the combination of the exchange rate and domestic prices, compared to those of trading partners. This shows how little the price structure of the five southern euro-area economies adjusted during the global financial crisis of 2007–10, compared to the euro area as a whole and to the United Kingdom, whose independent currency fell sharply.

Debt in the euro area grew steadily and debt deflation is at the heart of the problems in southern Europe. These economies have also not fared well in the economic openness criterion: too much of their output was in the non-traded sector at the onset of the crisis, leaving their economies hard-pressed to take advantage of improvements in competitiveness, such as they were. Fiscal integration was never part of the euro project and government attempts to balance the budget have made the contraction worse. The situation has been exacerbated further by the fact that business cycles were not synchronized. So out of the eight criteria for a successful common currency, only one – the ambivalent benefit of capital mobility – is clearly working in the euro area.

Fig. 8.4 Euro real effective exchange rate in the 2007–10 crisis

Source: BIS

If we imagine a common currency for the Gulf and apply the eight criteria, the picture is better. There is more likelihood of achieving adjustment through flexible wages and prices compared to the euro area. This is because employment in the private sector is dominated by expatriates, so wages and contracts can be reset more quickly than in the formal regulated labor market for Gulf nationals. Labor mobility is a tale of two workforces. It is good among expatriates who dominate the private sector and can choose which country to work in (those already in the GCC, however, have little mobility as they are tightly controlled by their sponsors), and the universal use of Arabic and English helps migrant labor to work in any country. Expatriates rarely move directly between countries and the main feature of the Gulf labor market is an almost flat labor supply curve, because labor can be imported or sent back home easily to adjust to business cycles. So labor mobility exists independently of any cross-border mobility within the GCC. In complete contrast, GCC nationals tend to work in the public sector and their labor market mobility is very low. Apart from Omanis and to some extent Bahrainis, GCC citizens rarely take jobs in neighboring countries.

There are few obstacles to moving capital around the area and there is a long tradition of shared cultural values which increases investors' comfort in putting money in another GCC member. Financial depth in these economies is much lower than in the euro area, so there is less borrowing by individuals and businesses, which makes the debt deflation problem less important. The sixth criterion – the degree of openness – is positive in one way but negative in another. The GCC economies are highly open but the main regional export is oil, which is not sensitive to changes in local wages. A big reason why SAMA did not follow Kuwait in moving the dollar peg was the central bank's conviction that commodity exporters cannot benefit from adjustments in local costs. If the GCC members continue to succeed in diversifying their exports, the significance of this factor will diminish, but at the moment it is important. As in Europe, there is also no fiscal integration in the GCC – each member runs its own budget and there is no GCC-wide budget – both factors which would impact negatively on a currency union.

The final criterion for a successful common currency – synchronized business cycles – needs more analysis. The Gulf economies do not have a business cycle in the same sense that the United States does, a repeating rhythm of economic expansion and contraction, because they depend more or less on the roulette wheel of the oil price. Nonetheless, it is in the GCC's favor that each member state has a similar economic structure, centered on hydrocarbons, so that the oil shocks should hit them in a similar way. This reduces the chances that the single monetary policy covering the whole of the GMU would be inappropriate for any one member – for example, too accommodating for an economy that is expanding while being too tight for another that is in trouble.

In fact, the business cycles of the Gulf economies before the drop in the oil price in 2014 were – surprisingly – not particularly well synchronized, at least compared to those of the euro area on the eve of unification, as shown in Figures 8.5 and 8.6. This observation is based on an empirical approach to measuring the underlying trends of the GCC economies.

Business cycles in the GCC members appear much less synchronized than was true of euro-area members on the eve of the euro's launch in 1999.

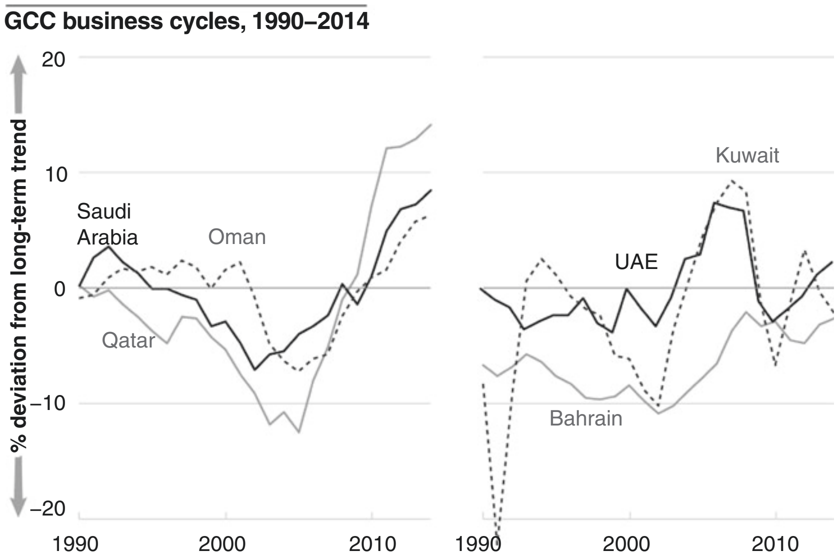

Fig. 8.5 GCC business cycles, 1990–2014

Sources: IMF WEO October 2015, authors

Note: The graph reports the cyclical component of Hodrick-Prescott-filtered real GDP indices. Specifically, it is the actual index minus the trended index. When the economy is below trend, the chart shows negative numbers.

Nonetheless, the Gulf economies come out well on six of the eight criteria for a successful common currency area – with the exception of fiscal integration and (possibly) synchronized business cycles. So the GMU project should work better than the euro area – in theory.

6 Practical Obstacles to GMU

While GMU might from a purely economic perspective appear easy, its benefits could be small. A common monetary policy is the main result of a common currency but it might make little difference to the Gulf economies. Their currencies are mostly linked to the dollar already, so interest rates are similar across the GCC. Monetary policy is already focused on maintaining the exchange rate as an anchor. Because there is little debt,

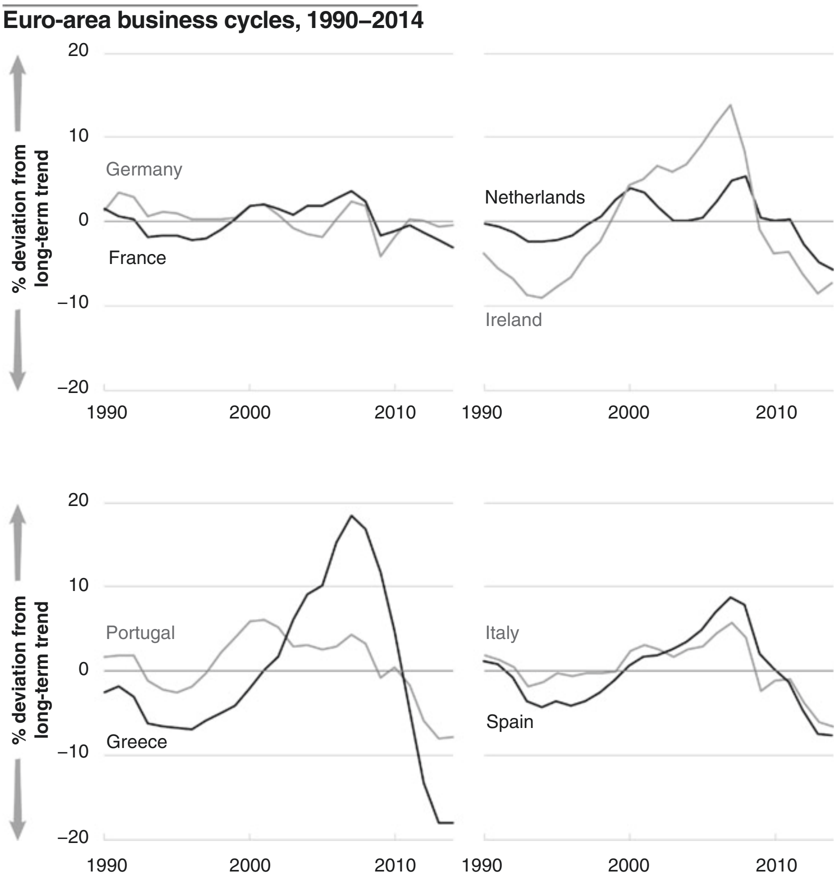

Fig. 8.6 Euro area business cycles, 1990–2014

Sources: IMF WEO October 2015, authors

Note: The graph reports the cyclical component of Hodrick-Prescott-filtered real GDP indices. Specifically, it is the actual index minus the trended index. When the economy is below trend, the chart shows negative numbers.

the effectiveness of monetary policy is much weaker than in the United States, where the Fed wields a powerful weapon. The absence of deep financial sectors with high levels of debt is one reason why overall financial deepening has not moved faster.

In actual fact, the main obstacles are not economic but rather political and cultural. One of the reasons why financial deepening has not taken place faster is that many Gulf citizens have a cultural problem with the notion of longer term investment as well as with interest or *riba*. Conventional bond markets – even sukuk or Shariah-compliant bonds – have failed to take root and most are private placements which appeal to the 'buy and hold' mentality of local financial institutions. Individuals prefer to invest in equities, adopting a short-term trader's approach that has deep roots in the region's history. So it is not likely that GMU would deepen capital markets with the benefits they can bring in terms of improving savings and investment decisions.

The regional dominance of Saudi Arabia is another factor which at first sight suggests the creation of GMU should be easier, but which in practice is an obstacle. The kingdom stands out among the Gulf states, containing 60 percent of the population and nearly half of GDP. It would dominate GMU far more than Germany does the euro area, almost regardless of whether the other five combined against it. If the economic benefits of GMU are not significant, the smaller countries can argue, why should we agree to lose our monetary independence, even if it is small? Currencies within the Gulf all operate on fixed rates against one another anyway.

This is a fair point, but a federal central bank will have other work to do, in particular on financial regulation where there are strong arguments for a common approach. For one thing, the global financial crisis showed the importance of agreeing on regulatory standards to prevent regulatory arbitrage or destabilizing cross-border capital flows. This will become a big issue again when the next global financial scare creates a strain on liquidity in emerging market borrowers. The global liquidity crunch caused by the failure of Lehman Brothers in 2008 ultimately led to the restructuring of Dubai's prominent borrower, Dubai World. Yet here too there is a problem. Despite its obvious benefits, the harmonization of the banking sector and the relinquishing of local authority that it would involve would most likely be unpopular. The UAE has made a push to be a center for financial innovation and will not want to give up its autonomy.

Strong feelings of nationalism will also make it difficult to achieve fiscal integration, which is one of the criteria for a successful common currency. One lesson to be learnt from the problems of the euro is the need for a much greater degree of centralized budgeting, which would mean cross-border taxation and spending authority. This would enable the federal authority to provide automatic fiscal stabilizers to help a depressed economy, as in the Oregon example above. The euro area has conspicuously avoided fiscal integration. The 2016 budget of the EU is only about 1 percent of GDP. The United States provides a better template for successful currency union. Take the example of federal transfers made to help Florida recover from its housing crash during the global financial crisis. Excluding subsidized health care and federal support of the secondary market for home mortgages, which make the actual number considerably larger, these amounted to over 4 percent of GDP. This fits with a 1977 report by the EC, which suggested that the budget for the central institution in a single-currency Europe would need to be 5 to 7 percent of the area's GDP.[2] Under GMU this money would have to be raised from local taxation, or by a transfer of oil revenues. There is no likelihood of the GCC members agreeing to this as it would add up to around $120 billion per year or about a fifth of all GCC budgets.

The final nail in the coffin of GMU is that each country would also have to lose some control over its own budget, in order to avoid the free-rider problem when a government borrows in the common currency but the consequences of these borrowings are shared by the currency zone as a whole. Let us go back to the Oregon example. If Oregon had free rein to borrow, it should do so – because the impact on the dollar would be borne across the whole of the American economy. In other words, if Oregon borrowed its weight in GDP to finance a high-speed rail system, that spending might push up inflation, which is suffered by everyone using the dollar. It might also depress dollar bond prices, which means higher interest rates for everyone borrowing in dollars. Oregon could also assume that the federal government would end up bailing it out if it looked as though it was going bankrupt. The several states in fact are almost entirely mandated by state constitutions to run balanced budgets. In order to make this work in the Gulf there would have to be rules on local spending, a much higher degree of transparency in fiscal accounts, and more uniformity in reporting and accounting than is currently the case. The bottom line is that monetary union needs to be accompanied by fiscal union or at least well-defined fiscal rules.

7 Awaiting Economic Diversification

The new King Abdullah Financial District in Riyadh already hosts the Tadawul – the Saudi stock exchange – and the Capital Market Authority, and in 2013 they were joined by the Gulf Monetary Council on King Fahd Road. It is across town from SAMA's headquarters on King

Saud Road, only a few miles as the crow flies, but in heavy traffic the journey can take up to an hour. The distance symbolizes the gap between the aspirations of GMU and the practical reality.

In the run-up to the creation of the euro many people believed it would never happen. Are current doubts over a Gulf currency similarly unwarranted? The project is far from abandoned. The convergence criteria have mostly been met. The in-tray for the Monetary Council is full – statistics harmonization, development of payments and settlement infrastructure and the standardization of banking regulation and supervision, to name just a few. However, the arrangements for the new body reveal the difficulty its members have in ceding sovereignty. Above all, decisions must be unanimous. This suggests that, for all the affirmations of a GMU, success is contingent on closer political cooperation and, by implication, trust.

It is worth rehearsing the two main arguments in favor of a Gulf banknote. The first is that the GCC states are almost there already. They already effectively have a common currency: bar Kuwait, they are all pegged to the dollar. This means their exchange rates with one another are fixed. And because they operate mainly open international financial accounts, it also means they have already given up their monetary independence. The fact that they (along with Hong Kong) have been doing this for the longest time among all high-income economies suggests they are adept at dealing with shocks. It should also be noted that currency pegging has become an established fact among economies with mid to high per-capita incomes in recent years, as Figure 8.7 shows. All GCC economies apart from Kuwait have been pegged to the dollar since 1986 – longer in some cases. By contrast, the currency pegs of the euro's forerunner system lasted less than 13 years (1979–92).

The second argument is that the Gulf economies have a lot to gain from a common currency. As we saw earlier, this is not at all clear when looked at only from within the region. But considered from an international perspective, a Gulf banknote would have a weighty economy and population behind it. Its issuer would be among the world's large international creditors. This would be a good position from which to launch a globally accepted currency. If it gained international acceptance, the currency would become a vehicle through which to invoice GCC trade. The Gulf states would be able to raise international funds in their own currency. This would free them from what economists call the curse of 'original sin' – the inability of most emerging economies to issue debt in their own currency. Being able to do this is useful because it insulates the

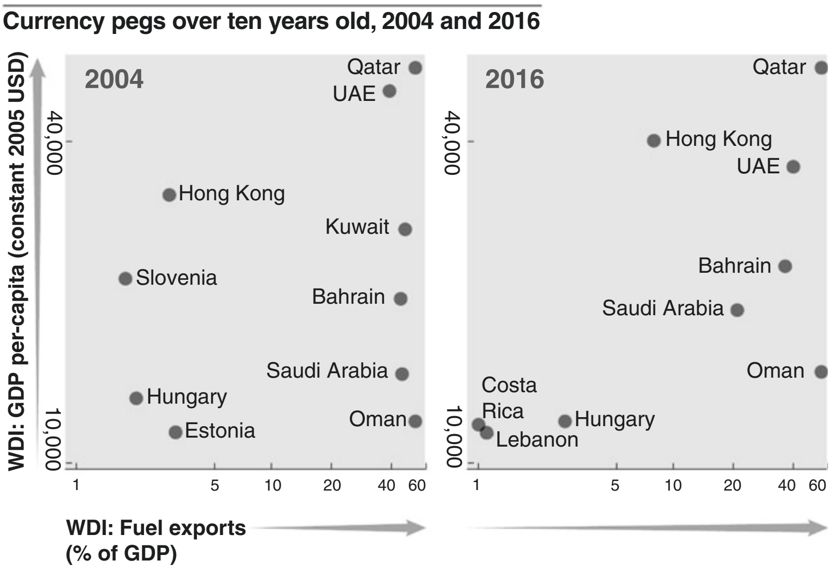

Fig. 8.7 Currency pegs over ten years old 2004 and 2016

Sources: World Bank WDI, IMF AREAER

borrower's balance sheet from changes in the exchange rate. The United States is currently the main gainer from this as the rest of the world is happy to buy Treasuries.

The Gulf's trading partners would only use and hold Gulf banknotes if they were liquid and there was a debt market in which to invest them. Currencies are obtainable externally only when the issuing economy runs an international payments imbalance. In the Gulf case, this means investment outflows, but these are already underway in the form of GCC sovereign wealth funds, which are invested widely in public and private assets. Liquidity in a Gulf-note-denominated debt market would be determined by the borrowing plans of the GCC members. In the near term, government debt will expand and contract as a function of counter-cyclical policy to smooth out fluctuations in hydrocarbons prices. In the long term, private and government debt should expand to finance financial deepening and economic diversification. The rise in external borrowing shown in Figure 8.8 could be in the Gulf note.

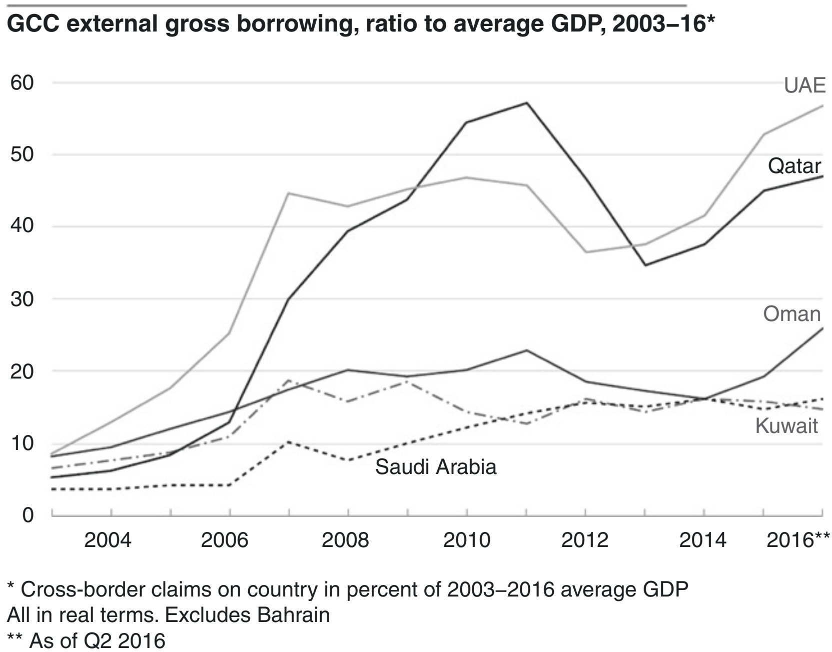

Fig. 8.8 GCC external gross borrowing, ratio to average GDP, 2003–16

Sources: BIS, World Bank WDI

The Gulf's history tells us that the most likely catalyst to move to action will be another geopolitical shock. Political exigency created the GCC, kept it together at crucial moments and will drive any progress toward a Gulf banknote. Right now the plan is stalled because it has copied the euro project, and the euro's travails since 2010 have given cause for caution. Looking at the euro too hard is probably a mistake. While the Gulf economies arguably have less convergence in business cycles than did the euro economies before unification, in other respects their characteristics suggest better prospects for unification. Politically, the GCC common currency area would be less of a jumble of creditor and debtor economies than the euro area.

Does the United States currency union offer any guidance? Only partially. Undoubtedly the most important feature of America's monetary union is the federalization of many of the fiscal roles of the state. Spending on defense, social security and infrastructure is all undertaken by the federal government.

When advocates of GMU can realistically say that this sort of unification is a prospect for the Gulf states, then GMU will be near, but this does not seem likely for many years. Yet GMU may become more, not less, difficult over time if economic diversification is successful and a single monetary policy has to be applied to both hydrocarbon-intense economies and more diversified economies. This is a recipe for the one-size-fits-all problems in the euro area. For GMU it may in fact be a question of – now or never.

Notes

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2. Cited in Amy Verdun, "The institutional design of the EMU: A democratic deficit?" *Journal of Public Policy* 18:2 (1998).

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CHAPTER 9

Foreign Exchange Reserves Management – SAMA's Experience

1 SAMA and Sovereign Wealth Funds

In the years before the global financial crisis, SAMA watched as the activities of some sovereign wealth funds (SWFs) triggered negative political reactions in Europe and North America. In 2006, Dubai Ports World was forced to sell the US ports it had bought after Congress moved to block the deal, and there was opposition when, in 2007 and 2008, SWFs bought stakes in the struggling financial giants Morgan Stanley, Merrill Lynch, Citigroup and UBS. The idea that wealth funds could dominate global investment was a chimera – in 2007 they accounted for only about 1 percent of global financial assets. But some mainstream politicians responded with veiled threats. Take, for instance, the remarks of European Trade Commissioner Peter Mandelson in March 2008:

> No one is worried about Norway's plans for global domination. Chinese investment vehicles and the Russian stabilization fund, on the other hand, are new investors, with huge reserves, backed by governments with mixed democratic credentials, substantial foreign policy projection and no track record as investors. This does not disqualify these countries and their funds. But there is a heightened need for reassurance.[1]

A. Banafe, R. Macleod, *The Saudi Arabian Monetary Agency, 1952–2016*, Financial Institutions, Reforms, and Policies in Muslim Countries, DOI 10.1007/978-3-319-55218-7_9

In the 1970s, there had been similar outcries against Saudi investment in the United States. Saudi Arabia was obviously not like China or Russia in that it had been a strategic ally of the US since the 1940s and had always participated fully in the Western financial system. The central bank saw no reason to raise its profile by labeling itself an SWF. This skeptical view was confirmed when a group – including Libya, China and Russia, as well as Abu Dhabi and Kuwait – collaborated on drafting a voluntary code of good practice, the Santiago Principles. The idea was that this would deflect criticism of their activities.

Saudi Arabia, unlike China, was not interested in deliberately building up export surpluses. Reserve accumulation in Saudi Arabia is a secular phenomenon as opposed to a policy decision, consisting of oil receipts rather than portfolio inflows (Fig. 9.1), and SAMA does not resort to active currency intervention. High oil prices lead to export surpluses and growth in SAMA's reserves, while low prices mean the reserves run down, so it is periodically necessary for SAMA to sell blocks of assets. Admittedly this practice does not necessarily contribute to stability in financial markets but the central bank could not in honesty promise not to do it any more. SAMA felt that the Santiago Principles infringed on its independence and on the sovereignty of the country. The particular Santiago Principle that SAMA found troublesome was GAPP 17: 'Relevant financial information regarding the SWF should be publicly disclosed to demonstrate its economic and financial orientation, so as to contribute to stability in international financial markets and enhance trust in recipient countries.'[2]

So SAMA has never labeled itself an SWF nor joined up to the group that signed the Santiago Principles. In this chapter, we will explain how

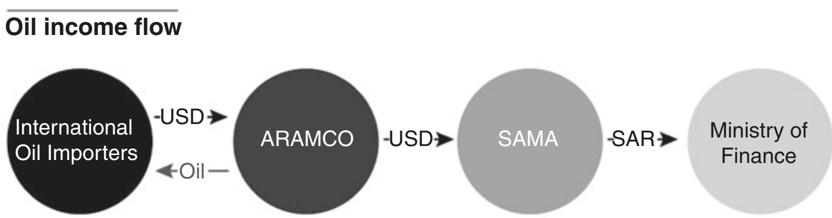

Fig. 9.1 Oil income flow

Source: Authors

SAMA manages its reserves and how it straddles the work of an SWF and a central bank. It covers the evolution of reserve management as well as SAMA's current practices and compares how SAMA's investment approach compares with Norway's sovereign wealth fund.

2 Evolution From the 1960s to the Present

SAMA has been managing foreign exchange reserves since its inception. Gold and silver were used as currency backing from 1952 onwards. The earliest data we have dates from 1958, when financial assets amounted to $2 million in dollar call accounts in New York. Half a century ago the central bank used passive, short-term investment strategies to preserve principal value and maintain maximum liquidity. Today by contrast, it makes active use of a broad range of instruments and benchmarks against which to measure performance.

SAMA resembles both a traditional central bank and an SWF, managing two portfolios: a reserve portfolio (which resembles the usual liquidity requirements of a central bank); and an investment portfolio (involving the deployment of surplus reserves over a medium-term horizon). From the late 1960s, when it began accumulating assets on a large scale, the central bank has maintained a strict separation between these two portfolios. SAMA's dual role means that it lays emphasis on liquidity as well as on return. Once liquidity has been created, any surplus is invested in longer term financial assets.

Before the first oil boom of 1973, SAMA's asset allocation consisted of spreading deposits among the major international banks. As the reserves accumulated, SAMA drew on the expertise of investment specialists from White Weld and Baring Brothers (WB) to assist in reserve management and staff training.[3] During this period global capital markets (with the exception of the United States) lacked liquidity and SAMA had to enter into memoranda of understanding with the United States, followed by Japan and Germany, in order to invest through direct acquisition of government bonds. As the markets gained liquidity, SAMA bought private placements in the G7 group of countries. It started investing in global equities at the same time, and in 2002 added emerging market debt. Since the financial crisis, it has moved into alternative investments as a way to enhance returns in an environment where traditional asset classes perform less well due to artificially depressed bond yields (in Chapter 7, Fig. 7.6 and 7.7 show the evolution of the reserves in recent years).

SAMA's history has shaped its investment culture. Above all, its approach is incremental and this stems from a low turnover of investment staff. There is no substitute for experience in the world of investment, which continuously displays unexpected changes and where fashions come and go. This means that senior SAMA staff with decades spent running money are available to train new joiners. They can explain how important risk-adjusted returns are, bearing in mind that a key to investment success is to avoid big losses. Continuity of personnel means that the team is highly resistant to investment fashions. This also explains SAMA's preference for recruiting staff either in-house or from university or business school so that they are trained up in the investment culture.

A second important human factor after continuity of staff is the unique location. The investment function has always been sited in only one place in an open-plan office where teams work together and ideas are shared at regular meetings. The in-house managers of deposits and shorter dated bonds sit alongside the teams dealing with the external managers and the people looking after the quantitative risk management function. A third historical factor making for continuity is the division between in-house management and external managers which goes back to the 1970s. Equities in particular have never been managed in-house.

SAMA's investment process is also highly centralized and is described in detail below. The job of forecasting returns on a currency-hedged basis is done by the different teams. This leads to a debate about investment weightings judged against the strategic benchmark. The final decision is made at the investment committee chaired by the Governor, although the Investment Department has tactical discretion over how to implement the changes, and if the investment environment changes in a marked way the decisions will be reviewed.

3 Structure of the Investment Deputyship

The Investment Deputyship has evolved over time but its form and functions have remained fairly constant. Until the 1980s, it was called the Foreign Department and was headed by a Director General. Its responsibilities included looking after investments, processing letters of credit and dealing with money transfers. Length of tenure is a feature of the department. In the nearly 50 years since 1970 it has had only six heads.[4]

The department has in the past also been called in to assist on other aspects of policy such as the design and management of the government debt program. Investment advisers report to the department head. In 2013, the department was upgraded to an Investment Deputyship with Ayman Al-Sayari as its Deputy Governor for Investment. The Investment Management Department (IMD) is one of two departments in the Deputyship, the other being Investment Accounts (and there is as well the separate function of Government Finances which deals with debt management).

The structure of the investment management function reflects the asset classes in which the central bank invests. There are five teams. Two of these oversee the external managers who run the bulk of the assets, both equities and fixed income. A third team looks after internal bond holdings and a fourth is responsible for foreign exchange and money market instruments (to provide liquidity for SAMA's role in supplying the banks with dollars). The fifth team is responsible for alternative investments. Accounting, liquidity and finance, performance and risk analytics, and operations are distinct and all report separately to the Deputy Governor (Fig. 9.2).

4 Investment Objectives, Philosophy and Process

SAMA has three objectives: to preserve capital; maintain liquidity; and achieve an investment return compatible with its risk appetite. In other words, it seeks to invest assets to achieve investment objectives without undue risk of loss.

The central bank's philosophy is to have a globally diversified portfolio using top-down asset allocation and combining in-house investment with external fund managers. The investment process is built around relative weightings versus the strategic benchmark (which SAMA does not disclose). SAMA is moving from its previous practice of allocating assets by class to allocation by risk according to asset roles. Long-term macro themes are developed by the investment teams (e.g., that emerging markets tend to have higher returns than developed ones) and converted into numbers by forecasting returns on bonds, equities and other asset classes on a currency-hedged and unhedged basis. This generates a matrix of expected returns and recommendations, which are forwarded to the Governor. SAMA's tactical asset allocation (TAA) is based on this total expected returns matrix, measured by its variation from the neutral position on the strategic benchmark, in other words whether it is over or underweight relative to strategic asset allocation (SAA). Between meetings

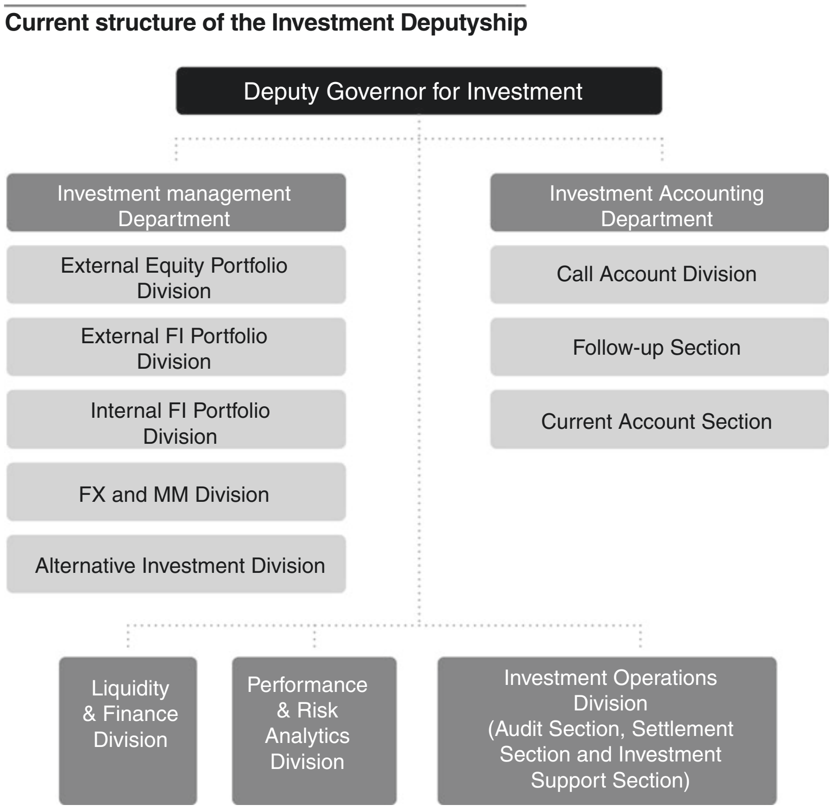

Fig. 9.2 Current structure of the Investment Deputyship

Source: SAMA

there is scope to take additional TAA decisions through investment program amendments, reflecting unpredictable market developments and price actions (Fig. 9.3).[5]

SAMA's investment style has been relatively conservative with emphasis on credit quality, liquidity of assets, a diversified portfolio and risk-adjusted return. The emphasis on diversification is not wholly due to the belief that future market returns are inherently unpredictable; rather it expresses SAMA's conviction of the wisdom of allocating risks to assets that have been uncorrelated in the past and are likely to remain so in the future.

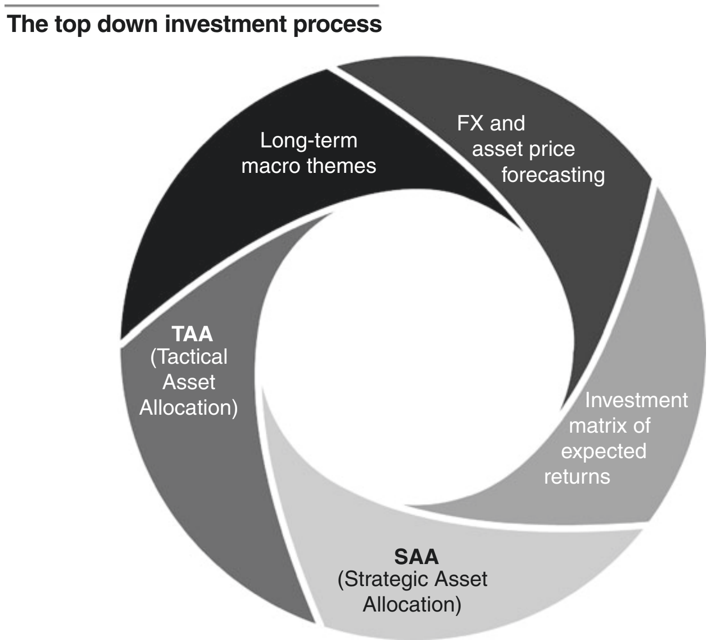

Fig. 9.3 The top-down investment process

Source: Authors

Benchmarks are selected on the basis of SAMA's risk tolerance for sovereign bonds, credit products and equities. Figure 9.4 demonstrates the division of assets between the more conservative reserve portfolio (RP) and the more aggressive investment portfolio (IP). Each one has specific investment guidelines and operational benchmarks. Given the RP's focus on liquidity, its performance is measured against short-term income generating investments. The IP's performance is measured against its composite policy benchmark. Given the objective of the IP (which includes taking active risk within the prescribed limits set by the Investment Committee), there is likely to be more volatility in its return over the short to medium run.

Measuring the adequacy of reserves: The central bank has always divided the portfolio into liquidity and investment segments, with the former intended to meet the needs of the Saudi banks for foreign exchange at the pegged exchange rate. Deposits are SAMA's first defense line, followed

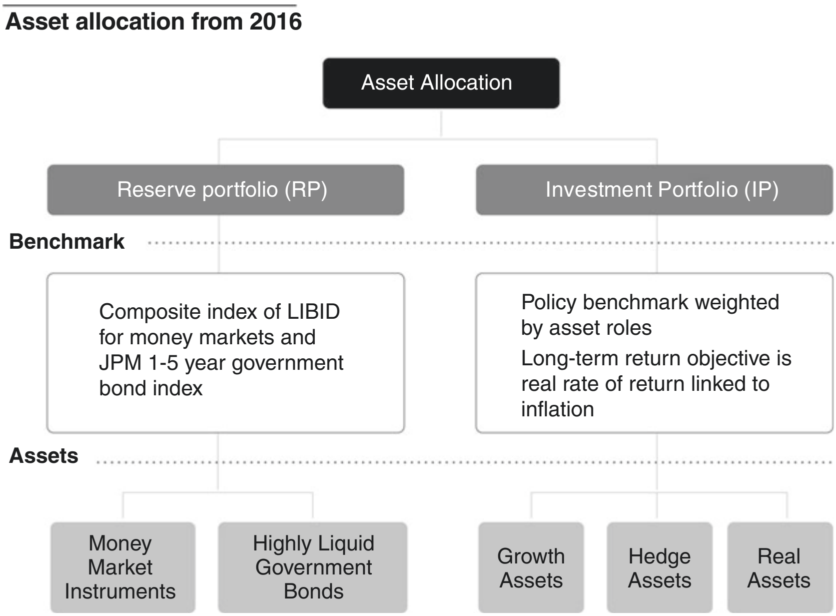

Fig. 9.4 Asset allocation from 2016

Source: Authors

by liquid marketable securities. The bulk of SAMA's assets are in the form of bonds and equities. This asset mix maintains the desired degree of liquidity and the diversification reduces risk and improves risk-adjusted return.[6]

SAMA has reviewed the allocation of money between the reserve and the long-term investment portfolios. It has adopted a comprehensive formula which captures potential sources of balance of payments outflows by taking into account the characteristics of the Saudi economy. For Saudi Arabia, where currency backing is mandatory and foreigners' remittances are significant, the formula for reserve adequacy is shown in Figure 9.5.

The implication is that the larger the reserve portfolio, the less could be allocated to the investment portfolio and the lower the investment returns would be over time.

Once every three months the Investment Deputyship runs a multi-stage process, as detailed below.

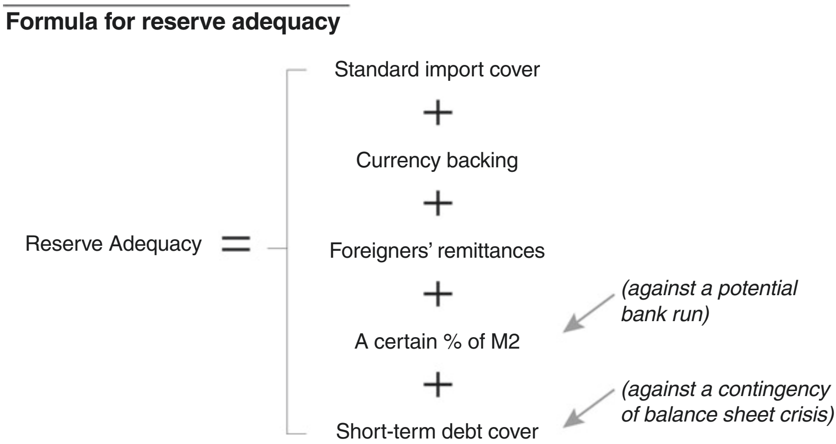

Fig. 9.5 Formula for reserve adequacy

Source: Authors

(1) Investment Matrix Meeting. Preparatory input for the investment matrix of expected returns comes from the Working Group headed by the IMD director. This matrix is based on macroeconomic themes, market trends and external forecasts for foreign exchange and asset prices.

(2) TAA Meeting. The IMD director and department unit heads participate in reviewing cash flow trends and forecasts for the RP allocation and their level of conviction for the IP position to enable them to propose a TAA for the IP. In summary, the TAA for the RP takes into account liability requirements for the target liquidity profile to meet dollar outflows, and for the IP expected investment returns over a one-year time horizon.

(3) Carve-up Meeting. The carve-up meeting is a two-day deliberation, headed by the Deputy Governor for Investment, and attended by the IMD director, senior investment adviser and department unit heads. At the carve-up a detailed agenda is reviewed covering global macro themes, market trends and outlook, investment matrix, foreign exchange flows, previous program execution, investment performance, custody and securities lending and performance of external managers for portfolio actions. The output of the carve-up is a set of investment program recommendations based on a broad-based consensus for discussion at the Investment Committee.

(4) Investment Committee Meeting (ICM). The ICM is a quarterly one-day gathering, headed by the Governor, and attended by the Vice-Governor, four Deputy Governors (Investment, Research and International Affairs, Supervision and Banking Operations), the IMD director, the IPRC (Investment Performance & Risk Control) director and the senior investment adviser. Invitees include the head of PRA (Performance & Risk Analytics), the head of external equities and the head of external fixed income to present their input.

(5) Program Execution Plan. Following the approval of the investment program, the Deputy Governor for Investment and senior colleagues discuss the execution strategy for both the RP and IP in terms of size, timing and cash flow coordination between the asset pools as well as an implementation plan for portfolio actions (funding or disfunding of external managers).

5 SAMA Portfolio Characteristics

Credit Criteria: Bank deposits are made according to the size of the bank's equity capital and a minimum Fitch viability rating of BBB minus, which is the lowest investment grade rating, as SAMA does not invest direct in sub-investment grade securities. Investment in sovereign paper and other fixed income securities is normally limited to 10 percent of the issue size. Investment in securities is restricted to sovereign, sovereign guaranteed, agencies, supranational and corporate obligations rated AAA or AA by two of the three leading rating agencies that SAMA uses (Moody's, Standard & Poor's and Fitch). Exceptions are made in some cases, such as emerging markets where sovereign credit rating is mostly lower than AA and for global fixed income portfolio managers who have the necessary expertise to invest in lower grade bonds. As the largest proportion of SAMA's exposure is in AAA and AA credits, default risk is virtually non-existent.

Asset Allocation: As stated earlier, asset allocation is determined by the recommendations that emerge from the total expected return matrix. Good asset allocation should generate 70–80 percent of return while the rest derives from a manager's skills and expertise (commonly called alpha generating capacity). Investment discipline is tight, and program execution must be within the parameters of what has been approved by senior management.

Currency Composition: The principle of diversification forms the basis of investment, and currency exposure is bound by the relevant operational benchmarks. The broad framework is determined by ongoing liquidity requirements to fund the foreign exchange outflow, which varies. The dollar is used as the base currency due to its importance to the kingdom's revenue and expenditure pattern and to international trade and finance, followed by a few other major currencies. The actual currency composition is the result of the sum of the benchmark weightings in each currency. Differences between trade and investment flows mean that currency allocations are not linked to trade flows but are rather driven by the availability of investments (as measured by market capitalization) and instrument liquidity.

Maturity Profile: Bank deposits are short-term up to one year. The internal bond duration benchmark for fixed income securities is less than five years. External portfolio managers, however, have greater flexibility on maturity and duration.

External Portfolio Management: Externally hired firms seeking maximum return while conforming to SAMA's guidelines manage a good proportion of SAMA's assets. SAMA has a blend of actively managed and indexed assets in global, regional and single-country portfolios, both in equities and fixed income. SAMA believes managers can bring added value through their research, expertise, weighting decisions, performance and sharing of experience. Overall, external managers' performance record has been mixed, including under and outperformance of indices over different time periods. Portfolios can be disfunded if they have underperformed or for asset allocation reasons (including raising liquidity).

The asset management industry has been impacted by increasing regulatory burdens and higher capital requirements. These have led to a significant deterioration in the cost/income ratio in the asset management industry and this has implications for SAMA. At the same time clients like SAMA have become more sophisticated in their demands, seeking a broader range of services from macro research and asset allocation ideas to solutions-based customized products, periodic portfolio reviews and staff training. Active managers are under increasing pressure to outperform their benchmarks as clients like SAMA increasingly favor passive mandates with lower fees. These are both cost-effective and less time-consuming, and tend to fare well in mature markets with fewer anomalies. Emerging markets, by contrast, offer more scope for active management and will become increasingly important due to their growth potential and relative undercapitalization.

SAMA debates these issues constantly and a comparison of the positives and negatives of active and passive management is shown in Figure 9.6.

SAMA believes the most efficient strategy is to adopt a blended combination of approaches depending on the market. The bulk of assets are

Active vs. passive portfolio management

Active	Passive
1 An inefficient market concept (it operates by taking advantage of market anomalies)	**1** An efficient market concept (prices reflect a security's true value)
2 Focuses on optimizing alpha returns (added value) by using skills and judgment to take advantage of market trends	**2** Focuses on cost effectiveness via low-cost market tracking index investments (reducing investment cost key to improving net returns)
3 Strategy well suited to inefficient/emerging markets	**3** Strategy appropriate to mature economies
4 Risk that closet indexers will underperform, along with the operational burden of frequently hiring/firing managers	**4** Generates auto risk adjusted returns as chosen indices reflect risk appetite (the downside of this is the problem of sectoral allocation risk in a crisis, as post-crisis the financial sector took a beating due to massive corrections)

Fig. 9.6 Active versus passive portfolio management

Source: Authors

allocated to cost-efficient passive investments designed to capture market returns by tracking a specific benchmark, primarily in the developed stock markets. The balance of the portfolio can then be invested in actively managed 'satellite' investments with the potential to boost returns and lower overall portfolio risk.

Swaps and Securities Lending: Swaps of securities are made to restructure a portfolio's maturity or credit quality and to improve overall liquidity. Since the global financial crisis, the internally managed portfolio has maintained a pool of highly liquid government bonds of shorter duration, reducing the need for swaps.

Hedging: SAMA largely runs its financial exposure unhedged, although external managers have the flexibility to use derivatives on an unleveraged or defensive basis if the underlying securities are within the portfolio guidelines.

Gold: Gold still represents more than 70 percent of the reserves of large developed countries. In contrast, the central banks of emerging economies hold on average less than 5 percent in gold and favor income-producing assets such as bonds. The advantages and problems of holding gold are well documented. It provides a hedge against inflation and has a value with no corresponding liability, unlike a bond. Gold held at home cannot be subject to asset freezing by another country. The negatives are that central banks can also use index-linked bonds as a hedge against inflation and, unlike bonds, gold has no inherent return.

Although SAMA has held gold as part of the backing for its currency for many years, it believes that with the end of the Gold Exchange Standard in 1971, this precious metal has gradually lost its usefulness as a reserve asset. Today gold has declined in importance for SAMA and currency issuance is instead backed by a pool of reserve currencies.

Call balances: The central bank keeps a target call balance to meet the day-to-day cash requirements of government and the dollar purchases of domestic banks. Foreign exchange is almost entirely generated in the public sector as a result of oil exports, making SAMA the sole provider of dollars to the private sector, through sales to domestic banks at a fixed dollar rate. Effectively, it intervenes passively in the domestic interbank market for spot settlement on a daily basis.

6 Performance Measurement and Risk Management

Performance: Portfolio benchmarking (also termed performance measurement) is used to evaluate both internally and externally managed portfolios. Performance is measured on a total return basis. SAMA's overall portfolio performance in the medium term is comparable with that of Norway's sovereign wealth fund. However, its relatively conservative approach has enabled it to outperform in downswings. For instance, in 2008, when large institutional investors suffered huge losses, the defensive positioning of SAMA's portfolio contained the crisis.

SAMA broadly applies the recommendations of Global Investment Performance Standards (GIPS) in measuring portfolio performance.

GIPS provide a set of standardized industry-wide principles that guide investment firms in calculating and presenting investment results.

Risk Management: Following the global financial crisis, risk management and compliance have assumed greater significance due to the changing nature of the markets and increased regulatory scrutiny. De-leveraging and caps on proprietary trading are part of the business of risk management and governance by central banks and commercial banks alike. But while regulation is necessary, it has to be managed sensitively and appropriately in order to ensure that innovation and initiative are not stifled.

SAMA's risk management parameters are covered below:

– Currency risk tends to be a function of asset allocation as no active currency bets are taken. While there are tactical opportunities only a very few of SAMA's managers take currency views and their returns are generally volatile.
– Credit criteria have remained conservative as preservation of capital comes ahead of return considerations.
– Liquidity risk is negligible because SAMA focuses on investments in liquid assets and markets.
– Market risk is primarily determined by the composition of benchmark portfolios, with limits on expected tracking error (a measure of how closely a portfolio follows the index to which it is benchmarked).
– Counterparty risk has become more pronounced since the Lehman collapse. SAMA policy and practice has been to deal only with reputable and financially strong institutions.
– Operational risk for internally managed portfolios is addressed through segregation between front and back office operations. The risk involved in externally managed portfolios is reduced by the separation of managers and custodians, and stringent guidelines and reporting requirements. Where there have been unintended breaches of guidelines, for example due to downgrades, managers are not forced to sell the securities immediately in order to avoid losses when prices are depressed.

SAMA divides the functions of risk control, compliance and performance measurement. Risk and compliance functions are centralized and report to the Vice Governor, but this excludes risks related to investments. For investment risks, SAMA has two specialized units. A Performance & Risk Analytics unit reports to the Deputy Governor for Investment.

Independently, an Investment Performance and Risk Control unit reports to the Vice Governor. The units liaise with each other and with the IMD to make sure that there is a high level of portfolio compliance with risk policies and guidelines, as well as monitoring, measuring, reporting and mitigating financial risks in a timely manner.

In short, risk management remains an area of active attention. As risk modeling is not reliable all the time, qualitative judgment should complement quantitative analysis.

7 Some Current Issues

The culture of investing at SAMA encourages continuous debate and discussion of financial issues. By way of example, here are two areas that are relevant at the moment.

Usefulness of Treasury Inflation Protected Securities (TIPS): Debate continues on the viability of TIPS or linkers (inflation-linked bonds) in the disinflationary environment which has followed the financial crisis. The yield on TIPS can be broken down into the real bond yield and a payment for expected inflation. In equilibrium, this should be the same as on a conventional bond that offers a nominal yield:

$$\textit{Nominal Bond Yield (NBY)} = \textit{Real Bond Yield (RBY)} + \textit{Inflation}$$

Immediately after the crisis, NBY and RBY were positively correlated since there was little market anxiety about inflation. In early 2009, breakeven inflation was almost zero with the result that NBY and RBY were identical. Linkers were then extremely cheap. In more recent times, when real rates have been extremely low and inflation expectations subdued, linkers have added little diversification value to a portfolio. The arguments SAMA has considered for and against them are outlined in Figure 9.7.

On balance it seems reasonable to conclude that while it is a close call on the return argument, TIPS remain useful as a way to diversify risk. The result is that SAMA has chosen to hold TIPS as hedges against inflation (as opposed to real estate which it does not hold). They are more liquid and real estate is not clearly protected by sovereign immunity in the way that financial assets are.

Treasury Inflation-Protected Securities TIPS: arguments for and against

For	Against
1 A global bond portfolio has the advantage of diversification and consequently of reduced volatility	**1** Real rates are too low to make a convincing case for TIPS.
2 Break-evens (nominal bond yield – TIPS yield) are fairly stable	**2** With inflation in most of the world logging behind CBK targets, and break-evens not compelling, the case for TIPS is weak
3 Offers the attractive prospect of a hedge against potential inflation	**3** Liquidity premium is an issue
4 Rising nominal bond yields (calculated as real premium) would appear to favor TIPS as they are less volatile	**4** In the absence of inflation or the expectation of inflation, rising nominal bond yields are equally as risky as TIPS
5 The TIPS return exceeds inflation over time	**5** Nominal bond yields also compensate investors for expected future inflation as well as volatility
6 If real rates were to decline nominal Treasuries and TIPS would both benefit	**6** TIPS issued with low real yields would be of concern to investors

Fig. 9.7 Treasury Inflation-Protected Securities (TIPS): arguments for and against

Source: Authors

Relevance of risk parity: The traditional approach for a balanced portfolio like SAMA's is to have 60 percent in bonds and 40 percent in equities, which are much riskier and generally have greater volatility. Risk parity allocates risk to a wider range of assets including property and commodities. The approach was popular following the crisis because of strong bond market performance, itself linked to the massive bond purchases made by the major central banks.

In simple terms, risk parity seeks to equalize weights or risks for bonds and equities by leveraging up the bond allocation. But risk parity allocation itself contains an element of risk. Focusing on risk alone ignores returns. How can an asset allocator such as SAMA have a plausible policy on asset allocation which is not based first and foremost on returns? For instance, if you leverage up a Treasury portfolio because it has historically offered stable returns, you can potentially incur big losses if bond yields rise. Whether or not this approach is superior to the traditional method of asset allocation by class is still the subject of considerable debate. SAMA's thinking is that strategic allocation by asset roles makes more sense than the traditional system of allocation by asset class or the novel idea of risk parity. The asset roles approach enables allocation to growth assets for maximum return, to hedge assets for stable income and to real assets for the preservation of value in an inflationary environment.

8 Norwegian Comparisons

SAMA has learned three enduring lessons over the decades: asset allocation remains an art rather than a science; there is no single strategy that performs in all market environments; and the key to a stable and positive return is investment in non-correlated assets.

There is no single successful formula for running money, but SAMA's approach is more akin to that of a central bank manager than a sovereign wealth fund, with outside managers being extensively used. The Norwegian sovereign wealth fund, run from within the central bank and known as Government Pension Fund Global (GPFG) has a culture of openness that allows the observer to make a number of interesting contrasts. The two funds are of broadly similar size: Norwegian staff numbers are much higher and their backgrounds are more diverse than the Investment Deputyship. It employs more than 500 staff from 35 countries, and although most of the senior people are Norwegian, the majority of staff are not, and have been trained elsewhere. Next, it is not in a single location: although headquartered in Oslo, it has offices in London, New York, Shanghai and Singapore.

GPFG, unlike SAMA, discloses its strategic benchmark, which is 60 percent equities, 35–40 percent bonds and up to 5 percent in real estate (the property side alone employs 100 people). The equity bias is not duplicated at SAMA, which does not own real estate (as we saw, it prefers to own financial instruments including inflation-protected bonds). SAMA continues to use external managers, while GPFG has changed to run most of its funds in-house since the financial crisis. By contrast, SAMA's approach has remained consistent. One other difference is that SAMA uses many passive external managers, mainly in developed stock and bond markets, while GPFG believes in active fund management. SAMA also has a bias toward emerging markets, reflecting a belief that their faster growth will translate into better investment returns.

It is difficult to determine the real nature of any investment process from the outside. GPFG, like SAMA, has a strategic benchmark against which investment decisions are measured, but since GPFG has multiple locations in different time zones and three chief investment officers for equities, real estate and asset allocation it seems there must be a lot of delegation, in particular for real estate. By contrast, the Investment Deputyship has a single head who carries out both the chief executive and investment tasks and it is part of the Governor's role to be ultimately responsible for investment decisions. Delegation takes place to external fund managers.

Notes

1. European Commission. "Press Release Database-Mandelson speech at OECD Conference, March 28, 2008." European Commission, http://europa.eu/rapid/press-release_SPEECH-08-155_en.htm?locale=en (Accessed on October 26, 2016). This was a typical comment from politicians in this period and formed the backdrop to the formation of the Santiago Group a few months later.
2. International Working Group of Sovereign Wealth Funds, "Generally Accepted Principles and Practices." IWG, http://www.iwg-swf.org/pubs/eng/santiagoprinciples.pdf (Accessed on October 26, 2016).
3. It is worth recording the names of all the WB team members who were seconded to SAMA from 1975 to 1988. On the American side (White Weld then Merrill Lynch when they acquired White Weld) were David Mulford, team leader, subsequently replaced by Steve Wilberding. Other team members were David Reid Scott, Tom Berger, Sam Forester, Bob Smith, Brian McKinley, David McCutcheon, David Hubert, Tom Latta, Bill Franks and Greg Ambrosio. The Barings team was initially led by Leonard Ingrams and subsequently by Jeremy Fairbrother and Bill Black. Other team members were Tony Hawes, Jonny Minter, Michael Baring, Robert Rice, Ian Cooper, Leslie Myers, Richard Comben and Rory Macleod. When the WB contract expired in 1988, Bill Black was retained by SAMA on a direct hire basis until 1997.
4. Ahmed Abdullatif in the 1970s was followed by Ahmed Al-Malik in the first half of the1980s. In 1985 Mohammed Omar Al-Khatib took over, followed in 1992 by Mohammed Al-Shumrani. He was succeeded by Khalid Al-Sweilem in 2004. Since 2012 the department has been run by Ayman Al-Sayari. Ayman is a member of SAMA's Investment, Risk, Monetary Policy,

Financial Stability and Senior Management Committees. He holds a BSc. in accounting from King Fahd University and an MBA in Finance & Investment from George Washington University. He is a CFA Charter Holder. He joined SAMA in 1999, and between 2003–07 worked at the International Finance Corporation in Washington DC. In 2007 he returned to SAMA as an adviser to the head of the Investment Department and subsequently as Director General of Investment and then Deputy Governor for Investment.

5. SAA = Strategic Asset Allocation, i.e. target allocation depending on risk tolerance, time horizon and investment objectives (periodically rebalanced back to the original allocation when the portfolio deviates significantly). It is derived from a mean variance optimizer model taking into account asset price correlations, volatility, return expectations, maximum drawdown and an element of qualitative judgment.
   TAA = Tactical Asset Allocation, i.e. a short-term tactical trading strategy to take advantage of market anomalies or opportunities
6. Fahad Alhumaidah, 'Asset-Liability Management for Reserves under Liquidity Constraints: The Case of Saudi Arabia,' *Procedia Economics and Finance* vol 29 (2015). This is a good practical examination of the problems of liquidity and unpredictable outflows. Dr. Alhumaidah is an Adviser in the Investment Department at SAMA.

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CHAPTER 10

Developing the Domestic Bond Markets

1 Building on Government Bonds

In 1987, when oil prices were weak and the government's account with SAMA was about to run out of money, Mohammed Omar Al-Khatib, who headed the Investment Department for many years, one hot afternoon told the story of how Abdullah Al-Suleyman had coped with a similar problem when he was Ibn Saud's right-hand man. He regularly borrowed from the rich Jeddah merchants against the king's good word, but sometimes they refused to lend him more. In the mid-1930s, when the state finances became so parlous that salaries could no longer be paid, Al-Suleyman gave out one-third in cash, one-third in provisions and kept the remaining third back as a compulsory loan.

Mohammed Omar said that when things got particularly bad and Al-Suleyman had no more money to pay the king's servants, he hit on a ruse and ordered the carpenters to come to his office early one morning. When he opened his doors a couple of hours later and the suppliants stepped in, they found the room filled with laborers and wood dust and the floor covered with shavings. Al-Suleyman waved his hands at them and pointed to the carpenters busy at work making wooden boxes. 'I cannot deal with you now,' he said. 'Fresh money has come in and we are busy building more chests to house the gold and silver. Return in two days' time and you will all be paid.' The suppliants filed out, only half-believing him, but willing to hold off for a couple of days at least.

A. Banafe, R. Macleod, *The Saudi Arabian Monetary Agency, 1952–2016*, Financial Institutions, Reforms, and Policies in Muslim Countries, DOI 10.1007/978-3-319-55218-7_10

Al-Khatib ended his story by sipping tea from his glass and took another puff on his cigarette: 'I am having some treasure chests made at this moment,' he joked.

Government revenue today is as unpredictable as when Al-Suleyman managed the finances because it depends on the flow of oil income. Since the oil price is effectively a random walk and shows no duration dependence, the government can hardly plan years ahead with confidence: it can however try to stabilize the economy by running budget deficits when the price is low (and cool it down through surpluses when the oil price is high). The deficits ultimately run down SAMA's foreign exchange reserves, but the government needs to acquire the riyals to spend even if oil income is inadequate. Apart from monetary financing direct from SAMA, the only way is to issue riyal bonds. So developing a government bond market is more important for Saudi Arabia than for most emerging economies, which have a predictable stream of government earnings. The other benefits of a bond market, including financing the non-oil economy, developing a savings market and encouraging the banks to invest in bonds, also matter.

The next year after Al-Khatib told his story, SAMA launched the first GDBs (Government Development Bonds) as an agent of the government. For the first time the state borrowed money in a systematic and organized way, and in the process it tried to create a bond market. The proceeds went to replenish the government's account with SAMA. Ever since then, SAMA and the banks have been engaged in exploring the best ways to develop a secondary market where bonds – both government and corporate – can be traded. This helps issuers because they can assess how to price their bonds by looking at what is happening in the market, while the liquidity the market provides should make it cheaper to issue bonds. New investors should also be drawn in, deepening the pool of funding. This is the theory at least, but, as in other emerging markets, it has not been easy to turn into practice.

In the first debt episode, which ran from 1988 for nearly 20 years, SAMA tried to make the bonds attractive to the banks. The central bank asked what kind of bonds they would like and met their needs with Floating Rate Notes (FRNs), which were particularly suited to the banks' balance sheet structure. It introduced a repurchase (repo) facility so the banks could borrow money against their bond holdings. It replaced multiple single issues with a few big ones on a tap basis in order to increase issue size, and it worked hard – but unsuccessfully – to get the banks to establish a network of primary dealers, where dealers would buy GDBs at issue and trade them afterward.

Government bonds became a successful financial product that the banks were prepared to buy. When the first debt episode came to an end and the banks needed more government assets, SAMA offered them central bank bills instead. But the banks were 'buy and hold' investors, who had no interest in trading and feared that a liquid secondary market, which made bonds more attractive to savers, would threaten their monopoly on taking deposits.

Even when the total value of GDBs was at its height around the turn of the century, there was hardly any secondary market in GDBs and corporate bonds, and matters have not changed much since. This is not just the consequence of a bank-centric culture or a 'buy and hold' mentality. It is costly for companies to issue bonds or sukuk (the Shariah-compliant equivalent). Changing these features of the market presents a considerable challenge for SAMA, working together with issuers, regulators, investors and other public sector bodies.

The question is whether the second debt episode that began in 2015 can trigger the growth of liquid debt markets and succeed where the first attempt failed. This chapter examines the role the central bank should play in the debt market in emerging economies like Saudi Arabia. It goes on to scrutinize the kingdom's debt markets, including the structure of the most commonly issued corporate bond, the sukuk, and to discuss their future development. Finally, it explains how – even though a Debt Management Office (DMO) was established inside the Finance Ministry in 2016 – SAMA continues to price government debt.

2 The Role of the Central Bank in Debt Markets

The central bank's role is largely confined to managing debt on behalf of the government in what is effectively a principal/agent relationship. The bank advises the government on investor appetite and sets bond features, such as average maturity and issue size as well as creating an efficient settlement infrastructure. It tries to organize issuance so that the government yield curve can serve as a benchmark for corporate bonds. Central bank repo facilities are used as a sweetener to encourage bond trading because the banks can swap the bonds for cash on a temporary basis.

But any central bank – and SAMA is no exception – will always want to go further and encourage an active secondary market for both government and corporate securities. It is about more than just lowering the cost of issuing the debt. The bond market is also vital in monetary policy operations for the sterilization of large capital inflows. In terms of developing the economy, the central bank wants to encourage other ways to expand credit in order to reduce the dominance of the banks. In the long run, a liquid bond market is also useful in ensuring the efficient allocation of capital.

When the first debt episode ended, the question arose of whether SAMA should step into the gap and issue its own debt as part of its efforts to promote bond markets. SAMA decided to limit the size and maturity of its new central bank bills to what was needed to manage system liquidity effectively. Table 10.1 shows how central banks in emerging market economies (EMEs) have come to the same conclusion. Only Chile out of a sample of 12 has a sizeable percentage of central bank bonds with a maturity of more than three years.

Central banks like SAMA accept they need to issue short-term Treasury-type bills but they are naturally reluctant to issue longer-dated bonds. To do so tends to interfere with monetary policy, as the central bank is tempted to keep its borrowing costs cheap rather than paying

Table 10.1 Central bank issuance of debt in emerging markets, 2010

	Total as percent of GDP, 2010	*Maturity below 1 year (percent)*	*Maturity between 1–3 years (percent)*	*Maturity above 3 years (percent)*	*Average maturity in years*
Asia					
China	10.3	70.3	29.7	0.0	
Hong Kong SAR	8.2	91.9	4.5	3.6	0.5
Korea	11.0	63.5	36.5	0.0	0.8
Thailand	23.6	68.0	26.0	6.0	1.0
Latin America					
Argentina	4.7	88.1	11.9	0.0	0.5
Chile	8.6	25.9	36.6	37.6	3.4
Mexico	2.7	61.0	36.0	3.0	1.1
Peru	0.8	100.0	0.0	0.0	0.3
Other					
Czech Republic	19.1	100.0	0.0	0.0	
Hungary	11.3	100.0	0.0	0.0	
Israel	18.4	100.0	0.0	0.0	0.5
South Africa	1.0	100.0	0.0	0.0	

Source: BIS

attention to its macroeconomic objectives – especially when that means raising interest rates. The full cost of any issuance will fall on the central bank profit and loss account unless it reinvests in debt with the same maturity. It also makes no sense for SAMA to borrow money in riyals and invest in overseas assets because it would simply be taking the dollars from its own foreign exchange reserves and missing out on the income it could have earned with them. From SAMA's viewpoint, issuing central bank bills when GDBs are not being issued has the effect of reducing foreign exchange outflow; because otherwise banks would buy foreign assets. In practice, SAMA simply accepts that there may be a cost if the return on foreign assets is lower than the cost of issuing bills (this is the so-called quasi-fiscal deficit).

Central banks are, however, responsible for system liquidity management, and in this context, they conduct open market operations and provide repurchase facilities against eligible collateral. SAMA's central bill issuance and repo facilities are well within the scope of central banking operations.

3 Structural Challenges in the Market for Corporate Debt

The government needs to adopt a less prescriptive approach toward borrowers. Issuing debt in the public market is more expensive and administratively burdensome than doing a private placement with a few investors, and this is one of the main reasons why companies prefer private placements.

Research shows that restrictions on who can issue bonds, high costs and the absence of local rating agencies have been identified as hindering the development of the debt market.

The result is that most private companies simply borrow from the banks, which of course suits the banks very well. Bond issuers tend to consist of the banks themselves (usually through private placements) or large companies with a government shareholding. The major issuers of bonds or *sukuk* are banks and publicly listed corporations which are generally partly owned by the state, accounting for 60 percent of the total. The private business sector only makes up 18 percent (Fig. 10.1).

Sukuk are by far the most popular local instrument as they appeal to a wider range of investors. Local demand means they can be issued

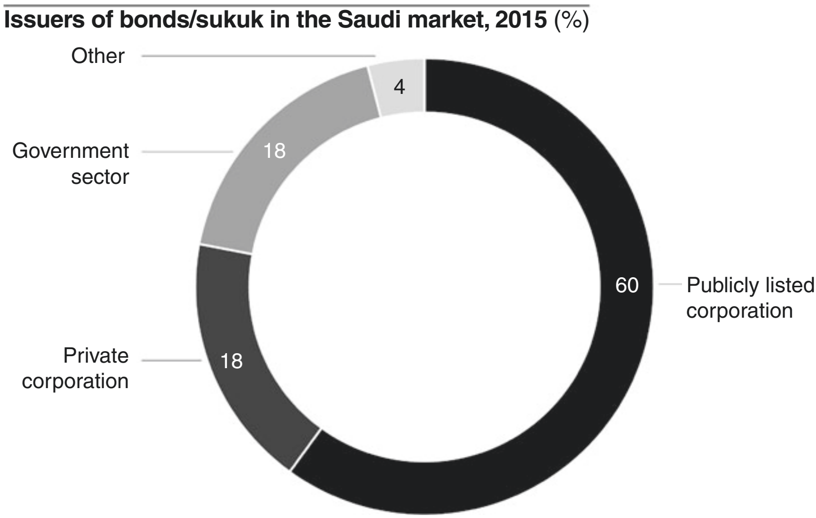

Fig. 10.1 Issuers of bonds/sukuk in the Saudi market, 2015

Source: Authors

at lower yields than conventional debt, so they are attractive to buyer and issuer alike. Since sukuk will likely remain the dominant form of bond financing, it is worth explaining how they work.

In the case of conventional bonds, the issuer has a contractual obligation to pay the bondholders interest and principal on specified dates. In contrast, under a sukuk structure the holders acquire fractional ownership in a common asset and receive a corresponding share in the profits. Any assets on which a sukuk is based must be Shariah-compliant. The most common structure involves physical assets such as land (*sukuk al-ijarah*). Other types involve the ownership of debts (*sukuk murabaha*), projects (*sukuk al-istisna*), businesses (*sukuk al-musharaka*) and investments (*sukuk al-istithmar*).

Let us take the example of a sukuk using an industrial project site as an asset. The borrower (known as the obligor), sets up a Special Purpose Vehicle (SPV), which acts as an intermediary and trustee. The SPV issues sukuk to the investors and uses the money to buy the site from the obligor. It then leases the site back to him, and uses the lease payments it receives to make payments to the sukuk holders. At maturity, the SPV sells the land back to the obligor and the sukuk's investors get their money back. The return on the sukuk should be affected by any change in the profitability of the asset and by any change in its value (in particular, if it falls in value then the sukuk will not pay back in full), so in theory the return on the sukuk is less certain than on a conventional bond. However, a sukuk is generally structured to minimize this risk. The obvious way is when the obligor agrees from the start to pay fixed lease payments and buy the project back at an agreed price, so guaranteeing the money the investors put in.

Major investors in bonds and sukuk are the banks, which hold over 50 percent of the total. Institutional investors, such as insurance companies, pension and investment funds in the public sector hold 26 percent of total issuance and mutual funds only 12 percent (Fig. 10.2).

The banks still play a dominant role within the kingdom's financial system, operating a wide variety of assets and liabilities. It is not in their interest to lose their grip on lending money. But the picture is beginning to change as they move into longer-term lending, especially into mortgage finance, because it does not match their short-term funding base. There is

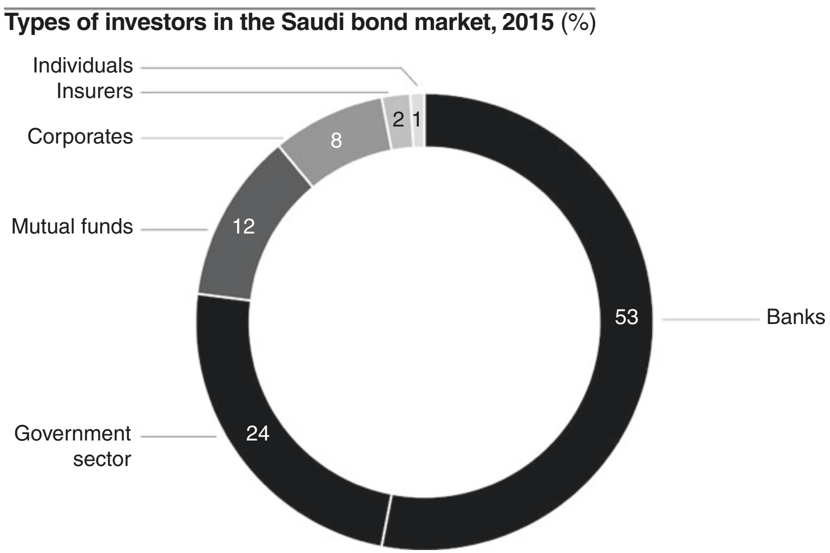

Fig. 10.2 Types of investors in the Saudi bond market, 2015

Source: Authors

an opportunity to provide finance through the capital markets to match the maturity of this lending. In well-developed markets, borrowers often raise funds by approaching investors directly, thus cutting the traditional banks out of the picture. This transformation, often called by the clumsy label 'disintermediation,' is hopefully the next stage in Saudi financial development.

Saudi banks focus on fee-generating business besides their traditional lending activities, which in turn is helping the cause of corporate bond issuance. But they have yet to be convinced that it is in their interests to provide secondary market liquidity. Unless new players enter the market and disrupt this situation, it is difficult to see why any of the banks would choose to commit capital to trading bonds. SAMA attempted to establish a dealer network in the 1990s by offering a variety of incentives, but with little success.

4 Developing a Secondary Bond Market: What Is Next?

Saudi Arabia has always suffered from a weak secondary market, even when it was actively issuing government debt in the first debt episode. The growth in the size of the primary market during the 1990s arose due to the necessity of funding the state budget deficit and the availability of adequate captive sources of funding from the banks and quasi-government funds, both of them being 'buy and hold' investors.

The failure of the secondary market in Saudi Arabia is a reflection of the kingdom's narrow investor base, its short-term investment culture and investment banks' reluctance or inability to promote secondary trading. Specifically, the lack of a market in government debt has become a significant policy problem for the central bank, now that a second debt episode is under way – since the existence of an efficient secondary market in government securities would facilitate future successful bond issuances. Other techniques of debt management, such as smoothing bond price movements and minimizing the impact of redemption payments, would also be eased if open market operations and interventions could all be conducted within the context of a well-functioning market in government debt. Faced with these impediments how then should SAMA go about developing a secondary market?

Three options – retail sales, compulsion and extending the repo facility beyond the banks – can be considered. Each might extend the market for debt but would do little to encourage liquidity. Encouraging retail distribution may be important from a public policy point of view, but in a Saudi context its contribution to developing a secondary market will prove negligible unless the culture of small investors changes. Even if they buy more, they are unlikely to want to start trading in bonds. Mutual funds are more likely to be traders.

Another suggestion is to compel financial institutions to invest in GDBs. It is possible that some sort of secondary market would spring up, as banks could offload their holdings, but that would hardly be desirable. Saudi Arabia has to date refrained from imposing investment directives on domestic financial institutions, because it is committed to the concept of the free market. Since compulsion did not happen during the first debt episode, it is unlikely to happen now.

The third option is for SAMA to extend its repurchase (repo) facility beyond the banks to public sector entities even when their bonds are not guaranteed by the government. The quasi-government and partly government-owned entities could be offered the same opportunity as the banks to raise money from the central bank against the security of their government bond holdings. Making GDBs more attractive to them would hopefully reduce the cost of borrowing. Also the repo facility could be extended to some other kinds of debt, especially that issued by corporations with large government shareholdings. This could encourage them to issue bonds.

But the disadvantage of this lop-sided repo facility for government-linked entities is clear. Extending the repo facility across the board would inhibit the development of an inter-bank repo market and would make it less likely than ever that a secondary bond market would develop. On balance, therefore, the idea of rolling out SAMA's repo facility beyond the banks should be viewed with caution.

One obvious way to boost the demand for bonds is to use the substantial pools of money in the form of the end of service benefits to employees (ESBs), which remains untapped due to the lack of a regulatory framework. ESBs, which are the deferred liabilities of employers, are not funded. Any social security reform should take the management of ESBs into account and ensure that they are fully funded. The pool so created would be available for long-term investment and the obvious place is in government and corporate bonds. But it is not clear how this would encourage a secondary market.

A final idea is for the Public Investment Fund (PIF) to energize the corporate bond market. A useful comparison here is Malaysia, where during periods of fiscal surplus the government investment arm, Khasanah Nasional, issued bonds on a regular basis to provide a quasi-sovereign benchmark curve. PIF could issue bonds and use the proceeds to finance some projects instead of tapping into its own resources. This would give the fund additional resources for investment. However, if it did not wish to leverage its balance sheet, it could securitize its outstanding loans and use the proceeds for long-term project financing. This would provide investment banking opportunities in structuring the deals. It remains to be seen what effect the PIF's recent activism will have on the domestic bond market.

Bond market development is a collective project, which warrants coordinated efforts involving government, including the DMO the central bank, the CMA and market participants (that is to say, both issuers and investors). SAMA has a long history of talking with market participants to resolve problems. There are easily identifiable obstacles, which could be dealt with by relaxing issue restrictions.

5 How SAMA Prices Government Bonds

A common complaint made against SAMA before the recent debt episode began was that there was no government bond yield curve that could provide a sovereign (in the jargon, 'risk free') benchmark in order to price bonds or sukuk efficiently. But this argument ignored the fact that in Saudi Arabia the currency peg leads to low and predictable spreads between riyal and dollar yields. The absence of a benchmark yield curve is not an impediment to issuing corporate bonds or sukuk, which can either be priced using US Treasury or swap yields, or linked to Saudi Interbank Offered Rate (SAIBOR).

The pegged exchange rate means that the dollar provides an anchor for the riyal, and US Treasuries provide exactly the same function for GDBs. Ever since 1988 the central bank has used the dollar yield curve to price GDB issuance (both the conventional and the Shariah-compliant form), and the same principle should apply to corporate bonds and sukuk. An alternative way of pricing private sector debt is to use the issues of government-owned corporations such as Saudi Basic Industries Corporation (SABIC).

A more valid criticism is that SAMA will not move to allowing the market to set the price of government bonds on issue through an auction system; but this is a contentious subject. The central bank needs to be confident that the banks would not collude to price the bonds at auction. If SAMA could persuade some financial institutions to commit capital as primary dealers in government bonds, this would be a positive move. The evidence is that bond markets in emerging economies need participants who are committed to providing liquidity. Primary dealers would be under an obligation to participate in auctions (so that bonds were not left unsold) and to provide liquidity by market-making in the secondary market.

Still today the banks want an auction system and SAMA wants primary dealers. The last attempt to strike an agreement, which would have involved a market-making arrangement, floundered. The result was that the banks showed no interest in primary dealing and the market-making side never took off. This is a classic chicken-and-egg problem. It seems unlikely that a liquid secondary market will develop as long as the banks will not commit capital to primary dealerships.

It is not yet clear how and when the DMO will move to play a role in the way government bonds are priced. But it is standard practice around the world for the DMO and the central bank to work closely together. Currently SAMA has an arrangement with the banks where they are given a price or yield range for each maturity of the bonds that are going to be issued, on the basis of a yield spread against Treasuries for the fixed rate bonds and against SAIBOR for the floating rate notes. Each bank responds by quoting the rates at which it would buy the bonds, and prices are set by SAMA on the basis of this information. In effect, it is a step toward market pricing and continues the tradition of a voluntary bond market.

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CHAPTER 11

Currency Regime and Monetary Policy

1 The Currency Peg

Before 1927 it could be argued that the 'dollar' was used in the Arabian Peninsula – but not the dollar we associate with the United States. It was the Maria Theresa riyal. Outside Arabia it was known mostly by its German name as the thaler, which is where the US dollar gets its name. By the 1950s, the Maria Theresa riyal was a distant memory, and Saudi Arabia was moving away from its coin-bullion standard. Pilgrim receipts started being used as a paper currency and then the first SAMA banknotes appeared. Since 1986, 3.75 riyals have bought one dollar – thanks to SAMA's commitment to trading with the banking system at this rate.

Saudi Arabia is an outlier even among oil exporters in being so highly dependent on a single commodity, with over 80 percent of exports coming from oil. If it became more diversified, then it could consider moving away from the peg and follow Norway or Chile toward a freely floating currency. But diversification of exports remains a distant prospect, even though this is the aim of 'Vision 2030,' a strategy announced in 2016 (Fig. 11.1).

The currency peg has delivered two things: a low interest rate differential between Saudi and US interest rates, and tracking of the US inflation rate. Even so, there are periods when the inflation differential is sizeable. The peg drags the Saudi inflation rate back toward the American one but only in the long run. Individual episodes of Saudi inflation and dis-inflation are generally

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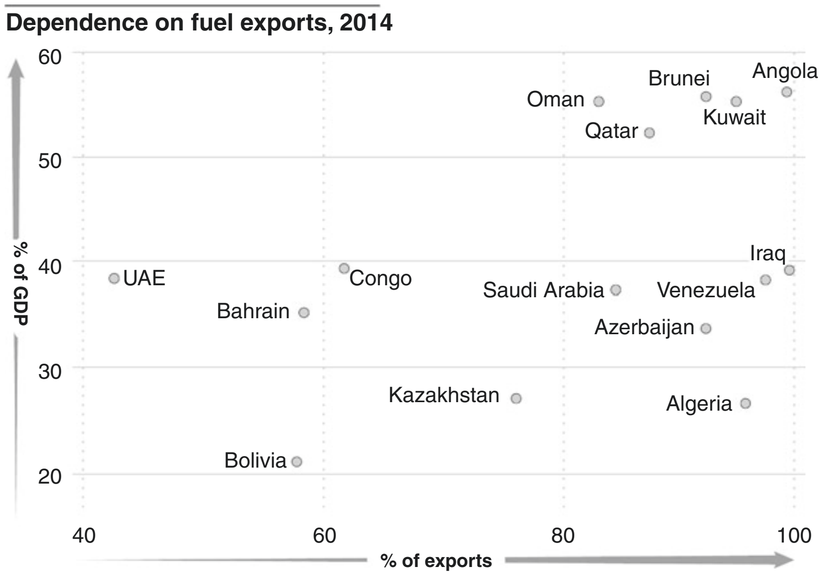

Fig. 11.1 Dependence on fuel exports, 2014

Source: World Bank WDI, IMF AREAER

driven by the domestic economy, which in turn is driven by the oil price. The 2008 episode when external factors caused inflation to rise over 6 percent was due to the high oil price which led to worldwide price rises in farming costs and so in foodstuffs ('energy in and energy out . . . ' as farmers say). And the peg means that SAMA has little discretion about where to set interest rates.

This chapter examines the riyal exchange rate regime, how it has developed and how SAMA runs monetary policy within its constraints. The central bank's mandate is to maintain internal and external price stability and it does this by keeping the value of the currency stable against the dollar. In common with central banks worldwide, SAMA faces what is known as the 'monetary trilemma' of being forced to choose to pursue only any two of three policies, namely: a fixed exchange rate; free capital movement; and an independent monetary policy. SAMA has chosen the first two and does not have the third. An independent monetary policy can only be achieved at the cost of leaving the peg. What might the first move away from a fixed peg look like? The final section discusses this aspect.

2 Currency Implications of the Oil Economy

In order to understand why Saudi Arabia has the currency peg we have to grasp the importance of oil and its dollar price to the economy. Even in 2015 as the oil price weakened, over 90 percent of Saudi exports came from oil and petrochemicals. This is not a position that one would choose to be in. A diversified economy is much more stable. Despite strenuous efforts to diversify, the Saudi economy remains stubbornly dependent on oil. But the world looks likely to use a lot of oil as far ahead as one can see.

The country is a low-cost producer of what is the most efficient way of powering any vehicle – less than 5 US cents worth of diesel will buy the same energy as $1,800-worth of lithium-ion battery.[1] Although oil's share of world energy consumption is falling, the actual consumption of oil has gone up fairly steadily following a sharp fall during the two oil shocks of 1973–74 and 1979–80. This is mainly due to its use in emerging economies, which have fast-growing populations and are less efficient in using energy (Fig. 11.2).

The oil price trumps all other candidates when it comes to explaining economic growth in Saudi Arabia since 1970. Specifically, a 10 percent rise in the real oil price is associated with a 2.2 percent rise in real Gross Domestic Product (GDP).[2] When oil prices go up the economy is strong, and vice versa (Fig. 11.3).

But oil prices do not feed directly into Saudi economic performance: their effect is filtered by fiscal and monetary policy. The counter-cyclical role played by the government is the key on the fiscal side. Historically, government expenditures have been more stable than oil revenues so they have smoothed the impact of variable oil prices on the economy. In lean years, the budget runs a deficit. In years of plenty, it runs a surplus. The balance of payments reflects this.

The currency peg does an analogous job on the monetary side. If the riyal floated freely then it would be stronger when the oil price was strong. This would make imports cheaper so more goods and services would come into the economy. Conversely, a weak oil price would lead to a weaker riyal and imports would cost more, so imports would drop at the same time that public expenditure was weak. The Saudi economy would go through a powerful boom-bust cycle. The currency peg dampens this volatility.

Let us consider the implications of changes in the oil price with a currency peg arrangement in a simple framework. SAMA's challenge is

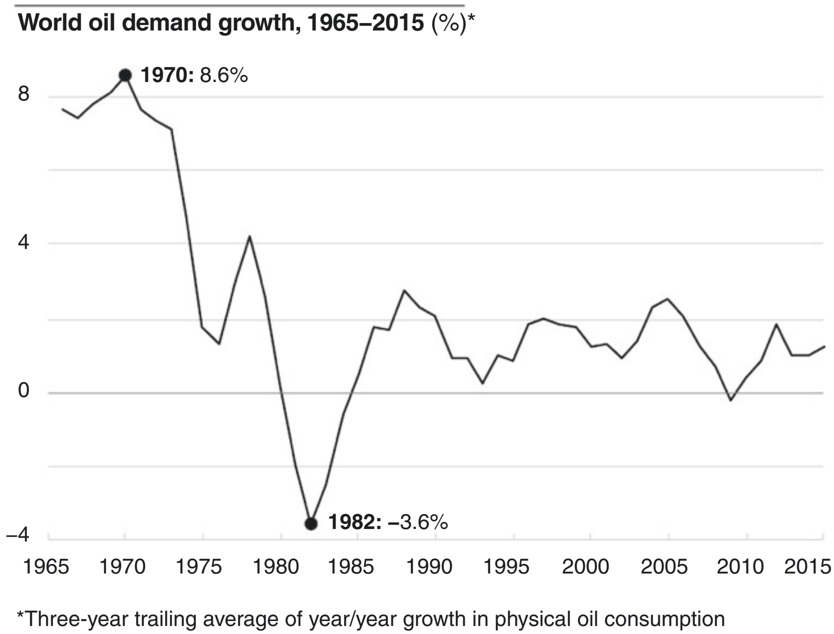

Fig. 11.2 World oil demand growth, 1965–2015

Source: BP Statistical Review 2016

to absorb export surpluses in dollars when the price is high and to pay for export deficits when prices fall. Neither is easy. Take a high oil price. This means that more riyals go into the economy as SAMA credits the government with the riyal equivalent of the dollars. Even when the government does not spend all the riyals right away (i.e., it runs a budget surplus) spending tends to rise with the oil price. Normally this is a recipe for inflation. Indeed, such episodes reliably prompt speculation that SAMA will revalue the peg and issue fewer riyals per dollar in order to reduce the inflationary pressure.

Consider next a weak oil price. Oil income falls, but import bills drop far less as the government seeks to maintain steady outlays in the interest of stabilizing the economy. Most of the kingdom's import needs remain constant because there are few domestic substitutes for goods, and most private sector services are provided by imported expatriate labor (that is to

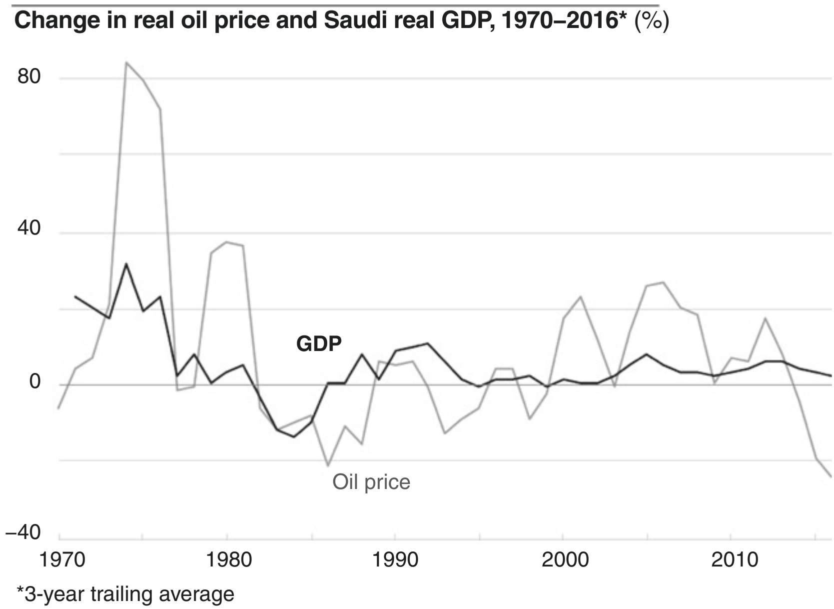

Fig. 11.3 Change in real oil price and Saudi real GDP, 1970–2016

Source: IMF, World Bank Penn World Table

say, there is a low price elasticity of demand for imports). Government responds to this by either drawing down riyals from the account it holds with SAMA or by issuing debt at home in riyals and abroad in dollars (which are credited to SAMA's foreign exchange reserves in return for the equivalent in riyals going into the government's account with the central bank). The deficit in dollars – the amount needed to pay the import bills minus the amount earned from oil and any external debt credited to SAMA's account – has to be provided by SAMA. If reserves fall far enough, this can give rise to speculation that the central bank will be forced to devalue the currency and issue more riyals per dollar to the government's account in order to make the dollar income go further in riyal terms.

This, in simple terms, is the currency problem facing SAMA – the riyal naturally has a tendency to rise when the oil price is high and to fall when it is low. But it does not necessarily follow that the problem must be solved through pegging the currency to the dollar. Other commodity-exporting economies have followed different paths. But before looking at this, it is useful to review how SAMA arrived at the hard currency peg.

3 Managing the Riyal: 1952–2016

The sixty-odd years of SAMA's history divide neatly into two periods: up to 1986 and afterward. In the first 30 years the central bank effectively shadowed the dollar, while in the last it has explicitly had a dollar peg which has not changed. Global oil contracts are denominated in dollars, so Saudi Arabia's export revenue is received in dollars. Prior to 1986, the dollar was given considerable weight in the management of the riyal, but was not always the sole consideration. In particular, the riyal was not allowed to follow a strong dollar to the top of its cycle. During the strong dollar period of 1981–86, the riyal was depreciated against it in several steps, leading to a final devaluation in 1986 (Fig. 11.4).

Riyal parity has not changed since and has ridden out subsequent gyrations in the dollar. Its international value has (in trade-weighted terms) risen and fallen along with the dollar, and for almost all that time the peg has been unquestioned. Over the years the riyal has weathered three episodes of currency speculation.[3] In 1993, speculators bet against it based on a combination of lower oil prices, widening budget deficits and falling foreign exchange reserves. They did so again in mid-1998, due to weak oil prices on the back of the Asian crisis. In both cases, SAMA responded by intervening in the forward exchange market, where speculators sold riyals. The central bank continued to provide dollars to the banks and injected liquidity through deposit placements. Foreign exchange intervention in the thinner forward market brought the forward rate back into line with the interest rate differential and inflicted losses on the speculators. The size of these unsterilized interventions was small, about $1.5 billion for the two episodes combined. SAMA made profits from both.[4]

The most recent attack occurred in 2007–08 when speculators wagered that the currency would be revalued. This assault lasted longer but was easier to manage. SAMA was already fighting inflation due to imported food price hikes and the domestic boom, and it would have raised interest rates, but this would have encouraged speculators to sell dollars and buy higher yielding riyals. Instead it raised reserve requirements, which drained system liquidity

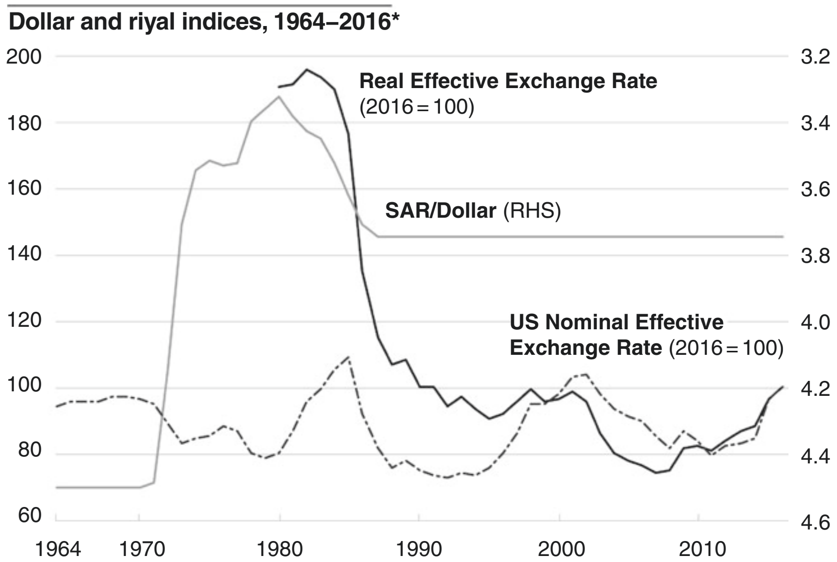

Fig. 11.4 Dollar and riyal indices, 1964–2016

Source: IMF IFS, SAMA, BIS

Notes: Up = appreciation. The real effective exchange rate (REER) is the average of all Saudi bilateral exchange rates against main trading partners, adjusted for any bilatereal difference in inflation rates. The nominal effective exchange rate (NEER) is the same for the USA, but without inflation adjusment. It is a measure of the dollar's international strength

and forced the banks to place money with SAMA on a non-interest bearing basis. SAMA's actions were successful and the riyal parity was unchanged.

In operational terms over several decades the peg has been highly successful, but despite this some commentators question its use. There are two distinct points: first, whether the peg is at the correct level today (which is really to question whether a varying or unvarying peg is better); and second, whether a peg is the best option. This analysis deals with the second point, and we will return to the point about varying the peg later. In principle there is no reason why the Saudi economy should be linked to a strong currency purely because the dollar has appreciated. In theory, a strong economy calls for a strong currency because strong economic growth generates inflationary pressures. These are easier to contain with a stronger currency: it dampens inflation in the import bill and, because imports become cheaper relative to domestic products, it diverts demand into imports and away from local output. When the economy is weak, a weaker currency can be helpful if it leads the non-oil sector to export more, thereby supporting growth and employment.

The question is how far these theoretical points apply to Saudi Arabia and the answer is – not much. While changes to the exchange rate may affect the price level, they would have almost no impact on output and employment. Demand for oil is price-inelastic and the country lacks a large non-oil sector to produce goods and services that can be sold abroad if they are cheap. Unlike emerging Asia, for example, Saudi Arabia does not make products such as textiles, where export demand is highly sensitive to changes in the cost of production. As for services, the country imports labor to run the private sector. The result is that changes in the exchange rate are unlikely to affect output and employment in either the oil or the non-oil economy. A weaker currency leads to higher import bills for those products that are not price sensitive. In other words, there is no price elasticity of demand for exports, and poor price elasticity of demand for imports. It can also leave Saudi borrowers insolvent if they have big exposure to foreign currency liabilities matched against domestic currency assets.

But the critics do have a valid point. The chances of getting the exchange rate and economic circumstances moving in the same direction are smaller if you are pegged to someone else's currency. Over the past decade those voices arguing that the currency peg's problems are growing have become more persistent, because, as Fig. 11.5 shows, the oil price and the dollar have moved in opposite directions: first the Saudi economy was strong while the dollar was weak and then more recently the economy has been weak and the dollar has been strong. This is not entirely a coincidence. A weak dollar is consistent with a higher oil price, insofar as more dollars are available globally to purchase oil. But rarely have the two series moved in such opposite directions over such an extended period. The strong-oil, weak-dollar combination created buoyant conditions for oil exporters like Saudi Arabia and it is arguable that without the global financial crisis that led to the dollar stabilizing in 2008, SAMA might have been forced to revalue. Since 2014 the danger has been the opposite one: of deflation as the strong dollar makes imports cheaper at the same time as the domestic dividend from oil is smaller. The point is that by having a currency that is pegged to the dollar, SAMA is forced to follow US monetary policy; which might suit Saudi Arabia's circumstances in some years, but not in others.

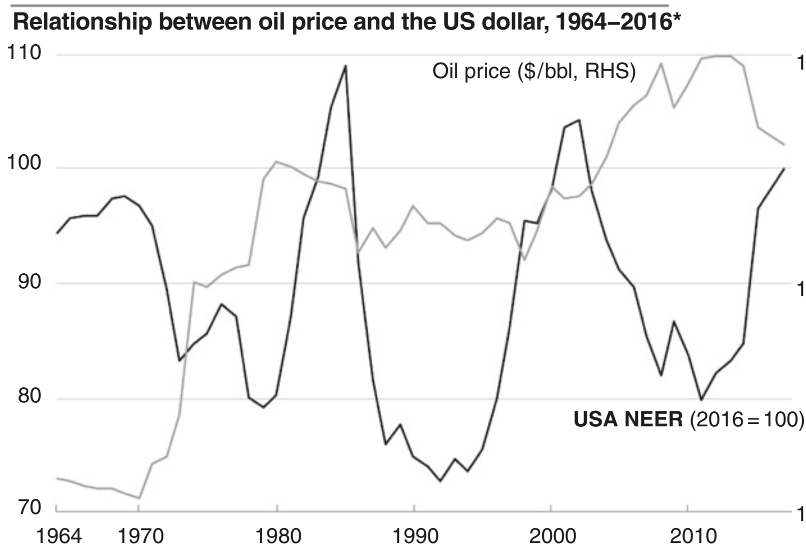

Fig. 11.5 Relationship between oil price and the US dollar, 1964–2016

Notes: *Right-Hand Scale (Oil Price is Logarithmic)

Source: BIS, World Bank

4 Operation of Monetary Policy

As then-Vice Governor Al-Jasser and Ahmed Banafe wrote in 2008:

> Saudi Arabia's monetary policy framework is firmly wedded to its fixed exchange rate policy. The operational target [of SAMA] is to manage system liquidity through the repo window and its intermediate target is stability of the riyal against the dollar, which is the anchor and intervention currency.[5]

Over time SAMA has built up a formidable armory to defend the peg within a financial system dominated by a small number of commercial banks. It is skilled at identifying and attacking speculators' positions. Every riyal in circulation is explicitly backed by foreign exchange[6] and, under Article Six of the Currency Law, currency issued cannot exceed foreign assets. SAMA pays more attention to the indicators for the financial economy, such as the exchange rate, swap points and interest rate spreads, credit growth and the stock market, than to those for the real economy. The reason is that the real economy is hugely influenced by the government's fiscal policy. The role of interest rate moves is relatively small because of the under-developed financial system and a private sector where borrowing levels are low (although there is some evidence that the sensitivity of the non-oil economy to availability of credit is increasing).

In a pegged regime like Saudi Arabia's, with an open capital account, other monetary levers are needed to manage liquidity. For example, when the balance of payments is unusually buoyant – perhaps due to a strong rise in export receipts, or possibly a rise in the oil price – it will be accompanied by a rise in domestic liquidity, since each dollar of export revenue that is spent domestically produces a rise in domestic currency. This surge in the money supply is often inflationary and the central bank can use several instruments in response. In particular, it can raise the repo rate to make it more expensive for the banks to acquire cash from the central bank or it can raise the required reserves of the banking system to reduce market liquidity. On a regular basis and separate from the debt issuance program renewed in 2015, the central bank issues its own short-term bills to mop up liquidity and has made noticeable changes in the volumes sold to either absorb or provide liquidity. There are prudential tools as well, in particular adjustment to the banks' loan-deposit ratio (LDR) and placing deposits with the banks to counteract a squeeze on system liquidity.

5 Choosing a Pegged Exchange Rate Regime

Critics of the peg also make a more specific point – that the actual level of the peg (as opposed to the concept) should be adjusted from time to time. The short answer is that by maintaining the peg at 3.75 riyals to the dollar since 1986, SAMA has earned a high degree of credibility and this, in turn, serves to reduce speculative pressures through a virtuous circle. But no exchange rate regime is perfect. Different oil exporters have adopted different measures to meet this challenge. The obvious step is indeed to change the exchange rate from time to time. However, since the exchange rate is held constant in a peg to the dollar, this would represent a significant departure from current practice. It would mean either abandoning the level of the peg or changing the exchange rate regime itself. This is in fact what Kuwait did in 2007, when it abandoned its peg to the US dollar and, instead, linked the Kuwaiti dinar to a basket of currencies, thus gaining the ability to allow an appreciation in the dinar against the dollar in a bid to soften inflationary pressures.

Instead of an exchange rate solution, Saudi Arabia took direct steps to ameliorate the worst effects of the buoyancy. For example, it introduced some food subsidies to help consumers deal with a rising food price basket. The logic was that the inflationary pressures would abate, whereas the damage a revaluation would inflict on the riyal's credibility would be long-lasting. This was wise, because it turns out that credibility is the key to the longevity of a pegged exchange rate.

Like any other asset price, the exchange rate is subject to speculation. Here the concept of credibility enters. Speculation is driven by the interplay of macroeconomic shocks and macroeconomic policy. If policy is inconsistent with the shock, then the exchange rate is in danger and it makes sense to speculate on how it might change. For example, imagine a situation in a theoretical economy where the exchange rate floats freely and the government stops the central bank from raising interest rates, even though inflation is bubbling up from some domestic or external source. This is because the government does not want a recession, which might result from higher interest rates. If the central bank does not hike rates, then inflation rises. This may lead to a falling currency because exports become less competitive and the current account deteriorates. Speculators who have bet against the currency by selling it and buying dollars instead can close their positions and take their profits.

Speculators can make even bigger profits by breaking a currency regime like Saudi Arabia's where no changes are allowed. The reason is that a pegged exchange rate allows a wider divergence between the market-equilibrium exchange rate and the actual pledged exchange rate. If the speculator succeeds in breaking the peg, the rewards are potentially great – the larger the divergence, the greater the reward.[7] The greater the credibility of a currency, the less likely it is to attract speculation. Hence the importance to Saudi Arabia of taking other steps to mitigate inflation, in order to avoid adjusting the peg.

Other oil exporters outside the Gulf Cooperation Council (GCC) have currency pegs as well but they tend to break down from time to time when the oil price falls. So the speculators have learned that the central bank lacks credibility and that a determined attack on the currency is likely to pay off – which makes the peg more liable to break.

Exchange rate regimes can be categorized by the amount of autonomy that a central bank has to alter the exchange rate. Indeed, by combining the degree to which market pressures affect the exchange rate most of the time and the extent to which the exchange rate system allows a central bank to act independently, it is possible to plot exchange rate regimes diagrammatically on two dimensions, as is done in Figure 11.6.

The Nigerian currency, the naira was until mid-2016 pegged with high central bank discretion to change the rate, but it was not a credible peg. The Saudi peg, by contrast, benefits from greater market confidence because it allows no central bank discretion. In the case of the free-floating Norwegian currency, the markets set the rate. The test of a successful peg is whether the exchange rate is stable over a number of decades. In this sense, the GCC is in a league of its own among oil producers (Fig. 11.7).

Most oil exporters pegged their currency before the latest oil price decline – with varying degrees of credibility. Some, like Ecuador, have gone so far as to adopt the dollar itself. Brunei has the strongest form of

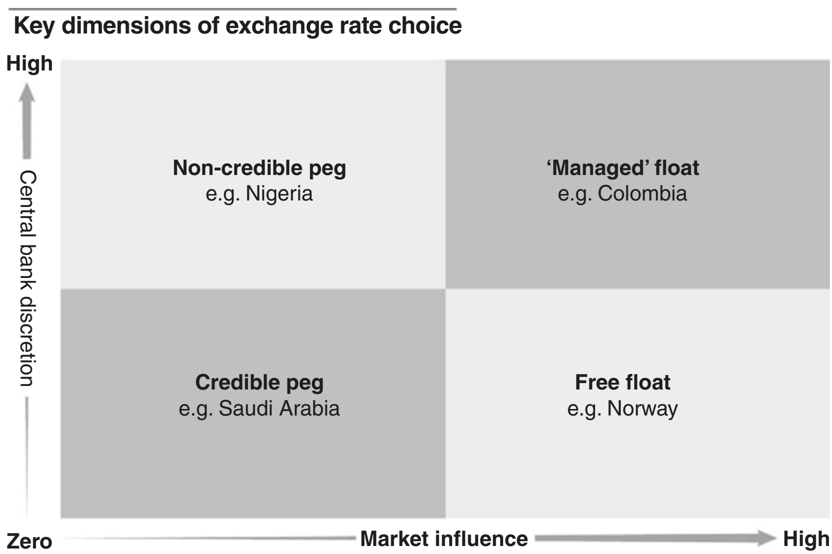

Fig. 11.6 Key dimensions of exchange rate choice

Source: Authors

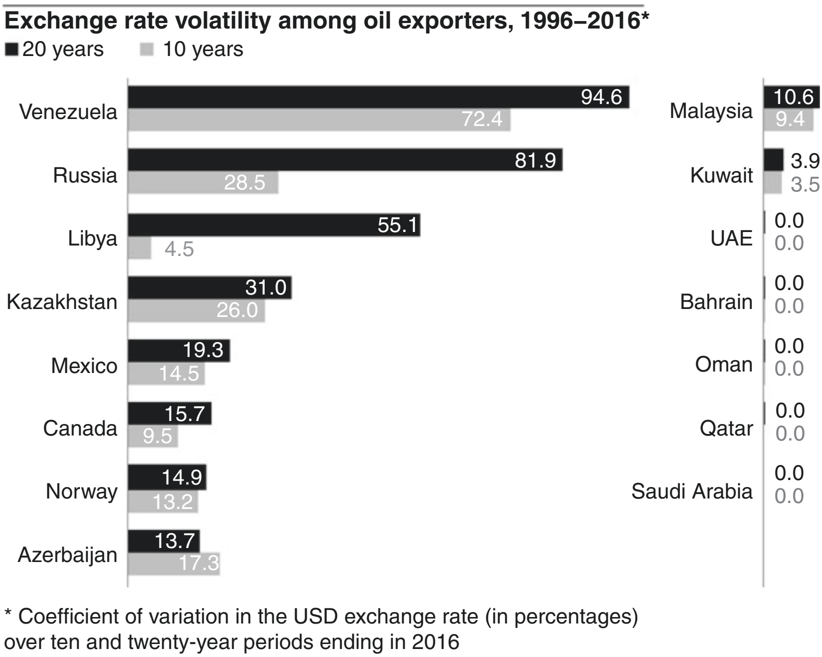

Fig. 11.7 Exchange rate volatility among oil exporters, 1996–2016

Source: Authors

peg (a currency board). The currency pegs of the GCC members are credible because of their longevity. By contrast, a handful of other middle-income oil exporters such as Mexico, Venezuela and Russia joined Nigeria in having unreliable currency pegs (but some were forced off them when oil prices fell after 2014).

For many emerging markets the decision to float the currency or accept that the peg is unreliable is not a free choice between policies – it represents a tacit acknowledgment that their foreign currency reserves are unlikely ever to be large enough to defend a peg against a potentially well-resourced currency attack. This is obviously not the case for Saudi Arabia. The combination of buoyant export earnings from the decade up to 2014 and a series of budget surpluses means that – even after several years of low oil prices – the kingdom's reserves are at a level where the riyal is for the time being immune from speculative attacks.

6 Alternative Currency Regimes to the Peg: Norway and Chile

A key feature of both Norway and Chile is that their commodities are much less important to them than oil is to Saudi Arabia. Norway is a diverse economy where oil income accounts for only 25 percent of government revenues while for Chilean copper the comparable figure is less than 30 percent. It is not a feasible option for Saudi Arabia to adopt their approaches at present, but there has been discussion of a Saudi version of their fiscal rules, so a short discussion of their exchange rate regimes is useful.

Norway is an oil exporter that does not peg and it can be joined by Chile where copper dominates export earnings. Norway runs a 'free float', a currency regime where the authorities do not buy or sell foreign exchange and the exchange rate is determined entirely by market forces. Anything else entails the central bank buying or selling in the foreign exchange market, whether infrequently (known as a 'dirty' or 'managed' float) or so extensively that the exchange rate is very nearly pegged. Chile is an instance of a managed float.

These countries can manage without relying on a peg because they have integrated their currency and budgetary policy by the application of a fiscal rule. In general terms, a fiscal rule sets a numerical target for what can be spent in the national budget. Applied in this context it means fixing the amount of money that can be transferred from commodity earnings to the government budget, according to an agreed assumption about the price of the commodity. When prices are high, the excess earnings go into a foreign currency fund. When prices are low, the fund is depleted to ensure that the rate of transfer to the government is maintained. In addition to the benefit of having a predictable stream of revenue into the government budget, the virtue of this arrangement is that it separates commodity prices from the exchange rate. In other words, a consistent amount of dollars is always converted into national currency, no matter what the commodity price. This means that the floating exchange rate will simply be determined by other forces acting on the balance of payments – primarily flows of investment and trade (other than the key commodity export which is subject to the fiscal rule).

The Norwegian government has limited access to oil income which goes into a sovereign wealth fund: it can, however, spend the assumed 4 percent return on the fund (proposed to be lowered to 3 percent in early 2017) and is allowed to apply this over a number of years. Norway's currency, the kroner, floats on the foreign exchange market, so the central bank is free to adjust the

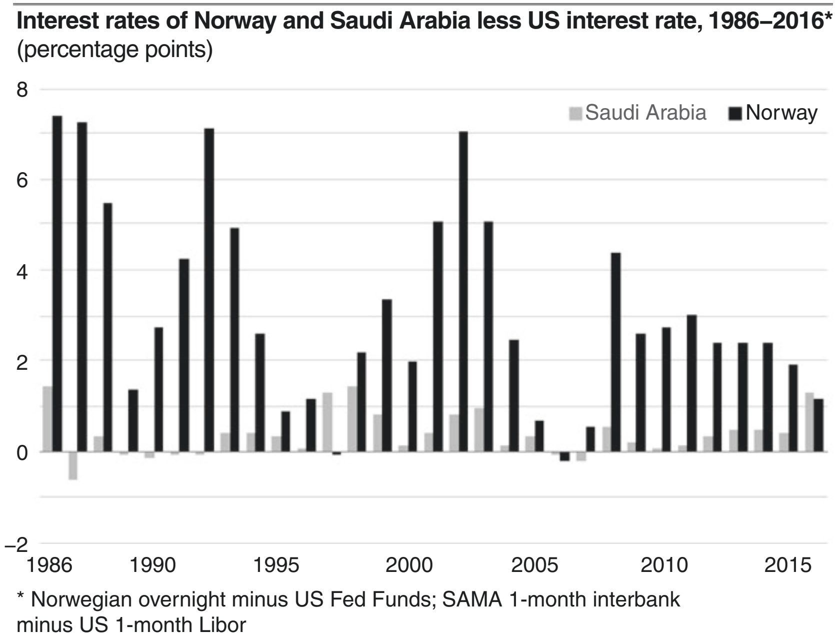

Fig. 11.8 Interest rates of Norway and Saudi Arabia less US interest rate, 1986–2016

Source: SAMA, Norges Bank, US Federal Reserve

interest rate to suit the economy's needs. Over the 20-year period ending in 2014 which covers a variety of oil price scenarios, kroner rates were very different to dollar rates for long periods and Norwegian short-term interest rate changes matched those of the United States for only 62 percent of the time; Saudi Arabia, by contrast, followed US rates nearly 100 percent of the time. This is a stark reminder of how little policy independence SAMA has (Figs. 11.8 and 11.9).

The floating regime in Chile was severely tested by the 2008–09 international financial crisis, and the exchange rate adjusted quickly, helping to stabilize the economy in the face of financial shock. It is worth contrasting Chile's fiscal arrangements with those of Saudi Arabia. Both countries' policies are counter-cyclical, but where Saudi policy is pragmatic and based on an estimated price for oil over the budget year, Chile's fiscal arrangement is less flexible: the budget is

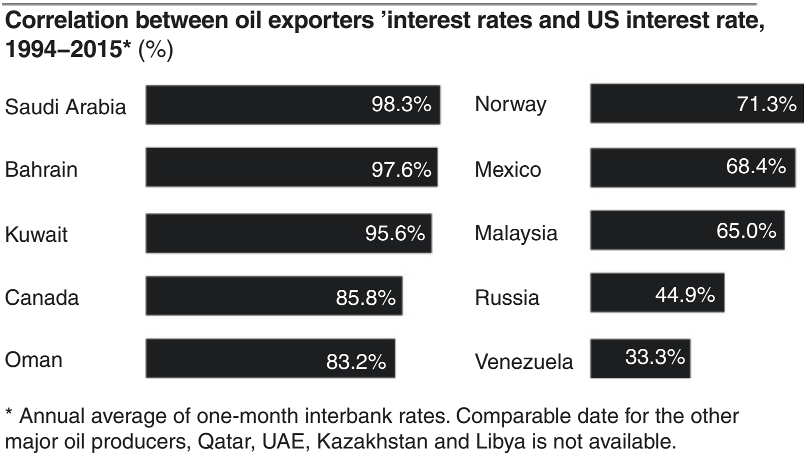

Fig. 11.9 Correlation between oil exporters' interest rates and US interest rate, 1994–2015

Source: Work Bank, US Federal Reserve, SAMA, Norges Bank

built around an assumed long-term price for copper. When the price is higher, the excess earnings go into a windfall fund (actually two funds – a stabilization fund and a pension reserve fund). When the price is lower, the budget is in deficit and draws on the stabilization fund.

Forecasts of the copper price are bound to be incorrect (or at least, correct only by happenstance) as copper too is subject to the random walk of commodity prices. Chile does its best, using an independent committee of experts. Because the price of copper has soared in recent years, the estimates turned out to be conservative. It remains to be seen how Chile will cope with copper prices that consistently fall below the long-term forecast. Copper peaked in 2011, and a 2013 review of the first ten years of this system concluded that while the predictions offered a better solution than those provided by alternative price stabilization mechanisms: 'in practice, the long-term copper price estimates vary substantially, since experts' projections tend to be influenced by current price levels.'[8]

Because Chile's earnings are in dollars, and the excess is diverted to a foreign-currency-denominated stabilization fund, the consequences for the foreign exchange market are benign. In other words, the Chilean peso is not subjected to the vagaries of the copper price. Copper-based inflows into the peso are predictable – and are consistent with the amount needed to fund the fiscal budget. This is approximately the same system that Norway uses. Partly as a result of this rule, the peso floats freely in the foreign-exchange market.

The central bank of Chile can focus almost exclusively on domestic conditions and it targets the inflation rate. There is no need to run down reserves to protect the currency. If there is downward pressure on the peso, it can fall. Crucially, the floating peso also leads to financial prudence on the part of domestic borrowers. They prefer to borrow in pesos since dollar borrowings run an exchange rate risk. And when they do so, they have an incentive to hedge their currency exposure. By contrast, the Saudi pegged regime is one in which the borrower is protected from the exchange rate by SAMA which underwrites any exchange rate risk itself.

7 A First Step Away From the Peg?

The Chinese sage Lao Tzu said that a journey of a thousand miles begins beneath one's feet. Reframing the currency discussion in terms of a journey instead of a destination makes sense. If we accept that the peg will not last forever, and that action is best taken while its reputation remains untarnished, what is the first step SAMA should take away from it? It may be helpful at this stage to review four key factors:

- First, the disappointment of diversification to date has put the country in a vulnerable position. Saudi Arabia cannot be sure how long the oil price will take to recover. It clearly makes sense to acquire more currency flexibility from a position of strength when the foreign reserves are high.
- Second, the very fact that the peg is so credible to the markets results in regular episodes of excess liquidity in the domestic market since the banks view the dollar and the riyal as interchangeable.
- Third, while the peg encourages foreign direct investment, it also means that domestic banks and businesses have grown comfortable with their assumption that there will never be a currency change and have consequently limited expertise in currency hedging.
- Finally, the peg to the dollar makes it tough for monetary policy to have much impact on inflation or deflation. As the financial system

> deepens – and the ratio of credit to non-oil GDP is rising steadily – interest rates will have a bigger role to play, especially as a means to cap any inflation that is driven by domestic demand, or stimulate prices when deflation results from cuts in government spending. Currently the central bank does not have the ability to deal effectively with this aside from using tools such as the loan to value ratio for mortgage lending and cash reserve ratios.

In this context, it is worth considering whether (since the Norwegian and Chilean paths are not practical at the moment) there might be a way for SAMA to retain the benefits of the peg while gaining some ability to control imported inflation? What would the problems be? As an illustration, Singapore's arrangements are set out below.

Singapore is a well-documented example, although its economy is very different to Saudi Arabia's. Singapore is a city-state with no natural resources and hosts a large offshore banking sector with assets far greater than those within the domestic banking system. Nonetheless, from the viewpoint of monetary policy, there are some strong similarities. Like the kingdom, Singapore relies heavily on imports, so inflation is strongly linked to the level of the exchange rate; and, like SAMA, the MAS (Monetary Authority of Singapore) can control the exchange rate directly through intervention in the money markets.

The ultimate target of the MAS's monetary policy is to preserve the purchasing power of the Singapore dollar but – again like SAMA – it has no numerical inflation target. The exchange rate is the intermediate target. This is similar to SAMA's objectives as stipulated in its charter, which are to stabilize the internal and external value of the currency. But where SAMA has targeted an exchange rate, the MAS uses what is known as a basket, band and crawl (BBC) system. This is similar to the system operated by Kuwait. The Singapore dollar tracks the performance of an undisclosed basket of currencies of the nation's major trade partners and is allowed to trade within an undisclosed band. This enables the MAS to apply a degree of constructive ambiguity to exchange rate management. Although the currency is not pegged, the MAS steers the exchange rate against the basket in a slow and predictable manner – this is the 'crawl.' The result is that the MAS can control the exchange rate without the markets being able to anticipate its actions, thus avoiding the problem that SAMA faced in the 1970s when it targeted the Special Drawing Right

(SDR) using a fixed band, only to find that the commercial banks were anticipating SAMA's moves and betting successfully against it.

A BBC arrangement gives the MAS a freer hand to deal with currency speculators because it can alter the daily rate at which it sells dollars to the banks, underpinning the currency if speculators short it, or holding the currency steady if they bet on a revaluation. As well as discouraging currency speculation, a BBC arrangement can be used to move the currency downwards when export prices are low, encouraging import substitution. When export prices rise, potentially a precursor of inflation, the central bank can adjust the crawl so the currency goes up, thus reducing imported inflation.

Over time a BBC arrangement makes it more difficult to anchor domestic interest rates to dollar rates. A similar story can be told on the currency side – under the hard currency peg SAMA only has to intervene occasionally, but the MAS monitors the currency continuously and can intervene before the band is breached.

In conclusion, Saudi Arabian policy-makers should explore alternative exchange rate regimes to fit in with the country's progress in diversifying its economy. In this respect, the MAS arrangements are one of the options worth studying to see if they are suitable or not. Actually saying goodbye to a credible peg is never a step to be taken lightly.

Notes

1. Calculations based in July 2016 on 1.75 dollars per liter of diesel, 500 dollars per kilowatt-hour of lithium-ion battery, 36 mega joules per liter of petrol and 0.277778 kilowatt-hours per mega joule.
2. This is allowing for the usual correlates of GDP growth, such as growth rates of population, education, the capital stock and overall productivity. OLS regression of Saudi real GDP on real oil price, population, human capital index (mainly comprised of educational attainment at age 25), the real capital stock and total factor productivity index, all in log-differences. R-squared is 0.640, *N*=41. Analysis by authors from data in the Penn World Tables version 8. University of Toronto, 'Penn World Tables,' University of Toronto. http://datacentre.chass.utoronto.ca/pwt/ (Accessed October 30, 2016)
3. There were also brief panics when Iraq invaded Kuwait and during the depths of the global financial crisis.
4. Unsterilized intervention in the foreign currency market is where the central bank does not attempt to offset the domestic money market impact of its actions. In a sterilized intervention a central bank which sells foreign exchange in the spot market (to increase the value of its own currency

when it is under pressure to devalue) and then buys domestic assets (such as central bank bills) has sought to neutralize (or 'sterilize') the balance-sheet impact of the initial intervention. The initial intervention withdraws domestic liquidity as the central bank buys local currency from the banks, and the monetary action injects liquidity back as the central bank buys bills in return for crediting the banks with local currency. SAMA's interventions were mostly in the forward market and so they remained unsterilized.

5. Muhammad Al-Jasser and Ahmed Banafe, 'Monetary Policy Transmission in Saudi Arabia,' BIS Papers 35 (2008):1.
6. Going back to Arthur Young's plan of 1952.
7. George Soros' successful attack on the British pound sterling, driving it off its shadowing of the Deutsche Mark in 1992, was a legendary 'short' of a quasi-pegged currency. The opposite maneuver – going 'long' with the currency and 'shorting' the dollar – will yield a profit if the currency is revalued upwards. This was what speculators against SAMA were trying to do in 2007–08.
8. Mario Marcel. 'The structural balance rule in Chile: Ten years, ten lessons,' *IADB Discussion Paper 289* (2013): 44.

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CHAPTER 12

SAMA and the International Monetary System

1 Great Bitter Lake

In February 1945, Ibn Saud met President Franklin Roosevelt for a four-hour discussion at the Great Bitter Lake in Egypt – the first time Ibn Saud had left his country. As the USS *Murphy* steamed from Jeddah toward the rendezvous, the king sat on his gilt throne facing the destroyer's bow. He prayed five times daily facing Makkah in the direction indicated by the ship's compass, and passed the nights on deck. He watched his first movie, a documentary about the Navy, while sipping coffee brewed on charcoal stoves, located perilously close to the ship's ammunition room. At meal-times, the king's cook slaughtered and roasted one of the sheep he had brought aboard with him. After a journey of two nights Ibn Saud boarded the President's warship, which had brought an ailing Roosevelt direct from the Allied summit at Yalta. Over the next few hours the king and the president talked. The outcome was an informal agreement under which the kingdom allowed the United States free access to its oilfields in exchange for security.[1]

Roosevelt needed this. America's oilfields had been seriously depleted by the effort of fighting World War Two. The president could foresee the time when his country and its allies would need the oil Aramco had discovered in eastern Saudi Arabia, and he wanted to befriend the king.

A. Banafe, R. Macleod, *The Saudi Arabian Monetary Agency, 1952–2016*, Financial Institutions, Reforms, and Policies in Muslim Countries, DOI 10.1007/978-3-319-55218-7_12

So did the British. Before sailing home on a British warship Ibn Saud met Winston Churchill in Cairo. The king found the Royal Navy's food unpalatable and their officers dull. As for Churchill, he complained, the prime minister had boasted about drinking alcohol and blown cigar smoke in his face.

The Great Bitter Lake meeting was the moment when America decisively eclipsed Britain in controlling the oil in the Gulf. In the decades that followed, Saudi Arabia would assume a prominence on the world economic stage far beyond anything Ibn Saud could possibly have imagined as he sat enthroned near the prow of the *Murphy*. In the aftermath of the 1974 price hike, during the reign of King Faysal, earnings from Saudi oil exports would provide liquidity to the international monetary system as they were recycled into petrodollars. SAMA would make the biggest loan in history to the International Monetary Fund (IMF) and in the oil bull market of the new millennium become a significant provider of finance to capital markets.

Today Saudi Arabia plays a vital part in the international monetary system that dates from the Bretton Woods agreement signed the year before Roosevelt and Ibn Saud met. But what of the years to come? How well positioned is the kingdom to deal with the challenges of the international money markets of the future? Imagine a Saudi Arabia that gets paid in Chinese currency for its oil and invests in China. What about a Saudi Arabia that prices oil in euros, or in a common Gulf currency? Against this background it is worth examining how the international monetary system has developed and what the kingdom's position in any new configuration of the system might be.

2 The Twin Roles of the Dollar

The foreign exchange market is by far the largest on the planet. At the last count, it turned over the equivalent of about $5.3 trillion each day – over twenty times the daily output of the global economy. But a monetary system is much more than just the sum total of vast quantities of financial transactions. It has to facilitate orderly transactions, make adjustments to any imbalances and provide access to liquidity to smooth over abrupt short-term shocks. It should have a numeraire or central currency in which traded goods – especially commodities – can be priced, and finally, it must facilitate capital markets.

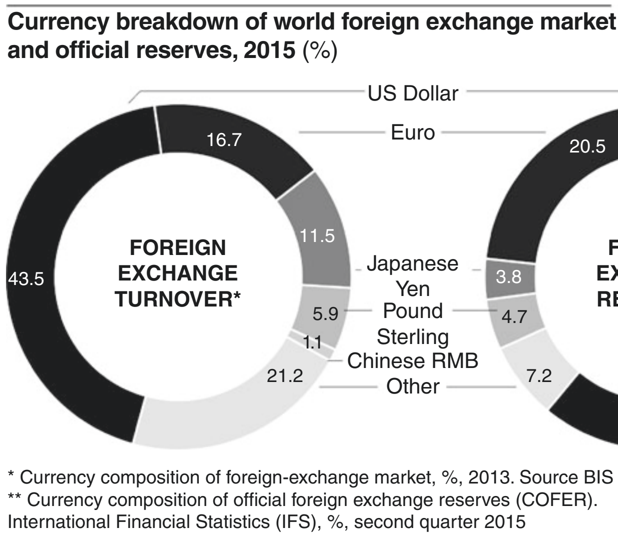

Fig. 12.1 Currency breakdown of world foreign exchange market and official reserves, 2015

Notes:
(1) Because foreign exchange market data involves two cuurrencies the BIS gives the data as adding to 200%: it has been reduced to 100% to be comparable with the COFER data. **(2)** The allocated share of foreign exchange reserves is only 58.2% of the total because many central banks (including SAMA) do not give currency details. **(3)** Totals may not add up due to rounding.

Figure 12.1 provides a snapshot of the dollar's significance, both as a traded currency and as a store of value in the reserves of central banks around the world. Interestingly the dollar is more dominant as a store of value than as a traded currency.

The dollar's roles are closely connected: for example, if the adjustment process to global imbalances damages liquidity, world prices will fall, creating a new set of problems. At the same time, global investment is critical for Saudi Arabia, which uses its foreign exchange reserves to support the countercyclical fiscal policy that stabilizes the economy. In addition, the numeraire – currently the dollar – must support rapidly growing cross-border asset flows in a world where opportunities for investment outside the United States are huge.

Any monetary system needs to include a mechanism to enable countries to adjust to persistent trade imbalances. Where there is consistently greater demand for foreign, rather than local currency, something has to give way. In the absence of some other mechanism this is the exchange rate. People who need foreign currency bid up its price so that greater quantities of local currency are needed to buy each unit of foreign currency. The primary advantage of this system is its simplicity – like daylight savings time the adjustment happens at a stroke.

Problems arise because the surplus country then needs to adjust wages and prices as well, leading to inflation. Although the central bank can prevent any inflationary consequence by 'sterilizing' the accumulation of foreign assets, inflation will still occur when there is no more spare capacity. The surplus economy then forces all of the adjustment onto the deficit economy, as we have seen in the euro area in recent times, with deflation in the south not matched by inflation in northern surplus members.

The Bretton Woods system, which underpinned the international monetary system until its collapse in the early 1970s, attempted to solve this problem by incorporating many of the advantages of fixed exchange rates without the defects of previous fixed-rate systems. The issue of a central currency was solved when the dollar was pegged to gold and other currencies were pegged to the dollar. But Keynes' idea of forcing a surplus economy to adjust as well was not agreed.

Although the rules drawn up at Bretton Woods worked well for many years, they never solved the problem of how to facilitate private capital flows – indeed, they tried to prevent them. Bretton Woods would never have coped with recycling the massive dollar flows that came to SAMA as a result of the two oil booms of the 1970s and early 1980s. But by then the system had already fallen apart after the dollar was devalued, the link with gold ended and currencies were allowed to float freely.

The Bretton Woods experience illustrates the concept of the monetary trilemma or 'impossible trinity,' the macroeconomic dictum that policy-makers can choose only two of the following three priorities: a fixed exchange rate; open financial flows; and policy independence. Bretton Woods jettisoned the second of these in order to enjoy the first and third. In the days of the international gold standard, by contrast, financial mobility was expressly embraced (indeed required); and policy independence was jettisoned. Saudi Arabia's dollar link is a version of this.

In today's international monetary system, capital movements and policy independence are central, while the system has abandoned the fixed exchange rate. But one link with Bretton Woods survives – the central role of the dollar.

The global market needs a standard unit of measurement, or numeraire, in which to quote and compare prices. The numeraire currency – today, the dollar – is the most important one, and central banks prefer to hold the majority of their reserves in dollars. Numeraire status is invaluable, since it means that other countries have to accept American promises to pay as the best promises in the system. To pre-empt a general tendency toward deflation, the world needs a growing supply of dollars to keep pace with the annual growth in global trade. If, however, the supply of dollars grows more slowly or indeed not at all, then there will be fewer dollars available per unit of world output – in a word, deflation. It follows that there is a tension between the dollar as a measurement unit and as a store of value. If more dollars are needed to finance world trade at stable prices, they may be too many for the dollar to play its store of value role. The increasing quantity of dollars necessitated by globalization will ultimately undermine its value if it grows faster than the American economy, since its status is underpinned by its exchangeability into American goods and services.

The slow decline in the dollar is a consequence of its global role, but institutions like SAMA can and do compensate by diversifying into other currencies such as the euro, yen and renminbi in an attempt to preserve the real value of their assets, while still holding enough US currency to meet payment for dollar- invoiced imports.

3 Capital Flows and Petrodollar Recycling: 1974–88

In 1974, the international monetary system and SAMA's role in it changed profoundly. Two new groups of players – oil exporters and developing economies – entered the scene and capital flows became more important as the former group financed the latter. During the decades that followed Bretton Woods, the prosperity of the developed economies had been underpinned by cheap oil. Western oil companies set the price in the numeraire of the system, the dollar, which was losing value thanks to US inflation. In 1969 a barrel of oil cost the equivalent of $10 (in 2010 terms) compared to $16 in 1952. But by January 1974, the price had soared to

$63 in 2010 prices, thanks to concerted action by the Organization of the Petroleum Exporting Countries (OPEC) cartel.

The price hike meant that, as with other oil exporters, the local Saudi economy was unable to absorb most of the increase in oil receipts. This had to be either given away or saved, and thus was born the petrodollar. Under King Faysal the Saudi surplus went into three places: part was donated as foreign aid; part was allocated to international financial institutions, such as the IMF; and the rest (by far the largest part) was invested in dollars. Because global capital markets were still relatively primitive, this meant in practice that the money was placed into bank deposits, leading to a spectacular increase in commercial bank lending to the middle-income countries.

The main borrowers were developing economies. The industrialized world was reeling from a recession, exacerbated but not caused by the 1974 price shock, and corporate borrowers had little appetite for additional debt (Fig. 12.2).

The first episode of petrodollar recycling constituted a massive change in international financial flows and it was initially feared that the system

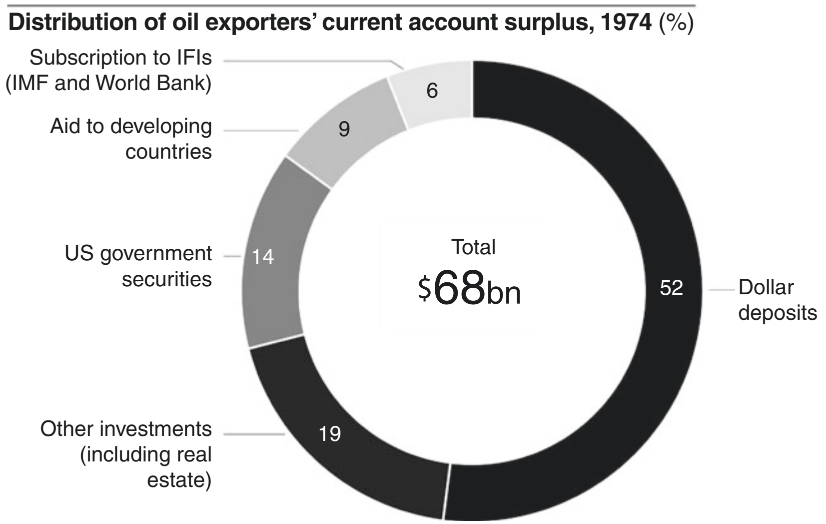

Fig. 12.2 Distribution of oil exporters' current account surplus, 1974

Source: ADB 1981

could not cope. But in the event, it caused no major disruption. By 1979, the OPEC surplus had virtually disappeared, as the oil exporters undertook massive investment programs, such as Saudi Arabia's giant Five Year Plan.

The Iranian Revolution of that year, when King Khalid was on the Saudi throne, released a frenzy of bidding for spot-market contracts. In 2010 terms, oil prices, which had fallen back to $42 a barrel from their 1974 high, more than doubled to $106.[2] Petrodollar loans again boomed as a result. But this time they were used to roll over debt that could no longer be serviced and to finance higher oil import bills. Another crucial development was the change in US monetary policy. After more than half a decade of inflation, by 1980 the US Federal Reserve was in inflation-fighting mode, together with other developed economies. Real interest rates on the commercial loans of developing countries shifted from negative to positive as dollar interest rates were hiked, while American demand for imports fell as the American economy went into recession. Saudi Arabia took advantage of high interest rates to invest in longer-dated bonds and equities. But this was a crushing burden for the developing economies, which had borrowed at variable rates. They were caught between the scissors of paying more dollars on their debt and getting fewer dollars for their exports.

By 1982, the year King Fahd came to the throne, the oil price had fallen and the petrodollar bonanza was over. OPEC members became net borrowers. The international banks, with a much diminished source of funds, could no longer roll over existing loans and the same year Mexico became the first of a number of Latin American countries to default.

SAMA has always been a strong supporter of the IMF. It joined in 1957 with what was known as Article 8 status, which required currency convertibility, the compilation and transparency of economic data and openness to the IMF regarding foreign exchange reserve practices. Soon after, the kingdom hosted an IMF mission to help reorganize its finances. It was during this visit that the leader of the mission, Anwar Ali, a Pakistani national, was invited to become Governor of SAMA. During the first oil boom, the kingdom loaned nearly $3 billion to the IMF. Subsequent loans resulted in Saudi Arabia being the largest contributor to several IMF lending facilities. It also donated its share of the proceeds from the IMF's sale of its gold reserves in 1978.[3] But the kingdom's most significant contribution came in the early 1980s, when the IMF's resources had shrunk dramatically relative to the size of world trade. It needed to find a generous lender if it was to allow enlarged access by borrowers. This was made possible primarily through a loan from Saudi Arabia amounting to almost $13 billion, the biggest loan in history at that time. Saudi Arabia gained a permanent seat on the Executive Board. As one of the IMF's major lenders already, it had till then been granted only a temporary seat. While its new permanent position still left it as the member with the lowest ratio of votes to dollars contributed, it represented a major development for both the international financial system and for Saudi Arabia. In one step the kingdom's role in the international capital markets was recognized.

4 A System Dominated by Bond Markets: 1989–2016

In the 1970s, SAMA lent mainly to banks, which then lent to borrowers. In other words, it was an intermediated system, in which the banks acted as the middlemen. After 1989, the international monetary system changed and became dis-intermediated as the banks lost ground to the markets. Bank deposits still played an important role for SAMA, but increasingly the central bank moved to lending directly to borrower nations through buying their bonds. The history of the monetary system now becomes a tale of boom and bust in the bond markets.

A new type of sovereign debt crisis started when Mexico defaulted once again in 1994. This was the first of a series of capital account crises that came about because the loans, in the form of bonds, were often financed by a new phenomenon of short-term speculative flows. Whereas problems in repaying debt owed to banks could be sorted out bilaterally or multilaterally, debt to the bond markets involved a myriad of private players. As a result, stresses – or fears of them – led quickly to major strains in the balance of payments through the capital account.

From the 1990s onward, foreign investors could reduce their exposure to an emerging market sovereign relatively easily by selling their bonds. As a result, the capital account crises of the 1990s were faster moving and less containable than the currency crises of previous decades. Mexico's default foreshadowed the key rupture – a wave of currency crises and debt defaults.[4] It is only by understanding the reaction to these crises that the current configuration of the international monetary system can be appreciated. Sovereign borrowers, starting with East Asia, resolved to manage their external accounts differently after 1998, keeping their currencies undervalued and accumulating large foreign reserves.

Oil exporters like Saudi Arabia had long held large foreign exchange reserves, which were accumulated involuntarily. But the new type of foreign exchange accumulation was voluntary and accompanied by the expansion of sovereign wealth funds that aimed to maximize return over the long run. One ramification of this was that the undervalued Asian currencies provided cheap exports. As a result, inflation stayed low in the developed economies and there was little pressure to raise interest rates. The new wealth funds came to hold a much wider range of assets – as Saudi Arabia had been doing for many years. In turn, this stimulated the capital markets to provide them with other investments – bonds from lower rated developed markets, emerging market bonds, securitized products, private and infrastructure capital products, and equities.

One barometer of the international monetary system is how foreign reserve 'adequacy' is measured. The capital account crises showed that reserves now had to cover flows on the capital account. The extent of potential flows can be represented by the supply of relatively liquid types of money, such as cash and bank balances which could, in theory, leave the economy. The central bank's ability to stabilize the exchange rate in the face of such an outflow is measured by its stock of foreign reserves relative to broad money. In 1952, Arthur Young had written into the founding deeds for SAMA that currency in circulation must be backed by foreign assets (in his day, gold and silver), which was an early version of this idea.

Figure 12.3 shows the ratio of foreign reserve holdings to broad money supplies (converted into dollars) for a selection of large foreign exchange reserve accumulators, including oil exporters. If even 50 percent reserve coverage of broad money is desirable, most of these countries could still accumulate many more reserves before they reach the comfort level of Saudi Arabia.

The role of Gulf oil exporters as providers of global savings was largely ignored amid the focus on reserve accumulation in East Asia – especially China. The Gulf nations came back into the picture due to the post-2002 recovery in oil prices. When King Abdullah took power in 2005, Saudi Arabia's surplus was about the same as China's. This time the method used to recycle oil earnings was different, reflecting a greater confidence and sophistication on the part of SAMA and others,

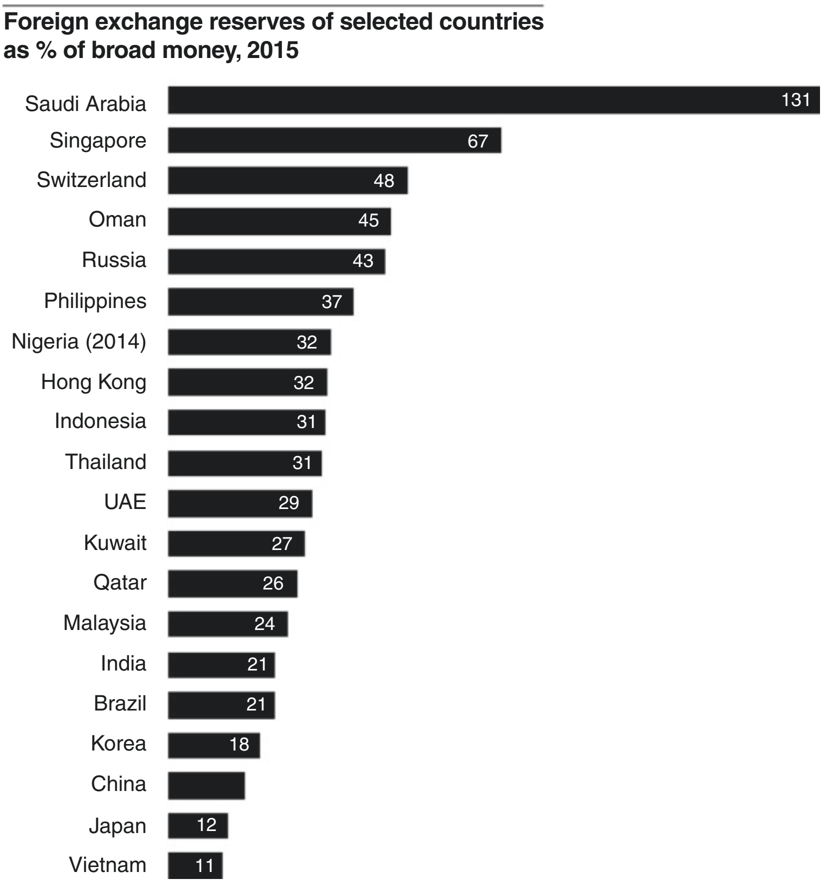

Fig. 12.3 Foreign exchange reserves of selected countries as percentage of broad money, 2015

Source: World Bank

as well as a wider menu of available investment products. During the 2007–09 global financial crisis for instance, Gulf interests acquired stakes in Western banks, most notably 30 percent of the British investment bank Barclays, while its competitors were forced into the arms of the state in order to avoid collapse. As the major central banks kept interest rates near zero in the years after the crisis, global capital flows became increasingly volatile as investors searched for extra yield. These strategic investment stakes are a useful counterweight to the short-term outlook of investment capital provided through the stock and bond markets.

5 Possible Futures for the International Monetary System and Saudi Arabia's Role

If one uses the trilemma concept as a framework for categorizing the international monetary system, it is clear that it has developed in a non-linear fashion. Configurations that seemed natural to earlier generations have been replaced by different ones and then re-embraced (Fig. 12.4).

Today's international monetary system has relinquished fixed exchange rates in order to enjoy free capital flows and policy independence.[5] Yet it is not a stretch to imagine the process working differently in future, especially in the euro area. Cyprus accepted temporary capital controls in exchange for monetary expansion in the form of bank rescues during the

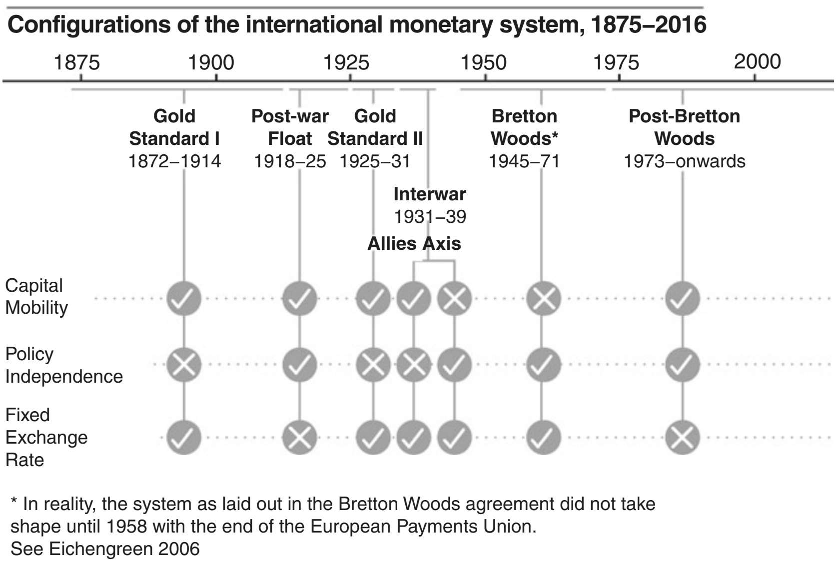

Fig. 12.4 Configurations of the international monetary system, 1875–2016

Source: Authors

banking crisis of 2013, setting a precedent for other troubled euro members. A future with many more fixed exchange rates and capital controls is certainly imaginable.

A force pushing in the same direction is China. The country's success in lifting its population out of poverty in two decades and its high growth rates, increased employment and living standards, set an example to other developing countries. China's capital controls – now being cautiously lifted – have always been a key component of its external policy, leading to other countries lower down the development chain accepting the need for greater state involvement in the control of financial flows.

In order to understand the implications for Saudi Arabia of possible changes in the monetary system, we need to look at the inflow of oil money and its investment by SAMA. When King Salman succeeded his brother in 2015, Asia had become even more important as a source of revenue. In 2005 Asia accounted for just over half of crude exports. Ten years later this had risen to over 60 percent. North America (mostly the United States), by comparison, took only 16 percent of Saudi oil. The story is the same for Western Europe which historically was a big user of Saudi oil, and the trend seems likely to continue as shale energy in the United States turns it into an energy exporter, Western Europe presses on with greater energy efficiency and the Chinese economy continues to flourish (Fig. 12.5).

For Saudi Arabia, a fast-growing Asia should mean a positive outlook for the oil price. The newly industrializing economies would provide new sources of demand. For the world as a whole, this scenario would lend itself to continued disinflation, low interest rates and faster growth than in recent years. But although the economic implications are attractive, SAMA would lose out if the Asian model of capital controls became more commonplace, because it depends on being able to invest freely. More importantly, it has to be free to sell its investments when the roulette wheel of the oil price means that it has to liquidate foreign assets.

But the disadvantages of Chinese capital controls should not rule out a consideration of the potential advantages of pricing oil in Chinese renminbi. Looking purely at the export numbers, it would seem to make sense to take payment for part of the oil in Asian currencies. SAMA's investment profile has also moved away from the dollar and today it invests around the world, in emerging as well as developed markets, and local currency bonds issued by Asian economies as well as in Treasuries.

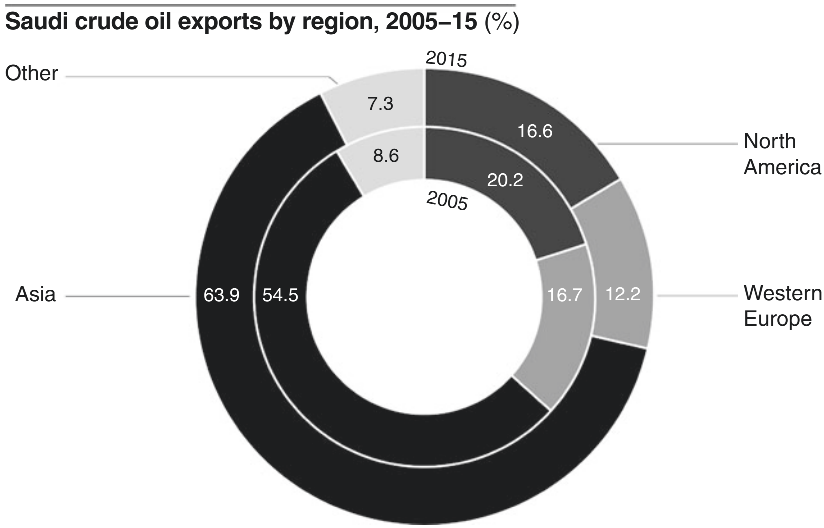

Fig. 12.5 Saudi crude oil exports by region, 2005–15

Source: SAMA Annual Reports

There is some discussion about whether the monetary system could work with more than one numeraire currency. The issuer of the dollar is privileged because the American economy enjoys lower interest rates due to the extraordinary demand for dollars for use in world trade. According to the 'network' view, there can be only one numeraire currency because its role is to simplify global transactions – the whole point being to express them in a common unit. However, according to the 'multipolar' view, there can be – and indeed have been in the past – multiple numeraire currencies. There is some debate about whether 'network effects' – such as the invoice quoting mentioned above – lead naturally to a single numeraire world. It may be that the smartphone and internet make uniformly quoted invoices a thing of the past, now that quotes can instantly be translated into a common standard electronically.[6]

Could the renminbi achieve numeraire status? China's leadership wants to make the renminbi an international currency that is accepted for payment globally and in which central banks would hold more of their reserves. The payment part of that aspiration is well underway with the renminbi joining the SDR. The renminbi's role as a store of value depends on Chinese capital markets. If the multipolar theory of numeraire status is correct, then the renminbi is extremely well positioned to achieve this goal. This is a view endorsed by the history of successive numeraires – the Dutch guilder, the pound sterling, the dollar – which indicates that a currency's international acceptance is determined by its importance in trade. China is the largest exporter of merchandise in the world.

Even a small diversification by the Chinese into foreign assets could give the world a lot of Chinese currency. A global supply of renminbi could be furnished when capital controls are fully lifted and Chinese households and businesses start to diversify their wealth. This scenario is consistent with a transition to a freely floating renminbi, since under a floating regime there is little point holding reserves – the United States holds almost none.

Saudi Arabia has a potentially vital role to play in any change of numeraire, through the invoicing of oil. As long as the kingdom continues to invoice primarily in dollars, any change away from the dollar – especially to the renminbi – is less likely. But in a world where the renminbi is a numeraire currency alongside the dollar, the invoicing of oil could fragment. The main consequence would be a further deepening in the economic relationship between China and Saudi Arabia. Chinese exports already find a ready market in the kingdom. Saudi investment in China would grow, just as China's economy starts running current account deficits as the private sector diversifies its savings.

The euro area has a bigger GDP than the United States. If the euro joined the renminbi and the dollar as a third numeraire, it would certainly be conceivable that Saudi Arabia could shift to selling oil in that currency – to the euro area at least. But only 13 percent of Saudi oil goes to Europe, compared to the 60 percent which goes to Asia. It is hard to see why Saudi Arabia should price oil in euros.

The final possible future to consider is the obvious one: a continuation of today's situation in which the US dollar stays as the key numeraire, and the kingdom's export receipts continue to be priced in dollars. Those who think this is unlikely point out that the United States changed from a net creditor to a net debtor in the 1980s. The rest of the world might refuse to add to its stocks of American assets given this debt. But this possibility seems remote. The international monetary system is now far more than a clearing mechanism for trade. The role of the dollar as a store of value has become just as – if not more – important than its job as an accounting device for trade transactions. The American capital markets offer foreign investors a range of assets combined with depths of liquidity that no other country can match.

The election of Donald Trump as President for 2017–21 is a signal that Washington DC might eventually refuse to countenance the large external deficits that the numeraire role involves. But as long as capital controls apply only at the fringes of the system as they do today, SAMA can exchange the oil dollars for other currencies and invest freely around the world. Its core investment market will remain the United States.

SAMA has no interest in seeing a major change in the international monetary system, which has served it well. It follows that the central bank should not encourage any change, as disruption to the system runs risks. The shift in numeraire from sterling to the dollar between 1914 and 1945 was not coincidentally accompanied by massive disruption of the international world order. But a gradual incremental change that reflects the underlying realities is another matter. As the United States declines slowly in relative terms and China rises, the renminbi could become an additional numeraire. And the quicker that China abandons capital controls, the more rapidly Saudi Arabia could increase investment in its fast-growing economy. It is also possible that technological change will make numeraires less important than in the past.

There is one final scenario to consider, in which Saudi Arabia and its Gulf neighbors invoice for their oil in a Gulf currency unit. This has some interesting implications. Because Saudi Arabia would be paid with Gulf notes, buyers would first need to acquire them. They could do this through a combination of Gulf payments for world exports – in Gulf notes – and through Gulf investment abroad. Gulf economies would find a ready demand for their common currency because their trade partners would compete to buy it, which would in turn reduce the cost of borrowing. This would appeal to a region that aspires to diversify into non-oil activities, a process that would be strongly assisted by a favorable cost of capital.

6 Going with the Flow

When the G-20 leaders met in Toronto in 2010, they asked a group of experts to analyze how the international monetary system could be reformed. Former SAMA Governor Al-Sayari was one of those consulted and the result was the Palais Royal Initiative of the following year. Its main call was for the IMF to set norms for national macroeconomic policies and impose sanctions of some kind on countries that broke them – essentially a reprise of Keynes' views at Bretton Woods. SAMA's views on the report are interesting. The central bank argued that it was too easy just to criticize the existing system. During the last 40 years, global GDP has increased more than four times in real terms, while world trade has grown consistently. This does not look like failure. Closer to home, it was skeptical of any suggestion that the IMF could target correct levels of exchange rates and set norms for countries to follow in terms of ensuring external balance. Because Saudi Arabia's external account swings from surplus to deficit, depending on the vagaries of the oil price, SAMA felt that it was misconceived for the IMF to tell the kingdom what its external balance should be.

The Palais Royal report tried to set the agenda for a bigger role for the SDR as part of a move away from the dollar. As a long-standing supporter of the IMF, SAMA favored this, but within the context of a natural evolution to a multipolar reserve system consisting of the currencies of the biggest economies, in which the dollar would be joined by the euro, the yen and the renminbi. But it could only support this evolution if the other currencies liberalized and deepened their capital markets to match the United States.

During the history of Saudi Arabia's involvement in the international monetary system, SAMA has been a team player, supporting key institutions. This was illustrated most dramatically by its loan in the early 1980s to help the IMF provide extra finance to the developing world. Its supportive role fits with the kingdom's orientation as a conservative power, which benefits from peace and stability and has an interest in maintaining it. Domestically, SAMA has always preferred to work with the market rather than being prescriptive – for instance, it has not used formal capital controls to restrict investment abroad for over 50 years. This market-oriented philosophy is reflected in the central bank's view on the international monetary system as well. Change should emerge from market forces rather than being mandated as part of a political agenda.

But can existing arrangements evolve to accommodate China – now the most dynamic buyer of the kingdom's oil? If history is any guide, change in the international monetary system proceeds by discontinuities rather than evolution. So far China's aim has been to integrate into the world's financial order rather than overturn it. If this continues, SAMA will not be forced to look for new solutions. Its attitude to the renminbi is that it is a good diversifier among other currencies. In other words, SAMA will go with the flow.

Notes

1. Thomas Lippmann, 'The Day FDR met Saudi Arabia's Ibn Saud,' in *The Link* vol 38/2 (2005), http://www.ameu.org/getattachment/51ee4866-95c1-4603-b0dd-e16d2d49fcbc/The-Day-FDR-Met-Saudi-Arabia-Ibn-Saud.aspx

(Accessed November 6, 2016). Lippmann was Middle East bureau chief for the *Washington Post*.

2. Calculations on the real oil price in 2010 terms are by the authors, using the IMF IFS dollar oil price and deflated using US consumer price index from US Bureau of Labor Statistics data.
3. This is the year the riyal was included in the 16-currency SDR basket, in use until 1981.
4. East Asia in 1997–78, followed by Russia (1998), Brazil (2000), Turkey (2001) and Argentina (2002).
5. The movement of 19 countries into the euro-area since 1999 is the prime instance of events going the other way, however. Capital flows freely in theory but policy independence has been sacrificed – exactly as under the gold standard.
6. Barry Eichengreen, Exorbitant Privilege: The Rise and Fall of the Dollar and the Future of the International Monetary System (Oxford: Oxford University Press, 2010). This is a good discussion about whether a multi-polar world means multiple numeraires. Eichengreen argues that three numeraires operated before the First World War and that three (dollar, euro, renminbi) are likely to operate in future.

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CHAPTER 13

The Saudi Banking System

1 SAMA's Authority and Its Limits

As is the case in other developing countries, the Saudi financial system consists of the private sector and the public sector. Commercial banks, insurance and finance companies and investment banks are in the private sector. In the public sector are two types of bodies. First are the autonomous government institutions (also called the quasi-government funds) such as the Public Investment Fund (PIF), which are investors in domestic and foreign assets.[1] Second are the providers of credit such as the Real Estate Development Fund (REDF).[2] They played a much larger role in the past when the private sector was less developed. As the private sector took over and the financial system became more complex, SAMA was over-burdened with mandates beyond its scope and expertise. Establishing the Capital Market Authority (CMA) to regulate the investment banks removed one role, but others remain with the central bank.

Insurance is an obvious example of a complex area, which should leave SAMA's remit. Insurance regulations were promulgated as long ago as 2004 but a moratorium was put on enforcing the Cooperative Insurance Regulations for four years until 2008. Today, the insurance business is still the subject of discussion due to a shortage of resources among several smaller companies and their lack of preparedness to serve clients efficiently. Insurance needs dedicated supervision by an independent entity that is not

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SAMA. This chapter deals only with the commercial banks, which are subject to SAMA's supervision.[3]

The central bank has from time to time been the target of blame. The perception among the Saudi public is that SAMA mostly sides with the banks. For example, it came in for heavy criticism for not doing enough to protect individual investors by preventing the bubble that preceded the 2006 stock market crash or containing it once it had occurred. On occasions in its history, SAMA has been approached by high net worth individuals and companies who have engaged in risky activities and lost money, and wanted the central bank to bail them out. This is a perennial issue, but SAMA obviously cannot overturn the contractual obligations of a bank and the client. Its roles as regulator and supervisor mean the central bank is the guardian of the financial system and its job is to maintain financial stability. That can mean restraining the banks' activities, such as restricting margin loans and cautioning them against excessive credit expansion. But there is little it can do to force the banks to improve their range of services to the public or to improve the way they deal with customers.

The Saudi banking system has been tested through thick and thin as the oil price has varied unpredictably. Although there have been issues with individual banks from time to time, as when Saudi Cairo Bank became technically insolvent in the 1980s after speculating in precious metals, there has never been anything close to the systemic banking crises that developed economies have suffered in the past few years. The resiliency of the system is due to the strong financial position of the domestic banks, the prudence of the banking control law and strong supervision.

2 Evolution of the Saudi Banking System

Broadly speaking, the development of the financial sector took place in two phases: the years before 1990 and those afterward. Table 13.1 illustrates the banking system's evolution. The first phase included the establishment of SAMA and the commercial banks and the development of a modern banking system. SAMA was founded in 1952 and the Banking Control Law was introduced in 1966; but the most rapid period of development occurred during the following decade when the country's economy was transformed by the oil boom and financial assets grew fast. As early as 1964, there was a banking problem when Riyad Bank had to be rescued by SAMA, and the public sector still maintains a strategic stake in it. The sector was gradually Saudized in the 1970s, as branches of foreign banks were converted into joint stock banks with majority Saudi participation. There were other changes in bank ownership, too, such as when United Saudi Commercial Bank was formed by the merger of the local branches of three foreign banks and subsequently took over Saudi Cairo Bank, which was in poor financial condition. SAMA made Saudization attractive by allowing the banks to raise fresh capital and open more branches.

The mid-1980s were a hard period for the banks when the number of non-performing loans (NPLs) rose. Quarrels with customers over impaired loans and non-payment of interest built up. The banks found it tough to enforce their claims through the Shariah court system. In 1987, troubles were so common that the central bank established a Committee for Settlement of Banking Disputes to resolve problems between banks and their customers. But the newly-Saudized banks invested more capital in their businesses, upgrading banking technology and installing automated teller machines (ATMs) so that customers could transact business at any time of day.

After 1990, the banks became more deeply intertwined with the financial affairs of the government when they helped finance the budget deficit. The result was a rapid rise in bank holdings of government debt. Restructurings and capital increases continued – in particular the combined United Saudi Commercial Bank was merged with Saudi American Bank (SAMBA), the old Citibank operation. Eventually, in the new millennium, the tough times that had persisted for almost 20 years gave way to a boom and the sector changed again. Sharia-compliant banking gained ground over the traditional Western model while investment banking was decoupled from its commercial counterpart and the new institutions that resulted were supervised by the CMA rather than the central bank. Finally, additional foreign banks were licensed in order to encourage competition.

During this period of rapid structural change, SAMA maintained its usual policy of tough regulation. It enforced compliance with the Basel II regulations across the board, and, in 2004, the Saudi Credit Bureau (SIMAH) began to enable information about bad credits to be shared among its member banks. While the economic background was positive, a couple of crises temporarily punctured confidence: the stock market crash of 2006 followed a couple of years later by the Saad-AHAB default.

The next few years of high oil prices were calmer, but the pace of progress has accelerated recently. Government debt was repaid and then, in 2015, it started to rise again due to the low oil price, which meant budget deficits. One outcome of the government's borrowing from the

Table 13.1 Evolution of the Saudi banking system (changes 1950–2020)

1950–60	• Establishment of SAMA • SAR peg to USD
1960–70	• Merger of Al-Watany Bank into Riyad Bank due to technical insolvency • Introduction of Banking Control Law
1970–80	• Conversion of foreign bank branches into joint stock banks with Saudi participation completed • Post-conversion rapid growth in banking assets • Merger of branches of Bank Melli Iran, National Bank of Pakistan and Banque de Liban et d'outre mer into United Saudi Commercial Bank, which then took over Saudi Cairo Bank
1980–90	• Increase in NPLs due to sharp decline in oil prices and economic stagnation • Capital increase in existing banks • IT upgrade and introduction of ATM
1990–2000	• Countercyclical fiscal policy via deficit financing • Rapid rise in banks' holdings of government debt • Bank restructuring and capital increases (merger of USCB into SAMBA)
2000–10	• Investment banking spun off from commercial banking into a separate entity • Licensing of additional foreign banks • Basel II implementation • Stock market collapse in February 2006 • Establishment of Saudi Credit Bureau • Domestic corporate credit event (Saad-AHAB) hitting banks' profitability
2010–20	• Enforcement of APR transparency • Encouragement of credit to SMEs • Issuance of SAMA bills for system liquidity management • Basel III implementation • NCB IPO • Deposit Insurance fund • Vision 2020 • Deficit financing resumes

Source: Authors

domestic market is that system liquidity and monetary conditions have become somewhat tighter. Because the banks are required to maintain prescribed liquidity ratios they tend to bid very competitively for customer deposits, which count for their loan to deposit ratio (LDR). SAMA keeps a close watch on system liquidity and takes action (such as injecting funds through repo operations and deposit placements) to make sure that conditions in the money markets do not get disorderly.

The central bank has taken on responsibility for customer protection and has enforced transparency on borrowing costs, insisting that the banks need to provide a more complete picture of the cost of funds, including compound interest calculations and fees. A deposit insurance fund was established in 2015 financed by a 0.05 percent levy on the eligible deposits of the banks covering deposits up to the equivalent of $53,000. The next set of Basel regulations – Basel III – is being implemented and financial institutions are being persuaded to lend more money to the small and medium-sized business sector in the hope that this will generate additional jobs. Mortgage lending on residential property is rising fast and SAMA is keeping a close eye on it to make sure borrowers act prudently – and acting on occasion to adjust conditions, as with the 70 percent loan to value ratio which was introduced in 2014 and relaxed to 85 percent two years later to keep access to mortgages open.

3 The Banking System Today

Saudi banks have been doing well – they are modestly leveraged, well capitalized, highly liquid, adequately provisioned and profitable. Their capital and reserves have doubled in size since 2007, they have high returns on equity and they exceed the Basel III minimum requirements on all points.

There are 24 banks in Saudi Arabia, 12 local and another dozen foreign institutions which have branches in the kingdom, but the sector remains dominated by a few big players: SAMBA, state-owned National Commercial Bank (NCB) and the Shariah-compliant Al-Rajhi are the largest. Al-Rajhi is the leading retail bank. Together these three have a share of just under half the banking market. In recent years, the growth of bank assets has been higher than that of both overall and non-oil Gross Domestic Product (GDP), which is strong evidence that financial deepening of the economy is finally taking place (Fig. 13.1).

Compared to overall GDP the provision of bank credit is about 57 percent, below most emerging economies, but similar to other Gulf Cooperation Council (GCC) economies (save for Kuwait), a reflection of the fact that the country's large oil sector does not need much borrowing. The ratio for the non-oil sector is higher at 79 percent and the figures for the GCC are rising (Fig. 13.2).

The banks continue to be conservative, largely because their core business – lending to the domestic market – is so rewarding. They are efficient at generating profitable business and their ratio of costs to income compares

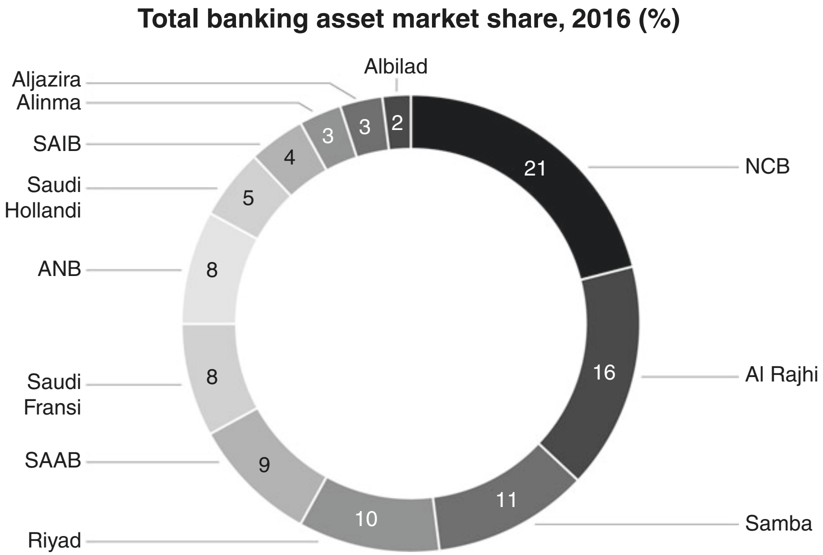

Fig. 13.1 Total banking asset market share, 2016

Source: Authors

very favorably with banks in developed markets. The main component of their profits consists of interest on private sector loans and they remain very cautious about using derivatives. Only about 14 percent of their assets are invested abroad. While lending to the private sector drives the banks' growth, their expansion nonetheless remains closely tied to the oil price and to government spending since many loans are to companies that rely on public contracts.

This is true of the retail sector as well, which is dominated by lending to Saudi civil servants. Consumer lending grew fast up to 2014 as banks benefited from the security provided by automatic loan repayments from public sector workers' salaries – although it has slowed since. To safeguard against consumers borrowing too much the central bank has prescribed debt service-to-income ratios that the banks cannot breach.[4] Real estate lending is also increasing very rapidly, underpinned by new mortgage laws. Yet it still represents a relatively small proportion of overall lending, with considerable capacity for growth over the coming years. Long-term loans[5] are an increasingly important part of the system, growing from 25 percent of the total in 2010 to 32 percent in 2015.[6]

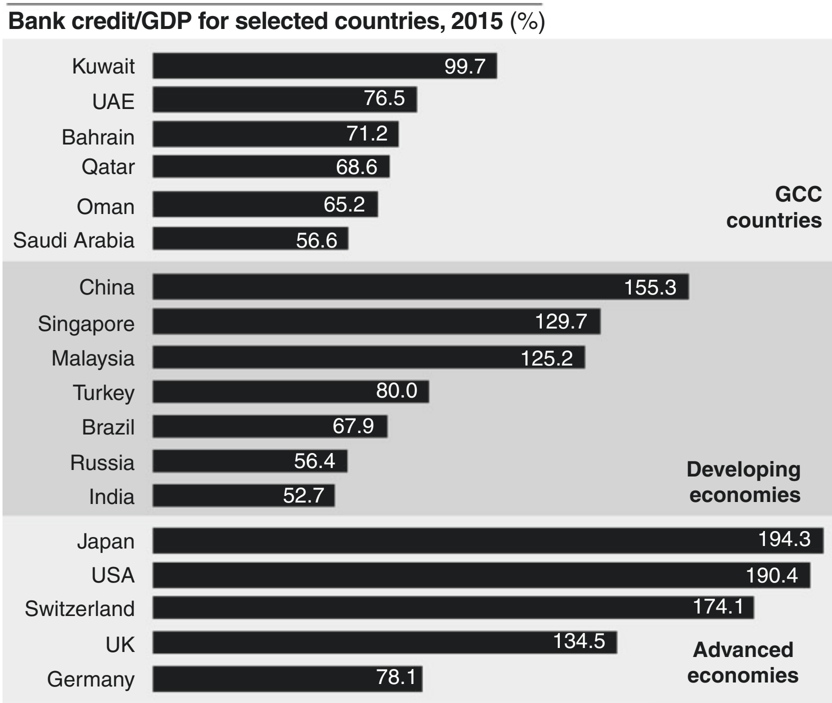

Fig. 13.2 Bank credit/GDP for selected countries, 2015
Source: World Bank (all data 2015)

The main source of funding for the banks continues to be short-term bank deposits, a considerable proportion of which do not pay interest. Almost all of these are for less than a year, resulting in a growing mismatch between loans and deposits. So far this has been manageable, but the banks will need to encourage time and savings deposits, on which interest is payable, as well as developing Shariah-compliant products in order to lock in longer-term funds from depositors to whom they pay no interest at the moment. Real estate lending is long-term and many consumer loans are for five years, so as these grow in importance the banks will have to find longer-term sources of money.

In mid-2015, liquidity in the banking system started to tighten and SAMA has had to take steps to counteract this. The main reason was that the government began to borrow from the domestic banks to fund its budget deficit. This is having an effect on bank lending to the private sector. The loan book of the banks grew by 14 percent in 2014 and slowed to 10 percent in 2015. As liquidity has tightened, the banks' cost of funds and Saudi Arabia Interbank Offered Rate (SAIBOR) have risen. SAMA began injecting liquidity in early 2016 and relaxed the loan to deposit ratio from 85 to 90 percent. Figure 13.3 shows the soundness of the banking system.

As well as being highly profitable, the banks are well capitalized under standard measures and they seem likely to remain so as loan growth slows. Their leverage ratio of 12 percent is well above the minimum 3 percent set under Basel III, which defines leverage as the percentage of total assets covered by capital and reserves. The Capital Adequacy Ratio (CAR) under Basel III regulations is a more complex variant of the leverage ratio and looks at how well supported bank lending is by the banks' own resources.[7] Some emerging economies – such as Russia, India and China – were in 2015 below the minimum level, but Figure 13.4 shows that the Saudi banks had a much stronger equity base at 16 percent (SAMA's own calculations put the number a bit higher than this). The banks are required to hold a minimum of 20 percent of their deposit liabilities in the form of high-quality liquid assets (HQLAs): government bonds, SAMA Bills and money market instruments maturing within 30 days. As at end-2016, the banks were comfortably positioned to meet or exceed this ratio.

The level of risk to which Saudi banks are exposed going forward seems to be low. Their operations are predominantly domestic. The banks rely heavily on net interest income for their revenue growth, followed by fees and commissions. Rising credit volumes either to the private or the public sector, the margin between cost of funds and lending rates and low NPLs are all key to the banks remaining healthy. During the global financial crisis, NPLs increased to around 3 percent of total lending due to the Saad-AHAB default. After peaking at 4 percent at the start of 2010, they dropped to 1.2 percent in the middle of 2016. But the appearance of stability and profitability can be deceptive. The impact of fluctuations in oil prices on government spending means that shifts in the economy occur rapidly and unpredictably.

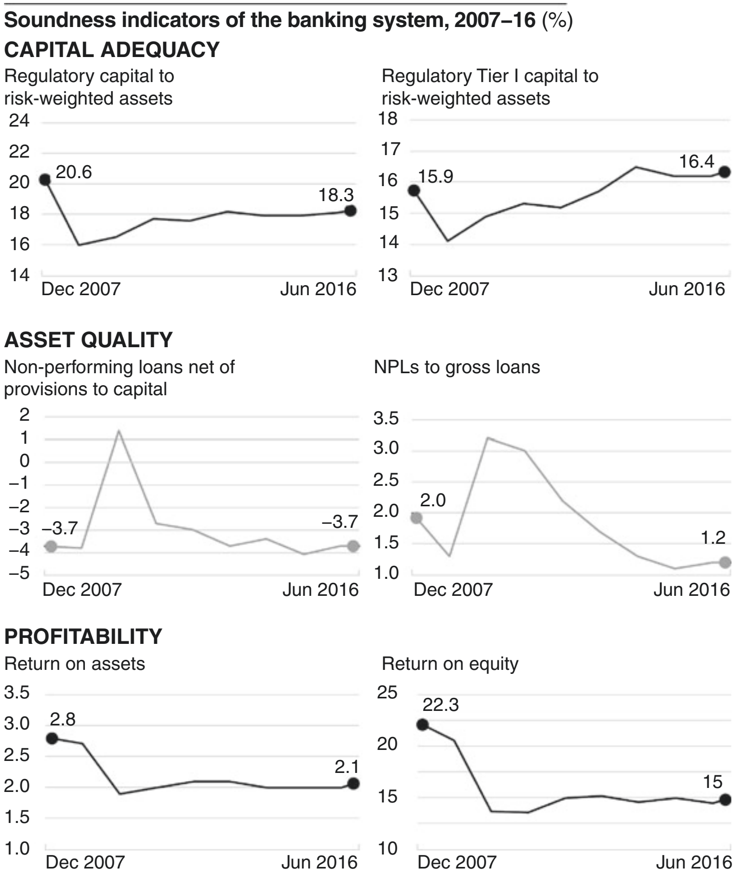

Fig. 13.3 Soundness indicators of the banking system, 2007–16

Notes:
Tier I= Common stock + reserves (shareholders' funds).
Non-performing loans = When payments of interest and principal are post due by 90 days or more.
Provisions = Money set aside to offset potential lossess from loans or other receivables (25% of NPL for 90 days, 50% for 180 days and 100% for 360 days. Additionally, there is a general provision of 1% of performing loans: IFRS provisions are different from SAMA calculations)

Source: Irish Stock Exchange (Saudi Arabia global medium-term note program 2016); SAMA Global Financial Stability Report 2015; IMF Financial Stability Assessment Reports.

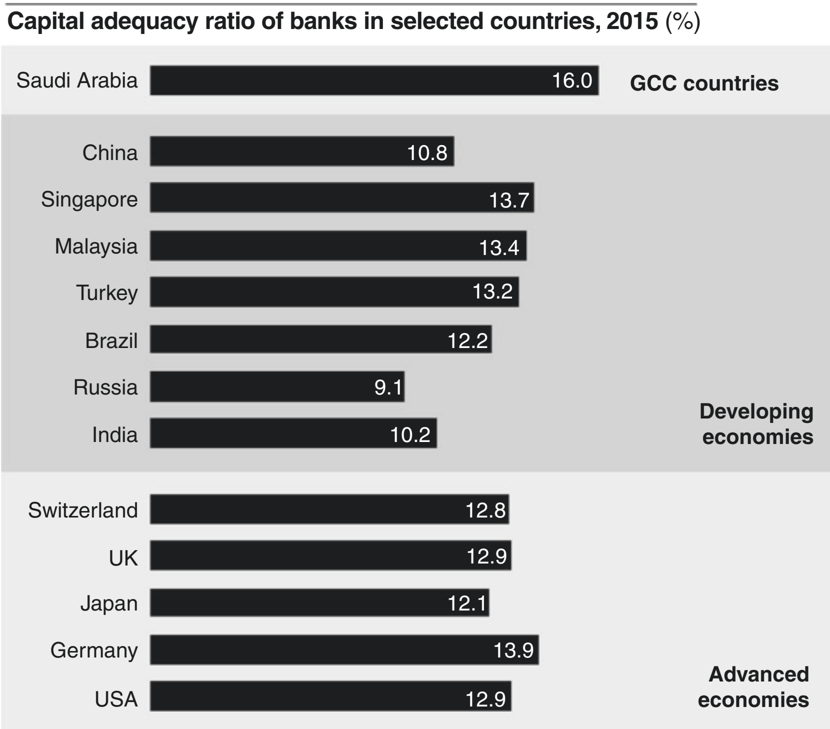

Fig. 13.4 Capital adequacy ratio of banks in selected countries, 2015

Notes: Tier 1 equity capital as a percent of risk-weighted assets (RWA). No GCC country data for 2015 available apart from Saudi Arabia.

Sources: IMF, Financial Stability Indicators

4 The Future of Saudi Banking

In the aftermath of the 2008 financial crisis, the global banking sector continues to change. Over-leveraged international banks are preoccupied with reshaping their balance sheets in the face of a barrage of new regulatory requirements concerning capital, liquidity and leverage. As a result, the big banks are withdrawing from some regions of the world. Problems for some are opportunities for others, and these changes offer the chance for smaller, more agile local banks to become regional leaders.

The overall winners in Saudi Arabia will be those banks which are best at controlling costs and investing in new technology.

In November 2014, the Saudi stock market saw the largest-ever privatization in the Middle East, when the Public Investment Fund sold a 25 percent stake in one of the jewels in its crown – the NCB. The announcement earlier in the year had caused great excitement in Riyadh – it was six years since any bank had come to the stock market. NCB was one of the first banks to be licensed in Saudi Arabia, just a year after SAMA was established, although it had been a moneylender for many years before that, and its history shows how the fortunes of the Saudi banks vary with the oil price. In the 1980s and 1990s, several banks – including NCB – suffered because they had lent money to businesses during the boom years of high oil prices, loans that could not be repaid when the economy went into recession due to the fall in the oil price. SAMA helped their balance sheets and profit and loss accounts by placing 'soft deposits' with them, that is to say deposits at interest rates lower than the market rate (so-called 'off market rates'). NCB did not publish accounts for several years in the late 1980s. In 1999, the bank got into trouble again and the government acquired a majority stake, nursing it back to health. Today NCB is highly profitable and the largest bank in the kingdom in terms of assets. The sale raised over $6 billion for the government, whose stake has fallen to 55 percent, and was massively over-subscribed, a development that was interpreted as a vote of confidence in the state of the country's banks by the ordinary investor.

The success of the NCB flotation underlines the fact that the future for the banks is promising. There are three main reasons. First, banks' modestly leveraged balance sheets allow them room to grow in tandem with the region's economic activity. Second, they operate in an under-exploited market. Banking services in the kingdom still have a long way to go before they can provide most citizens with the range of products that a customer in North America takes for granted. The starting point is to expand lending to small businesses. In the developed economies, this type of loan amounts to more than a quarter of total bank lending, but for banks in the GCC that figure is a meager 2 percent. Supporting small businesses is an obvious way of increasing employment levels among young Saudis, an objective that is supported by the government's Kafala Program, which guarantees part of any loan a bank makes to a small business.

The third advantage that Saudi banks have is that they benefit from the fact that a significant proportion of deposits do not pay interest. But these funds are short term – 61 percent of total deposits at end-2015 were demand deposits that can be withdrawn without any notice, and the level is even higher when government deposits are taken out of the picture. A mismatch between assets and liabilities is a fact of life in banking and Saudi banks are no exception; but the problem could grow as banks bid aggressively for deposits to maintain their market share. A particular danger that has emerged is that government bonds are in competition with the banking system for savers' funds. The answer for the banks is to diversify their sources of funding and to develop mutual funds that invest in government debt rather than let businesses and high net worth individuals buy the bonds direct. This is an area where SAMA can encourage the banks to start marketing longer-term savings products and develop the local debt markets.

These are the challenges on the funding side; but – apart from their exposure to an economy still driven by public spending – the biggest challenge the banks face is concentration risk in lending. They need to distribute their lending among more than just a few big borrowers and reduce their exposure to sectors that are highly subject to fluctuations. But this means they need to improve their assessment of the credit-worthiness of borrowers. Concentration risk has been a big issue in the past when bank borrowing was dominated by a few family-owned groups and local conglomerates, and quasi-government entities. A case in point was the banks' exposure to the Saad-AHAB groups which defaulted in 2009. SAMA has imposed stricter maximum exposure limits from the start of 2016. The maximum exposure a bank is allowed to have to a single borrower (including all entities within a business group) has been cut from 25 percent of their Tier 1 capital to 15 percent. This will apply to all new lending and the banks were given three years until 2019 to apply it to their lending books.

The central bank is working to provide clarity on the future of banking in the kingdom by drawing together stakeholders and experts to ensure that bankers are able to keep up with international best practice. Under the umbrella label *Banking Vision 2020*, it has set out some areas where improvement is needed. Investing in technology, for instance, is crucial to capturing more customers without incurring the costs involved in an extensive branch network. Other emerging markets are already utilizing mobile phone technology for accessing bank accounts and transferring money. Providing a wider range of financial products also presents new opportunities.

Generally, the banks need to be more competitive in terms of servicing and pricing. Saudi banks operate under conditions of monopolistic competition with little customer awareness of the prices, terms and conditions of their products. In 2013, SAMA assumed responsibility for consumer protection and used its powers to promote transparency in customer relations. Its requirements are set out in the Consumer Protection Principles that banks have to give to all new and existing customers. These include, for instance, strict rules about how banks can advertise their products, the protection of customers' personal data and procedures for the approval and explanation of fees and charges.

Finally, the financial sector needs to push the Saudization of personnel in order to improve job opportunities. SAMA facilitated the transfer of bank ownership into local hands a long time ago, but 40 years later there are still non-Saudis in senior positions.

5 Fundamentals of Banking Supervision

Banking supervision in Saudi Arabia has long adhered to the Basel standards, which aim to promote international financial stability by building a system of common international standards to avoid the repetition of a crisis on the scale of 2008–09. One of its principal concerns is leverage, because many of the banks that failed during the crisis had high leverage levels. The nature of their work means that banks necessarily borrow more than non-financial companies.

International banks are today highly regulated because they are highly leveraged. Basel III's minimum leverage ratio is 3 percent, meaning that banks may not have assets and commitments exceeding 33.3 times their capital. Even the much lower leverage ratio of Saudi banks at 12 percent (8.5 times capital) is an order of magnitude greater than a non-financial business would ever consider. Some claim that limiting leverage inhibits lending, because equity is more costly for banks than debt, and that imposing limits on leveraging inevitably increases costs which then have to be passed on to borrowers. However, SAMA's experience of managing the domestic banks on a conservative basis has made it skeptical about the need for very high levels of leverage.

The central bank is fully engaged with the work program of Basel III, and in a consistency assessment test in 2015 Saudi Arabia was listed as compliant or largely compliant across a wide range of areas – including operational risks, shadow banking and counterparty risk. SAMA has also followed the Basel plans by identifying six systemically important financial institutions within the domestic banking system – the so-called Domestic Systemically Important Banks (D-SIBs). The D-SIBs will be required to hold higher amounts of capital and liquidity, which are more commensurate with their risk profiles. SAMA selected four factors including size, interconnectedness and complexity as it addressed several big questions in deciding which banks should be considered as D-SIBs and what should be done about them. How big is big enough? No financial firm should be thought too big to fail, but big banks can be more resilient than smaller ones. What is the best balance between size and complexity – if it is not adequate to mainly rank banks just by size, then how does the complexity of their business model enter into the judgment? Are Shariah-compliant banks really more secure than conventional ones? Can more intense supervision, risk-based payments into deposit insurance schemes and more regulatory capital eliminate the chances of a major financial institution going under?

Stability and resilience will require a combination of factors. The difficulty is to promote healthy change in the financial system while recognizing that many developments, which were in the past praised by central bankers as being positive – such as the explosion in the use of derivatives – turned out to be dangerous errors.

6 Macroprudential Policy

Before the crisis, central banks thought the financial system was self-regulating because financial institutions had a vested interest in avoiding bad and greedy behavior. It was believed that a central bank's role was restricted to maintaining price stability and providing 'light touch' regulation. SAMA was always skeptical about this. Its own experience of regulation showed it was a mistake to assume banks would avoid bad behavior, but in line with the prevailing philosophy, it believed that tough supervision of each bank would lead to a stable financial system. The crisis demonstrated the inadequacy of this approach and showed that microprudential policy (looking at each bank individually) has to be matched by macroprudential policy (looking at the system as a whole). Although in practice SAMA had been operating elements of macroprudential policy for many years, the idea only came into common usage after 2008. The objectives of macroprudential policy are to limit systemic risk by focusing on the financial system as a whole with specific policy tools.

The concept can best be understood by taking the analogy of a boat that springs a leak. It is logical for each passenger to run to the opposite end of the boat from the end that is sinking. But if all the passengers do that, then the boat will capsize and throw them into the sea. Individual self-interest often does not aggregate into collective self-interest. Somebody has to stop the passengers rushing to one end of the boat. If all banks in the system have exactly the same risk management strategy that is optimal for each one, it will not be optimal for the system as a whole if they all behave in the same way. In the United States, for instance, as the crisis deepened, the banks followed their own risk models – which in many cases were very similar – in dumping illiquid assets and refusing to lend to other banks, which just made matters worse. From a macroprudential viewpoint, it would make more sense for banks to have different risk models, even if some were inferior to others. The same argument applies to banks' plans to expand their businesses. If they all rush into the same area of lending together, the danger is that a credit bubble will form, for instance in the housing market. These are complex issues which central banks are just starting to address.

Just as there is a tension between microprudential and macroprudential policy within a single economy, there is a tension between the macroprudential policies of individual nations and the interests of the global financial system as a whole. For instance, developing economies may find that their own financial systems benefit from making it more difficult for 'hot money' to enter; but for the system as a whole, the growth of informal exchange controls is not likely to be beneficial if it prevents the free flow of capital.

The implementation of macroprudential measures is still in its infancy, and each central bank must choose the most appropriate ones for its own financial system. For SAMA, the challenge is to take ideas that have been developed in the advanced economies and apply them to Saudi Arabia. A particular problem, as the Saad-AHAB affair showed, is that many businesses are privately owned, so there is less information about them available in the public domain to enable banks to assess their creditworthiness. There is a tendency to rely on personal guarantees and reputation, rather than on thorough financial audits. In one way, however, SAMA's practical

Table 13.2 SAMA's macroprudential toolkit

Instrument	*Regulatory Requirement*
Capital Adequacy Ratio	Basel requirements of minimum 10.5% (including the conservation buffer)
Provisioning	General: 1% of total loans Specific: Minimum 100% of NPLs
Leverage Ratio	Deposits less than or equal to 15x Capital and Reserves
Cash Reserve Ratios	7% of Demand Deposits 4% of Time/Savings Deposits
Loan to value [LTV]	Mortgage loans at or below 85% of residential real estate value
Debt Service to Income [DTI]	Monthly repayments at or below 33% of employed salary and 25% of retired person
Statutory Liquidity Reserves	Liquid assets not less than 20% of deposit liabilities
Liquidity Coverage Ratio [LCR]-Basel III	100% by 2019 [already fulfilled]
Net Stable Funding Ratio [NSFR]-Basel III	100% by 2019 [already fulfilled]
Counterparty Exposure	Individual Exposure at or below 25% of bank capital (lowered to 15% by 2019)
Foreign Exposure	SAMA approval needed before foreign lending [qualitative measure]

Note: The Basel III metrics, the Liquidity Coverage Ratio [LCR] and Net Stable Funding Ratio (NSFR) are designed to improve the liquidity of the banks and reduce the insolvency risk. Under the LCR, high quality liquid assets must be available to exceed the net cash outflows expected for the next 30 days. Similarly, the NSFR promotes banks' resiliency over a longer period. Banks resources must exceed their long-term commitments.
Source: Authors

experience has been helpful because the currency peg means the effectiveness of monetary policy is limited, with the result that it has long relied on providing informal and formal guidance to the banks as a means to steer the availability of credit.

There is some overlap between micro and macro measures. For instance, larger capital buffers mean less risk of a bank running out of capital, which on a system-wide basis will prevent contagion from one bank to another. SAMA's current toolkit consists of measures that it applies to individual banks (Table 13.2).

SAMA possesses its own macroprudential measures as well as the standard ones. It uses changes in capital ratios, provisioning and reserve requirements counter-cyclically. Cash dividend payments by the banks are subject to SAMA approval to prevent too many reserves being paid out to shareholders, so that strong capital buffers are maintained. SAMA has also developed a series of early warning indicators to identify the triggers that could guide the use of countercyclical capital buffers. A number of the ratios are set out below.

Loan to value ratios (LTVs). Since November 2014 it has used an adjustable LTV ratio to prevent leverage-encouraged speculation on domestic property.

Debt service to income ratios. These, likewise, apply to individual borrowers and in this form they date back to 2005. In 2014 SAMA set a cap on the total amount of consumer lending made by an individual bank.

Dynamic countercyclical loan loss provisions. Dynamic provisioning is simply the idea that banks should build up their reserves when their profits are growing, so that they can draw them down when they face an economic downturn. After the Saad-AHAB affair, Saudi banks' total provisions against NPLs dipped below 100 percent. SAMA insisted that they should increase them and by 2016 provisions were well above NPLs.

True macroprudential measures would go beyond individual banks to economy-wide rules, and central banks are engaged in discussing several ideas, including credit to GDP limits, controlling inflation in asset prices and temporary foreign exchange controls. Their application in Saudi Arabia is not easy. Credit to GDP is not as relevant for oil-reliant economies as it would be in more diversified economies, because overall GDP for oil economies tends to be more volatile. A better alternative for Saudi Arabia would be to target credit to non-oil GDP. But even this approach is problematic since the non-oil economy is still linked to the oil price. Foreign exchange controls are becoming more acceptable in the international monetary system, provided controls are applied temporarily in order to smooth capital flows. But this runs contrary to the free market philosophy that SAMA has always nurtured since its own bad experience with exchange controls in the 1950s.

As part of its macroprudential effort, in 2015 SAMA published its first Financial Stability Report, which analyzed the existing state of the financial system and listed the work that needed to be done, such as identifying D-SIBs and monitoring whether credit risk in real estate had slowed down as a result of the LTV ratio. A shadow banking sector hardly exists in Saudi Arabia, but SAMA is keeping a particularly close watch on lending by the new real estate lending companies and on the new field of microfinance lending to small businesses.

At about the same time as this report was published, SAMA ran the first of an annual series of stress tests including one designed to see how the banks would respond to a rise in global interest rates. In fact, their profitability was robust, partly because of the non-interest bearing deposits, which means that when rates rise the banks' lending margins actually go up. They are also resilient to a drop in oil prices. If there was a run on their deposits they would all survive for at least one working week. As well as these top-down tests SAMA requires each bank to run its own stress tests twice a year and report the results.

SAMA fulfils the role of both financial regulator and central bank. In many other jurisdictions these roles are split and periodically it is suggested that some of the central bank's functions should be run by new bodies. It was noted above that some jobs, for instance regulation of insurance companies, probably do not belong under SAMA's umbrella, but the central bank's experience is that it can do each job better because it is doing the other one as well. Separating functions can cause problems – a case in point was the Northern Rock debacle in Britain where the financial supervisor completely missed the market risks of the bank's model of funding illiquid mortgages through the interbank money market.

There are three main arguments for splitting up regulation. First, a separate regulator will have more specialized skills. This point has some force in a mature economy with high skill levels. But in an emerging economy like Saudi Arabia skills are at a premium and are more likely to be available in an existing institution such as the central bank, rather than in a new regulatory body, which has to start from scratch.

Second, a formalized split between central bank and regulator with sufficient checks and balances will ensure that each organization does its own job without the risk of overlap. But SAMA's experience is that such overlap is vital, not least since the lender of last resort (LOLR) function is the final step in preserving financial stability. On several occasions, SAMA has been able to act swiftly to support banks by providing them with cheap deposits after its regulatory side identified a problem.

The final argument for splitting responsibilities is that it facilitates the adoption of diverse approaches toward the banks, which promotes healthy innovation and competition. In practice, the regulator and the central bank all too often end up passing the buck and avoiding responsibility for a failure, something that is less possible within a single institution. Separate agencies will also have different goals and priorities, requiring increased coordination, while it is far easier for insights and information to be shared within a single institution. For example, when SAMA took on the task of regulating the Shariah-compliant banks, it continued with its traditional approach of keeping things simple, and Shariah-compliant banks are subject to the same regulatory and supervisory tests it carries out on conventional banks.

7 Shariah-Compliant Finance

Islamic finance is extremely important in Saudi Arabia with four of the 12 local banks carrying out only this type of activity – Al Rajhi is the biggest and the others are Al-Jazeera, Albilad and Alinma. It is difficult to estimate the overall market share of Shariah-compliant business. These four banks account for 24 percent of banking assets but Shariah-compliant activity is likely over 50 percent when the Islamic arms of the conventional banks are included. In SAMA's view it complements conventional financial services rather than replacing them. Shariah-compliant banking is a system based on the principles of Islamic law (the relevant Shariah rules are known as *Fiqh al-Muamalat*) and guided by Islamic economics. Payment and acceptance of interest are prohibited, while the sharing of profits and losses is permitted. The sector is characterized today by rapid development and Saudi banks offer many different types of Shariah-compliant products, each of which has to be approved by the bank's own Shariah board. Most transactions are either *Murabaha* (for retail customers and small businesses) or *Sukuk* (for large companies). The list that follows – which includes both standard banking services and investment products – is not comprehensive.

Murabaha (cost-plus sale) or ***Bai Muajjal*** (deferred payment sale). The bank buys an asset from a supplier and sells it to the customer at a higher price. The customer is made aware of the cost price and agrees to pay the higher price in deferred payments. Common usage includes trade finance, vehicle finance and term finance. *Murabaha* were issued by SAMA to the Shariah-compliant banks as an alternative to SAMA Bills and when debt financing resumed in 2015, a Shariah-compliant alternative to conventional government bonds was used with an asset (typically an industrial commodity like copper) in the tripartite arrangement.

Musharakah (partnership or joint venture). The bank and the customer enter into a business venture together on a common basis. At the end of the project, the profit or loss is shared between them based on an agreed ratio.

Mudarabah (profit sharing). The bank invests capital provided by the customer on a profit sharing basis, but the customer assumes all the capital risk himself.

Istisna* or *Bai Salam (order sale, or trust sale for future delivery). Typically used for infrastructure or housing projects. The bank agrees with the customer to produce the asset which is sold to the customer at an agreed price on completion of the project. The project work is typically carried out by a construction company acting as agent for the bank. Payment to the bank by the customer may be in stages. When the property is complete the customer has the right to refuse it if it has not been built according to his specifications

Qardh* or *Qardh al Hasan (interest free or benevolent loan). The bank gives the customer money for a short period. The customer repays the money plus a service fee.

Wakala Fi Isthithmar (agency for investment). The bank invests the customer's money to generate an indicative return for a fee. If the actual return is below the indicative return, the bank reimburses and also compensates the customer. If the actual return is higher the bank retains the surplus.

Profit-Sharing Investment Accounts (PSIAs). These differ from conventional deposits in being more like non-discretionary wealth management accounts, and are treated by Shariah-compliant banks as off-balance sheet funds under management. In Saudi Arabia, mutual funds are widely used. With 60 percent of Islamic bank funding from PSIAs, the way these accounts are managed has a major impact on the soundness and capital requirements of Islamic banks.

Shariah-compliant finance presents challenges for the central bank, especially exposure to liquidity risk. Within the unified regulatory system, Islamic banks are not treated differently from conventional ones. But because Shariah-compliant products are based on what is approved by their Shariah Boards' interpretations of Islam – and because the scholars differ in their interpretations with one another – the products have proliferated and lack standardization. This makes regulating them much harder – for example, making sure that the banks maintain adequate provisions against loss.

The fragmentation of Islamic products as well as structural issues mean they are less liquid than conventional instruments. This is a concern since, in the event of a banking crisis, it would be harder for a bank to raise money from its Islamic products than from its conventional ones. PSIAs are a big unknown factor because there are no guaranteed returns on them, in contrast to a rate of interest that would be offered on conventional bank deposits. So in a crisis Islamic depositors may rush to pull their money out for fear of losses. The counter-argument is that because the returns are not fixed, depositors would have to share in the bank's losses (a form of bailing-in).

Finally, there is a lack of Islamic hedging instruments that would enable the banks to hedge the risks on their Shariah-compliant exposure. At the moment there is no robust infrastructure – in particular no liquid money markets – to allow the Shariah-compliant banks to manage their risk. High-quality liquid assets that meet Shariah-compliant guidelines are in short supply. As banks around the world are meeting the tough Basel III regulatory standards, Islamic banks face a source of uncertainty that could prove expensive for them. To deal with this problem, Shariah-compliant banks need to go beyond their traditional role and develop asset and risk management capabilities.

8 Developing a System for Bank Failures

The banks face one old challenge and one new one. They continue to be dependent on levels of government spending because these support the rest of the economy. Saudi Arabia is a highly cyclical economy. The banking sector is exposed to the tougher operating environment caused by lower oil prices, restrained government spending and debt issuance, and a slowdown in corporate activity. If the oil price stays low, eventually bad loans will rise and bank profits will be less strong. The new challenge is that the non-interest bearing deposits on which the conventional banks have relied for decades are unlikely to grow and will probably shrink as customers look for a return – especially if that can be provided by a Shariah-compliant product. The banks that prosper in this environment will be the ones that are nimble in developing technology and find new ways to link borrowers and savers.

The single regulator approach has served Saudi Arabia well and there is no prospect of it changing in a major way. Activities will continue to be hived off and the insurance area is an obvious candidate. SAMA is looking at ways to improve its supervision, including amending the way that SAIBOR is set and stopping the use of structured derivative products (which can be used to speculate against the currency). One obvious gap in the system is that SAMA lacks a formal system for dealing with bank failures. In the past SAMA handled bank problems pragmatically, but now there is draft legislation under study to strengthen the resolution and recovery regime for banks. This will focus on a way of netting off banks' claims on each other, debt restructuring and a transparent method of resolving conflicts between banks (which was a criticism of SAMA's handling of the Saad-AHAB affair).

In general, to use a footballing metaphor, SAMA believes in setting the rules of the banking game and enforcing them toughly, but it does not run onto the playing field to reorganize the teams. Requests from banks to merge and combine their activities will continue to get a sympathetic hearing if they are genuinely based on economies of scale and exploiting synergies, but SAMA has no grand plan for the banking system since it believes that market forces work best.

Notes

1. Also the Public Pension Authority (PPA), General Organization for Social Insurance (GOSI) and Saudi Fund for Development (SDF).
2. Others include the Saudi Industrial Development Fund, (SIDF) the Saudi Credit and Savings Bank (SCSB) and the Agricultural Development Fund (ADF)
3. The rapidly growing area of finance companies, also supervised by SAMA, only amounts to 1 percent of banking assets.
4. The 2014 regulations stated that maximum monthly deductions cannot exceed one-third of salary and loan repayment must be made within five years.
5. In Saudi Arabian banking circles 'long term' means three years or more.
6. Short-term loans for less than a year still make up 50 percent of bank lending
7. The narrowest definition compares the banks' Tier 1 equity base (the highest quality of capital consisting of shareholders' equity and disclosed reserves) with their assets, which are weighted according to degrees of risk of loss. Tier 2 capital comprises other forms of reserves and some types of debt. Basel III requires a combined Tier 1 and Tier 2 capital adequacy ratio of at least 10.5 percent (including the 2.5 percent capital conservation buffer in periods of stress).

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CHAPTER 14

SAMA and the Future

1 Modernizing and Reorganizing

SAMA's seventh Governor, Fahad Almubarak (2011–16), employed the management skills he had honed in the financial markets to modernize SAMA. He instituted a rapid program of change – in three areas in particular: recruitment, remuneration and organization.

Traditionally, progress up the pay scale at the central bank came through seniority. Almubarak introduced the private sector practice of rewarding merit. He established new pay scales and grades, abolishing overtime and other special payments, but emphasizing performance-related bonuses. SAMA is now very different to what it was before the financial crisis, and its investment in people and resources has been huge. Evidence of this is that the central bank's running costs and expenses rose by three-fourths between 2011 and 2015 (from $315 million to $550 million). In the key area of managing the foreign exchange reserves, Almubarak put Ayman Al-Sayari, nephew of the former Governor Hamad Al-Sayari, in charge, with the title of Deputy Governor for Investment. The Investment Department became a Deputyship with a new status, while the new pay arrangements made it easier to recruit and retain talent.

A. Banafe, R. Macleod, *The Saudi Arabian Monetary Agency, 1952–2016*, Financial Institutions, Reforms, and Policies in Muslim Countries, DOI 10.1007/978-3-319-55218-7_14

Again, a lot changed in the recruiting base, because Almubarak brought in outsiders from the private sector to fill key positions on a contract basis. He broke through the gender barrier by hiring Saudi women, and at the end of 2016 there were around 130 female staff at the central bank. Almubarak's insistence on getting the right person for the job meant that it took him a year to fill the post of number two at the central bank after the departure of Al-Hamidy. Finally, Abdulaziz Alfuraih, a senior executive at Riyad Bank, was appointed SAMA Vice Governor in 2014. Alfuraih, a distinguished scholar in the field of accounting, had an insider's view of the strategic challenges facing the Saudi financial sector.

As well as bringing in new blood, Almubarak engaged with the private sector in other ways. Some of his views were shaped while he served on the Shura Council during the battle against inflation, when he formed the opinion that SAMA needed to open up more to the outside world. He believed that it was not enough merely to learn from the experiences of others, but that a central bank also had a responsibility to explain what it was doing. So the Institute of Banking was rebadged as the Institute of Finance (IOF) and included other disciplines such as insurance. This hosts workshops in Riyadh with outside participants from the International Monetary Fund (IMF), the academic world and the financial sector.

At the same time as SAMA's resources were being upgraded, the central bank continued to acquire new responsibilities. In particular, its regulatory role for consumer protection in the financial services industry was expanded. This meant that SAMA had to engage with the millions of people who used the banks and other financial institutions, explaining their rights to them and handling their complaints. The stage was set for a major organizational change.

In 2013, Almubarak, who had used external consultants to analyze all aspects of SAMA's activities, carried out a flattening of the structure of the central bank. He decided to organize it around three functions: governance, procedural and general support.

- **Governance** related to SAMA's policy leadership in establishing strategy, resolving crises and working with international agencies and setting standards of operational efficiency.
- **Procedural** covered the traditional central bank functions of making sure that monetary and financial stability was maintained, and serving

the rest of the government. Achieving monetary stability included setting and implementing monetary policy, and reserve and currency management. The financial stability aspect involved licensing and regulating the financial sector, while the task of serving the rest of the government covered debt management, providing investment services to other bodies and evaluating the banking services they received.
- **Support** included the new job of protecting the consumer as well as educating and training employees in the financial sector and sharing information with the public.

The global financial crisis had already made it clear that SAMA needed to revise its activities in the areas of risk management, governance at the commercial banks and macro-prudential supervision. The decentralized structure established in 2013 has five deputy governors responsible for investments, supervision, banking operations, research and international affairs and administration. New departments include consumer protection, risk and compliance and investment performance and risk control (Fig. 14.1).

SAMA's 50th Annual Report for the year 2014 celebrated the growth in the central bank's responsibilities since the first report came out under Anwar Ali. When it was set up, SAMA's job was to act as the government's bank and to stabilize the domestic and external value of the riyal. In its early years, it moved to a paper currency, introduced limited banking supervision and conducted a simple form of monetary policy. Now it supervises the finance and insurance sectors, exchange dealers, looks after credit information companies including the Saudi Credit Bureau (SIMAH), advises on managing the public debt[1] and is responsible for protecting the financial consumer.

SAMA works with the Capital Market Authority (CMA) and the Tadawul (Stock Exchange) at home, while abroad it works with a range of international institutions including the IMF and the BIS. From Anwar Ali's first slim report of 22 pages, the annual report and the revamped and comprehensive SAMA website have become the best source of financial and economic information about the country.

In the coming years, SAMA's operations will continue against the background of potentially ground-breaking changes in the structure of the kingdom's economy and financial system. In May 2016, the government launched an initiative aimed at eventually ending the kingdom's

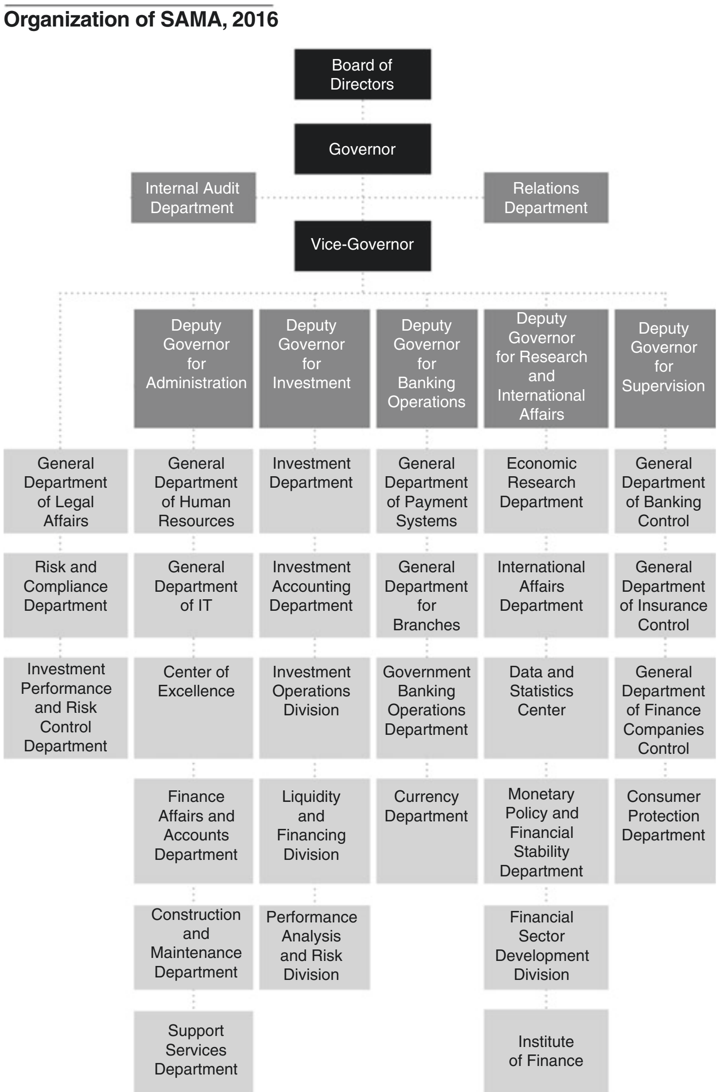

Fig. 14.1 Organization of SAMA, 2016

Source: SAMA

dependence on oil revenue. 'Vision 2030' involves, among other things, selling part of the state energy firm, Aramco. The proceeds are set to go to the Public Investment Fund (PIF) which could turn into one of the world's largest sovereign wealth funds with several trillion dollars in assets. At the same time, subsidies on fuel and water will gradually be lifted. The Saudi government plans to raise non-oil revenue from the $43 billion collected in 2015 to $267 billion by 2030, with the aim of becoming one of the 15 largest economies in the world by that date. The degree to which 'Vision 2030' will require changes in SAMA's traditional roles remains to be seen.

The backdrop to 'Vision 2030' is that Saudi Arabia is confronted today by three major financial challenges: the implications of the shift of its oil sales toward Asia; the impact of substantial population growth, which will inevitably place further strain on the nation's budget; and the question of how to get a grip on state spending.

2 Tilting Toward Asia

A tilt to Asia in the years ahead is inevitable since the region consumes the bulk of Saudi Arabia's oil. This challenge is probably manageable without much difficulty. The important thing is that Asia is likely to want more oil every year. It is easy to focus on the downside of being primarily an oil exporter. The kingdom's economy is highly exposed to the oil price, and the oil price is subject to myriad shocks, a recent one being a supply shock in the form of the entry of the United States to the market as a supplier of shale oil. All of these weigh down on the oil price. Demand shocks include global recessions like the one triggered by the financial crisis in the advanced economies. These too push down prices. And finally there is geopolitics in the Middle East, which tends to push prices up.

But even with the unpredictability of the oil price, the calm and secure possession of lots of oil to sell still looks like a good problem to have. The world's consumption of energy rose 40 percent in the period 2000–15 and demand for oil rises almost every year. Although oil's share of overall energy consumption has fallen, making room for renewables and for a big increase in coal, real demand growth since the financial crisis is around 1 percent annually. This is due to the growth of emerging markets (Fig. 14.2).

While Western Europe and North America combined used 65 percent of the world's oil output in the1960s, their share had shrunk to less than 40 percent by 2015, as Fig. 14.3 shows. China's consumption share at 13 percent is second only to that of the United States.

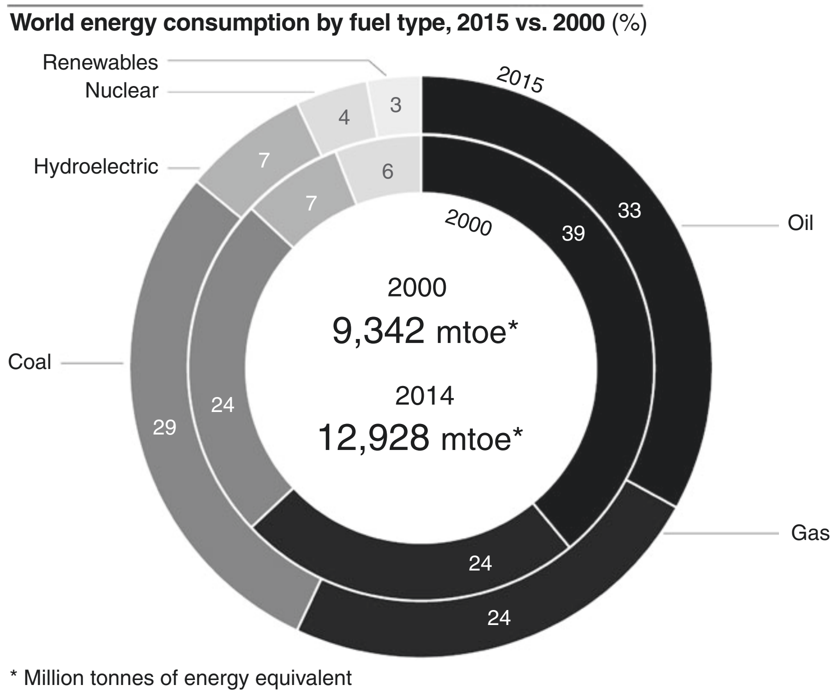

Fig. 14.2 World energy consumption by fuel type, 2015 vs. 2000

Source: BP Statistical Review 2016

Exports to Asia already account for over 60 percent of Saudi Arabia's sales of oil, a figure that is rising by about 1 percent a year. The challenge is how to reflect this by making a gradual shift in orientation away from the relatively declining Western powers, especially the United States, and toward Asia, especially China. For SAMA, this will involve increasing the investment focus on emerging markets and managing the changes to the international monetary system, in particular the status of the renminbi. Although the country's interests continue to be well served by the central role of the dollar, investment returns are what have always driven SAMA's orientation, and China seems likely to provide higher – if more volatile – returns than Europe and North America.

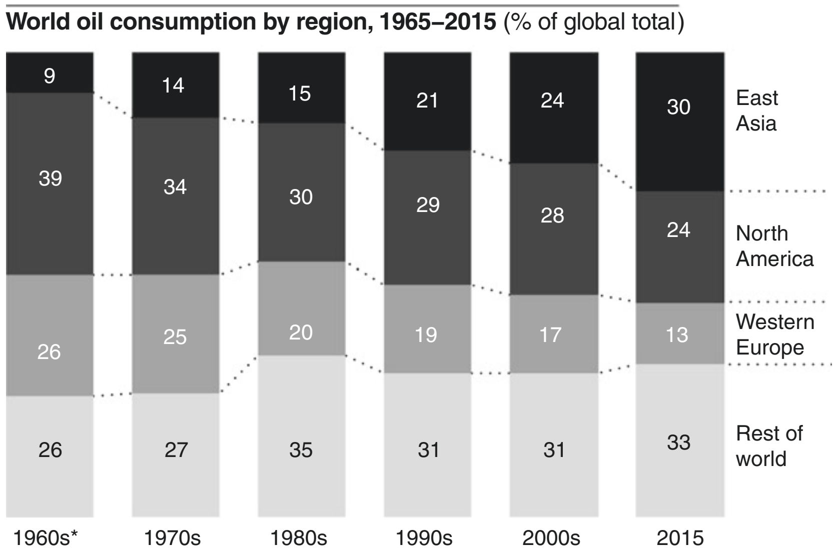

Fig. 14.3 World oil consumption by region, 1965–2015

Source: BP Statistical Review 2016

Note: 'Rest of world' includes Japan. *= 1965–1969

3 The Challenge of Population

It is likely that anxieties about the future of oil are being overplayed, and that reorientation rather than crisis management is probably the correct response. The problem of population growth in the kingdom is, however, inescapable. Saudi citizens have done well out of oil income. In inflation-adjusted terms, Gross Domestic Product (GDP) per head (including foreign residents) has risen from about $4,600 in 1970 to about $21,000 today.[2] This nearly fivefold increase occurred in the context of a quadrupling of the population of Saudi nationals, from 5 million to around 20 million. No other large country (i.e., with a population of at least 20 million) comes close to posting Saudi Arabia's combination of population growth and per-capita output gains. But how much longer can it go on for?

The rise in the 'break-even' oil price for Saudi Arabia, that is the price at which budget revenues meet spending, is being driven by the growth in the population. In 2016, the local population was estimated at 20.1 million, with foreign residents increasing that number by a further 11.7 million. Recent SAMA estimates are that the local population will reach 40 million by 2050. Doubling the population in a generation sounds apocalyptic, but it actually represents a big slowdown in growth. The average Saudi female had seven children in 1950 when Ibn Saud was on the throne, a figure that has dropped to three today. If the country follows the demographic path of other middle-income nations as they move up the prosperity ladder, the population will eventually be stable or falling by mid-century. But population forecasts are inevitably tentative and this is anyway a long way off.[3]

To maintain static living standards for a population that is growing by over 2 percent each year, the country will either have to pump a lot more oil for export, or the oil price will have to rise by a corresponding 2 percent each year, and by far more if the population is to be better off than today.[4] This is a massive gamble because Saudi Arabia cannot control the oil price. If population growth slows unexpectedly in the big oil importing countries, or if energy efficiency and use of renewable sources grow, the picture will be even gloomier. This is the logic of 'Vision 2030'. The economy must reduce its dependence on oil exports.

A particular problem is the diversion of oil into the domestic market where it has been used extravagantly. Just as an obvious answer to this is to carry on increasing local energy prices, an answer to the population question is to replace the expatriate workers. But neither answer is easy to implement. Diversification projects have been under-pinned by promises of guaranteed cheap energy. The country has been trying to wean itself away from dependence on foreign labor for many decades. The problems are well known: foreigners remit their earnings home, draining foreign exchange from SAMA; they do not invest in the local economy and they occupy jobs and run businesses which could be done in theory by young Saudi men and women. If Saudization of the labor force, one of the goals of 'Vision 2030', could be achieved, this would radically improve the budget at a stroke.

'Vision 2030' moots the idea of an American-style Green Card scheme to attract higher quality expatriate workers, who would invest money in the country rather than send it home. The current attempt to encourage Saudization is known as *Nitaqat* (meaning 'ranges'). Private companies are classified in a hierarchy of Premium, Green, Yellow and Red. The idea is to encourage firms to move up the ladder by making it very difficult for them to recruit new foreign workers unless they also increase the percentage of Saudis in their workforce. The problem is that Saudis are often unable or unwilling to carry out what they see as menial or service jobs. But over time, *Nitaqat* and a Green Card system should have some effect.

4 Should There Be a Fiscal Rule for Saudi Arabia?

Saudi Arabia has in the past managed the unpredictability of the price of oil (the 'roulette wheel') not by having hard and fast rules on public spending but rather by a flexible approach to the budget. When the budget is announced at the start of every year, it is understood that it will be adjusted if there is a substantial change in oil income. But there is often a lag of a year or more before a change in oil price is reflected in public spending, as happened after 2014.

Saudi Arabia's fiscal stance is countercyclical in the sense that the government runs surpluses in good times and deficits in bad times, but there is also a countervailing force because spending tends to over-run more in good times than it is cut back in bad times. Particularly in the first years of the new millennium when the good times lasted for a number of years, there was a tendency to increase government spending too fast rather than build up the reserves. In retrospect, the big budget surpluses should have been even bigger so that SAMA's reserves could have grown faster.

The difficulty that Saudi Arabia faces today is that the price of oil must keep rising in order to meet the demands of its growing population. In other words, the 'break-even' price must go up. But there is no reason to suppose that the oil price will actually rise in line with the country's needs. If it does not, then the buffer of SAMA's reserves will be drawn down to meet the demand for imports, but these reserves will not last forever. There is no escaping the fact that, unless the oil price rises to new highs, the lifestyles of ordinary Saudis will have to change in the near future to reduce foreign exchange outflows – either through a change in the exchange rate regime which will bring about inflation in imported products and erode living standards that way, or directly by fiscal tightening, or a combination. This is not necessarily bad for the quality of life of Saudis, if part of the answer is to replace the expatriate workforce with local labor, particularly women, many of whom are economically inactive and can hopefully enjoy a fulfilling career in the workplace. It would be even better if the economy could improve its productivity record and diversify its exports.

The intention to raise local revenues (in particular, to have a sales tax) is good and necessary. But the gap between non-oil revenue and government spending has grown sharply in the last decade, as shown in Figure 14.4, and increases in local taxes will impact adversely on living standards.[5]

Fiscal tightening is essential to reduce the impact of the '90 percent solution' where spending not financed by local taxes flows out as foreign exchange. Raising the price of fuel in particular would have two positive effects: it would increase government revenue and by reducing consumption it would boost oil exports. In December 2015, gasoline prices at the pump were lifted. Nevertheless, fuel remains cheap and is used wastefully. According to BP's annual energy review, Saudi Arabia consumes about the same amount of oil as the whole of the African continent. That is a high number, even accepting that a lot of energy is used in downstream petrochemical activities.

This is the background to the discussion about whether to apply a fiscal rule to limit the government's flexibility on spending. The idea is that when oil prices are strong, less will be spent and more of the money will go into foreign reserves to support spending when oil prices are weak. Commodity exporters – such as Chile and Norway – have a fiscal

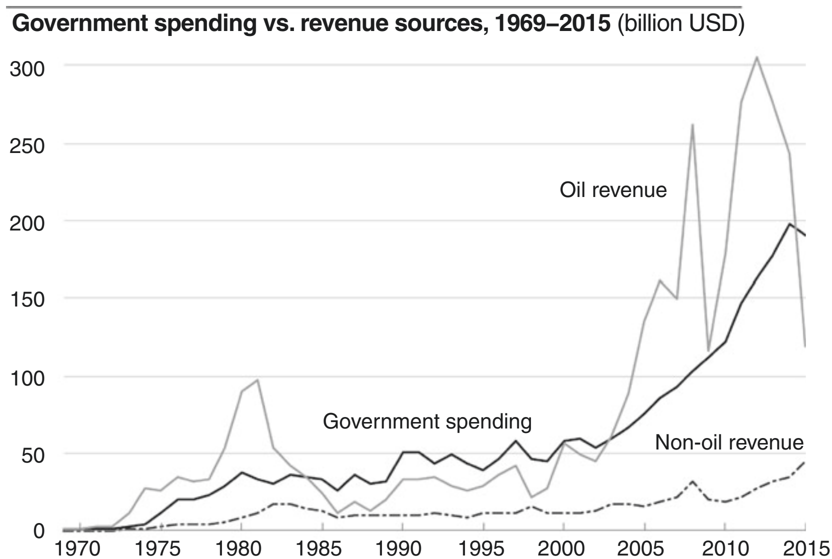

Fig. 14.4 Government spending vs. revenue sources, 1969–2015

Source: SAMA Annual Statistics

rule, a mechanical device that decouples current spending from the vagaries of commodity prices. But the instruments both countries employ – to forecast commodity prices or investment returns – are not reliable due to the random walk effect on commodity and market returns. The IMF has suggested a third approach to Saudi Arabia, which is that rather than attempting to forecast future oil prices or the rate of return on foreign investments, it should simply limit the extent to which spending can rise or fall versus the previous year. Applied to oil income, which is the overwhelmingly important element, the Fund proposed that instead of working on an annual projection of oil prices and export volumes as at present, the Finance Ministry uses moving averages based on historical data. For the oil price it would be the average of the previous five years, and for export volumes the average of the previous three years. Once again the random walk applies, for the use of historical oil prices as a predictor is flawed. Export volumes are much more stable.

It is easy to criticize any fiscal rule for commodity exporters by referring to the principle of no duration dependence for commodities and investment returns. The question is whether it is better than not having one at all. On balance, having a fiscal rule is better because it promotes certainty in planning – even if it does not always hold up. The advantages are that it strengthens the hand of the Finance Ministry in negotiations with the spending departments because it provides a spending baseline by relating the current year's expenditure fairly closely to the previous year's. Also – coupled with a medium-term spending framework – it means that capital projects can be pursued systematically. The temptation in cutting spending is always to rein in long-term projects and keep up current spending, such as civil service salaries and social benefits. Implementing a fiscal rule would likely accompany cuts in spending, which would preserve SAMA's foreign exchange reserves for longer, reducing the pressure on the exchange rate peg.

This analysis of fiscal rules has been kept simple. In practice, there will always be get-out clauses, especially if an economic downturn is unexpectedly sharp, resulting in strong pressure not to cut spending. Realistically, the rule will always be interpreted with some flexibility. But it is better to have a rule tying the course of future spending to the past – even if it can be bent – rather than to continue with the current situation where, largely owing to demographics, there is an impetus to raise spending year on year.

5 The Watchwords of Pragmatism, Stability and Continuity

It is appropriate to end this work with some short general comments. What became the kingdom of Saudi Arabia was founded in 1902 by Ibn Saud, the greatest figure to emerge in Arabia for over a thousand years. King Salman bin Abdulaziz Al Saud is his son, and the seventh king. All transfers of power have taken place smoothly within the royal family, and the next generation has been lined up to take over.

Saudi Arabia is known for its cultural cohesiveness. The royal family commands domestic support on the twin bases of respect for monarchy and for Islam as it protects the Holy Places. The country has been extremely fortunate to have been blessed with mineral resources, and the dividends from oil earnings are distributed widely through economic development initiatives and social benefits. Loyalty and competence in key positions are valued and rewarded. The challenges caused by oil as well as its blessings are well understood throughout the government and they are starting to be addressed.

SAMA is one of the most powerful institutions in the kingdom, dating back over 60 years. Its authority is set by laws promulgated by Royal Decree, issued by the King who heads the Council of Ministers. The critical tasks of the central bank are to stabilize the internal and external value of the currency and maintain financial stability. SAMA has wide discretion on how to act. The laws governing SAMA can be changed, but it is rarely done.

In many ways, thinking about Saudi Arabia as a giant privately owned business faced with the problem of unpredictable demand for its main product, and comparing the way it operates with other businesses, gives a more useful framework than lumping it together with other commodity exporting nations, especially in the developing world. These countries often have bitter colonial legacies, dysfunctional political ideologies, can be plagued by coups and civil wars, and at best often have unpredictable policy changes. The watchwords of Saudi Arabia are ones any businessman would applaud – pragmatism, stability and continuity. Under Governor Ahmed Alkholifey, SAMA will do what it has done since its creation, and throughout a range of regional crises that have thrown neighboring countries into turmoil. It will bring these qualities to bear on the challenges of the future, whether they be a change in the currency regime, supporting economic diversification and reducing dependency on oil, or other challenges as yet unknown.

Notes

1. The Debt Management Office in the Finance Ministry works very closely with SAMA.
2. Measured in 2010 US dollars.
3. Hussein Abusaaq, 'Population Aging in Saudi Arabia,' SAMA Working Paper WP/15/2 (2015): 7.
4. Assuming that diversification does not make a lot of difference over the next few years.
5. The ratio that really matters is not non-oil revenues against overall revenues, but local taxes and charges versus spending (i.e., deducting investment income from non-oil revenues, and investment income has risen sharply in recent years). Local revenues defined in this narrower way were 15 percent of spending in 2016. There is a long way to go.

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Appendix: Inside SAMA

SAMA has been fashioned by the vision of its eight governors. As one or the other of the authors has known personally every governor since the third, Anwar Ali (who took office only six years after SAMA was founded),[1] we thought it would be interesting to set out some account of what each man brought to the job, our view of their major achievements and in particular what they were like as human beings.

Anwar Ali (1958–74) was a roving spirit, born a British subject in colonial India and later becoming a Pakistani national before moving to the US. He arrived in Jeddah on a temporary assignment from the IMF and fell in love with the country. He effectively re-founded SAMA and championed its independence. As early as 1962, he explained in SAMA's first Annual Report the mechanism by which the non-oil economy depended on oil income through the multiplier process. Ali established the system of financial supervision that endures to this day. He was a hard worker, and his only relaxation was playing the card game of bridge which requires a strong capacity for mental arithmetic and calculating odds. Ali died in office in 1974, as did his mentor, King Faysal, the following year. Ali relied on his secretary to translate Arabic documents into English, and SAMA's Investment Deputyship still uses English as well as Arabic as an operational language.

When Ali died of a heart attack while visiting Washington DC, the US Embassy in Jeddah sent Secretary of State Kissinger a fine epitaph:

A. Banafe, R. Macleod, *The Saudi Arabian Monetary Agency, 1952–2016*, Financial Institutions, Reforms, and Policies in Muslim Countries, DOI 10.1007/978-3-319-55218-7

> [Anwar Ali's] long tenure was based on two facts: his close personal relationship with King Faisal and his ability to balance pressures from the Saudi royal family in a period of massive growth of Saudi financial resources against the need to modernize the kingdom's social and other services. He was a world-class financier, highly conservative in his approach to placement of Saudi surpluses which were growing at an unprecedented rate when he died . . . The US particularly and the Western world in general owe a great debt to Anwar Ali for his refusal to switch Saudi Arabian financial resources from one market to another for political or other reasons without due regard for monetary effects on other nations.[2]

Anwar Ali's successor, the first Saudi to fill the top job, was Abdulaziz Al-Quraishi (1974–83). He was born in Al-Ahsa in the Eastern Province, where his father worked for the Ministry of Finance, and took an MBA in the US. Before he came to the central bank, he had a number of government roles (including that of minister of state) and gained a reputation as a problem solver. He ran the Saudi railroad system and Dammam port, and then was appointed President of the Civil Service Bureau. In person, he was a skilled manager of people and treated his staff in a friendly way.

Al-Quraishi certainly had enough problems to solve. His time in office spanned both oil shocks and saw a huge rise in the size of the foreign exchange reserves. He implemented within a few weeks Ali's plan to bring in a team of Western advisers and pushed ahead with buying US Treasuries direct, followed by similar deals with the Japanese and Germans. He also began investing in equities worldwide. Critically, in 1981, he extended the maturity of bond investments at a time when interest rates were at a post-war high and he diversified out of the dollar. These were bold and creative decisions that were to generate big profits in the decade that followed.

His job went beyond dealing with crises and he was committed to empowering Saudis to take charge of their economy. He pushed hard for the Saudization policy that handed the management of the banks over to local investors. His personnel skills showed when he built a housing compound ahead of SAMA's relocation to Riyadh, which meant that key staff became good neighbors and family friends as well as colleagues. SAMA's culture today owes him a lot. Al-Quraishi stepped down in 1983 to join his family business, Al-Quraishi Brothers.

Hamad Al-Sayari (1983–2009) hails from Dhurma, about fifty miles from Riyadh, and is SAMA's longest-serving governor, spanning the reigns of King Fahd and King Abdullah. He had a charming manner and a habit of listening carefully to what was being said, preferring to act cautiously and exert influence behind the scenes. He had an opinion about everything, but was a man of few words and when he spoke, he did so without affectation, simply and directly. At such moments, watching him as he twirled his reading glasses in his hand, his audience would recall that the fifth Governor had begun his career as a teacher.

Al-Sayari took a master's in economics from the University of Maryland. Before specializing in finance and economics, he gained a degree in Sharia. He shifted from a teaching job to work with the quasi-government organizations that invested public money. Following Al-Quraishi's resignation in 1983, Al-Sayari moved up from Vice Governor to the top job, as Acting Governor until his appointment as SAMA Governor in September 1985. In the 1980s, oil had slumped in price, the government was running budget deficits and the foreign reserves were falling fast. Al-Sayari pegged the riyal firmly to the dollar – a peg that has endured for three decades – and handled the decline in assets adroitly.

Undoubtedly the biggest crisis that Al-Sayari had to tackle during his term in office was the Iraqi invasion of Kuwait in 1990. It is testimony to the caliber of the fifth Governor that his swift and effective response to this regional catastrophe led to what was certainly SAMA's outstanding international achievement. After Al-Sayari had restored confidence in the wake of the original panic in Riyadh, he turned SAMA into an effective substitute for the Central Bank of Kuwait (then in exile) by organizing his fellow governors in the Gulf to defend the Kuwaiti dinar exchange rate and provide whatever liquidity was needed. This violent episode was followed by the long grind of the 1990s when Al-Sayari was responsible for handling both a foreign assets rundown and fast-rising government debt. He stuck firmly to the principle that the bond market should be voluntary, insisting that SAMA should not coerce the banks into buying bonds.

When oil prices finally began to turn, a new and more complex set of problems emerged for Al-Sayari – those stemming from Saudi Arabia's growing involvement in the global financial system. The 9/11 terrorist attacks against US targets in 2001, the stock market crash in 2006, inflation and the start of the global financial crisis all occurred on his watch. He was also the first head of SAMA to grapple with the relatively new phenomenon of critical public opinion. Al-Sayari tried to manage the bubble in stocks by limiting consumer loans; but the market collapse triggered a blame game.

He found himself in a similar situation with inflation due to higher oil and food prices, but he stood firmly against popular demand to revalue the riyal and fought a long defensive battle which kept the peg in place.

Managing matters is less challenging in affluent times, but in a difficult economic climate, decision makers' skills are truly tested. Al-Sayari stood up to that test. By the end of his term, his skill at crisis management and his pragmatism in quietly transforming Saudi Arabia's financial infrastructure had made him an internationally respected figure – leading to him being awarded the title of central banker of the year for 2008 by *The Banker* magazine.

Al-Sayari's successor, Muhammad Al-Jasser (2009–11), a stickler for punctuality, had a commanding demeanor and was active and decisive in everything he did. A fast walker, his staff found difficulty in keeping up with the pace he set. In discussions, Al-Jasser was a sharp talker, who, while he relished dominating the debate, was nevertheless delighted when a good point was made by somebody else. He was stern but could be utterly charming when he chose and enjoyed the complete loyalty of those with whom he worked closely.

Al-Jasser hails from the middle of the kingdom, Bureydah, a desert oasis 200 miles north of Riyadh. He and his successor Almubarak were the first governors to see as children how the oil wealth had helped to transform their country into a modern state. Al-Jasser's early career was peripatetic. After completing his bachelors and master's program in the USA, he worked at the Saudi Ministry of Finance (1981–83) and then went to the US to get a PhD in economics (1983–86). After only two years back at the finance ministry in Riyadh, he returned to Washington DC to become ultimately an executive director at the IMF. Finally, at 40, he returned permanently and became Vice Governor of SAMA (1995–2009). Al-Jasser's other achievements in those years included working on the Gas Initiative (1997–98) and serving as a member of the Saudi negotiating team when the country entered the WTO.

Within four months of taking office in the middle of the global financial crisis, Al-Jasser had to deal with the biggest corporate failure in the country's history when the Saad-AHAB groups failed. He moved swiftly to contain the problem, making sure the banks had no other problem loans. By 2010, credit growth had resumed and no Saudi banks had collapsed. Al-Jasser also worked hard on implementing Gulf Monetary Union, but he was probably well aware that politics would make it hard to achieve a common Gulf currency.

Al-Jasser made a powerful contribution to the debate about what had gone wrong with the world's financial system. He was not afraid to be blunt and controversial, and to explain to the media how the Saudi economy really worked. He vehemently defended Saudi Arabia's macroeconomic policy framework, arguing strongly about the benefits of countercyclicality to economic growth and the cost of the text book approach for exchange rate management (in short, why switch certainty for uncertainty?). Al-Jasser had a good rapport with Al-Sayari when they were at the helm of SAMA affairs, and this lent credibility to the central bank's decision making on various policy and operational issues. Al-Jasser was in favor of transparency where appropriate, agreeing to provide the IMF with gross figures for SAMA's foreign exchange reserves, as they were counted toward quota allocation. In 2010, Al-Jasser was honored with the ABANA (Arab Bankers Association in North America) Achievement Award. At the end of 2011, after one thousand days in office and after having piloted the banking system through the greatest world financial crisis in seventy years, he was promoted to become Minister of Economy and Planning.

Fahad Almubarak, the next governor (2011–16), had an unconventional background. Unlike his six predecessors, he was neither an economist nor a public servant by training, but earned his doctorate in business administration in the United States. He comes from the Eastern Province and rose to wealth and prominence in the world of Saudi finance. After founding his own investment firm, he merged it into Morgan Stanley Saudi Arabia, holding the positions of chairman and managing director. He later moved on to chair the Saudi Stock Exchange (Tadawul) and served for six years on the Shura Council.

Almubarak's appointment reflected the government's wish to infuse private sector discipline and culture into the public-sector bureaucracy and he focused on mentoring and training SAMA's staff. Almubarak often said his aim was to move SAMA from good to great, influenced by Jim Collins' book of the same name. The seventh governor's style was to encourage staff to interact across the central bank rather than work in separate silos. His aims included the achievement of good time management and staff cohesiveness. Almubarak encouraged emailing for ease and speed rather than sticking to an old-fashioned paper trail for approvals, emphasizing internal meetings and presentations with wider participation. A whirlwind of change after his appointment saw a redesign of SAMA's organizational structure, overall strategy, governance, risk management mechanisms and information systems, along with a complete refining of the investment procedures and processes. Almubarak was a hard worker who put in long hours and had little time for the relaxed habits of some old-timers. He liked to relax by going on long walks.

The current Governor, Ahmed Abdulkarim Alkholifey, is the eighth to head the central bank. He succeeded Almubarak in May 2016 in a round of senior changes in the government (which included the appointment of Muhammad Al-Jasser as an adviser to the General Secretariat of the Cabinet). In contrast to his predecessor Alkholifey is a long-time central banker who has served in a variety of positions at SAMA since 1995. A native of Riyadh, he took a degree in law from King Saud University in 1987 and joined the Ministry of Petroleum and Natural Resources as a legal adviser. He studied in the US, gaining a Master's degree in Economics in 1993 and two years later he joined SAMA as an economist. He continued his academic work and was awarded a PhD and MBA in Business Administration and Economics in 2000. By 2002 he had a director-level post at SAMA and was appointed Director General of Research and International Affairs in 2010. The following year Alkholifey was seconded to the IMF as an executive director, and when he returned to SAMA in 2013 he became a Deputy Governor.

Alkholifey served on the technical committee of the Islamic Financial Services Board until 2015, so he is familiar with issues concerning Islamic finance. He has also been on the board of SAGIA for some years. Under him the Research Department started to publish academic papers on subjects such as exchange rate policy, inflation forecasting, the output gap in the economy, the Saudi stock market and challenges to economic diversification. His main priority is to maintain monetary stability and make sure that SAMA fully supports the ideas of 'Vision 2030.'

SAMA Board of Directors 2016

1. Governor Dr. Ahmed Alkholifey, Chairman
2. Vice Governor Abdulaziz Alfuraih, Vice Chairman
3. Hamad Al-Sayari
4. Abdulaziz Al-Athel
5. Khalid Al-Juffali

N.B. At the time of SAMA's creation in 1952, the Minister of Finance was the Chair of SAMA's Board, and his deputy was the Vice Chair. SAMA's charter was amended in December 1957 to emphasize its autonomy, making the Governor and the Vice Governor the Chair and Vice Chair of the Board. At the same time, the Board was assigned the appropriate powers to ensure sound management.

SAMA Senior Management at the end of 2016
Governor: Dr. Ahmed Alkholifey (appointed May 2016)
Vice Governor: Abdulaziz Al-Furaih, (July 2014)
Deputy Governor for Investment: Ayman Alsayari (May 2013)
Deputy Governor for Banking Operations: Hashem Alhekail (May 2013)
Deputy Governor for Administration: Dr. Fahad Aldossari (May 2016)
Deputy Governor for Research & International Affairs: Dr. Fahad Alshathri (July 2016)
Deputy Governor for Supervision: Ahmed Alsheikh (November 2016)

Former Governors
Fahad Almubarak 2011–16
Muhammad Al-Jasser 2009–2011
Hamad Al-Sayari 1983–2009
Abdulaziz Al-Quraishi 1974–1983
Anwar Ali 1958–1974
Ralph Standish 1954–1958
George Blowers 1952–1954

Former Vice Governors
Post vacant 2013–2014
Abdulrahman Al-Hamidy 2009–2013
Muhammad Al-Jasser 1995–2009
Ibrahim Al-Assaf July 1995–Oct 1995
Ahmed Al-Malik 1988–1995
Post vacant 1983–1988
Hamad Al-Sayari 1980–1983
Khalid Al-Gosaibi 1972–1980
Junaid Bajunaid 1963–1972
Abed Sheikh 1958–1963
Ma'touk Hasanain 1954–1958
Ralph Standish Sept 1954–Nov 1954
Rasem Al-Khalidi 1952–1954

Notes

1. The first two Governors were George A. Blowers (1952–54) and Ralph Standish (1954–58).
2. From Jeddah Embassy to Secretary of State: SAMA after Anwar Ali, November 13, 1974. Canonical ID: 1974jidda06631_b. Accessed online at www.wikileaks.org.

Glossary

Active investment management Use of human or computer skills with the aim of out-performing after fees the return on an investment benchmark.

Appreciation A gradual increase in the value of a currency in response to market demand rather than by an official revaluation.

Arbitrage The process of taking advantage of the existence of different prices for the same product (or its substitute) at the same time but in different markets.

Basel Standards Recommendations on banking regulations to improve the regulation, supervision and risk management of financial institutions issued by the Basel Committee on Banking Supervision operating under the auspices of the Bank for International Settlements (BIS). In historical sequence they are known as Basel I, Basel II and Basel III.

Basis Point 1/100 of 1 percent i.e. 100 basis points = 1 percent.

Bear Market A declining market or a period of pessimism when declines in market prices are anticipated.

Benchmark Return In investment management, performance of active and passive fund managers is typically measured against the percentage return on an index of securities over the same period to see whether they are beating it or not. This is the return on the benchmark.

Bid Rate The price at which the quoting party is prepared to purchase a currency or deposit.

Black Market An unlicensed market, normally illegal, in an asset or currency. Black market prices will normally be substantially higher

A. Banafe, R. Macleod, *The Saudi Arabian Monetary Agency, 1952–2016*, Financial Institutions, Reforms, and Policies in Muslim Countries, DOI 10.1007/978-3-319-55218-7

than official prices for the same asset, but while the asset is rationed in supply at the official rate, it is freely available on the black market.

Bond A security which shows the liability of the issuer to pay the bearer or holder. It is usually a negotiable instrument with a fixed interest rate and fixed maturity date which is longer than a year.

Bretton Woods Conference A meeting of representatives of non-Communist countries in Bretton Woods, New Hampshire in 1944. The participants agreed on the characteristics of the international monetary system which prevailed through 1971. This system is generally known as the 'Bretton Woods System' and was a fixed exchange rate system with some capital controls.

Broker A broker brings buyers and sellers together for a fee for this service. Brokers do not take a position as principals themselves, they only arrange for transactions among other parties as agents.

BSDA Bankers Security Deposit Account. A now-defunct non-negotiable instrument issued by SAMA to the banks as a way of mopping up excess liquidity. Now replaced by Central Bank Bills.

Budget Deficit A situation in which the government budget spends more than it receives as revenues in the same timeframe. Also called a fiscal deficit.

Budget Surplus A situation in which the government budget spends less than it receives as revenues in the same timeframe. Also called a fiscal surplus.

Bull Market A rising market or a period of optimism when increases in market prices are anticipated.

Capital Controls The regulation of foreign exchange transactions by a government or central bank to avoid an excessive expansion of the local money supply or depletion of the country's foreign exchange reserves. Such controls are usually imposed when a country has undesirably large capital inflows or outflows. Also known as Exchange Controls.

Central Bank A bank acting on behalf of a country's government with the right to issue the country's currency and with responsibilities which may include the management of the country's money supply, short-term interest rate levels and availability of credit. It may also manage the level of the country's foreign exchange reserves and seek to control the external value of its currency.

Central Bank Bill A negotiable instrument, typically with a maturity of less than a year, issued in domestic currency by the central bank to the banks in the financial system. The aim is to mop up excess liquidity on issuance, and to adjust liquidity by varying the issuance size or allowing the banks to temporarily sell the bills back (see Repurchase Facility). The Saudi version is a SAMA Bill.

Central Bank Swap The practice of central banks using foreign exchange swaps with each other to inject or withdraw liquidity from the domestic money market without creating a net exchange position.

Competitive Devaluation Devaluation in excess of the estimated equilibrium rate to gain price advantages in export markets.

Collateralized Debt Obligation (CDO) A structured financial product that pools together cash flow-generating assets and repackages this asset pool into discrete tranches carrying different risks that can be sold to investors. Synthetic CDOs add derivatives.

Credit Rating A measure of the riskiness of an instrument or a borrowing entity from the point of view of the investor not receiving interest payments and principal in full and on time. The principal rating agencies are Standard and Poor's (S&P), Moody's and Fitch. An S&P rating of AAA is the highest possible rating for a bond or currency, and credit quality declines as the alphabet is gone through.

Credit Risk In lending operations, the likelihood that a borrower will be unwilling or not able to repay the principal or pay the interest on time.

Currency Peg A situation where the central bank holds the foreign exchange value of a currency constant against another currency (such as the dollar). The typical arrangement is a promise by the central bank to buy or sell unlimited amounts of the dollar for the local currency.

Current Coupon (1). In fixed rate securities, a coupon rate which approximates the current yield to maturity level for similar securities. Straight bonds with current coupons have prices close to par (100). (2). In floating rate securities, the rate of interest on the current interest period.

Current Yield The ratio of a coupon on a security to its market price, expressed as a percentage (coupon/price x100).

Debt Service Payments of interest and principal on total borrowings which must be made by the borrower in the period during which its debt is outstanding. A debt service ratio compares these payments to some other number (e.g. exports in the case of a sovereign borrower in foreign currencies, or to GDP in the case of a sovereign borrower in its own currency).

Depreciation A gradual decline in the value of a currency in response to trends in demand and supply rather than by an official devaluation.

Developed Market A developed country, in which investment would be expected to achieve lower returns than in an emerging market but accompanied by lower risk. The United States and United Kingdom are developed markets.

Dollarization The process by which the money supply in a country comes to consist of a mixture of local currency and foreign currency (typically, dollars) because economic participants in the country prefer not to take the risk of devaluation of the local currency. In an extreme situation, the dollars are used exclusively.

Duration Dependence Positive duration dependence is when the likelihood of an event is more likely given the passage of time. Negative duration dependence is when the event is less likely given the passage of time. No duration dependence is when the likelihood of an event is independent of the passage of time. See also Random Walk.

Duration-Neutral Switch or Swap Duration measures the amount by which the price of a bond changes in response to a given move in its yield to maturity (YTM). A duration-neutral switch is where the buyer sells a bond with a given YTM and buys a bond of the same credit risk but with a higher YTM, thus improving his return to maturity.

Emerging Market A developing country, in which investment would be expected to achieve higher returns than in a developed market but be accompanied by greater risk. Saudi Arabia is an emerging market.

Eurodollars Dollars belonging to non-residents of the United States invested in external money markets. Eurodollar settlements are made over a banking account in the US and form part of the US money supply. Eurodollars are not subject to the same regulations as domestic dollars.

External Managers External money managers are third parties to whom an institution such as a central bank out-sources the management of its assets, such as equities and bonds.

Equity Investments Generally refers to the buying and holding of shares or stock on a stock market by individuals and firms in anticipation of income from dividends and capital gains, as the value of the stock rises. Equities are known in the United States as stocks and in the United Kingdom as shares.

Forward Foreign Exchange Transaction A foreign exchange settlement later than spot rate (i.e. at a settlement date beyond the nearest settlement date) at a pre-determined rate, calculated as the spot rate, plus the swap rate based on the interest rate differential between the two currencies.

Government Development Bond (GDB) A coupon-paying bond in local currency issued by the government of Saudi Arabia.

Hedging An arrangement whereby the risk of holding an asset is offset by holding another asset. For instance, an American investor holding a Japanese bond while engaged in a forward foreign exchange transaction by selling the yen forward against the dollar is hedging the currency risk in the yen.

Internal Managers Staff directly employed by an institution such as SAMA to manage its assets such as equities and bonds.

Investment Benchmark A standard against which the performance of an actively managed security portfolio or investment manager can be measured. Generally, broad market and market-segment stock and bond indexes are used for this purpose. Under-performance of a benchmark means the portfolio has returned less in the time period; out-performance means it has returned more than the benchmark.

Leverage The use of financial instruments or borrowed capital, such as margin, to increase the potential return of an investment. In business terms, the amount of debt used to finance a firm's assets. A firm with significantly more debt than equity is considered to be highly leveraged. A bank is typically very highly leveraged compared to other businesses.

Liquidity In a multicurrency-investment portfolio the liquidity of a given foreign currency has to be viewed in terms of exchange liquidity and instrument liquidity. Exchange liquidity depends upon the ease with which a currency can be converted into and out of another major currency. Instrument liquidity depends upon the ease with which a negotiable instrument denominated in that currency can be purchased and sold without noticeably affecting the market rate for that instrument. Both types, exchange liquidity and instrument liquidity, determine the overall liquidity of a given currency in a portfolio. In domestic currency situations, liquidity generally refers to instrument liquidity. The global financial crisis of 2008–09 was notable for the drying up of liquidity across a range of markets.

Loan to Deposit Ratio (LDR) A measure of the ratio of the volume of lending (assets on the balance sheet) by a bank compared to the deposits (liabilities on the balance sheet). An LDR below 100 percent indicates loans are less than deposits. A regulatory tool used by SAMA on the basis that the lower the LDR the less risky are the operations carried out by the bank.

London Interbank Offered Rate (LIBOR) This rate is often used as the basis for pricing Eurodollar loans. The lender and the borrower agree to a markup over LIBOR, and the total of LIBOR plus the market makes the effective interest rate for the loan. Typical LIBOR periods are one month and six months.

Long 'Going long' is the market term for buying a security or currency in anticipation of a rise in price. An example is buying a currency in anticipation of an upward revaluation.

Money Market Instruments The most popular money market instruments are Treasury Bills (or the equivalent in local currency), commercial paper, bankers' acceptances and certificates of deposit.

Offer Rate The price at which a quoting party is prepared to sell or provide a currency or loan. See LIBOR.

Offshore Banking Unit (OBU) A financial institution (typically a bank) located outside the jurisdiction of the central bank in whose currency it is doing business. An offshore market is a market outside this jurisdiction

Open Market Operation Open market operations are conducted by a central bank in its domestic money market to reduce or increase the money supply and hopefully alter liquidity conditions. For instance, the purchase of government bonds will increase the money supply and improve liquidity as the central bank pays cash for the instrument. Conversely, selling government bonds will decrease the money supply and reduce liquidity in the market. Repurchase agreements are typical money market operations.

Optimal Currency Area A geographical region in which it would improve economic efficiency to have the entire region share a single currency.

Passive Investment Management Holding a basket of securities to closely follow the return on a benchmark without using human or computer skills.

Project Finance The financing of long-term infrastructure, industrial projects and public services based upon a non-recourse or limited recourse financial structure, in which project debt and equity used to finance the project are paid back from the cash flow generated by the project. Toll highways are typical examples of projects financed in this way.

Purchasing Power Parity (PPP) Theory A theory which assumes that in the long run exchange rates adjust to reflect the relative inflation rates of the two currencies. PPP can be difficult to estimate when the goods and services consumed are very different in each country.

Random Walk The observation that many asset prices such as oil follow no discernible pattern or trend. See also Duration Dependence.

Repurchase Agreement Commonly known as a Repo. A contract in which the seller of an asset (say, a government bond) agrees to buy it back on a specific date. This technique is frequently used for raising short term liquidity by dealers to finance their positions. Central banks use repos to increase liquidity in the financial system.

Reserve Requirement An amount of money (usually a percentage of deposits) which commercial banks in a country are required to keep on deposit with the central bank. Originally, these requirements were designed to protect the solvency of banks. Today, central banks, especially in emerging markets, adjust requirements principally as a tool to affect the money supply and the liquidity of the banking system, that is, its ability to make loans.

Reverse Repurchase Agreement Also known as a Reverse Repo. A contract in which the buyer of an asset agrees to sell it back on a specific date. Central banks use reverse repos to withdraw liquidity from the financial system.

Riyal The Saudi riyal (SAR) is the currency of Saudi Arabia.

Saudi Arabia Interbank Offered Rate (SAIBOR) The riyal equivalent of LIBOR for the dollar.

Shariah-Compliant A term applied to financial instruments that are deemed not to breach the rules of Shariah, principally a prohibition on paying or receiving interest. Often known as Islamic instruments.

Short 'Going short' is the market term for selling a security or currency which the investor does not hold, typically promising to deliver it at a future time. An example is selling a currency short in anticipation of a devaluation.

Short Covering The purchase by an investor of a security or currency previously sold short in order to deliver them and thereby to close out his short position.

Soft Deposit Money placed by the central bank with a local bank on concessionary terms (generally by a lower interest rate than the market interest rate) in order to help the bank through a difficult period.

Sovereign Risk The risk that the government of a country may interfere with the repayment of a debt. This can happen when a borrower is willing to repay a loan in local currency, but the government may not allow him to repay the loan to a foreign bank because of a lack of foreign exchange or for political reasons. The sovereign government itself may be unwilling to pay a loan taken out in foreign currency. In an extreme situation, there may be a sovereign debt default.

Special Drawing Right (SDR) A composite fiduciary reserve asset (only an entitlement to credit) created in 1970 by the IMF as a supplement to existing reserve assets. The SDR is worked out from a basket of fixed amounts of major trading currencies. The SDR has failed to make headway outside official institutions such as central banks. As at end-2016 the currencies in the SDR basket were the dollar, euro, yen, sterling and the yuan (or renminbi).

Sukuk A Shariah-compliant bond instrument.

Swap Line See Central Bank Swap.

Treasury Bond A term typically applied to a bond issued by the US Treasury as a direct obligation of the government of the United States and known as a US Treasury, or simply as a Treasury. Instruments that are shorter than a year and pay no coupon are Treasury Bills. See Zeros.

Yield Curve A graphical representation of yield to maturity (YTM) for each different maturity of a financial instrument, such as US Treasuries. Time to maturity is represented on the horizontal axis. When longer maturities have higher YTM than shorter maturities, which is the situation most of the time, the curve is called a positive, upward-sloping or normal curve. The opposite type of curve is called a negative, downward-sloping or inverted yield curve. When YTM is the same for all maturities it is called a flat yield curve.

Yield to Maturity (YTM) The return a security earns on the price at which it was purchased if it is held to maturity. This is a function of coupon rate, reinvestment rate and accrual or amortization of premium. The critical assumption is that all coupon payments are reinvested at the same YTM (i.e., the calculation is an unrealistic one).

Zero Coupon Instruments Commonly known as Zeros. Securities issued in the primary market at a discount to their principal amount at final maturity, with no coupons attached and paying no income, so that the return consists entirely of capital gain with no reinvestment risk. Treasury and Central Bank Bills are zeros, normally with a maximum maturity of one year.

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